Statistics and Econometrics with R

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Welcome

This book is the result of several years of teaching introductory statistics and econometrics at Azim Premji University. It was shaped not only by lectures and classroom discussions, but also by countless questions from students who wanted to understand *why* statistical methods work, rather than simply *how* to apply them.

Those questions made me rethink the way statistics is usually introduced.

Many introductory courses place considerable emphasis on mathematical derivations, definitions and problem-solving techniques. These are important, and this book does not seek to diminish their value. Yet, in my experience, many students complete such courses having learned the mechanics of statistical procedures without fully appreciating the logic that makes them work. They know the steps involved in carrying out a hypothesis test or estimating a regression model, but the underlying ideas — the beauty of reasoning from a sample to an entire population — often remain hidden beneath the mathematics.

This book begins from a different premise. Before introducing notation, formulas or proofs, it asks a simple question: **what problem are we trying to solve?** Once that question is clear, intuition follows naturally. The mathematics is then introduced not as something to be memorised, but as a precise language for expressing ideas that the reader has already encountered through examples, simulation and reasoning.

The philosophy of the book can be summarised in five words:

> **Question** $\rightarrow$ **Intuition** $\rightarrow$ **Rigour** $\rightarrow$ **Computation** $\rightarrow$ **Interpretation**

Every chapter follows this progression. We begin with a real question, usually drawn from economics or public policy. We build intuition through examples and simulation. We then develop the mathematical foundations carefully and without unnecessary shortcuts. Finally, every idea is implemented in R using real data so that the transition from theory to empirical analysis becomes seamless.

This integration of intuition, mathematical reasoning and programming reflects the way statistics is practised today. There is no separate chapter devoted to software. Programming is not an optional add-on; it is part of the process of learning statistical thinking. Every table, figure and numerical result in this book is generated directly from R, allowing readers to reproduce every analysis for themselves.

The examples throughout the book are grounded, wherever possible, in the Indian economy. Rather than relying on unfamiliar datasets and examples, we study questions involving wages, education, poverty, agriculture, inflation, employment, health and public policy. My hope is that students will not only learn statistical methods but also develop a deeper understanding of the economic and social issues that shape contemporary India.

This emphasis is particularly important because empirical economics has changed dramatically over the past two decades. Modern applied research increasingly seeks to answer causal questions rather than merely document statistical relationships. Although this book develops the foundations of frequentist statistics, estimation and regression analysis, it does so with this broader objective in mind. The methods introduced here are intended not simply as computational tools, but as the foundations upon which modern empirical economics is built.

This book also draws extensively on the outstanding textbooks that have shaped the teaching of statistics and econometrics over many decades. I have learned enormously from those works, and this book should be viewed as a companion to that tradition rather than a replacement for it. Its contribution lies not in presenting new statistical methods, but in presenting familiar ideas through a pedagogical approach that places intuition, conceptual understanding and empirical application at its centre.

The book is written primarily for undergraduate students in India who are studying quantitative social science — economics, but equally public policy, sociology, political science, development studies and related disciplines.

I want to be explicit about one group of readers in particular. Many students arrive in these programmes without a mathematics training in school, and find statistics and econometrics genuinely difficult as a result. In my experience the difficulty is almost never a lack of ability. It is that most textbooks are written for readers who already have the background, and a student without it is left trying to learn the ideas and the notation simultaneously, from a book that explains only the second.

This book assumes no such background. Every idea is introduced before the symbols for it, mathematical steps are shown rather than left to the reader, and nothing is described as obvious. Motivated readers should be able to work through it independently.

My hope is that readers finish the book not only knowing how to perform statistical analyses in R, but also understanding the logic of frequentist inference and developing the habits of mind needed to think critically about empirical evidence.

I owe a considerable debt to my colleagues in economics at Azim Premji University. Good pedagogy is treated here as a serious professional undertaking rather than an afterthought, and the care that colleagues take — day after day, and largely unremarked — to become better teachers and better practitioners has been invaluable. That environment has done a great deal to bring this project to light.

Finally, this book would never have been written without my students. Every question asked after class, every thoughtful challenge to an explanation, every mistake that forced me to rethink the way I taught, and every conversation about data and economics has left its mark on these pages. I owe a sincere debt of gratitude to all the students who have taken my courses in introductory statistics and econometrics over the years. Your curiosity, patience and willingness to ask difficult questions have continually pushed me to become a better teacher. This book is the result of that shared journey, and I hope it helps future students discover the same excitement for statistics that you brought into my classroom.

How to use this book

Statistics is not a subject that rewards passive reading. Understanding comes from asking questions, working through examples, making mistakes and revisiting ideas. This book is written with that philosophy in mind, and you will get the most out of it if you engage with it actively.

Read the book in order

The first few units are designed as a sequence rather than a collection of independent topics. Ideas introduced early — such as variation, repeated sampling and statistical inference — form the foundation for everything that follows. While later chapters can be used as references, I strongly recommend reading the book from beginning to end the first time.

Work through the mathematics

Do not skip the derivations because they look unfamiliar. Every mathematical result is developed gradually, with intermediate steps shown. Try to work through each derivation yourself before reading ahead. The objective is not to memorise proofs, but to understand where formulas come from and why they work.

Think in words before thinking in mathematics

One piece of advice has stayed with me throughout my academic career. During my PhD, my advisor once told me:

> If you can explain the problem clearly in simple English, you have already solved half of it. The mathematics is just a matter of practice.

Over the years I have found this to be remarkably true, and nowhere more so than in statistics and econometrics.

Students often assume that success in these subjects depends primarily on being good at mathematics. Mathematical reasoning is undoubtedly important, but most difficulties arise much earlier than the algebra. They arise because we lose sight of the question we are trying to answer. Before writing down a formula, ask yourself: what am I trying to learn? What information do I have? Why should this method work at all? If you can answer those questions clearly in words, the mathematics usually becomes far easier to follow.

Throughout this book I encourage you to explain ideas to yourself in plain English before expressing them in symbols. Statistical notation is a powerful language, but it is still a language. Its purpose is to communicate ideas precisely, not to replace logical reasoning.

Whenever you find yourself stuck in a derivation or confused by a formula, pause and return to the underlying question. More often than not, the mathematics becomes much less intimidating once the logic is clear.

Write the code yourself

Every analysis in this book is accompanied by R code. Resist the temptation to copy and paste it. Typing the code yourself forces you to pay attention to every line and every argument. Once the code works, experiment with it. Change a parameter, modify the data, or deliberately introduce an error. Some of the most valuable lessons come from understanding why something fails rather than why it succeeds.

Think before calculating

Whenever a new method is introduced, pause before looking at the computation. Ask yourself what you expect the answer to be, and why. Developing statistical intuition is just as important as obtaining the correct numerical result.

Attempt the exercises first

The exercises are an essential part of the book, not an optional supplement. Attempt every question before consulting the solutions, even if you are unsure how to proceed. Struggling with a problem is often where the deepest learning takes place. The solutions are intended to explain your reasoning, not replace it.

Use the book as a reference

Once you have worked through the material, the book can also serve as a reference. Use the search function at the top of the page to revisit concepts, definitions, examples or R commands. You are not expected to remember every formula or every line of code. What matters is understanding the underlying ideas and knowing where to find them when needed.

Learn by connecting ideas

Throughout the book, try to ask not only how a method works, but why it was introduced in the first place. Every chapter begins with a question and develops the statistical tools needed to answer it. If you understand the question, the mathematics that follows will feel much more natural.

Above all, remember that statistics is a way of thinking rather than a collection of techniques. The goal of this book is not simply to help you perform statistical analyses in R, but to help you reason carefully about evidence, uncertainty and data. If you finish the book asking better questions than when you began, then it will have achieved its purpose.

A note on the data

Every dataset used in this book is either built into R, supplied with the book, or generated directly by code that you can read and run yourself. This means that every table, figure and numerical result can be reproduced by running the accompanying script from beginning to end.

Reproducibility is a fundamental principle of statistical work. If an analysis cannot be recreated from the original data and the accompanying code, its results cannot be independently verified. Good statistical practice therefore means leaving behind a complete record of every step that produced the final result.

> **REMEMBER THIS**
>
> A result that cannot be reproduced by running a script from top to bottom is not yet a result.

One dataset deserves a special mention. The class survey used throughout the book was collected from students enrolled in this course. Before it was included here, all identifying information was removed and each student was assigned an anonymous identification number. Protecting the privacy of participants is one of the first responsibilities of anyone working with data about people.

Later in the book we return to questions of sampling, data collection and research ethics. For now it is enough to remember that every dataset represents real individuals, and good statistical practice begins with treating their information responsibly.

What you need

This book requires two free programs, which should be installed in the following order.

1. **R** — the statistical programming language, available from cran.r-project.org.
2. **RStudio Desktop** — an integrated development environment that provides a more convenient interface for working with R, available from posit.co/download/rstudio-desktop.

Several chapters use additional R packages. These need to be installed only once, by running the following command in the RStudio console:

```r
install.packages(c("tidyverse", "BSDA", "wooldridge", "AER",
                   "stargazer", "sandwich", "lmtest"))
```

No other software is required, and everything used in this book is freely available.

Mathematical background

This book assumes a basic acquaintance with introductory probability and descriptive statistics. Specifically, it will help to have met:

- basic probability rules and notation;

- random variables and their probability distributions;
- probability mass functions, probability density functions and cumulative distribution functions.

Everything else the book relies on is built up within it. Measures of central tendency and dispersion are reviewed in Unit 1 before they are used, and random variables, expected values and variances are developed there from first principles rather than assumed. If those terms are unfamiliar, that is where they are introduced.

The focus throughout is statistical inference and data analysis rather than probability theory itself. Where an idea from probability is needed, it is recalled briefly at the point of use.

A word to readers who are uncertain whether they have enough mathematics for this. Almost everything in the list above is notation and vocabulary rather than technique, and the difficulty of statistics lies elsewhere — in understanding what a question is asking and what the data can answer. If you can follow an argument carefully and are willing to work through the derivations rather than skip them, you have what this book requires. Begin at Unit 1 and consult a probability text only if something there genuinely does not yield.

Licence and citation

If you spot an error, the source of every chapter is on GitHub and corrections are welcome as pull requests.

Corrections from students are genuinely welcome; several of the sharper points in this book exist because someone asked why.

To cite:

> Kulkarni, K. (2026). *Statistics and Econometrics with R.* Azim Premji University. https://kedarkulkarni1991.github.io/stats-econometrics-r/ Licensed under CC BY 4.0.

Chapter 1

Towards Inferential Statistics

You already know how to calculate the average of a sample.

In this unit, you will learn why that average is almost certainly wrong.

At first glance, this seems like a strange claim. After all, there is only one way to calculate a sample mean, and if the arithmetic is correct, the answer must be correct as well. Yet the number you obtain from a sample is almost never exactly equal to the quantity you ultimately wish to know — the mean of the population from which that sample was drawn.

Suppose you wish to estimate the average income of households in India. Since surveying every household is impossible, you select a sample of 2,000 households and calculate its mean income. The calculation is straightforward, but the result immediately raises a question. How close is this sample mean to the true population mean?

The difficulty is that the sample you have drawn is only one of many that could have been observed. Had a different 2,000 households been selected, the sample mean would almost certainly have been different. Draw another sample, and the answer changes again. Every statistic calculated from a sample therefore reflects two things at once: the characteristics of the underlying population, and the randomness introduced by the sampling process.

This simple observation marks the boundary between descriptive and inferential statistics.

Until now, our objective has been to describe the data that we observe. Measures of central tendency, such as the mean and the median, summarise where observations are concentrated. Measures of dispersion, such as the variance and standard deviation, describe how widely they are spread. Histograms, bar charts and density plots reveal the overall shape of a distribution and help us identify skewness, outliers and other important features. Together, these tools provide a concise description of a dataset and are the natural starting point of every empirical investigation.

Most empirical research, however, is concerned not with the sample itself but with the population from which it was drawn. Economists survey a few thousand households to understand millions. Polling agencies interview a sample of voters to estimate national voting intentions. Medical researchers evaluate new treatments on a limited number of patients before recommending them to the wider population. In every case, the sample serves as evidence about a population that cannot be observed directly.

The central problem of inferential statistics is therefore straightforward to state: **how can we use the information contained in a sample to learn about the population from which it came?**

Throughout this course we adopt the **frequentist** approach to answering this question. Rather than treating the observed sample as unique, we imagine drawing many samples from the same population and calculating the same statistic repeatedly. Some sample means will be larger than others, some smaller. Understanding how statistics vary across these hypothetical samples allows us to assess how much confidence we should place in the estimate computed from the one sample we actually observe.

Before we can understand why, we need to understand where the uncertainty comes from. The answer lies in randomness.

1.1 Randomness and variation

> *Where does the uncertainty come from?*

The uncertainty that surrounds every statistical estimate has a single source: randomness.

If we were to repeat the same study with a different sample, we would almost certainly obtain a different average. Inferential statistics is concerned with understanding this variability. Without randomness, every sample would be identical, every estimate would equal the population value, and there would be no uncertainty to quantify.

The idea is easiest to see with a die.

Suppose we roll a fair die once.

```r
set.seed(1)
sample(1:6, size = 1)
```

```
#> [1] 1
```

The outcome is a one. Beyond that, there is little to say. A single observation tells us almost nothing about the process that generated it. Was the die fair? Would the next roll also be a one? Is one an unusually common outcome? One observation cannot answer these questions.

Now imagine repeating the experiment ten thousand times.

```
rolls <- sample(1:6, size = 10000, replace = TRUE)

round(prop.table(table(rolls)), 3)
```

```
#> rolls
#>     1     2     3     4     5     6
#> 0.169 0.162 0.160 0.170 0.173 0.166
```

Each individual roll remains unpredictable, but something remarkable happens collectively. The outcomes no longer appear as isolated events. They form a stable pattern. Every face occurs with almost the same frequency, close to one-sixth of the time, exactly as we would expect from a fair die.

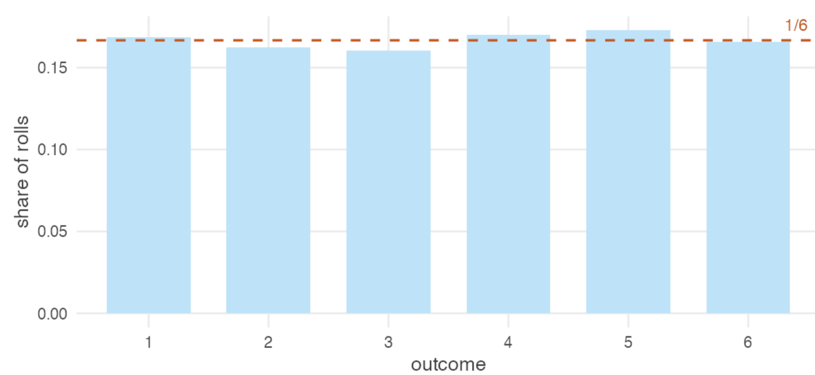

Figure 1.1: Ten thousand rolls of a fair die. No single roll was predictable. The pattern across all of them is.

The important point is not that the die turned out to be fair. It is that a pattern exists at all.

To see why that matters, imagine removing the randomness altogether. Suppose every face of the die is labelled three.

```
loaded <- rep(3, 10000)

table(loaded)
```

```
#> loaded
#>     3
#> 10000
```

```
sd(loaded)
```

```
#> [1] 0
```

Every roll now produces exactly the same outcome. There is no variation, no uncertainty and no pattern waiting to emerge through repetition. The average is always three. The standard deviation is zero. Repeating the experiment a million times teaches us nothing that we did not already know after the first roll.

The contrast between the two dice illustrates a fundamental idea. Randomness makes individual outcomes unpredictable, but repeated observations reveal stable patterns. Statistics is built on this coexistence of uncertainty and regularity. If outcomes never varied, there would be no need for inference; if they varied without any underlying regularity, inference would be impossible.

Dice provide a simple illustration, but the same principle governs every empirical investigation. Household surveys interview different families. Election polls reach different voters. Clinical trials recruit different patients. Agricultural experiments observe different harvests. Every study could have produced a different dataset. The central challenge of inferential statistics is to understand what can be learned from **one realisation of a process that could have produced many others**.

1.2 Describing variation

> *Once variation exists, how do we describe it?*

Variation is what makes statistics necessary. The next question is equally natural: how do we describe it?

A die has only six possible outcomes, so its behaviour is easy to summarise. Real datasets are rather different. Consider the annual earnings of twenty thousand Indian workers.

```
workers <- read.csv("data/wages-india-synthetic.csv")

nrow(workers)
```

```
#> [1] 20000
```

```r
head(workers$annual_earnings)
```

```
#> [1]  17773  60847 115739  34834  30784   2068
```

This dataset is simulated, but it closely resembles the earnings distribution observed in large household surveys such as the India Human Development Survey (IHDS). Throughout this book, we will use it to illustrate statistical ideas and methods.

Twenty thousand numbers are too many for anyone to understand by inspection. We need ways of reducing the data without losing its essential features. The first step is simply to look at it.

```r
ggplot(workers, aes(annual_earnings / 1000)) +
  geom_histogram(bins = 60, fill = book_palette$fill, colour = "white",
                 linewidth = 0.15) +
  scale_x_continuous(labels = scales::comma) +
  labs(x = "Annual earnings (₹ thousand)",
       y = "Number of workers") +
  theme_book()
```

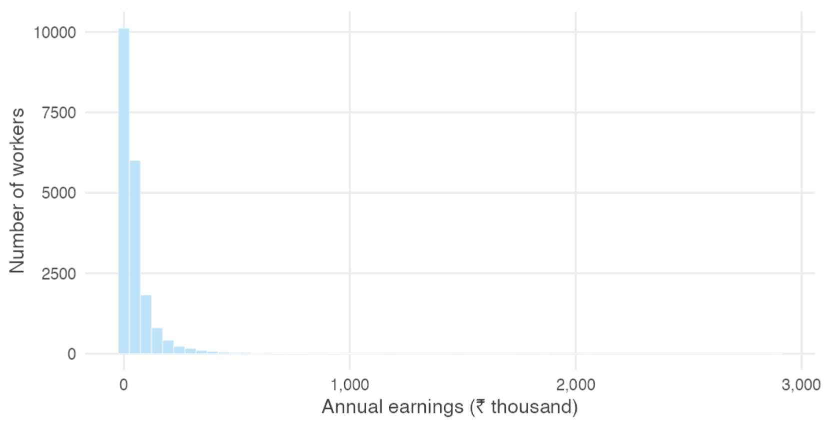

Figure 1.2: Annual earnings of 20,000 Indian workers.

The histogram immediately reveals two important features. Most workers earn relatively modest incomes, while a much smaller number earn substantially more. The distribution is therefore concentrated on the left with a long tail extending to the right — a pattern known as **right skewness**.

One picture already tells us a great deal, but we can describe the same distribution more precisely using a few numerical summaries.

```r
c(mean   = mean(workers$annual_earnings),
  median = median(workers$annual_earnings))
```

```
#>   mean median
#>  54181  24220
```

The mean annual earnings are more than twice the median. This difference reflects the long right tail visible in the histogram. A relatively small number of high earners pull the mean upwards, whereas the median simply divides the distribution into two equal halves and is much less affected by extreme values.

Variation also concerns how widely observations are dispersed.

```r
c(sd  = sd(workers$annual_earnings),
  iqr = IQR(workers$annual_earnings))
```

```
#>     sd    iqr
#> 100141  46756
```

The **standard deviation** measures the typical distance of observations from the mean, while the **interquartile range** measures the spread of the middle fifty per cent of the distribution. Together they indicate that earnings vary considerably across workers.

Sometimes we are interested not only in the overall distribution but also in how it differs across groups. A density curve provides a smooth representation of the same data and makes such comparisons easier.

```r
ggplot(workers, aes(annual_earnings / 1000, colour = residence)) +
  geom_density(linewidth = 0.8) +
  scale_colour_manual(values = c(Rural = book_palette$ink,
                                 Urban = book_palette$accent),
                      name = NULL) +
  coord_cartesian(xlim = c(0, 400)) +
  labs(x = "Annual earnings (₹ thousand)", y = NULL) +
  theme_book() +
  theme(axis.text.y = element_blank(),
        legend.position = "top")
```

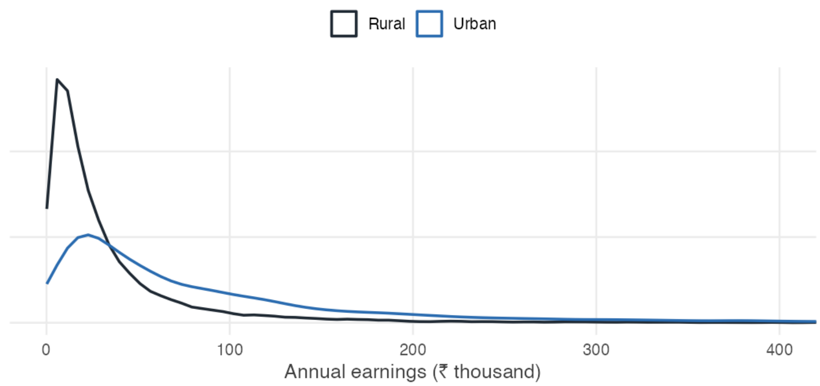

Figure 1.3: Density curves for rural and urban workers.

The entire urban distribution lies to the right of the rural distribution, indicating that urban workers generally earn more than rural workers. The density curve makes this comparison immediately visible without requiring us to examine thousands of individual observations.

By this point, twenty thousand observations have been reduced to a handful of informative summaries: a picture of the distribution, measures of its centre and spread, and a comparison across groups. This is the purpose of **descriptive statistics**. Rather than examining every observation individually, we summarise the data in ways that preserve its most important features.

Descriptive statistics is always the starting point of a statistical analysis. Before asking whether a result is statistically significant or whether it can be generalised beyond the data at hand, we must first understand what the observed data look like. Only then can we ask the central question of this unit: if we had collected a different sample, would these summaries have been different?

1.3 The limits of descriptive statistics

> *How far can description take us?*

Descriptive statistics tells us everything we could want to know about the data we have. The difficulty is that the data we have are rarely the data we ultimately care about.

The twenty thousand workers in our dataset were not selected individually because they were special. They are one sample drawn from a much larger population of Indian workers. Had a different set of workers been surveyed, the dataset would have been different — and so too would the summaries we calculated in the previous section.

To see this, let us treat our dataset as the population and repeatedly draw samples of two thousand workers.

```
set.seed(1)

s1 <- workers$annual_earnings[sample(nrow(workers), 2000)]
s2 <- workers$annual_earnings[sample(nrow(workers), 2000)]
s3 <- workers$annual_earnings[sample(nrow(workers), 2000)]
```

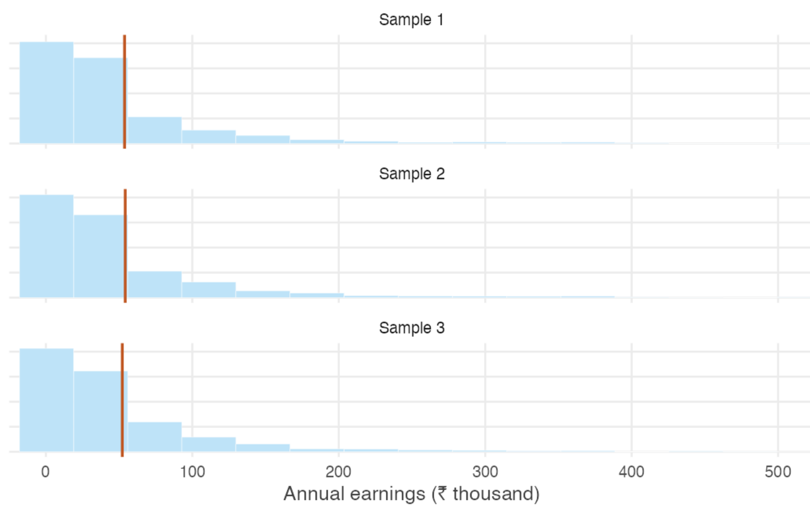

Figure 1.4: Three samples of two thousand workers drawn from the same population.

The three histograms look remarkably similar. Each has the same broad shape: most workers earn relatively little, while a small number earn much more. Nothing suggests that one sample is “better” than another.

Yet the numerical summaries are not identical.

```
c(sample1 = mean(s1),
  sample2 = mean(s2),
  sample3 = mean(s3))
```

```
#> sample1 sample2 sample3
#>   53727   54185   52176
```

Each sample produces a different mean.

The same is true of the median.

```
c(sample1 = median(s1),
  sample2 = median(s2),
  sample3 = median(s3))
```

```
#> sample1 sample2 sample3
#>   24974   25238   24599
```

And of the standard deviation.

```
c(sample1 = sd(s1),
  sample2 = sd(s2),
  sample3 = sd(s3))
```

```
#> sample1 sample2 sample3
#>   99376   94857   86327
```

Every statistic changes because the sample itself has changed. None of the calculations is incorrect, nor has anything gone wrong with the survey. The differences arise simply because different individuals happened to be included.

The exercise above highlights a distinction that lies at the heart of statistical inference. The **population** is the complete collection of units about which we wish to draw conclusions, while the **sample** is the subset that we happen to observe. Because different samples contain different observations, the statistics computed from them also differ. This inherent variability is known as **sampling variation**.

Sampling variation is neither a mistake nor evidence of poor data collection. It is an unavoidable consequence of drawing inferences about a population from a sample. Even when a survey is carefully designed and correctly implemented, a different sample would generally produce different numerical summaries.

This observation also marks the boundary between descriptive and inferential statistics. Descriptive statistics summarises the sample accurately: the sample mean is the mean of the observed data, the sample median is its median, and so on. What descriptive statistics cannot tell us is how closely these sample summaries reflect the corresponding characteristics of the population, or how much they would differ had another sample been drawn.

Inferential statistics addresses precisely these questions. By recognising that the observed sample is only one of many that could have been selected, it provides a framework for assessing the uncertainty associated with sample-based conclusions and for drawing conclusions about the underlying population.

1.4 What if we repeated the survey?

> *If our conclusions depend on the particular sample we happened to observe, how can we learn anything about the population?*

The previous section ended with a simple but unsettling observation. A different sample produces a different mean. The same is true of the median, the standard deviation and every other statistic we compute.

One possibility is to collect another sample.

Imagine repeating the survey. A new team visits a different set of households, following exactly the same sampling design. They compute the average annual earnings and obtain a value that differs slightly from the first survey.

Now imagine repeating the exercise again.

And again.

Every new survey produces a new sample, and every sample produces a new mean. None of these means is "correct" or "incorrect". Each is simply the result of observing a different subset of the population.

Of course, conducting thousands of nationwide surveys would be prohibitively expensive. Fortunately, we do not need to. Since we are treating our dataset as the population for the purposes of this chapter, we can recreate this thought experiment on a computer.

Suppose we repeatedly draw random samples of 2,000 workers from our population. For each sample we calculate the mean annual earnings and store the result. After repeating this process one thousand times, we have one thousand sample means.

```r
set.seed(1)

sample_means <- replicate(
  1000,
  mean(sample(workers$annual_earnings,
              size = 2000,
              replace = TRUE))
)
```

What do these one thousand means look like?

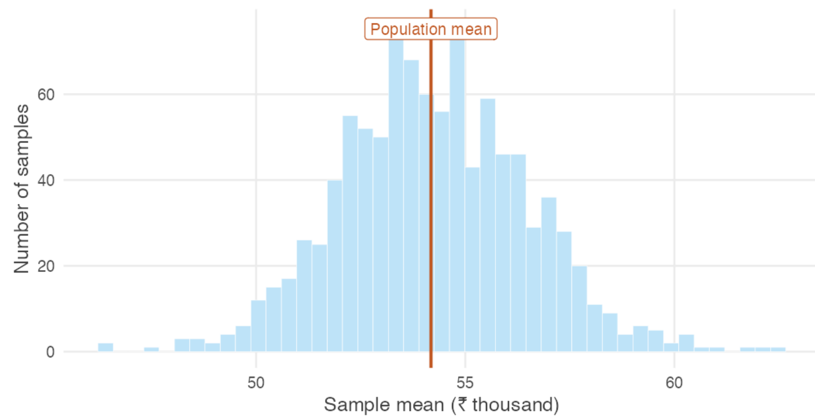

Figure 1.5: Distribution of the sample mean based on 1,000 samples of 2,000 workers.

Several features stand out immediately.

First, the sample means are not all identical. Even though every sample was drawn from the same population using the same procedure, each survey produced a slightly different average.

Second, the variation is not arbitrary. Most of the sample means cluster within a relatively narrow range, while only a few lie far from the centre. Extreme values are possible, but they are uncommon.

Finally, the distribution itself has a remarkably regular shape. Although the earnings of individual workers are highly skewed, the distribution of the sample means is much more symmetric and concentrated.

The thousand means are not just a collection of numbers. They form a distribution of their own. Notice what this distribution describes. It is not the distribution of workers' earnings — that was the distribution we began with. Instead, it is the distribution of the sample mean obtained by repeating the survey many times. The object of interest has changed. We are no longer studying individual workers; we are studying the behaviour of the sample mean.

This distinction lies at the heart of inferential statistics. The original distribution describes the variability in earnings across workers. The new distribution describes the variability in the sample mean across repeated samples. It is this second source of variation that determines how much confidence we should place in the average computed from a single survey.

The distribution formed by a statistic over repeated samples is called its **sampling distribution**. Every statistic has a sampling distribution: the mean, the median, the standard deviation, and many others. In this chapter we focus on the sampling distribution of the sample mean because it provides the foundation for much of statistical inference.

Two features of Figure 1.5 are too striking to ignore. First, the sample means are tightly concentrated around the population mean. Second, despite the strong skewness of individual earnings, their sampling distribution is almost bell-shaped. Neither feature is a coincidence. Both are consequences of two fundamental results in probability theory: the Law of Large Numbers and the Central Limit Theorem. Together, these results explain why a single sample can provide reliable information about an entire population.

1.5 Why do larger samples give more reliable answers?

> *What determines how much they vary?*

Figure 1.5 answered one important question. Repeating the survey many times does not produce wildly different sample means. Instead, the sample means cluster around the population mean.

A natural question follows. What determines how much they vary?

One possibility is the size of the survey itself. Intuition suggests that a survey of twenty workers should produce averages that vary much more from one survey to the next than a survey of two thousand workers. But how much difference does sample size really make?

To find out, let us repeat the experiment from the previous section several times. Instead of fixing the sample size at 2,000 workers, we consider surveys of different sizes. For each sample size we draw one thousand independent samples and compute the average annual earnings.

```r
set.seed(3)

earn  <- workers$annual_earnings
sizes <- c(20, 100, 500, 2000)

by_size <- do.call(rbind, lapply(sizes, function(n) {
  data.frame(n = factor(paste("n =", n), levels = paste("n =", sizes)),
             m = replicate(4000, mean(sample(earn, n, replace = TRUE))))
}))
```

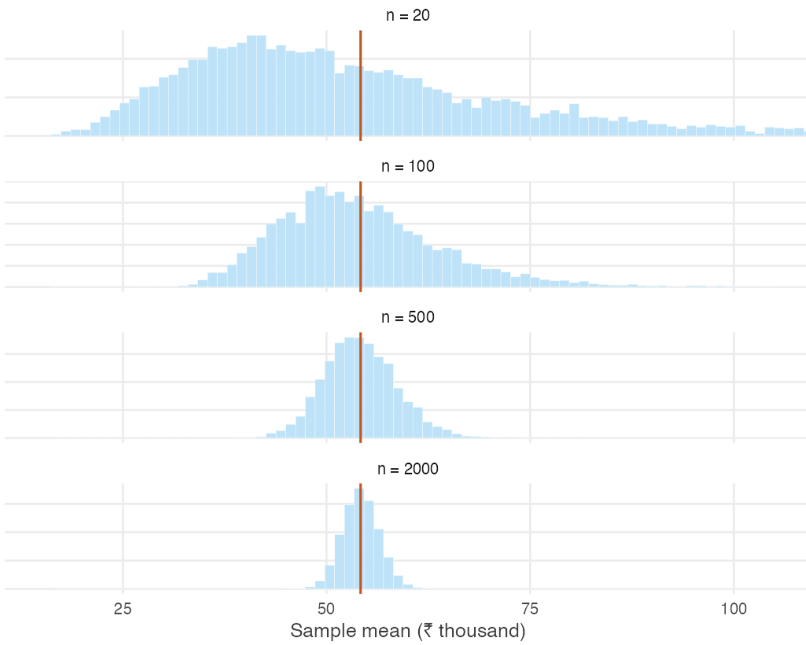

Figure 1.6: Sampling distributions of the sample mean for different sample sizes.

The pattern is unmistakable.

When the sample contains only a small number of workers, the sample means are widely dispersed. Some surveys produce averages well above the population mean, while others fall well below it. The averages vary substantially from one survey to the next.

As the sample size increases, the sampling distribution becomes progressively narrower. The sample means remain centred at the population mean, but they fluctuate much less. Large departures from the population mean become increasingly uncommon, while averages close to the population mean become increasingly likely.

```r
round(tapply(by_size$m, by_size$n, sd))
```

```
#>   n = 20  n = 100  n = 500 n = 2000
#>    22998     9973     4433     2252
```

This behaviour reflects a simple idea. Every sample contains chance variation. Some samples happen to include more high-income workers than average, while others include more low-income workers. In a small sample these random differences have a noticeable effect on the average. As the sample grows, unusually high and unusually low observations increasingly offset one another, and the sample mean settles closer to the population mean.

The behaviour we have just observed is not merely an empirical regularity. It is one of the most fundamental results in probability theory and forms the foundation of statistical inference. It is known as the **Law of Large Numbers**.

In informal terms, the Law of Large Numbers states that as the sample size increases, the sample mean gets closer and closer to the population mean. Large samples do not eliminate randomness, but they reduce its influence. Although no single survey is guaranteed to produce the correct answer, larger surveys are increasingly likely to do so.

The Law of Large Numbers explains why surveys, opinion polls, clinical trials and economic censuses can provide reliable information about populations without observing every individual. The key is not that they observe everyone, but that they observe enough.

EVIDENCE FROM THE REAL WORLD

Real demonstrations of the Law of Large Numbers are surprisingly hard to find. The law concerns what happens when the same random process is repeated a great many times, and few processes in economic life are repeated identically enough, often enough, for the convergence to be visible.
The cricket toss is an exception. Before every international match a coin is tossed, and over nearly a century and a half some teams have accumulated thousands of these tosses while others have played only a handful.

```r
toss <- read.csv("data/cricket-toss.csv")
keep <- c("team", "span", "matches", "pct_toss_won")

# the most and least fortunate teams, and the most experienced
rbind(head(toss[order(-toss$pct_toss_won), keep], 2),
      head(toss[order( toss$pct_toss_won), keep], 2),
      head(toss[keep], 3))
```

```
#>            team      span matches pct_toss_won
#> 90       Greece 2019-2024      15        80.00
#> 88 Cook Islands 2022-2025      17        70.59
#> 94    Africa XI 2005-2007       6        16.67
#> 91        China 2023-2024      11        27.27
#> 1       England 1877-2025    2124        49.25
#> 2     Australia 1879-2025    2116        51.23
#> 3         India 1932-2025    1926        49.58
```

Greece has won 80% of its tosses and Africa XI barely one in six, while

England, after 2,124 matches, sits at 49.2%. Nobody is luckier than anybody else. The teams with extreme records are simply the teams that have played few matches.
That is the Law of Large Numbers, and the agreement with theory is close. If the toss is fair, a team with n matches should have a toss-win percentage scattered around 50 with a standard deviation of $50/\sqrt{n}$ percentage points. Restricting attention to teams above a given number of matches:

```r
lln_toss <- t(sapply(c(0, 50, 100, 200, 500), function(m) {
  x <- toss[toss$matches >= m, ]
  c(min_matches = m,
    teams       = nrow(x),
    observed_sd = sd(x$pct_toss_won),
    predicted   = sqrt(mean((50 / sqrt(x$matches))^2)))
}))

round(lln_toss, 2)
```

```
#>      min_matches teams observed_sd predicted
#> [1,]           0    94        8.46      7.91
#> [2,]          50    49        4.92      4.77
#> [3,]         100    27        3.03      3.16
#> [4,]         200    16        2.10      2.05
#> [5,]         500    10        0.85      1.32
```

The spread of toss-win percentages falls from 8.5 percentage points across all teams to 0.9 among the ten most experienced — and at every step it tracks the value the theory predicts. No survey was designed and no experiment was run. The convergence is simply there in the record of the game.

Yet Figure 1.6 reveals another feature that the Law of Large Numbers does not explain. As the sample size increases, the sampling distributions not only become narrower — they also become increasingly bell-shaped. This is surprising because the underlying distribution of earnings is strongly skewed.

Why should averages of skewed data look approximately normal?

The answer is provided by an even more remarkable result: the **Central Limit Theorem**.

1.6 The Central Limit Theorem

> *Why do sample means look bell-shaped?*

The Law of Large Numbers explains why the sample mean moves closer to the population mean as the sample size increases. It tells us where the sample mean is centred, but not how it is distributed for any given sample size. Without knowing that distribution, we cannot quantify the uncertainty associated with a single survey.

The Central Limit Theorem answers precisely this question. Its remarkable feature is its generality.

> REMEMBER THIS
>
> **Central Limit Theorem.** For sufficiently large samples, the sampling distribution of the sample mean is approximately normal, regardless of the shape of the population from which the sample is drawn.

The theorem will be stated more precisely once we have developed the necessary notation. For now, its essential idea is already visible in Figure 1.6. Even when the population itself is far from bell-shaped, the distribution of the sample mean becomes increasingly bell-shaped as the sample size grows.

The strength of this result is easy to underestimate. The population may be binary, highly skewed, or concentrated in only one part of its range. It makes remarkably little difference. Average enough observations together, and the distribution of those averages approaches the familiar bell-shaped curve.

A simulation

The easiest way to appreciate the theorem is to watch it happen. Figure 1.7 compares the sampling distributions of the sample mean for four populations with very different shapes, using sample sizes of one, five and thirty.

```
set.seed(3)

draw_mean <- function(parent, n) {
  switch(parent,
    "Bernoulli" = mean(rbinom(n, 1, 0.3)),
    "Poisson"   = mean(rpois(n, 2)),
    "Uniform"   = mean(runif(n)),
    "Earnings"  = mean(sample(earn, n, replace = TRUE)))
}

parents <- c("Bernoulli", "Poisson", "Uniform", "Earnings")
```

```
sizes   <- c(1, 5, 30)

clt <- do.call(rbind, lapply(parents, function(p)
  do.call(rbind, lapply(sizes, function(n)
    data.frame(parent = p,
               n      = factor(paste("n =", n), levels = paste("n =",
                ↪ sizes)),
               m      = replicate(4000, draw_mean(p, n)))))))

# Standardise within each panel, so all twelve are on one scale and the
# comparison is purely about shape.
clt$z <- ave(clt$m, clt$parent, clt$n,
             FUN = function(x) (x - mean(x)) / sd(x))
```

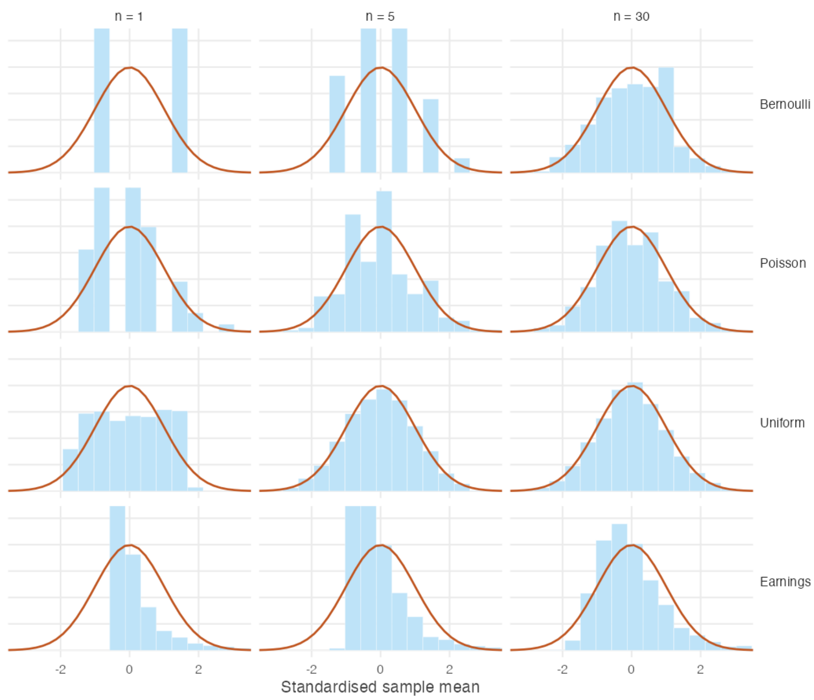

Figure 1.7: Standardised sampling distributions of the sample mean for four populations, against the standard normal curve.

The first column shows the four underlying populations. None resembles a normal distribution. By the third column, however, the distributions of the sample means are all much closer to the familiar bell-shaped curve.

The theorem guarantees that this convergence eventually occurs, but it says nothing about how quickly it happens. Populations that are already fairly symmetric require only modest sample sizes, whereas heavily skewed populations may require substantially larger samples before the normal approximation becomes accurate. In Figure 1.7, the uniform population is already close to normal when $n = 5$, while the earnings distribution remains visibly asymmetric even when $n = 30$.

An intuition

Why should averaging observations produce a bell-shaped distribution?

A simple example provides the intuition. Roll a fair die once and every outcome from one to six is equally likely. Now roll two dice and add the results.

```
sums <- outer(1:6, 1:6, "+")

table(sums)
```

```
#> sums
#>  2  3  4  5  6  7  8  9 10 11 12
#>  1  2  3  4  5  6  5  4  3  2  1
```

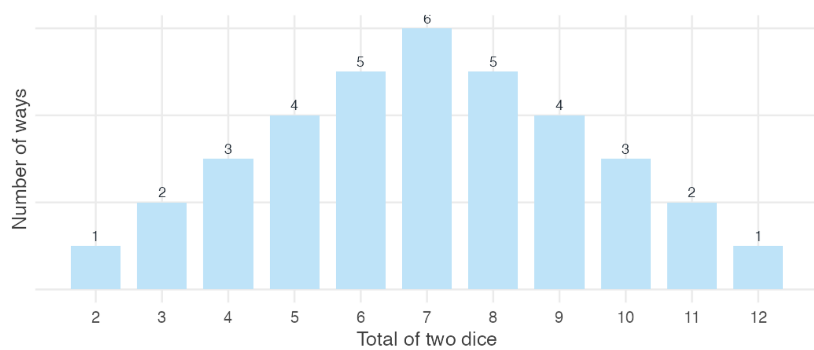

Figure 1.8: The number of ways each total can be obtained when two dice are rolled.

Seven can be obtained in six different ways: $1 + 6,\ 2 + 5,\ 3 + 4,\ 4 + 3,\ 5 + 2,\ 6+1$. By contrast, a total of two can be obtained in only one way. The middle values have many more possible combinations than the extremes, so they occur more frequently.

The same intuition extends far beyond dice. To obtain an unusually large average, most observations must themselves be unusually large. Likewise, an unusually small average requires most observations to be unusually small. Both situations are relatively rare. A value close to the middle, however, can arise from countless combinations of larger and smaller observations offsetting one another.

As more observations are averaged together, the number of ways to obtain values near the centre grows much faster than the number of ways to obtain values at the extremes. The result is the familiar bell-shaped distribution.

How large is "large enough"?

Many textbooks suggest that a sample size of at least thirty is sufficient for the Central Limit Theorem to provide a good approximation. This is a useful rule of thumb, but it is not a mathematical guarantee.

How quickly the approximation becomes accurate depends largely on the shape of the population. Populations that are approximately symmetric require relatively few observations, whereas heavily skewed populations often require much larger samples.

```r
set.seed(9)

skewness <- function(x) mean((x - mean(x))^3) / sd(x)^3

sizes <- c(5, 30, 100, 500)

sk <- c(population = skewness(earn),
        sapply(sizes, function(n)
          skewness(replicate(2000, mean(sample(earn, n, replace =
          ↪  TRUE))))))
names(sk)[-1] <- paste0("n=", sizes)

round(sk, 2)
```

```
#> population        n=5       n=30      n=100      n=500
#>       7.32       2.99       1.20       0.76       0.27
```

The earnings distribution used throughout this chapter has a skewness of approximately 7.3. When the sample size is five, the sampling distribution remains strongly right-skewed. Even at thirty observations, some asymmetry is still visible. By one hundred observations, however, the distribution is much closer to symmetric.

This matters because many economic variables — including incomes, firm sizes, land holdings and city populations — are themselves highly skewed. For such variables, the normal approximation may still be poor when the sample size is only thirty. In these situations, larger samples or simulation-based checks are often preferable to relying mechanically on a rule of thumb.

Why this matters

The Central Limit Theorem, together with the Law of Large Numbers, explains why statistical inference is possible.

The Law of Large Numbers tells us that repeated samples become increasingly centred on the population mean as the sample size grows. The Central Limit Theorem tells us that the distribution of those sample means approaches a normal distribution. Together, these results reveal that although different samples produce different answers, those answers are not arbitrary. They vary in systematic and predictable ways.

> **REMEMBER THIS**
>
> This insight lies at the heart of inferential statistics. We observe only one sample, but by understanding how the same statistic would behave across many hypothetical samples, we can quantify the uncertainty associated with the sample we actually observe.

So far we have developed these ideas largely through intuition, simulation and visualisation. The next section introduces the notation and concepts needed to express them more precisely and to build the methods of statistical inference developed in the remainder of this book.

1.7 A language for randomness

> *Why do we need a mathematical language?*

Throughout this chapter we have described repeated sampling almost entirely in plain English. We have spoken about rolling dice, drawing surveys, computing averages and repeating experiments thousands of times. That language has helped us build intuition, but it is no longer sufficient. To develop statistical inference, we need a concise and precise way of expressing these ideas.

The first concept we need is that of a random variable.

Random variables

Some quantities are known before we observe them. If you measure your height today, it is a fixed number. It may be unknown to you, but it is not changing from one repetition to the next.

Other quantities are different. Roll a die and you do not know what number will appear. Randomly select a worker from a population and you do not know that person's earnings. Draw a sample of one hundred households and you do not know what the sample mean will be until the sample has actually been collected.

Statisticians describe such quantities as **stochastic**. A stochastic quantity is one whose value is determined by the outcome of a random process.

The mathematical object used to describe a stochastic quantity is called a **random variable**. We usually denote random variables by capital letters such as X, Y and Z. For example, X may denote the earnings of a randomly selected worker, Y the annual rainfall in a particular district, and D the outcome of rolling a fair die.

The sample mean deserves special attention. Every time we draw a different sample, we obtain a different average. The sample mean is therefore itself a random variable. Before observing the data, we denote it by $\bar{X}$.

> **REMEMBER THIS**
>
> This observation is fundamental to inferential statistics. Since $\bar{X}$ is a random variable, it has a probability distribution. That distribution tells us how the sample mean behaves across repeated samples. Once we understand its centre and spread, we can quantify the uncertainty associated with the single sample we actually observe.

Realisations

Eventually every random variable takes a particular observed value.

Suppose X represents the outcome of rolling a fair die. Before the die is rolled, X could take any value from one to six. After the die lands showing four, we have observed one **realisation** of the random variable, written as $x = 4$.

The same distinction applies to the sample mean. Before collecting a sample, the sample mean is the random variable $\bar{X}$. After collecting the data and computing the average, we observe one realisation of that random variable, written as $\bar{x}$.

Throughout the remainder of this book, capital letters denote random variables, while lower-case letters denote their observed values.

The long-run average

Every random variable has a centre. The centre is called its **expected value** and is written as

$$E(X) = \sum_i x_i \, P(X = x_i).$$

The term *expected value* can be misleading. It is not the value we expect to observe on any particular repetition. Rather, it is the average value the random variable would take over a very large number of repetitions.

For a fair die,

$$E(X) = \frac{1+2+3+4+5+6}{6} = 3.5.$$

Of course, no single roll can ever produce 3.5. It is a theoretical average, not an individual outcome.

This distinction becomes clearer when we compare theory with data.

```
mean(rolls)              # average of 10,000 simulated rolls
```

```
#> [1] 3.514
```

```
sum(1:6 * rep(1/6, 6))     # theoretical expected value
```

```
#> [1] 3.5
```

The close agreement between these two numbers is exactly what the Law of Large Numbers predicts. The expected value is calculated from the probability model, while the sample average is calculated from observed data. As the number of observations grows, the latter approaches the former.

When the random variable represents a randomly selected member of a population, its expected value is simply the population mean. For this reason we denote both by the same symbol,

$$E(X) = \mu.$$

Measuring variation

Knowing the centre of a distribution is not enough. Two random variables can have the same expected value but very different amounts of variation.

The spread of a random variable around its expected value is measured by its **variance**,

$$\mathrm{Var}(X) = E\big[(X-\mu)^2\big],$$

and the square root of the variance is the **standard deviation**,

$$\sigma = \sqrt{\mathrm{Var}(X)}.$$

The expected value tells us where a random variable is centred. The variance and standard deviation tell us how much it fluctuates around that centre.

Parameters and statistics

Statistics is concerned with learning about populations from samples. It is therefore essential to distinguish between quantities that describe a population and quantities computed from a sample.

A **parameter** is a numerical quantity describing the population. The population mean is written μ, the population variance σ^2, and the population standard deviation σ. These quantities are fixed — they do not change — but they are usually unknown, because we rarely observe the entire population.

A **statistic**, by contrast, is computed from a sample. The sample mean, sample variance and sample standard deviation are all statistics. Unlike parameters, statistics change from one sample to the next, because they depend on which observations happen to be selected.

The entire objective of inferential statistics can now be expressed in a single sentence:

> **REMEMBER THIS**
>
> Use statistics computed from a sample to learn about unknown parameters describing the population.

Everything that follows in this book — estimation, confidence intervals, hypothesis testing and regression analysis — is an extension of this single idea.

A summary of notation

The notation introduced in this section will be used throughout the book.

Symbol	Meaning
X, Y, Z	random variables — quantities not yet observed

Symbol	Meaning
$x,\ y,\ z$	realisations — values actually observed
$\bar{X}$	the sample mean, before the sample is drawn
$\bar{x}$	the sample mean, once the sample is observed
n	the sample size
$E(X)$	expected value: the long-run average of X
$\mathrm{Var}(X)$	variance: the spread of X about its expected value

And, separating what describes the population from what is computed from a sample:

Quantity	Population (parameter)	Sample (statistic)
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard deviation	σ	s

The left-hand column is fixed and unknown. The right-hand column is observed and changes with every sample. Almost all of statistics is the effort to say something about the first using only the second.

1.8 Before any mathematics can help

> *What must be true of the data before any of this applies?*

Everything in this unit assumes that the sample carries information about the population we care about. That assumption is not guaranteed by any formula, and when it fails, no amount of statistical sophistication will announce the failure. The arithmetic runs perfectly on data that cannot answer the question.

Four ways it fails, none of which any calculation will warn you about.

The sample is not from the population you mean. The Annual Survey of Industries covers *registered* factories. Most Indian manufacturing employment is unregistered. Conclusions about "Indian manufacturing" drawn from that source are conclusions about a formal sliver of it, however large the sample.

The sample is not random. The class survey in Section 1.2 has thirty-two heights. They are not a random sample of students at the university, still less of young Indians: they are the people who took the course and attended that day. A confidence interval computed from them is arithmetically correct and tells you nothing about anyone else.

The data are measured badly. Incomes reported from memory are systematically understated. Rainfall attributed to a district may have been recorded at a station a hundred kilometres away. National accounts are revised for years after first publication. A sampling distribution describes how an estimate varies across samples; it says nothing about whether the quantity was measured correctly in the first place. Careful statistics performed on poor measurements remain poor statistics.

The question is causal but the data are not. Suppose states with more industry have higher literacy. Does industry raise literacy? Does literacy attract industry? Does something else drive both? The correlation is identical in all three cases, and no statistical procedure separates them without an argument about how the data came to exist.

> **REMEMBER THIS**
>
> These are not edge cases. They are the ordinary condition of applied economics, and recognising them is a skill quite separate from computation. Good inference begins with good data and thoughtful study design, not with clever mathematics.

1.9 Summary

> **REMEMBER THIS**
>
> 1. **Randomness creates variation, and variation is the raw material of statistics.** Where nothing varies there is no distribution, no uncertainty, and nothing to infer.
> 2. **Descriptive statistics** summarises the data in hand. **Inferential statistics** uses that data to make claims about a population it did not observe, and reports how uncertain those claims are.
> 3. Different samples give different answers. That is **sampling variation —** unavoidable, and not a defect in the data.
> 4. A quantity whose value changes from one repetition to the next is a **random variable**, written X. Its observed value is a **realisation**, written x. The sample mean is $\bar{X}$ before the sample is drawn and $\bar{x}$ after.

5. $E(X) = \sum_i x_i P(X = x_i)$ is the **expected value** — a long-run average, not a value you expect to see. For a population draw, $E(X) = \mu$.
6. $\text{Var}(X) = E[(X-\mu)^2]$ measures spread, and $\sigma = \sqrt{\text{Var}(X)}$ is the standard deviation.
7. **Parameters** (μ, σ^2, σ) describe the population and are fixed but unknown. **Statistics** ($\bar{x}$, s^2, s) are computed from a sample and change with every sample. Inference uses the second to learn about the first.
8. The distribution of a statistic across repeated samples is its **sampling distribution**. Every statistic has one, not only the mean.
9. **Law of Large Numbers:** as n grows, the sample mean becomes increasingly likely to lie close to μ. It does not say any particular large sample is correct, and it offers no comfort to a gambler.
10. **Central Limit Theorem:** for sufficiently large n, the sampling distribution of $\bar{X}$ is approximately normal whatever the shape of the population. The rule $n \geq 30$ is a rule of thumb, not a guarantee — a heavily skewed population needs far more.
11. No formula warns you when the sample is drawn from the wrong population, drawn non-randomly, measured badly, or asked a causal question it cannot answer.

Taken together, these amount to a single change of perspective. We stopped asking what a particular sample says, and began asking how a statistic behaves across the many samples that might have been drawn. Everything that follows — estimation, confidence intervals, hypothesis testing, regression — rests on that shift.

1.10 Practice problems

YOUR TURN

These are for learning. Work each one before opening the solution; the answer is worth very little if you have not first tried to produce it yourself. The question bank that follows is for testing yourself and has no answers at all.

Concept checks

Short questions. A sentence or two is enough.

1.1 Why does repeating the same survey produce a different average each time?

SOLUTION

Because a different set of people is surveyed each time. Every average is computed from whoever happened to be selected, and selection involves chance. The population has not changed and no arithmetic error has occurred — only the membership of the sample.

1.2 A die with a three painted on all six faces is rolled ten thousand times. What is the distribution of the outcome? Why is this example in the chapter at all?

SOLUTION

There is no distribution to speak of: every outcome is three, and the standard deviation is zero. The example is there to show that statistics requires variation. Where outcomes never differ, there is nothing to describe, nothing to be uncertain about, and no inference to perform.

1.3 Distinguish between $\bar{X}$ and $\bar{x}$.

SOLUTION

$\bar{X}$ is the sample mean regarded as a random variable, before the sample is drawn: it has a distribution, a centre and a spread. $\bar{x}$ is the single number obtained once a particular sample has been collected — one realisation of $\bar{X}$.

1.4 A survey of 400 households gives a mean monthly income of ₹18,000. Is ₹18,000 a parameter or a statistic? What is the parameter here?

SOLUTION

₹18,000 is a statistic: it was computed from a sample and would change with a different sample. The parameter is the mean monthly income of the whole population, μ, which is fixed but unknown.

Explain

One paragraph each. These reward clarity rather than calculation.

1.5 Explain to a friend who has never studied statistics why the Law of Large Numbers does not mean that a large sample is guaranteed to give the right answer.

SOLUTION

The law says that as the sample grows, the sample mean becomes increasingly likely to lie close to the population mean — not that it will equal it. Any particular large sample may still be unlucky. What changes with sample size is the *probability* of being far off, which shrinks; the possibility never disappears. A large sample makes a badly wrong answer improbable, not impossible.

1.6 A friend argues that because a coin has landed heads six times running, tails is now more likely. Using the Law of Large Numbers, explain why this is wrong.

SOLUTION

The coin has no memory, and the law promises nothing of the kind. What converges is the *proportion* of heads, and it converges by dilution rather than by correction. The six extra heads are never cancelled out; they simply become a smaller and smaller share of a growing total. The absolute surplus of heads typically grows with the number of tosses even as the proportion approaches one half.

1.7 Two researchers study earnings in the same district. One surveys 100 households, the other 2,500. Both report a mean. Which should be trusted more, and what would you need to know before answering?

SOLUTION

Other things equal, the larger survey has the smaller standard deviation of the sample mean — by a factor of five, since $\sqrt{2500/100} = 5$. But this holds only if both samples were drawn from the population of interest and drawn at random. A carefully designed survey of 100 households is worth more than a badly drawn survey of 2,500. Sample size governs precision; sampling design governs whether precision is about the right quantity at all.

Numerical problems

Work each of these with a pen first. Each ends by asking you to check your answer by simulation — the point being that theory and simulation should agree, and that when they do not, one of them is wrong.

1.8 The earnings data used throughout this chapter has a population mean of about ₹54,200 and a population standard deviation of about ₹100,100.

Suppose a survey team draws random samples of (a) 50, (b) 100, (c) 200 and (d) 1,000 workers.

i. Find the expected value of the sample mean in each case.
ii. Find the standard deviation of the sample mean in each case.
iii. What happens to that standard deviation as the sample size increases, and why?
iv. Verify your answers by simulation.

SOLUTION

The expected value of the sample mean equals the population mean whatever the sample size, so $E(\bar{X}) = ₹54{,}200$ in all four cases. Sample size affects precision, not location.
The standard deviation of the sample mean is $\sigma/\sqrt{n} = 100{,}100/\sqrt{n}$:

```r
sigma <- 100100
n     <- c(50, 100, 200, 1000)

round(data.frame(n, expected = 54200, se = sigma / sqrt(n)))
```

```
#>      n expected    se
#> 1   50    54200 14156
#> 2  100    54200 10010
#> 3  200    54200  7078
#> 4 1000    54200  3165
```

It falls in proportion to $1/\sqrt{n}$. Each additional worker contributes information, so the average over many of them is pulled more tightly around the population value. Note that raising the sample from 50 to 1,000 — twenty times the data — reduces it only by a factor of $\sqrt{20} \approx 4.5$.
Because we hold the whole population, the answer can be checked directly:

```r
set.seed(4)
earn <- read.csv("data/wages-india-synthetic.csv")$annual_earnings

sim <- sapply(n, function(k)
  sd(replicate(2000, mean(sample(earn, k, replace = TRUE)))))

round(data.frame(n, theoretical = sd(earn) / sqrt(n), simulated =
↪  sim))
```

```
#>      n theoretical simulated
#> 1   50       14162     14219
#> 2  100       10014      9740
#> 3  200        7081      7204
#> 4 1000        3167      3120
```

The two columns agree closely. The small discrepancies are themselves sampling variation: each simulated standard deviation is estimated from 2,000 samples rather than infinitely many.

1.9 An auto-rickshaw driver's daily earnings vary from day to day, with a mean of ₹800 and a standard deviation of ₹200. In a particular month the driver works 30 days.

i. What are the mean and standard deviation of total earnings for the month?
ii. Approximate the probability that the month's total exceeds ₹25,000.
iii. Approximate the probability that it falls below ₹22,000.
iv. Approximate the probability that it lies between ₹22,000 and ₹25,000.
v. What assumption does this calculation require, and is it plausible here?
vi. Check your answers by simulating 10,000 months.

SOLUTION

The monthly total is a sum of 30 daily earnings. Its mean is $30 \times 800 =$ ₹$24{,}000$, and because variances add for independent quantities, its standard deviation is $200\sqrt{30} =$ ₹$1{,}095$.
By the Central Limit Theorem the total is approximately normal, so:

```r
mu_day <- 800; sd_day <- 200; days <- 30

mu_tot <- mu_day * days
sd_tot <- sd_day * sqrt(days)

c(mean = mu_tot, sd = sd_tot,
  above_25000 = pnorm(25000, mu_tot, sd_tot, lower.tail = FALSE),
  below_22000 = pnorm(22000, mu_tot, sd_tot),
  between     = pnorm(25000, mu_tot, sd_tot) - pnorm(22000, mu_tot,
  ↪   sd_tot))
```

```
#>        mean          sd above_25000 below_22000     between
#> 24000.00000  1095.44512     0.18066     0.03394     0.78540
```

So roughly an 18% chance of clearing ₹25,000, a 3% chance of falling below ₹22,000, and a 79% chance of landing between the two.
The assumption is independence. The calculation treats each day's earnings as unrelated to every other day's. That is questionable. A week of heavy monsoon suppresses earnings on consecutive days; a festival week raises them together. When days are positively correlated, the true standard deviation of the monthly total is *larger* than $200\sqrt{30}$, and the probability of an unusually bad month is correspondingly higher than this calculation suggests.

This is worth dwelling on. The arithmetic is not wrong; the assumption is. No amount of care in the computation would reveal the problem. Simulating under the stated assumption:

```
set.seed(5)
months <- replicate(10000, sum(rnorm(days, mu_day, sd_day)))

c(mean = mean(months), sd = sd(months),
  above_25000 = mean(months > 25000),
  below_22000 = mean(months < 22000),
  between     = mean(months >= 22000 & months <= 25000))
```

```
#>        mean          sd above_25000 below_22000     between
#>  23987.5096   1090.6065      0.1788      0.0321      0.7891
```

The simulated proportions match the normal approximation closely — as they must, since the simulation was itself built on the assumption of independent daily earnings. Simulation confirms the arithmetic; it cannot confirm the assumption.

R exercises

1.10 Simulate the Central Limit Theorem starting from an exponential distribution, which is strongly right-skewed. Draw 5,000 sample means for sample sizes of 1, 5, 30 and 200 from `rexp(n, rate = 1)`, and plot the four distributions. At which sample size does the result begin to look normal?

SOLUTION

```r
set.seed(1)

sizes <- c(1, 5, 30, 200)
sim <- do.call(rbind, lapply(sizes, function(n)
  data.frame(n = factor(paste("n =", n), levels = paste("n =",
  ↪  sizes)),
             m = replicate(5000, mean(rexp(n, rate = 1))))))

ggplot(sim, aes(m)) +
  geom_histogram(bins = 50, fill = book_palette$fill, colour =
  ↪  "white",
                 linewidth = 0.1) +
  facet_wrap(~ n, scales = "free") +
  labs(x = "Sample mean", y = NULL) +
  theme_book() +
  theme(axis.text.y = element_blank())
```

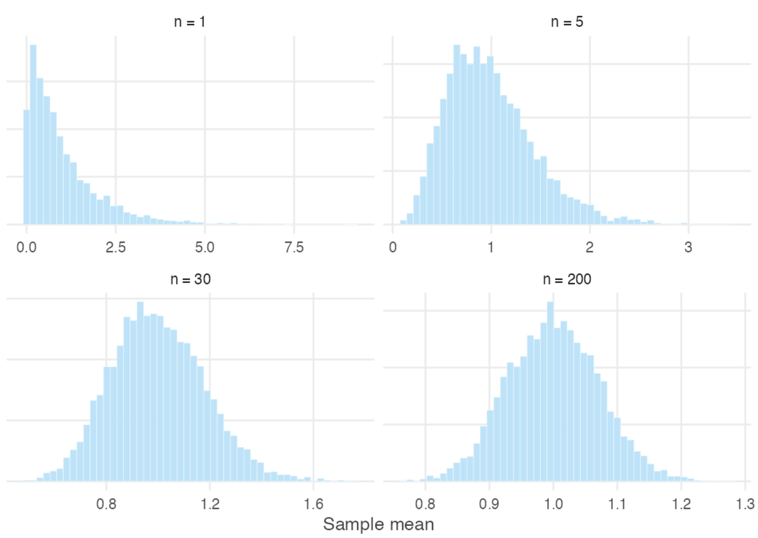

Figure 1.9: Sampling distributions of the mean from an exponential population.

At $n = 1$ the picture is the exponential distribution itself — a sharp peak at zero with a long right tail. By $n = 5$ it has a recognisable mound but remains clearly skewed. By $n = 30$ it is close to symmetric, and by $n = 200$ it is difficult to distinguish from a normal distribution by eye.

The exponential distribution has skewness 2, which is milder than the earnings data used in this chapter. That is why $n = 30$ works reasonably well here and less well there.

1.11 Using `data/cricket-toss.csv`, plot each team's toss-win percentage against its number of matches. What shape does the scatter take, and why?

SOLUTION

```
toss <- read.csv("data/cricket-toss.csv")

band <- data.frame(n = 5:2200)
band$hi <- 50 + 2 * 50 / sqrt(band$n)
band$lo <- 50 - 2 * 50 / sqrt(band$n)

ggplot(toss, aes(matches, pct_toss_won)) +
  geom_ribbon(data = band, aes(x = n, ymin = lo, ymax = hi),
              inherit.aes = FALSE, fill = book_palette$fill, alpha =
               ↪  0.45) +
  geom_hline(yintercept = 50, colour = book_palette$reject,
   ↪  linewidth = 0.6) +
  geom_point(alpha = 0.7, size = 1.6, colour = book_palette$ink) +
  scale_x_log10(labels = scales::comma) +
  labs(x = "Matches played (log scale)", y = "Toss-win percentage") +
  theme_book()
```

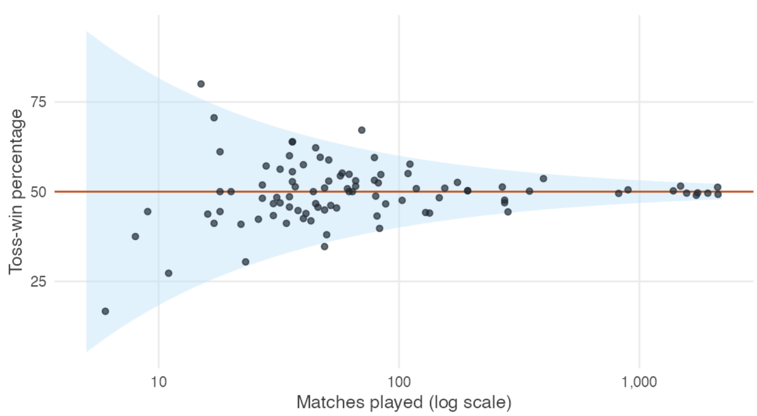

Figure 1.10: Toss-win percentage against number of matches played, with the theoretical limits of two standard deviations.

The scatter forms a funnel: wide on the left, where teams have played few matches, and narrowing towards 50% as the number of matches grows.

The shaded band shows two standard deviations either side of 50, computed as $50/\sqrt{n}$ — the same formula that governs the sample mean. Almost every team falls inside it.
This is the Law of Large Numbers made visible. The teams with extreme records are not lucky; they are simply the teams with few observations.

1.11 Question bank

WATCH OUT

No answers are given for this section, by design. These questions are for testing yourself once the practice problems are complete. There are no hints and no worked solutions — exactly as in an examination.

Multiple choice

QB 1.1 Which of the following is a random variable? *(select all that apply)*

a. The number showing after a die is rolled
b. The number showing on a die with a three painted on every face
c. The mean earnings of a sample of 500 workers, before the sample is drawn
d. The mean earnings of a sample of 500 workers, after the sample is drawn

QB 1.2 A population has mean $\mu = 40$ and standard deviation $\sigma = 12$. For samples of size 36, the standard deviation of the sample mean is

a. 12
b. 2
c. 0.33
d. 6

QB 1.3 Which statements about the Law of Large Numbers are correct? *(select all that apply)*

a. As the sample grows, the sample mean becomes more likely to be close to the population mean
b. As the sample grows, the sample mean becomes certain to equal the population mean
c. It applies only when the population is normally distributed

d. It explains why a survey need not interview everyone

QB 1.4 A researcher doubles the size of a survey from 500 to 1,000 households. The standard deviation of the sample mean falls by a factor of approximately

a. 2
b. 4
c. 1.41
d. It does not change

QB 1.5 The Central Limit Theorem states that

a. Large samples are normally distributed
b. The population becomes normal as the sample grows
c. The sampling distribution of the sample mean is approximately normal for sufficiently large samples
d. Any variable is approximately normal if enough observations are collected

QB 1.6 Which of the following would make the normal approximation to the sampling distribution *less* accurate at a given sample size? *(select all that apply)*

a. A heavily right-skewed population
b. A symmetric population
c. A population with a small number of extreme outliers
d. A larger population standard deviation

QB 1.7 A statistic differs from a parameter because

a. A statistic is always correct
b. A statistic is computed from a sample and changes from sample to sample
c. A parameter is computed from a sample
d. A parameter changes when a different sample is drawn

Short reasoning

QB 1.8 A newspaper reports that average household income in a district is ₹32,000, based on a survey of 800 households, and describes this as "the income of a typical household". Give two separate objections to that description.

QB 1.9 An analyst has a sample of 50,000 observations and concludes that because the sample is very large, the results must be reliable. Under what circumstances would this reasoning be wrong?

QB 1.10 The **gambler's fallacy** is the mistaken belief that chance events "balance out" in the short run.

A batsman has scored 0 runs in each of his last five innings. A commentator says:

> He is due for a big score today.

Explain why this may be an example of the gambler's fallacy, and what the Law of Large Numbers does and does not say about it. Under what circumstances could the commentator's statement nevertheless be reasonable?

QB 1.11 Explain why the sampling distribution of the sample mean is narrower than the distribution of the individual observations, using the idea of offsetting variation rather than any formula.

Numerical

QB 1.12 The weights of rice bags filled by a machine have mean 25 kg and standard deviation 0.6 kg. A quality inspector weighs 36 bags.

(a) What is the expected value of the sample mean weight?
(b) What is its standard deviation?
(c) How many bags would need to be weighed to halve that standard deviation?

QB 1.13 Daily wages of agricultural labourers in a district have mean ₹350 and standard deviation ₹90. For a random sample of 100 labourers, approximate the probability that the sample mean wage

(a) exceeds ₹360, (b) falls below ₹340, (c) lies between ₹340 and ₹360.

QB 1.14 A population is strongly right-skewed with mean 12 and standard deviation 15. A student proposes to use the normal approximation for the sampling distribution of the mean with a sample of 20 observations. Comment on whether this is advisable, and describe how you would check.

1.12 Looking ahead

So far we have treated the sample as though it were the population. The average age, average income or average height was simply computed from the data in front of us.

In practice the sample is only one of many that might have been drawn.

Suppose you collect a random sample of fifty households and find an average monthly income of ₹24,000.

Should ₹24,000 be your estimate of the population average? Why not ₹23,800, or ₹24,500? What makes one estimate better than another?

You might think the answer is obvious: use the sample mean.

But why the sample mean? Why not the median? Why not the largest observation, or some weighted average of the fifty?

Before we can measure uncertainty, we need to understand why the sample mean deserves to be called an estimate at all. That is the subject of the next unit.

Chapter 2

Estimation: Why the Sample Mean?

Every empirical study begins with a population and ends with a sample.

We would like to know the average income of every household in a country, the average yield of every farm, or the average height of every adult. Measuring an entire population is almost never possible. Instead we observe a sample and use it to learn about the population it came from.

That raises an immediate question.

> If the population average is unknown, how should we estimate it from the sample?

Many answers are possible. We could use the sample mean. We could use the median. We could report the largest observation, or pick one household at random and use its income.

Clearly some of these are better than others. This unit is about how we judge the difference.

The sample mean turns out to have remarkable properties. Across repeated samples it gets the right answer on average. As the sample grows it becomes more precise. And its behaviour can be described mathematically, which lets us say how uncertain any particular estimate is.

Those ideas extend far beyond the sample mean. Every estimator in this book — the sample proportion, a regression slope, a difference between two groups — is judged by exactly the same questions.

REMEMBER THIS

By the end of this unit you should be able to ask three things of any estimator.

1. Does it get the right answer on average?
2. How much does it vary from sample to sample?
3. How does that variability change as the sample grows?

The sample mean is simply our first estimator.

2.1 The problem of estimation

> *We want to know something about a population we cannot observe. What exactly are we going to do?*

The previous unit described how a statistic *behaves*: repeated samples produce different averages, and those averages have a distribution of their own.

It did not tell us what to *do*.

Consider the position of anyone conducting a real survey. There is a quantity they want to know — the average annual earnings of Indian wage workers, the proportion of households with piped water, the average yield per hectare of paddy in Punjab. That quantity is a fixed number. It exists. It is simply not observed, because nobody has surveyed all 1.4 billion people, and nobody ever will.

What they have instead is a sample, and a decision to make: which number computed from that sample should be put forward as the answer?

REMEMBER THIS

A **parameter** is a fixed, unknown feature of the population, such as μ or σ^2.
An **estimator** is a rule for computing a guess at a parameter from a sample. Because it is computed from a sample, it is a random variable, and we write it with a capital letter: $\bar{X}$.
An **estimate** is the number that rule produces from the sample actually drawn: $\bar{x} = 54{,}300$.

The distinction between the last two is worth pausing on, because the words are often used interchangeably in ordinary speech. An estimator is the rule: take the observations, add them together and divide by n. An estimate is the number produced after the rule has been applied to a particular sample.

The difference matters. An estimate can never be judged directly, because we almost never know the true population mean. If a survey estimates average monthly income to be ₹50,000, we do not know whether the truth is ₹49,800, ₹50,100 or ₹55,000. The truth is precisely what we are trying to discover.

The estimator, however, can be judged. Although we never learn whether one particular estimate is close to the truth, we can ask how the estimator behaves across many hypothetical samples. Does it systematically overestimate the population mean? Does it systematically underestimate it? Does it produce very different answers from one sample to the next? These are questions about the procedure, not about any single outcome.

> **REMEMBER THIS**
>
> Since the parameter is never known, an estimator cannot be judged by any single estimate. It is judged by how it behaves across the many samples that might have been drawn.
> Evaluating procedures rather than outcomes is the idea that drives the whole of this unit.

There are many possible estimators of the population mean. We could use the sample median. We could take the midpoint of the largest and smallest observations. We could even ignore the data entirely and always report ₹50,000. Each is an estimator of μ, because each provides a rule for producing an estimate from a sample.

Why, then, has statistics settled almost universally on the sample mean?

That is the question for the rest of this unit.

2.2 When can a sample represent a population?

> *Before comparing estimators — can a sample tell us anything about a population at all?*

There is an objection to deal with first. Everything above assumed that a sample carries information about the population it came from. That is not automatic. It depends on how the sample was collected, and it can fail badly.

Rather than assert this, we can watch it happen. Below is a population of 20,000 households, spread across 200 localities. Sixty of those localities are urban, and incomes there are higher. Because we build the population ourselves, we know its mean exactly.

```r
set.seed(2)
```

```r
n_loc <- 200                 # localities
per   <- 100                 # households in each

urban    <- rep(c(TRUE, FALSE), times = c(60, 140))
loc_mean <- 18000 + 9000 * urban + rnorm(n_loc, sd = 4000)

pop <- data.frame(
  locality = rep(1:n_loc, each = per),
  urban    = rep(urban,   each = per),
  income   = round(rnorm(n_loc * per, rep(loc_mean, each = per), sd =
   ↪  5000))
)

mu    <- mean(pop$income)
sigma <- sd(pop$income)

c(mu = mu, sigma = sigma, se_if_random = sigma / sqrt(100))
```

```
#>           mu        sigma se_if_random
#>      20724.8       7974.4        797.4
```

The population mean is 20,725. Each survey below collects 100 households, and is repeated a thousand times. Only the *method of collection* differs.

```r
# Survey A: 100 households drawn at random from the whole population
# Survey B: 100 households drawn at random, but only from urban localities
# Survey C: five localities chosen at random, 20 households in each
# Survey D: five *urban* localities chosen at random, 20 households in each

cluster_sample <- function(pool) {
  chosen <- sample(pool, 5)
  unlist(lapply(chosen, function(l) sample(pop$income[pop$locality == l],
   ↪  20)))
}

urban_pop <- pop$income[pop$urban]
urb_loc   <- which(urban)

set.seed(11)
designs <- list(
  A = replicate(1000, mean(sample(pop$income, 100))),
  B = replicate(1000, mean(sample(urban_pop,  100))),
  C = replicate(1000, mean(cluster_sample(1:n_loc))),
  D = replicate(1000, mean(cluster_sample(urb_loc)))
)

round(t(sapply(designs, function(x) c(centre = mean(x), spread = sd(x)))))
```

```
#>   centre spread
```

```
#> A  20702    788
#> B  27503    667
#> C  20774   2843
#> D  27538   2140
```

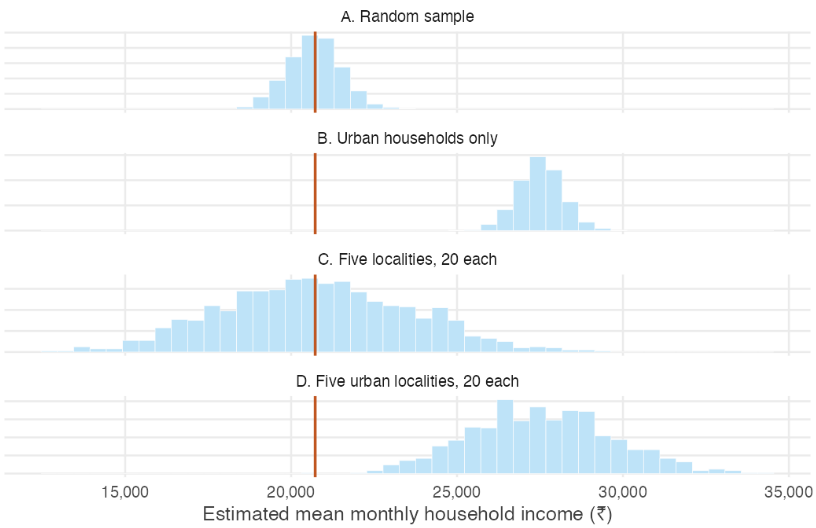

Figure 2.1: One thousand surveys under each design. The vertical line is the true population mean. Only the first design is both centred on it and as precise as theory promises.

Read the four rows carefully, because they fail in different ways.

Survey A works. The thousand estimates are centred on the population mean, and their spread is almost exactly the $\sigma/\sqrt{n}$ predicted in the previous unit.

Survey B is wrong, and looks excellent. Every estimate is far too high — the surveyor never visits a village, so village incomes cannot appear in the answer. Notice the second column: the spread is *smaller* than in Survey A. Urban households resemble one another, so the estimates agree closely with each other while agreeing on the wrong number. Consistency is not accuracy. This is the more dangerous failure, because a narrow spread is easily mistaken for a reliable result.

Survey C is centred correctly but far too variable. Choosing five localities and surveying twenty households in each is cheap and common — an interviewer travels to a place and works through it. But households in a locality resemble one another, so the second household adds much less new information than the first. The sample has 100 rows and nothing like 100 households' worth of information. The spread is over three times what $\sigma/\sqrt{n}$ predicts, so a researcher using that formula would report confidence they have not earned.

Survey D fails both ways at once, which is what most convenience samples do in practice.

> *What did the first survey have that the others did not?*

Two things, and each failure above corresponds to losing one of them.

> **REMEMBER THIS**
>
> **Identically distributed** — every observation is drawn from the same distribution, and it is the distribution we want to learn about. Survey B violated this: urban households are drawn from a different distribution than the population of Indian households.
> **Independent** — knowing the value of one observation tells us nothing about another. Survey C violated this: neighbours share a locality, a labour market and often an occupation.
> Together the two are abbreviated **i.i.d.**

These assumptions are not mathematical decoration. Sections 2.4 and 2.5 derive the two properties that justify using the sample mean, and each derivation uses exactly one of them — in one identifiable line. Break the assumption and the property it supports disappears, which is precisely what the four surveys show.

> **WATCH OUT**
>
> No amount of mathematical sophistication repairs a badly collected sample. If the households you surveyed are not drawn from the population you are asking about, there is no formula that recovers the ones you missed.
> Good inference begins with study design, not with statistics.

2.3 How can we judge an estimator?

> *If μ is never known, how can we ever tell whether an estimator is any good?*

We now know what a sample must satisfy. The original question remains: of the many rules available, why the sample mean?

The obstacle is the one identified at the start. To see whether an estimator lands near the truth we would have to know the truth, and in any real application we do not. If we survey Indian households we do not know true average household income; if we sample firms we do not know true average productivity. There is nothing to compare against.

The way around this is to build a laboratory. We construct one artificial population whose properties are known exactly, and study repeated sampling from it. Because the true mean is known in advance, every sample mean can be measured against it. Nothing is being discovered about the world here — the point is to watch a procedure work under conditions where its performance is visible.

The population

Take a deck of playing cards, scored A = 1, J = 11, Q = 12, K = 13, and 2 through 10 at face value.

```
deck <- rep(1:13, times = 4)

c(N = length(deck), mu = mean(deck), sigma = sd(deck))
```

```
#>      N     mu  sigma
#> 52.000  7.000  3.778
```

This population contains 52 values. Its mean is exactly $\mu = 7$ and its standard deviation is $\sigma = 3.78$. Both are known with certainty, which is exactly what real data never offers.

One sample from it

Draw ten cards and compute their average.

```
set.seed(7)
mean(sample(deck, size = 10, replace = TRUE))
```

```
#> [1] 5.7
```

The result is not 7. Draw again and it would differ again — sampling variation, as before. What is new is that we can see how far this sample strayed, because we know where it should have been.

> **WATCH OUT**
>
> Note `replace = TRUE`: each card is returned to the deck before the next draw.

> That is what makes the draws **independent**, in the sense of the previous section. Drawing without replacement changes what remains after each card is removed, so the draws are dependent and the sample mean varies slightly less than theory predicts.
> It is also the better model of what we usually mean. Surveying 2,000 people does not meaningfully deplete a population of a billion.

A thousand samples

One sample tells us little about a procedure. Repeating the exercise a thousand times shows the whole sampling distribution — and, for the first time, shows it alongside the truth it is supposed to be tracking.

```
set.seed(7)
deck_means <- replicate(1000, mean(sample(deck, size = 10, replace =
↪  TRUE)))

c(mean_of_sample_means = mean(deck_means),
  sd_of_sample_means   = sd(deck_means))
```

```
#> mean_of_sample_means   sd_of_sample_means
#>                6.932                1.201
```

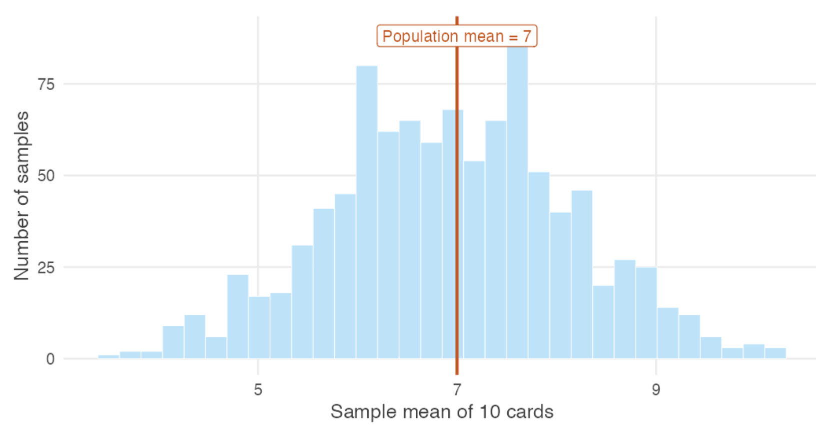

Figure 2.2: One thousand sample means, each computed from 10 cards drawn from a population whose mean is exactly 7.

Two features of this picture are worth separating, and the next two sections take them one at a time: where the distribution sits, and how wide it is.

2.4 Is the sample mean centred on the truth?

> *The sample means cluster around 7. Is that a coincidence of this simulation?*

No individual sample produced 7. Yet the average of the thousand sample means is 6.932 — essentially the population value.

It is not a coincidence, and it does not depend on the deck. It follows from the definition of the sample mean and one line of algebra.

WHERE THIS COMES FROM

$$\begin{aligned} E(\bar{X}) &= E\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) \\ &= \frac{1}{n}\Big(E(X_1) + E(X_2) + \cdots + E(X_n)\Big) \\ &= \frac{1}{n}\underbrace{(\mu + \mu + \cdots + \mu)}_{n \text{ times}} \\ &= \frac{n\mu}{n} \;=\; \mu \end{aligned}$$

The third line is the one that matters. It replaces every $E(X_i)$ with the same μ, and it is entitled to do so only because the observations are **identically distributed** — each drawn from the population we are asking about. Averaging n of them then leaves μ unchanged.

Look back at Survey B. Its observations were identically distributed among themselves, but not drawn from the population of interest, so the μ substituted in that third line was the urban mean rather than the national one. The algebra was never wrong; it answered a question nobody had asked.

This is the first of the two properties promised earlier, and it now has a name.

REMEMBER THIS

An estimator is **unbiased** if its expected value equals the parameter it estimates. The sample mean is unbiased for the population mean:

$$E(\bar{X}) = \mu$$

Unbiasedness is a statement about the procedure, not about any particular estimate. It says the rule is not systematically too high or too low. It does not say your $\bar{x}$ is close to μ — every one of the thousand samples above missed.

2.5 Why do larger samples fluctuate less?

> *Unbiasedness cannot be the whole story. What does a larger sample actually buy?*

Unbiasedness alone is worth remarkably little. Consider the rule "take the first observation and ignore the rest." It is perfectly unbiased — $E(X_1) = \mu$ — and nobody would use it, because it never improves. Collect a million households and it still reports one household's income.

So the second property has to concern spread. Run the deck experiment across a range of sample sizes.

```r
set.seed(7)

sizes <- c(5, 10, 20, 40, 80)
results <- t(sapply(sizes, function(n) {
  m <- replicate(2000, mean(sample(deck, size = n, replace = TRUE)))
  c(n = n, mean = mean(m), sd = sd(m), theoretical_sd = sd(deck) / sqrt(n))
}))

round(results, 3)
```

```
#>       n  mean    sd theoretical_sd
#> [1,]  5 6.932 1.688          1.690
#> [2,] 10 7.022 1.165          1.195
#> [3,] 20 7.009 0.845          0.845
#> [4,] 40 7.019 0.597          0.597
#> [5,] 80 6.996 0.415          0.422
```

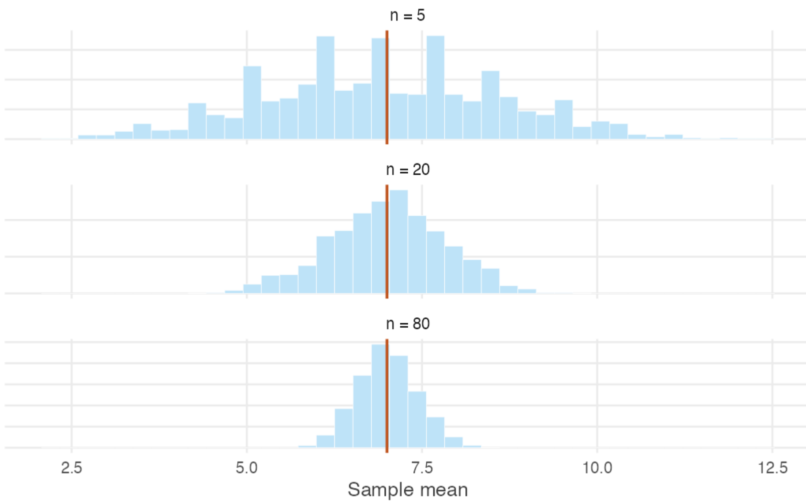

Figure 2.3: Sampling distributions of the mean for growing sample sizes. The centre never moves. The spread collapses.

The second column confirms the previous section: every sample size is centred on 7, and even samples of five are centred correctly. Centring does not improve with n, because it was never wrong.

The third column is what changes. At $n = 5$ the sample means have a standard deviation of about 1.69; at $n = 80$ it is about 0.42. Compare it against the fourth column: the observed spread matches $\sigma/\sqrt{n}$ almost exactly at every sample size.

WHERE THIS COMES FROM

$$\begin{aligned}
\mathrm{Var}(\bar{X}) &= \mathrm{Var}\left(\frac{1}{n}(X_1 + \cdots + X_n)\right) \\
&= \frac{1}{n^2}\,\mathrm{Var}(X_1 + \cdots + X_n) \\
&= \frac{1}{n^2}\,\underbrace{(\sigma^2 + \sigma^2 + \cdots + \sigma^2)}_{n \text{ times}} \\
&= \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}
\end{aligned}$$

The second line uses $\mathrm{Var}(aX) = a^2\mathrm{Var}(X)$. The third line is the one that matters: the variance of a sum equals the sum of the variances only when

> the observations are **independent**. This is the single point in the entire argument at which independence is required.

Survey C is now explained. Its households were drawn from the right population, so it stayed centred — but they were not independent, the third line above did not hold, and the true spread was more than three times what $\sigma/\sqrt{n}$ claimed. A researcher who applied the formula anyway would have reported a precision the data did not contain.

The square root of this variance is used so constantly that it has its own name.

> **REMEMBER THIS**
>
> The **standard error** of the sample mean is the standard deviation of its sampling distribution:
>
> $$\mathrm{se}(\bar{X}) = \sqrt{\mathrm{Var}(\bar{X})} = \frac{\sigma}{\sqrt{n}}$$
>
> It measures how much the sample mean moves from sample to sample. A small standard error means that whatever answer we happened to get, a different sample would have given something similar.

The word *error* is unfortunate, and worth guarding against. It does not measure a mistake, and it is not the distance between your estimate and the truth — that distance is unknowable. Like unbiasedness, it is a property of the procedure.

2.6 Why does the sample mean become more reliable?

> *Two properties, established separately. What do they amount to together?*

We can now answer the question this unit opened with.

The sample mean is used almost universally because its sampling distribution sits in the right place and tightens as the sample grows. It is centred on μ at every sample size, and the width of that distribution, $\sigma/\sqrt{n}$, falls towards zero as n increases. A distribution centred on the correct value and becoming ever narrower must concentrate on that value. Large samples are therefore increasingly likely to produce averages close to μ.

That last sentence is the Law of Large Numbers, which Section 1.5 introduced as a description of observed behaviour. It is no longer a description. Averages settle down because the procedure that produces them is unbiased and increasingly precise, and there is nothing else to the result.

Both halves are load-bearing, and it is worth seeing why neither would do alone.

An estimator that was centred but whose spread never shrank would remain permanently unreliable — the "use the first observation" rule from the previous section, which stays honest and stays useless however much data you give it.

An estimator whose spread shrank around a value that was not μ would be worse still. It would become *reliably wrong*, converging with increasing confidence on the incorrect answer, and every additional observation would harden the mistake rather than expose it. Survey B is exactly this: had the urban-only surveyor collected ten thousand households instead of a hundred, their estimates would have clustered even more tightly around a number that was still ₹6,800 too high.

> **REMEMBER THIS**
>
> The sample mean earns its place through two properties together:
>
> $$E(\bar{X}) = \mu \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}$$
>
> Centred on the truth, and increasingly precise. Neither is sufficient on its own, and both depend on assumptions about how the sample was collected.

2.7 Why is precision expensive?

> *If larger samples are better, why does anyone stop collecting data?*

One practical consequence deserves its own section, because it shapes how survey work is actually done.

Return to the square root. Doubling the sample size does not halve the standard error; it divides it by $\sqrt{2} \approx 1.41$. To halve the standard error, the sample must grow *fourfold*.

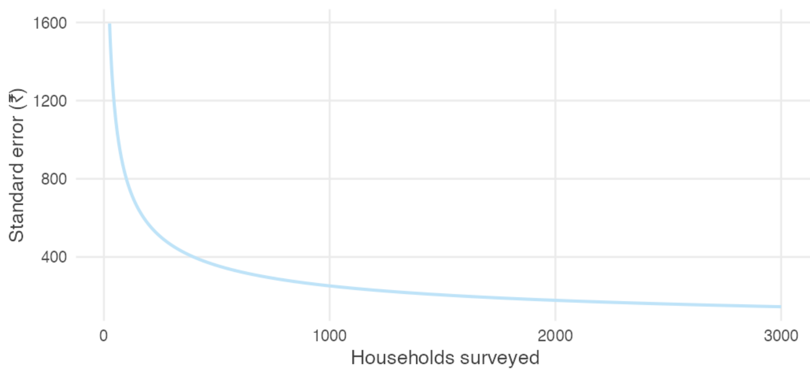

Figure 2.4: The standard error of the sample mean against sample size. Almost all of the available precision is bought in the first few hundred observations.

Precision is therefore expensive, and grows more expensive the more of it you buy. Each additional household costs the same to interview as the last, but contributes less than the last. Going from 100 households to 400 halves the standard error; going from 400 to 1,600 halves it again, at four times the cost.

This is why national surveys run to a few thousand households rather than a few hundred thousand. Past a certain point the additional precision is not worth the additional money, and the money is better spent on the things no sample size can fix — better questions, better interviewers, better coverage of the households that are hardest to reach.

That last point is worth holding onto. The failures in Section 2.2 were all failures of design, and none of them would have been helped by a larger sample. Survey B with ten thousand urban households is still wrong. When a result is disappointing, the instinct is to ask for more data; often the honest answer is that more of the same data will not help.

2.8 Summary

REMEMBER THIS

- A population has fixed but unknown **parameters**.
- An **estimator** is a rule; an **estimate** is the result of applying it to one sample.
- Since the truth is never observed, estimators are judged by their behaviour across repeated samples, not by any single estimate they produce.

- A sample speaks for a population only if it is drawn **independently** and from **the population of interest**. Neither assumption is decoration: break the first and the spread is far wider than any formula predicts, break the second and the estimates are centred on the wrong number while often looking more precise than the honest ones.
- Under random sampling the sample mean is **unbiased**: $E(\bar{X}) = \mu$.
- Its **standard error**, $\sigma/\sqrt{n}$, falls as $1/\sqrt{n}$ — so larger samples give more precise estimates, at a rising price per unit of precision.
- Centred and increasingly precise together are what make the sample mean trustworthy, and together they are the Law of Large Numbers.

Everything in this unit has been about choosing a good estimator. The next unit asks how to use that estimator to make statements about an unknown population from a single observed sample.

2.9 Practice problems

YOUR TURN

These are for learning. Work each one before opening the solution; the answer is worth very little if you have not first tried to produce it yourself. The question bank that follows is for testing yourself and has no answers at all.
Several problems treat `data/wages-india-synthetic.csv` as a *population* of 20,000 wage earners. This is artificial, and deliberately so — it is the same device as the deck of cards, at realistic Indian magnitudes. The dataset is simulated but calibrated to IHDS-II estimates, so the returns to schooling and the urban premium in it are close to the real ones.

Concept checks

Short questions. A sentence or two is enough.

2.1 A survey of 800 households reports mean monthly income of ₹18,400. Which part of that sentence is the estimator, and which is the estimate?

SOLUTION

The estimator is the sample mean — the rule “add the 800 incomes and divide by 800.” The estimate is ₹18,400, the number that rule returned

from this particular sample.
The estimator exists before any data are collected. The estimate exists only afterwards.

2.2 Why can a statistician never check whether a particular estimate is close to the truth?

SOLUTION

Because checking would require knowing the population mean, and if the population mean were known there would have been no reason to conduct the survey. The truth is precisely what the estimate is trying to discover.
This is why the subject evaluates procedures rather than results.

2.3 Consider the estimator "ignore the data entirely and always report ₹50,000." Is it unbiased? What is its standard error?

SOLUTION

Its standard error is zero — it returns the same number from every possible sample, so it does not vary at all.
It is unbiased if and only if the population mean happens to be exactly ₹50,000, which we have no way of knowing. In general it is biased by whatever amount μ differs from ₹50,000.
The problem is worth doing because it separates the two properties completely. Perfect precision is compatible with being arbitrarily wrong, and a small standard error is therefore no evidence that an estimate is any good.

2.4 In Section 2.2, Survey B produced estimates with a *smaller* spread than Survey A. Why is that alarming rather than reassuring?

SOLUTION

Because the small spread came from surveying households that resemble one another, not from surveying more of India. The estimates agreed closely with each other while agreeing on a number ₹6,800 above the truth.
A researcher who saw only their own survey would observe a tight, well-behaved sample and conclude that the estimate was reliable. Spread measures agreement among samples drawn the same way; it says nothing about whether that way was the right one.

2.5 A report states: "mean earnings were ₹54,000 with a standard error of

₹10,000, so the true figure is within ₹10,000 of our estimate." What is wrong with the second half of that sentence?

SOLUTION

The standard error is not the distance between the estimate and the truth. That distance is unknowable, since the truth is unobserved.
The standard error describes the *procedure*: it says that repeating this survey would produce sample means with a standard deviation of about ₹10,000. Any particular estimate may lie much closer to μ than that, or much further from it.

Explain

A paragraph each.

2.6 Explain why unbiasedness alone is not a good enough reason to use an estimator. Give an example of an unbiased estimator nobody would use.

SOLUTION

Unbiasedness says the estimator is right *on average across repeated samples*. It says nothing about how far any single sample strays, and a rule can be centred correctly while being wildly variable.
The clearest example is "use the first observation and ignore the rest." Since $E(X_1) = \mu$, it is exactly unbiased. It is also useless, because its standard error is σ regardless of sample size. Collecting a million households does not improve it by any amount at all.
Usefulness requires a second property: the spread must fall as the sample grows. Unbiasedness places the distribution correctly; the variance is what makes the placement worth anything.

2.7 An interviewer travels to six villages and surveys thirty households in each. Which i.i.d. assumption is threatened, which of the two derivations in this unit stops working, and what happens to the results?

SOLUTION

Independence is threatened. Households in a village share a labour market, a cropping pattern, and often an occupation, so once the first few are surveyed the rest carry much less new information.
The derivation that fails is the variance one. Its third line replaces $\mathrm{Var}(X_1 + \cdots + X_n)$ with $\sigma^2 + \cdots + \sigma^2$, and that step requires independence. Nothing goes wrong with $E(\bar{X}) = \mu$, whose third line uses only identical distribution.

So the estimates remain centred on the population mean — the design is not biased — but they vary far more from survey to survey than $\sigma/\sqrt{n}$ predicts. A researcher who applied that formula would report a precision the data does not contain. In Survey C the true spread was more than three times the formula's claim.

2.8 Explain why the Law of Large Numbers needs *both* properties of the sample mean. Describe what would go wrong if each were missing in turn.

SOLUTION

The Law of Large Numbers says large samples are increasingly likely to produce averages close to μ. That requires the sampling distribution both to sit in the right place and to narrow.

Without shrinking variance, the estimator stays centred but never improves. "Use the first observation" is centred on μ forever and never gets closer to it, so no sample size makes it likely to land near the truth.

Without correct centring, shrinking variance actively makes matters worse. The estimator becomes *reliably wrong* — it converges with increasing confidence on the incorrect value, and every extra observation hardens the error rather than exposing it. The urban-only survey is exactly this case: ten thousand urban households would have produced a tighter cluster around a number still ₹6,800 too high.

Numerical problems

Pen and paper. Calculators are fine; R is not needed until the next section.

2.9 In the population of wage earners used below, annual earnings have standard deviation $\sigma = ₹100{,}141$.

i. Find the standard error of the sample mean for $n = 100$, $n = 400$ and $n = 2{,}500$.
ii. How large a sample is needed for the standard error to fall below ₹5,000?
iii. A colleague proposes to improve an $n = 400$ survey by adding 100 households. By how much does the standard error fall?

SOLUTION

(i) se $= \sigma/\sqrt{n}$:

$$n = 100: \quad \frac{100{,}141}{10} = ₹10{,}014$$
$$n = 400: \quad \frac{100{,}141}{20} = ₹5{,}007$$
$$n = 2{,}500: \quad \frac{100{,}141}{50} = ₹2{,}003$$

(ii) Require $100{,}141/\sqrt{n} < 5{,}000$, so $\sqrt{n} > 20.03$ and $n > 401.1$. A sample of **402** suffices.
(iii) At $n = 500$ the standard error is $100{,}141/\sqrt{500} = ₹4{,}479$, down from ₹5,007 — a fall of about ₹528, or roughly 11%, for 25% more fieldwork. The returns are already thin at this sample size.

2.10 A researcher with n observations proposes the estimator $T = \frac{1}{2}(X_1 + X_n)$: average the first and last household only.

i. Show that T is unbiased.
ii. Find $\text{Var}(T)$.
iii. Using $\sigma = ₹100{,}141$, compare the standard error of T with that of the sample mean of all 100 observations.
iv. What does this show about unbiasedness?

SOLUTION

(i) $E(T) = \frac{1}{2}(E(X_1) + E(X_n)) = \frac{1}{2}(\mu + \mu) = \mu$. Both observations are drawn from the same population, so each has expectation μ.
(ii) Using $\text{Var}(aX) = a^2\text{Var}(X)$ and independence,

$$\text{Var}(T) = \tfrac{1}{4}(\sigma^2 + \sigma^2) = \frac{\sigma^2}{2}$$

which is σ^2/n with $n = 2$ — unsurprising, since T *is* the sample mean of two observations.
(iii) $\text{se}(T) = 100{,}141/\sqrt{2} = ₹70{,}810$, against $100{,}141/10 = ₹10{,}014$ for the full sample mean. T is about seven times more variable.
(iv) That unbiasedness is a low bar. T is exactly as unbiased as $\bar{X}$ and vastly worse, because it discards 98 of the 100 observations collected. Choosing between unbiased estimators is decided on variance.

2.11 A survey costs ₹900 per household in fieldwork. The current design surveys 400 households.

i. What does the current survey cost?

ii. What would it cost to halve the standard error?
iii. The funder instead offers to double the budget. By what factor does the standard error fall?

SOLUTION

(i) $400 \times$ ₹$900 =$ ₹$360{,}000$.
(ii) Halving the standard error requires four times the sample, so 1,600 households at a cost of ₹1,440,000 — an extra ₹1,080,000 to buy one factor of two.
(iii) Doubling the budget buys 800 households. The standard error falls by a factor of $\sqrt{2} \approx 1.41$, a reduction of about 29%.
Twice the money does not buy twice the precision, and this gets worse the larger the survey already is. It is often a better use of the second ₹360,000 to improve coverage of hard-to-reach households than to double the sample — because the errors that arise from missing those households are not reduced by sample size at all.

R exercises

2.12 Treat `data/wages-india-synthetic.csv` as a population of 20,000 wage earners. Compute its true mean and standard deviation. Then draw 2,000 samples of 100 workers each, and check both properties of the sample mean against theory.

SOLUTION

```r
wages <- read.csv("data/wages-india-synthetic.csv")
earn  <- wages$annual_earnings

mu_pop    <- mean(earn)
sigma_pop <- sd(earn)

set.seed(3)
sim <- replicate(2000, mean(sample(earn, 100, replace = TRUE)))

round(c(population_mean = mu_pop,
        centre_of_estimates = mean(sim),
        theoretical_se = sigma_pop / sqrt(100),
        observed_se = sd(sim)))
```

```
#>     population_mean centre_of_estimates      theoretical_se
 ↪  observed_se
#>               54181               54072               10014
 ↪  9939
```

The estimates are centred on the population mean, and their spread is close to $\sigma/\sqrt{n}$. Both properties hold.
This population is severely right-skewed — its mean is 54,181 against a median of 24,220 — yet neither property needed the population to be well behaved. Unbiasedness followed from each observation having expectation μ, and the variance result from independence. Shape never entered either derivation.

2.13 Repeat the previous exercise sampling only from urban workers, as an interviewer working exclusively in cities would. Report the centre and spread, and say which assumption has been broken.

SOLUTION

```
urban_earn <- earn[wages$residence == "Urban"]

set.seed(3)
sim_urban <- replicate(2000, mean(sample(urban_earn, 100, replace =
 ↪  TRUE)))

round(c(population_mean = mu_pop,
        centre_of_estimates = mean(sim_urban),
        observed_se = sd(sim_urban)))
```

```
#>     population_mean centre_of_estimates         observed_se
#>               54181              101270               15099
```

The estimates are centred near ₹102,000 against a true population mean of about ₹54,000 — almost double. The assumption broken is *identically distributed*: urban workers are drawn from a different distribution than Indian wage earners as a whole, so the μ appearing in the third line of the $E(\bar{X})$ derivation is the urban mean, not the national one.
Note also that no amount of extra data would help. The estimator is converging efficiently on the wrong number.

2.14 Draw 2,000 sample means for $n = 25, 100, 400$ and $1{,}600$ from the same population, and plot the observed standard error against the theoretical $\sigma/\sqrt{n}$. Does the $1/\sqrt{n}$ rule hold?

SOLUTION

```
set.seed(3)
ns <- c(25, 100, 400, 1600)

se_tab <- data.frame(
  n = ns,
  observed = sapply(ns, function(n) sd(replicate(4000,
  ↪   mean(sample(earn, n, replace = TRUE))))),
  theory   = sigma_pop / sqrt(ns)
)

round(se_tab)
```

```
#>      n observed theory
#> 1   25    20143  20028
#> 2  100    10024  10014
#> 3  400     4961   5007
#> 4 1600     2510   2504
```

```
ggplot(se_tab, aes(n)) +
  geom_line(aes(y = theory), colour = book_palette$reject, linewidth
  ↪   = 0.9) +
  geom_point(aes(y = observed), size = 2.6, colour =
  ↪   book_palette$fill) +
  labs(x = "Sample size", y = "Standard error (₹)",
       subtitle = "Line: theoretical. Points: simulated.") +
  theme_book()
```

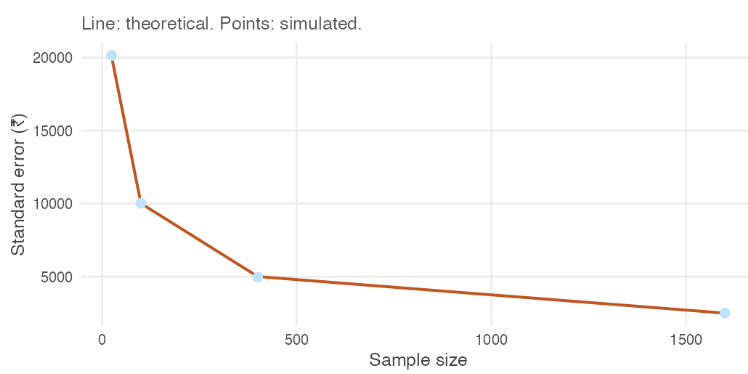

Figure 2.5: Observed and theoretical standard errors for growing sample sizes.

The points sit on the curve at every sample size, agreeing with theory to

within about one per cent. Quadrupling n halves the standard error each time, exactly as $1/\sqrt{n}$ requires.
Note the `replace = TRUE`, which matters more here than it did with the deck. Drop it and the observed spread at $n = 1{,}600$ comes out roughly five per cent *below* theory, because 1,600 is eight per cent of this population of 20,000 — sample a substantial fraction of a finite population without replacement and the draws are no longer independent, so $\sigma/\sqrt{n}$ stops being right. Real surveys of India are nowhere near this fraction, which is why the assumption is harmless in practice and not in a simulation.

2.15 Do the two properties depend on the population being well behaved? Compare the sampling distribution of the mean at $n = 10$ and $n = 400$ for this heavily skewed earnings population, and comment on what changes and what does not.

SOLUTION

```
set.seed(3)
sk <- do.call(rbind, lapply(c(10, 400), function(n)
  data.frame(n = factor(paste("n =", n), levels = c("n = 10", "n =
  ↪  400")),
             m = replicate(4000, mean(sample(earn, n, replace =
              ↪  TRUE))))))

ggplot(sk, aes(m)) +
  geom_histogram(bins = 60, fill = book_palette$fill, colour =
  ↪  "white",
                 linewidth = 0.1) +
  geom_vline(xintercept = mu_pop, colour = book_palette$reject,
             linewidth = 0.8) +
  facet_wrap(~ n, scales = "free") +
  scale_x_continuous(labels = function(x) format(x, big.mark = ","))
  ↪  +
  labs(x = "Sample mean of annual earnings (₹)", y = NULL) +
  theme_book() +
  theme(axis.text.y = element_blank())
```

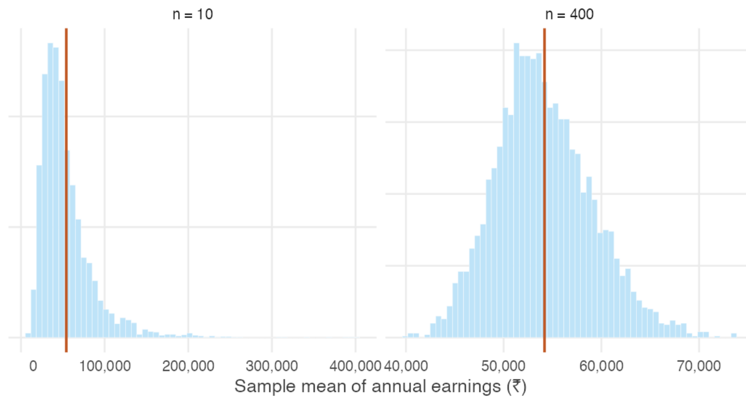

Figure 2.6: Sampling distribution of mean earnings at two sample sizes. The red line is the population mean.

Both panels are centred on the population mean. Unbiasedness holds at $n = 10$ just as at $n = 400$, and it required nothing of the population's shape — the derivation used only that each observation has expectation μ.

What changes is the *shape*. At $n = 10$ the sampling distribution inherits the population's long right tail and is clearly skewed. At $n = 400$ it is close to symmetric. That is the Central Limit Theorem from Section 1.6 at work, and it is a reminder that "large enough" depends on the population: for earnings data, $n = 30$ would not have been remotely sufficient.

2.16 Using the same population, compare three estimators of μ across 2,000 samples of 50: the sample mean, the sample median, and the midpoint of the smallest and largest observations. Which would you use, and why?

SOLUTION

```r
set.seed(3)
midrange <- function(x) (min(x) + max(x)) / 2

comp <- replicate(2000, {
  s <- sample(earn, 50, replace = TRUE)
  c(mean = mean(s), median = median(s), midrange = midrange(s))
})

round(cbind(centre = rowMeans(comp),
            bias   = rowMeans(comp) - mu_pop,
            spread = apply(comp, 1, sd)))
```

```
#>          centre   bias spread
#> mean      53896   -285  14550
#> median    24653 -29528   5399
#> midrange 250648 196467 172624
```

Only the sample mean is centred on μ. The median sits far below it, because this population is strongly right-skewed and the median estimates the population *median*, not the mean — it is an excellent estimator of a different parameter. The midrange is wildly variable, since it depends entirely on the two most extreme observations drawn and discards the other 48.

The sample mean is the only one of the three that is unbiased for μ and tightens dependably as the sample grows.

Worth noting, though: if the question of interest were "what does a typical wage earner make?", the median would be the better summary. Choosing an estimator requires first being clear about which parameter you actually want.

2.10 Question bank

WATCH OUT

No answers are given for this section, by design. These questions are for testing yourself once the practice problems are complete. There are no hints and no worked solutions — exactly as in an examination.

Multiple choice

QB 2.1 Which of the following is an estimate rather than an estimator?

a. $\bar{X}$
b. The rule "take the median of the sample"
c. ₹18,400
d. $\sigma/\sqrt{n}$

QB 2.2 An estimator is unbiased if

a. every estimate it produces equals the parameter
b. its expected value equals the parameter
c. its variance falls as the sample size grows

d. the sample is drawn at random

QB 2.3 A population has $\sigma = 60$. For the standard error of the sample mean to equal 2, the sample size must be

a. 30
b. 120
c. 900
d. 3,600

QB 2.4 A survey samples only households with telephones. Which is the most accurate description of the consequence?

a. The estimates will be centred correctly but too variable
b. The estimates will be centred on the wrong value, and a larger sample will not fix it
c. The estimates will be centred on the wrong value, but a larger sample will fix it
d. Nothing goes wrong provided the sample is large

QB 2.5 The variance derivation Var$(\bar{X}) = \sigma^2/n$ uses independence at exactly one step. Which one?

a. Taking the constant $1/n$ outside the variance
b. Writing Var$(aX) = a^2$Var(X)
c. Replacing the variance of the sum with the sum of the variances
d. Cancelling n against n^2

QB 2.6 Two surveys estimate the same quantity. Survey X has a standard error of ₹4,000; Survey Y has a standard error of ₹1,000. Which statement follows? *(select all that apply)*

a. Survey Y's estimate is closer to the truth
b. Survey Y's estimates vary less across repeated sampling
c. Survey Y used a larger sample, if both were drawn the same way
d. Survey Y is unbiased

QB 2.7 Doubling a survey's sample size reduces its standard error by a factor of

a. 4
b. 2
c. $\sqrt{2}$
d. It depends on σ

Short reasoning

QB 2.8 A student writes: "the standard error tells us how much our estimate differs from the population mean." Correct the statement, and explain in one sentence why the original cannot be right.

QB 2.9 Explain why an estimator with zero variance is not necessarily a good estimator. Give an example.

QB 2.10 A researcher collects 500 households by standing outside a government hospital on a weekday morning and interviewing whoever passes. Identify at least two distinct problems with this sample, and state for each whether it affects the centre of the estimates, their spread, or both.

QB 2.11 Two rules for estimating μ from n observations are proposed: the sample mean, and the average of the first $n/2$ observations. Both are unbiased. Explain how you would decide between them, and what the answer is.

QB 2.12 An economist reports that increasing a survey from 1,000 to 1,200 households "substantially improved the precision of our estimates." Evaluate that claim quantitatively.

QB 2.13 Section 2.2 showed a badly designed survey producing estimates with a *smaller* spread than a well-designed one. Explain how this is possible, and what it implies about using a reported standard error to judge whether a study is trustworthy.

Numerical

QB 2.14 A population of farm households has mean annual income $\mu =$ ₹84,000 and standard deviation $\sigma =$ ₹$36{,}000$.

i. Find the standard error of the sample mean for a survey of 144 households.
ii. What sample size would be needed to reduce the standard error to ₹1,500?
iii. Fieldwork costs ₹1,200 per household. What is the additional cost of the sample in (ii) relative to the one in (i)?

QB 2.15 For a population with variance σ^2, consider the estimator $W = \frac{1}{3}X_1 + \frac{2}{3}X_2$ computed from two independent observations.

i. Show that W is unbiased.
ii. Find Var(W).
iii. Compare it with the variance of the ordinary sample mean of the same two observations. Which is preferable, and what general principle does this illustrate?

QB 2.16 A state has 40 districts. A survey selects 4 districts at random and interviews 250 households in each, reporting a standard error of $\sigma/\sqrt{1000}$.

i. State the assumption this calculation requires.
ii. Explain why it is unlikely to hold here.
iii. Say whether the reported standard error is likely to be too large or too small, and why that direction is the dangerous one.

QB 2.17 Using `data/wages-india-synthetic.csv` as a population, write R code to estimate how large a sample is needed before the standard error of mean annual earnings falls below ₹4,000. Verify your answer by simulation as well as by formula, and explain any discrepancy between the two.

2.11 Looking ahead

We opened this unit by asking why statistics has settled on the sample mean, and the answer is now in hand. It is centred on the truth, its variability shrinks as the sample grows, and larger samples therefore tend to produce estimates closer to the population mean.

But every argument in this unit relied on something we never have in practice.

We drew a thousand samples from the same population. We watched the sampling distribution emerge. We could see exactly where our estimate sat among all the other estimates that might have been obtained.

Real research is nothing like that.

A survey is conducted once, often at considerable expense, and produces exactly one sample. There is no second sample to compare against, no visible sampling distribution, and no way to know how far the estimate lies from the population value.

Yet researchers routinely write things like

> The average household income is between ₹23,500 and ₹24,500.

How can anyone say that after observing a single sample?

The next unit answers that question. It shows how the sampling distribution, although invisible, can still be used to measure the precision of an estimate and to state a range of population values consistent with it.

Chapter 3

How Much Can One Sample Tell Us?

The previous unit established that the sample mean is a good estimator of the population mean. Across repeated samples it is centred on the truth, and larger samples produce more precise estimates.

There is a practical problem with how we established it.

Researchers do not observe repeated samples. A survey is conducted once, producing exactly one sample and one estimate. The sampling distribution that justified the sample mean is invisible from inside a single dataset.

So the question of this unit is the one in its title.

> How much can a single sample tell us about an unknown population?

This unit develops the tools for answering it. We begin by separating an estimate from the uncertainty surrounding it — a distinction the previous unit made informally and this one makes precise. We then show how the standard error measures the precision of an estimate, and how a confidence interval turns that precision into a range of plausible values for the population parameter.

> **REMEMBER THIS**
>
> By the end of this unit you should be able to say not only *what* a sample suggests about a population, but *how confident* that suggestion deserves to be — and why an estimate reported without a measure of its uncertainty is an incomplete answer.

The unit stops short of one thing. Evaluating a specific claim about a population — testing whether the average really is ₹20,000 — is the subject of Unit 4. What we build here turns out to be most of the machinery that unit needs.

3.1 One sample, one number

> *We have one sample and one number. What are we entitled to say?*

The previous unit answered an important question: why do statisticians use the sample mean to estimate the population mean? We found that it has two desirable properties. First, it is unbiased — over repeated samples it is centred on the true population mean. Second, its variability decreases as the sample size grows. Larger samples therefore tend to produce estimates closer to the truth.

There was, however, one feature of that argument which is impossible in practice.

To judge the behaviour of the sample mean we repeatedly drew thousands of samples from a population whose true mean we already knew. By watching the sampling distribution emerge, we could see that the sample mean was centred correctly and became more precise as the sample grew.

Real surveys do not work that way.

A labour force survey, a household expenditure survey, a school attendance audit — each is conducted once. We observe one sample, calculate one estimate, and the data collection ends. We never see the thousands of alternative samples that might have been drawn.

Suppose a survey of students reports an average attendance of 26.1 classes.

Is the population mean exactly 26.1? Almost certainly not. How close is it? The data alone do not tell us.

Simply reporting the estimate is therefore incomplete. We already know that every sample mean varies from sample to sample; some will be larger than the population mean and others smaller. Without a measure of that variability, the estimate has no context.

The natural response is to report the estimate together with its standard error. For the sample mean we found that

$$\mathrm{se}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

This is a useful result, and it immediately raises another difficulty. The formula contains the population standard deviation σ, which is unknown. If we are estimating the population mean because it is unknown, we can hardly assume we know the population standard deviation.

One missing quantity has become two.

Nor are all questions about population means. Many surveys seek to estimate proportions instead.

- What proportion of households live below the poverty line?
- What proportion of workers are employed in the informal sector?
- What proportion of children are fully vaccinated?

These require estimators of a different kind.

Finally, even once we can calculate both an estimate and its standard error, we still face the central question of statistical inference. Suppose two surveys both estimate average attendance at 26.1 classes. One has a standard error of 0.4; the other has a standard error of 3.2.

How should those numbers change our confidence in the estimate? How uncertain is "uncertain"?

Answering that requires moving beyond a single number. Rather than reporting only a point estimate, we will construct **interval estimates** — ranges of plausible values for the unknown parameter, together with a statement of how reliable such ranges are.

> **REMEMBER THIS**
>
> This unit develops the three ingredients needed for statistical inference.
>
> 1. How to estimate unknown population variability.
> 2. How to extend estimation from means to proportions.
> 3. How to combine point estimates and standard errors into confidence intervals.

By the end of the unit we will be able to replace a statement such as

> The average attendance is 26.1 classes.

with one that is considerably more informative:

> We estimate the average attendance to lie between 24.4 and 27.8 classes, with 95% confidence.

That is the central goal of this unit.

3.2 Estimating the standard error

> *The standard error is the number we want. Where is it supposed to come from?*

At the end of the previous section we identified the first obstacle to statistical inference. The standard error of the sample mean is

$$\mathrm{se}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

but the population standard deviation σ is unknown.

This should not surprise us. Every quantity in that formula except n belongs to the population, while all we possess is a sample. If the population mean must be estimated, the population standard deviation must be estimated too.

The obvious strategy is the one that already worked. Replace the unknown population quantity by its sample analogue. For the mean this gave

$$\mu \quad \longrightarrow \quad \bar{X}$$

For the variance we might therefore try

$$\sigma^2 \quad \longrightarrow \quad \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

It is an appealing estimator. It measures the average squared distance of the observations from their centre, exactly as the population variance does.

Unfortunately it is wrong. Not disastrously wrong, but systematically wrong: it underestimates the population variance.

Before trying to understand why, let us see the problem.

A statistical laboratory

As in the previous unit, we need a population where the truth is known. A fair die is ideal, because its variance is known exactly:

$$\mu = 3.5 \qquad \sigma^2 = \frac{35}{12} = 2.9167$$

Draw samples of ten rolls, compute the sum of squared deviations from the sample mean, and divide it by four different numbers. If any of these is a good estimator, its average across thousands of samples should land on 2.9167.

```r
set.seed(11)
n <- 10

ss <- replicate(20000, {
  d <- sample(1:6, n, replace = TRUE)
  sum((d - mean(d))^2)                    # the sum of squared deviations
})

divisors <- c(n, n - 1, n - 2, n - 3)

tab <- data.frame(
  divisor      = divisors,
  avg_variance = sapply(divisors, function(k) mean(ss / k)),
  avg_sd       = sapply(divisors, function(k) mean(sqrt(ss / k)))
)

round(tab, 4)
```

```
#>   divisor avg_variance avg_sd
#> 1      10        2.630  1.601
#> 2       9        2.922  1.688
#> 3       8        3.287  1.790
#> 4       7        3.757  1.914
```

```r
c(true_variance = 35 / 12, true_sd = sqrt(35 / 12))
```

```
#> true_variance       true_sd
#>         2.917         1.708
```

Notice what the table shows. Dividing by n is always too small. Dividing by $n-2$ is too large, and $n-3$ is worse. Only one divisor consistently lands on the truth, and it is not n.

It is $n-1$, which gives 2.9219 against a true 2.9167.

That observation raises two questions, and the rest of this section answers them in turn.

1. Why does dividing by n underestimate the variance?
2. Why is the correction exactly $n-1$, rather than $n-2$ or some other number?

The second question is the one that matters. Any downward adjustment would move the estimate in the right direction; only one gets the size right.

Why dividing by n fails

The reason is that $\bar{x}$ is computed from the very observations whose spread we are measuring.

The sample mean sits, by construction, in the middle of the sample. It is the value that makes $\sum(x_i - \bar{x})^2$ as small as it can possibly be — no other number, including the true μ, gives a smaller total. So when we measure the spread of the sample around its own mean, we measure it around the most flattering point available, and the answer comes out too small.

THE IDEA IN PLAIN WORDS

Once $\bar{x}$ is known, the deviations are not free to take any values. They must sum to zero:

$$\sum_{i=1}^{n}(x_i - \bar{x}) = 0$$

Given the first $n-1$ of them, the last is determined. Only $n-1$ of the n deviations carry independent information about the spread.
The quantity $n-1$ is called the **degrees of freedom**: the number of observations, less the number of quantities estimated from them along the way.

Why the correction is exactly n − 1

The argument above explains the direction of the error. It does not explain its size, and the size is what distinguishes $n-1$ from $n-2$. For that we need to take an expectation.

WHERE THIS COMES FROM

Write $\widetilde{S}^2$ for the divide-by-n version, and simplify it first. Expanding the square,

$$
\begin{aligned}
\widetilde{S}^2 &= \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2 \\
&= \frac{1}{n}\left(\sum_{i=1}^{n} X_i^2 - 2\bar{X}\sum_{i=1}^{n} X_i + n\bar{X}^2\right) \\
&= \frac{1}{n}\left(\sum_{i=1}^{n} X_i^2 - 2n\bar{X}^2 + n\bar{X}^2\right) \qquad \text{since } \textstyle\sum X_i = n\bar{X} \\
&= \frac{1}{n}\sum_{i=1}^{n} X_i^2 \;-\; \bar{X}^2
\end{aligned}
$$

Now take expectations:

$$E(\widetilde{S}^2) = \frac{1}{n}E\left(\sum_{i=1}^{n} X_i^2\right) - E(\bar{X}^2)$$

Both terms follow from rearranging the definition of variance, $\text{Var}(X) = E(X^2) - [E(X)]^2$, into $E(X^2) = \text{Var}(X) + [E(X)]^2$.
For a single observation, $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$, so

$$E(X_i^2) = \sigma^2 + \mu^2 \qquad \Longrightarrow \qquad E\left(\sum_{i=1}^{n} X_i^2\right) = n(\sigma^2 + \mu^2)$$

For the sample mean, the previous unit established $E(\bar{X}) = \mu$ and $\text{Var}(\bar{X}) = \sigma^2/n$, so

$$E(\bar{X}^2) = \frac{\sigma^2}{n} + \mu^2$$

Substituting both,

$$
\begin{aligned}
E(\widetilde{S}^2) &= \frac{1}{n}\,n(\sigma^2 + \mu^2) - \left(\frac{\sigma^2}{n} + \mu^2\right) \\
&= \sigma^2 + \mu^2 - \frac{\sigma^2}{n} - \mu^2 \\
&= \sigma^2\left(1 - \frac{1}{n}\right) \;=\; \sigma^2\,\frac{n-1}{n}
\end{aligned}
$$

So the divide-by-n estimator is too small by exactly the factor $(n-1)/n$ — not approximately, and not only for dice. Multiplying it by $n/(n-1)$ removes the factor, and multiplying by $n/(n-1)$ is the same as dividing by $n-1$.

> **REMEMBER THIS**
>
> The **sample variance** and **sample standard deviation** are
>
> $$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \qquad\qquad s = \sqrt{s^2}$$
>
> s^2 is an unbiased estimator of σ^2: $E(s^2) = \sigma^2$. The adjustment from n to $n-1$ is called **Bessel's correction**.
> In R these are `var()` and `sd()`, both of which already use $n-1$.

Check the theory against the simulation. With $n = 10$ the predicted shortfall is $(n-1)/n = 0.9$, so dividing by n should return $0.9 \times 2.9167 = 2.625$. The simulation returned 2.6297.

When the population mean is known

Occasionally μ is known even though σ^2 is not — a machine calibrated to a specification, or a process with a target value. Then no degrees of freedom are spent estimating the centre, nothing is flattered, and the divisor is n:

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$$

Note that this uses μ, not $\bar{x}$. Since $\bar{x}$ minimises the sum of squared deviations, the total measured from μ is *larger* than the total measured from $\bar{x}$ — which is precisely why it can be divided by the larger n and still come out right.

> **THE IDEA IN PLAIN WORDS**
>
> The rule is easier to remember as a question than as two formulas. How many quantities did I have to estimate from this sample before I could measure its spread?
> Estimated the centre from the data: divide by $n-1$. Knew the centre already: divide by n.

Back to the standard error

We can now return to where the section began. The standard error of the sample mean is $\sigma/\sqrt{n}$, and with σ unknown we substitute s:

$$\widehat{\text{se}}(\bar{X}) = \frac{s}{\sqrt{n}}$$

This is the quantity reported by every statistical package. Note the hat: it is itself an estimate, depending on the random sample through s, and therefore carrying sampling variability of its own.

That substitution looks harmless. It is not, and Section 3.5 measures the damage.

> **GOING A LITTLE FURTHER**
>
> s^2 is unbiased for σ^2, but s is **not** unbiased for σ. Taking a square root does not preserve unbiasedness.
> The evidence is already in the table above. Look at the `avg_sd` column: the $n-1$ row gives 1.6876 against a true σ of 1.7078 — close, but low, and it stays low however many samples are run. On that column no divisor in the table is right; the best would fall between $n-1$ and $n-2$, and would depend on the distribution.
> The bias is small, shrinks quickly with n, and is not worth correcting in practice. What is worth taking away is that unbiasedness attaches to a specific estimator of a specific parameter, and does not survive being passed through a nonlinear function.

3.3 How do we estimate a proportion?

> *What proportion of Indian wage earners work in urban areas? That is not an average. Do we need new machinery?*

Many important quantities are proportions. The share of households below the poverty line, the share of children enrolled in school, the share of a workforce that is female, the unemployment rate. None of these is obviously an average, and it would be reasonable to expect a separate theory.

There is none, and the reason is a small trick.

Write down, for each person in the sample, a variable that equals 1 if they have the characteristic and 0 if they do not.

```r
wages <- read.csv("data/wages-india-synthetic.csv")

is_urban <- as.numeric(wages$residence == "Urban")
head(is_urban, 20)
```

```
#>  [1] 0 0 1 1 0 1 0 0 0 1 0 1 0 1 1 0 0 0 0 1
```

The *average* of that column is the *proportion* of people with the characteristic. Adding up a column of ones and zeros counts the ones, and dividing by n turns the count into a share.

```r
c(mean_of_indicator = mean(is_urban),
  proportion_urban  = sum(wages$residence == "Urban") / nrow(wages))
```

```
#> mean_of_indicator  proportion_urban
#>            0.2684            0.2685
```

> **REMEMBER THIS**
>
> A proportion is a mean. The **sample proportion**
>
> $$\hat{p} = \frac{x}{n}$$
>
> where x is the number of observations with the characteristic, is exactly the sample mean of a variable coded 1 and 0.
> Everything established about $\bar{X}$ therefore applies to $\hat{p}$ unchanged. It is unbiased, its variability falls with $\sqrt{n}$, and the Central Limit Theorem makes it approximately normal in large samples.

This is worth pausing on, because it is the first time in the book that a result has been obtained for free. We did not prove anything new. We noticed that a question we had not met was a question we had already answered, wearing different clothes.

How much does a sample proportion vary?

One thing does simplify. For a general population we need σ from somewhere. For a 0/1 variable, σ is determined by p itself.

> **WHERE THIS COMES FROM**
>
> Let X_i equal 1 with probability p and 0 with probability $1-p$. Then
>
> $$E(X_i) = 1 \cdot p + 0 \cdot (1-p) = p$$
>
> so the population mean of the indicator *is* the population proportion. For the variance, note that $X_i^2 = X_i$, since $0^2 = 0$ and $1^2 = 1$. Hence $E(X_i^2) = p$ and

$$\text{Var}(X_i) = E(X_i^2) - [E(X_i)]^2 = p - p^2 = p(1-p)$$

Applying $\text{Var}(\bar{X}) = \sigma^2/n$ from the previous unit,

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n} \qquad \text{se}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

In practice p is unknown — it is what we are estimating — so we substitute $\hat{p}$:

$$\widehat{\text{se}}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

```r
set.seed(8)
sample_urban <- sample(is_urban, 500)

p_hat <- mean(sample_urban)
se_p  <- sqrt(p_hat * (1 - p_hat) / 500)

round(c(p_hat = p_hat, se = se_p, true_p = mean(is_urban)), 4)
```

```
#>  p_hat     se true_p
#> 0.2820 0.0201 0.2684
```

THE IDEA IN PLAIN WORDS

The quantity $p(1-p)$ is largest at $p = 0.5$ and falls away towards zero at either end.

That has a sensible meaning. A population evenly split is the hardest to pin down, because samples can land either way. A population where 99% share a characteristic is easy — almost every sample says 99%. Rare and near-universal things are estimated precisely; genuinely contested ones are not.

The consequence for survey design is that $p = 0.5$ is the worst case, and a sample large enough for a 50-50 split is large enough for anything.

3.4 From a point estimate to an interval

> *Our best guess is a single number. How often is a single number right?*

Everything we have estimated so far has been a **point estimate**: $\bar{x}$ for the population mean μ, s^2 for the population variance σ^2, and $\hat{p}$ for the population proportion p. Each is a single number extracted from the sample and presented as our best estimate of an unknown population quantity.

A point estimate, however, is only a guess. It is the best guess the sample can provide, but it is almost certainly not exactly equal to the population parameter.

This is worth stating explicitly because it runs against our intuition. Suppose a survey reports that students attend an average of 26.1 classes. It is tempting to conclude that the population mean attendance is 26.1. In reality that conclusion is almost certainly wrong. The estimate is not poorly calculated or misleading — it is simply unlikely to equal the true population mean exactly. The real question is not whether the estimate is correct, but **how far from the truth it is likely to be**.

To explore that question, let us return to a setting where we know the truth. Treat the full dataset of 680 students as the population, whose mean attendance is 26.147 classes. We repeatedly draw random samples of 40 students and compute the sample mean for each one. Comparing these estimates with the known population mean allows us to see how much a point estimate typically differs from the truth.

```r
students <- read.csv("data/attendance-grades.csv")
att      <- students$attendance
mu_pop   <- mean(att)

set.seed(2)
guesses <- replicate(10000, mean(sample(att, 40, replace = TRUE)))

error <- abs(guesses - mu_pop)

c(exactly_right    = sum(guesses == mu_pop),
  typical_miss     = median(error),
  average_miss     = mean(error),
  missed_by_over_1 = mean(error > 1))
```

```
#>    exactly_right     typical_miss     average_miss missed_by_over_1
#>           0.0000           0.5779           0.6895           0.2464
```

Not one of the ten thousand estimates was exactly right. The typical estimate missed the population mean by about 0.58 classes, and roughly a quarter of them were off by more than a full class.

None of this indicates that anything has gone wrong. Each of those ten thousand estimates was computed correctly, from an honestly drawn sample, using an estimator we spent the whole of the previous unit justifying. Missing the truth is simply what point estimates do.

> *So how do we say something that stands a chance of being true?*

By widening the guess.

Instead of naming a single value, we name a *range* and claim the truth lies inside it. A range can be right, and how often it is right is something we can control: the wider we make it, the more often it succeeds.

Notice that this is not a retreat into vagueness. We are not saying "somewhere around 26" because we have given up on precision. We are making a **calculated guess** — calculated in the literal sense, because the previous units told us exactly how much $\bar{X}$ varies from sample to sample. It is $\sigma/\sqrt{n}$. Knowing that, we can widen the guess by a specific, defensible amount and state in advance how often such a range will contain the truth.

> **REMEMBER THIS**
>
> A **point estimate** names one value. It is the best single guess available, and it is essentially never exactly correct.
> An **interval estimate** names a range, together with the proportion of the time such ranges contain the parameter. It trades precision for the ability to be right.
> The standard error is what makes the second possible. Without a measure of how much the estimate moves from sample to sample, there would be no principled way to choose the width.

The rest of this section works out how wide the range must be.

Building the interval

Suppose a survey of 40 students reports an average attendance of 26.4 classes.

This is our point estimate of the unknown population mean. It is the single best guess the sample can provide.

But how much faith should we place in that guess?

From the previous unit we know that if the survey were repeated many times, each sample would produce a slightly different mean. Some would fall above the population mean and others below it. The point estimate we happen to hold is only one of many values that could have been observed.

Fortunately we also know something about how those sample means behave.

The Central Limit Theorem tells us that, for reasonably large samples, the sampling distribution of the sample mean is approximately normal. Its centre is the population mean μ, and its spread is the standard error $\sigma/\sqrt{n}$.

Suppose previous studies tell us the population standard deviation is $\sigma = 5.45$ classes. With $n = 40$,

$$\text{se}(\bar{X}) = \frac{5.45}{\sqrt{40}} = 0.86$$

So although we do not know the value of the population mean, we do know how much sample means typically fluctuate around it.

The normal distribution turns that fluctuation into a probability. Its probabilities are determined entirely by distance from the centre: about 68% of values lie within one standard deviation, about 95% within 1.96, and about 99.7% within three.

The middle one is what we need.

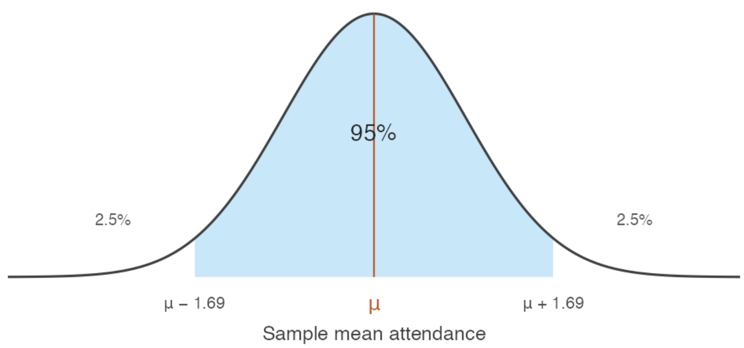

Figure 3.1: The sampling distribution of the sample mean. The shaded region holds 95% of all sample means that could be drawn.

The shaded region represents the 95% of all possible sample means lying within 1.96 standard errors of the population mean. If we repeatedly drew samples of 40 students, roughly 95 out of every 100 sample means would fall inside this region and about 5 would fall outside.

The relationship between distance and probability is worth exploring rather than memorising.

Notice what happens as the distance grows. Widening the region always captures more of the distribution, but never all of it, and the gain slows sharply. Moving from 1.96 to 2.58 standard errors buys the last four percentage points, and no finite distance reaches 100%.

Returning to our survey: with a standard error of 0.86, the distance corresponding to 95% is

Table 3.1: How much of the distribution lies within a given number of standard errors of the centre. The web edition of this book presents the same relationship as a slider.

from centre	inside	outside
0.500	38.29%	61.71%
1.000	68.27%	31.73%
1.500	86.64%	13.36%
1.960	95.00%	5.00%
2.000	95.45%	4.55%
2.500	98.76%	1.24%
2.576	99.00%	1.00%
3.000	99.73%	0.27%
3.500	99.95%	0.05%

$$1.96 \times 0.86 \approx 1.69 \text{ classes}$$

Reversing the question

Now comes the key idea.

If almost all sample means lie within 1.69 classes of the population mean, then the sample mean we actually observed is usually no more than 1.69 classes away from the truth. So instead of asking

> How far is the sample mean from the population mean?

we ask

> Given the sample mean we observed, which values of the population mean are consistent with it?

The answer is obtained by taking the observed sample mean and moving 1.69 classes in each direction:

$$26.4 - 1.69 \qquad \text{to} \qquad 26.4 + 1.69$$

which gives

$$(24.71,\ 28.09)$$

Rather than offering a single best guess, we now report a range of plausible values for the unknown population mean.

WHERE THIS COMES FROM

The reversal above was stated in words. Here it is as algebra, which is where both the $\pm$ formula and the number 1.96 come from.

For large enough samples the sample mean is approximately normal with mean μ and standard deviation $\sigma/\sqrt{n}$. Standardising it — subtracting its mean and dividing by its standard deviation — gives a quantity with a distribution containing no unknowns at all:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

This is the step that makes everything else possible. Whatever the population, whatever its units, whatever n, the standardised sample mean has the *same* distribution — so a single number read off that distribution serves every problem of this kind.

That number is the solution to $P(|Z| \le c) = 0.95$, which is $c = 1.96$. In R it is `qnorm(0.975)`, the value with 2.5% of the distribution above it.

Now unwind the standardisation. Each line is the previous one with the same operation applied to all three parts of the inequality:

$$\begin{aligned}
P(|Z| \le 1.96) &= 0.95 \\
P\left(-1.96 \le \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \le 1.96\right) &= 0.95 \quad \text{def. of } Z \\
P\left(-1.96\frac{\sigma}{\sqrt{n}} \le \bar{X}-\mu \le 1.96\frac{\sigma}{\sqrt{n}}\right) &= 0.95 \quad \times\,\sigma/\sqrt{n} \\
P\left(-\bar{X}-1.96\frac{\sigma}{\sqrt{n}} \le -\mu \le -\bar{X}+1.96\frac{\sigma}{\sqrt{n}}\right) &= 0.95 \quad -\bar{X} \\
P\left(\bar{X}-1.96\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{X}+1.96\frac{\sigma}{\sqrt{n}}\right) &= 0.95 \quad \times\,(-1)
\end{aligned}$$

The last step multiplies through by -1, which reverses both inequality signs and therefore swaps the two endpoints.

Nothing has been added and nothing has been assumed beyond the Central Limit Theorem. The same statement now has μ in the middle instead of $\bar{X}$, which is what turns a fact about how sample means scatter into an interval around the one we happened to observe.

WATCH OUT

The second line above must be read as a **single event** — Z lying between two bounds — and not as two separate probabilities added together. It is tempting to split $P(|Z| \leq 1.96)$ into $P(Z \leq 1.96) + P(-Z \leq 1.96)$. Those two probabilities are each 0.975, and they sum to 1.95, which is not a probability at all. The two events overlap almost entirely, and adding them counts the overlap twice.

Only the constant changes for other confidence levels. Replacing 1.96 by $z_{\alpha/2}$, the value with $\alpha/2$ of the distribution above it, gives the general result at once — 1.645 for 90%, 2.576 for 99%.

This interval is called a 95% **confidence interval**, and the remaining question is what that phrase actually means — a question the next section answers.

REMEMBER THIS

A **confidence interval** for μ, when σ is known, is

$$\bar{x} \ \pm \ z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}}$$

The quantity $z_{\alpha/2}\,\sigma/\sqrt{n}$ is the **margin of error**. The confidence level is $1-\alpha$.

Confidence	α	$z_{\alpha/2}$
90%	0.10	1.645
95%	0.05	1.960
99%	0.01	2.576

In R these come from `qnorm(1 - alpha/2)` — there is no need for a printed table.

```
x_bar <- 26.4
sigma <- 5.45
n     <- 40

for (level in c(0.90, 0.95, 0.99)) {
  z  <- qnorm(1 - (1 - level) / 2)
  ci <- x_bar + c(-1, 1) * z * sigma / sqrt(n)
  cat(sprintf("%2.0f%%  z = %.3f   [%.2f, %.2f]   width %.2f\n",
              100 * level, z, ci[1], ci[2], diff(ci)))
}
```

```
#> 90%  z = 1.645   [24.98, 27.82]   width 2.83
```

```
#> 95%  z = 1.960   [24.71, 28.09]   width 3.38
#> 99%  z = 2.576   [24.18, 28.62]   width 4.44
```

Higher confidence buys a wider interval, and this is the central trade-off of the whole method. A 99% interval is more likely to contain μ precisely because it commits to less. An interval certain to contain the truth would run from minus infinity to infinity and tell us nothing at all.

What does 95% actually refer to?

The interval either contains μ or it does not. Nothing is random about it once the sample is drawn — μ is a fixed number, and so are the two endpoints sitting in front of us.

The 95% describes the **procedure**. If we drew sample after sample and constructed an interval each time, about 95% of those intervals would contain μ. Since we cannot know whether ours is among them, we report the interval and its coverage rate together.

We can watch this happen. Treat the 680 students as a population, so μ is known, and construct 100 intervals from 100 samples.

```r
students <- read.csv("data/attendance-grades.csv")
att      <- students$attendance

mu_pop    <- mean(att)
sigma_pop <- sqrt(mean((att - mean(att))^2))

set.seed(1)
ci_sim <- t(replicate(100, {
  x <- sample(att, 40, replace = TRUE)
  mean(x) + c(-1, 1) * 1.96 * sigma_pop / sqrt(40)
}))

covers <- ci_sim[, 1] <= mu_pop & ci_sim[, 2] >= mu_pop

d <- data.frame(id = 1:100, lo = ci_sim[, 1], hi = ci_sim[, 2],
                covers = ifelse(covers, "Contains μ", "Misses μ"))

ggplot(d, aes(y = id, xmin = lo, xmax = hi, colour = covers)) +
  geom_errorbarh(height = 0, linewidth = 0.45) +
  geom_vline(xintercept = mu_pop, linewidth = 0.7, colour = "grey20") +
  scale_colour_manual(values = c(book_palette$fill, book_palette$reject),
                      name = NULL) +
  labs(x = "Mean attendance", y = NULL) +
  theme_book() +
  theme(axis.text.y = element_blank(), legend.position = "top")
```

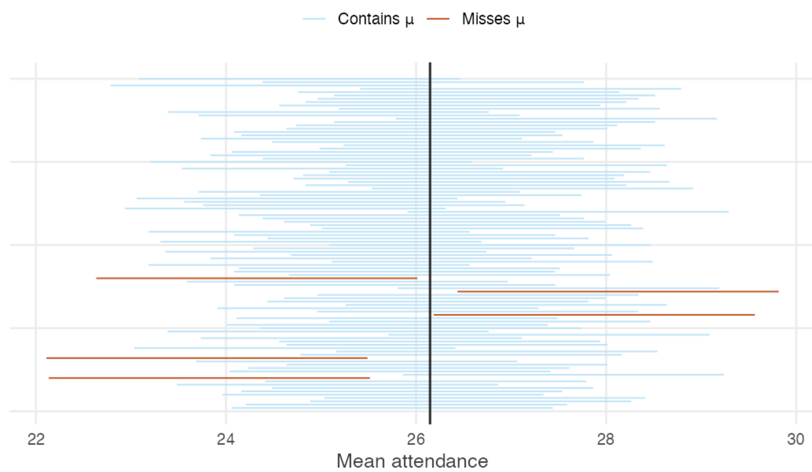

Figure 3.2: One hundred 95% confidence intervals, each from a fresh sample of 40 students. The vertical line is the true population mean.

Of these 100 intervals, 95 contain the population mean and 5 do not. The misses are not mistakes. They are the 5% the method openly budgets for, and a procedure that never missed would be one that had stopped saying anything.

Note also that the misses are not near-misses in any special sense — they are simply the samples whose means happened to fall furthest from μ. Nothing about such a sample looks wrong from the inside. A researcher holding one of those five would see a perfectly ordinary set of forty students.

3.5 What changes when σ is unknown?

> *Every interval so far used σ. We never know it. Can we just use s instead?*

In the previous section we constructed confidence intervals assuming that the population standard deviation σ was known. This allowed us to calculate the standard error exactly,

$$\mathrm{se}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

and to use the normal distribution to determine how far the interval should extend on either side of the sample mean.

In practice the population standard deviation is almost never known. This is not a new problem — Section 3.2 showed how to estimate it with the sample standard deviation s. So a natural question arises.

Can we simply replace σ with s and continue as before?

At first sight the answer appears to be yes. We replace $\sigma/\sqrt{n}$ with $s/\sqrt{n}$ and construct the interval exactly as before.

Unfortunately, that interval is no longer a 95% confidence interval.

Why does the interval change?

The interval developed in the previous section relied on one important fact: the only random quantity in it was the sample mean. The width was fixed, because σ was assumed known.

Once σ is replaced by s, the sample determines both the centre of the interval **and** its width.

Most samples produce reasonable estimates of the population standard deviation, but some do not. Occasionally a sample happens to contain observations that are unusually similar, giving a value of s that is too small, and the resulting interval is then narrower than it should be. At the same time, the sample mean may lie further from the population mean than usual.

These two errors can occur together, and when they do the interval is both badly placed and too narrow to reach the truth. The result is that the interval misses more often than we intended.

How serious is the problem?

The easiest way to find out is by simulation.

Suppose the population is normal with mean 26 and standard deviation 5.45. We repeatedly draw samples of size $n = 10$, estimate the standard deviation with s, and construct intervals in two ways: the first using the familiar normal critical value of 1.96, the second using the appropriate critical value from the t distribution.

```
set.seed(6)
n <- 10

cover <- replicate(50000, {
  x  <- rnorm(n, mean = 26, sd = 5.45)
  se <- sd(x) / sqrt(n)
  c(using_1.96       = abs(mean(x) - 26) < 1.96 * se,
    using_t_critical = abs(mean(x) - 26) < qt(0.975, df = n - 1) * se)
```

```
})

round(rowMeans(cover), 4)
```

```
#>      using_1.96 using_t_critical
#>          0.9181           0.9495
```

The results are striking. Using the normal critical value, the interval captures the population mean only about 91.8% of the time rather than the advertised 95%. One statement in twenty was supposed to be wrong; nearly one in twelve is.

Simply replacing σ by s has made the interval too optimistic.

The t distribution

To restore the intended coverage we must make the interval slightly wider. This is achieved by replacing the normal critical value with one from **Student's t distribution**, which has heavier tails than the normal.

> **REMEMBER THIS**
> When σ is unknown, the confidence interval for μ is
>
> $$\bar{x} \pm t_{n-1,\alpha/2}\frac{s}{\sqrt{n}}$$
>
> where $t_{n-1,\alpha/2}$ is the critical value from the t distribution with $n-1$ degrees of freedom.
> In R: `qt(1 - alpha/2, df = n - 1)`.

Notice that the same quantity $n-1$ appears here as in the definition of the sample variance. Both arise for the same reason: the sample standard deviation is itself estimated from the data, and estimating the centre used up one observation's worth of information.

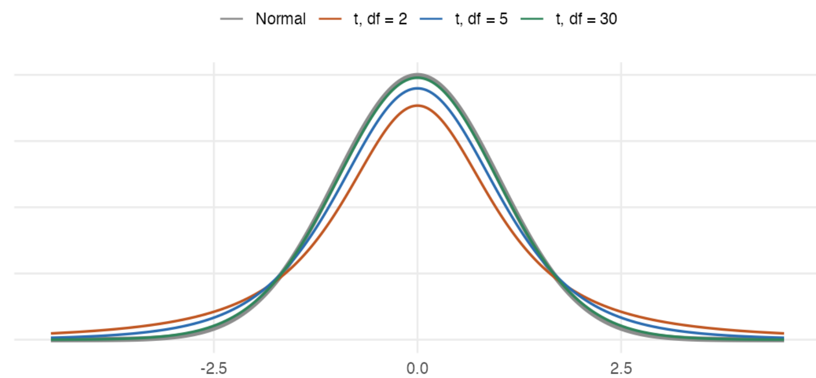

Figure 3.3: The t distribution against the normal. Lower degrees of freedom put more probability in the tails, which is what widens the interval.

The heavier tails are the whole mechanism. Because more of the t distribution's probability sits far from the centre, reaching 95% of it requires travelling further — and travelling further is exactly what widens the interval.

How much difference does it make?

The t distribution depends on the sample size. Small samples carry greater uncertainty about the population standard deviation and therefore need wider intervals. As the sample grows, s becomes a more reliable estimate of σ, the t distribution approaches the normal, and the difference between the two intervals gradually disappears.

```
ns <- c(5, 10, 20, 30, 60, 120)
round(data.frame(n = ns, df = ns - 1,
                 t_critical = qt(0.975, ns - 1),
                 z_critical = 1.96,
                 pct_wider  = 100 * (qt(0.975, ns - 1) / 1.96 - 1)), 2)
```

```
#>     n  df t_critical z_critical pct_wider
#> 1   5   4       2.78       1.96     41.66
#> 2  10   9       2.26       1.96     15.42
#> 3  20  19       2.09       1.96      6.79
#> 4  30  29       2.05       1.96      4.35
#> 5  60  59       2.00       1.96      2.09
#> 6 120 119       1.98       1.96      1.03
```

For a sample of five the correction is substantial — the interval is over 40% wider than the normal one. By the time the sample reaches around a hundred observations the two critical values are almost identical, which is why the distinction is rarely important in large samples and matters a great deal in a study with thirty.

One final point

The t distribution corrects only for the fact that the population standard deviation is unknown.

It does not correct for departures from normality.

If the underlying population is strongly non-normal and the sample is very small, even the t interval may fall short of its advertised coverage. Repeat the coverage experiment above on the real attendance data rather than a normal population, and at $n = 10$ the t interval achieves about 90% rather than 95%. That remaining shortfall has nothing to do with estimating σ. It is the sampling distribution of the sample mean not yet being well approximated by a normal distribution.

> **WATCH OUT**
>
> Two separate issues are in play, and only one of them has been fixed here.
>
> - **Estimating an unknown standard deviation** — addressed by the t distribution.
> - **Approximating the sampling distribution by a normal** — addressed by the Central Limit Theorem, and only when n is large enough for the population in question.
>
> Keeping these distinct is essential for knowing when a confidence interval can be trusted. A t interval computed from twelve observations of a heavily skewed variable is not made trustworthy by the t correction; the correction was never aimed at that problem.

3.6 How large a sample do we need?

> *A survey has not yet been run. How many households should it cover?*

Everything so far took the sample as given. In practice the question usually arrives earlier, from someone who has to budget for the fieldwork.

The margin of error for a mean is

$$E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

which can be solved for n:

$$n = \left(\frac{z_{\alpha/2}\, \sigma}{E} \right)^2$$

The squaring is the $\sqrt{n}$ rule from the previous unit, arriving from the other direction: halving the margin of error quadruples the sample.

For a proportion the same rearrangement gives

$$n = \frac{z_{\alpha/2}^2\, p(1-p)}{E^2}$$

with one difficulty. It requires p, which is what the survey exists to find out. The standard resolution is to use the worst case. Since $p(1-p)$ is maximised at $p = 0.5$, setting $p = 0.5$ gives a sample size large enough whatever the truth turns out to be.

```
sample_size_prop <- function(margin, level = 0.95, p = 0.5) {
  z <- qnorm(1 - (1 - level) / 2)
  ceiling(z^2 * p * (1 - p) / margin^2)
}

data.frame(margin_of_error = c("5%", "3%", "2%", "1%"),
           n_at_95 = sapply(c(0.05, 0.03, 0.02, 0.01), sample_size_prop),
           n_at_90 = sapply(c(0.05, 0.03, 0.02, 0.01), sample_size_prop,
                            level = 0.90))
```

```
#>   margin_of_error n_at_95 n_at_90
#> 1              5%     385     271
#> 2              3%    1068     752
#> 3              2%    2401    1691
#> 4              1%    9604    6764
```

These numbers explain something otherwise puzzling about published surveys. A national opinion poll covering 1.4 billion people typically samples one or two thousand — and the table shows why that is sufficient rather than negligent. The population size does not appear anywhere in the formula. What determines precision is the number of people asked, not the fraction of the country they represent.

WATCH OUT

This calculation answers a narrow question: how large a sample is needed for the standard error to reach a target, *assuming the sample is drawn properly*.
It says nothing about the failures of Section 2.2, and the sample size it returns will not repair any of them. A survey of two thousand urban households is not a survey of India, however carefully the two thousand was computed.

Intervals for a proportion

Collecting the pieces, an interval for a population proportion is

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Note that z appears here rather than t. The t correction exists because σ was estimated separately from μ; for a proportion, $\hat{p}$ determines the standard error on its own, so there is no second estimate to account for.

Suppose that in a random sample of 1,000 secondary school teachers in Karnataka, 518 are women.

```r
x <- 518
n <- 1000

p_hat <- x / n
se    <- sqrt(p_hat * (1 - p_hat) / n)

round(c(p_hat = p_hat, se = se,
        lower = p_hat - 1.96 * se,
        upper = p_hat + 1.96 * se), 4)
```

```
#>  p_hat     se  lower  upper
#> 0.5180 0.0158 0.4870 0.5490
```

So we report that between 48.7% and 54.9% of secondary school teachers in Karnataka are women, with 95% confidence.

That interval contains 0.5, which is worth noticing. The sample proportion is above half, but the data are also consistent with women being a minority of the profession. A headline reading “most secondary school teachers are women” would be going beyond what a sample of a thousand can support.

3.7 When only one direction matters

> *Sometimes we care about one end only. Why pay for the other?*

A confidence interval spends half its error budget at each end. Often only one end is of interest.

A regulator checking whether a pollutant exceeds a limit does not care how low it might be. A funder asking whether attendance is at least 24 classes does not care how high it might be. In these cases the whole of α can be spent on the side that matters, which buys a tighter bound.

REMEMBER THIS

A **one-sided confidence bound** uses z_α rather than $z_{\alpha/2}$:

$$\text{upper bound:} \quad \bar{x} + z_\alpha \frac{\sigma}{\sqrt{n}} \qquad\qquad \text{lower bound:} \quad \bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}}$$

At 95% confidence, $z_\alpha = 1.645$ rather than 1.960.

```r
x_bar <- 26.4; sigma <- 5.45; n <- 40

round(c(two_sided_lower = x_bar - qnorm(0.975) * sigma / sqrt(n),
        two_sided_upper = x_bar + qnorm(0.975) * sigma / sqrt(n),
        one_sided_upper = x_bar + qnorm(0.950) * sigma / sqrt(n),
        one_sided_lower = x_bar - qnorm(0.950) * sigma / sqrt(n)), 3)
```

```
#> two_sided_lower two_sided_upper one_sided_upper one_sided_lower
#>           24.71           28.09           27.82           24.98
```

The one-sided bound is closer to the estimate than the two-sided one at the same confidence level. Nothing has been gained for free: we have given up any claim about the other direction entirely.

WATCH OUT

The choice between one-sided and two-sided must be made **before** looking at the data, on the basis of what the question is.
Computing a two-sided interval, noticing it just fails to exclude some value of interest, and switching to a one-sided bound to obtain a tighter result is not a calculation. It is a way of arranging to be right, and the stated confidence level no longer describes what was done.

3.8 What a confidence interval does not mean

> *Four statements about the interval we just built. Which of them are correct?*

By now you know how to construct a confidence interval and how to interpret it through repeated sampling. Yet confidence intervals are among the most misunderstood ideas in statistics, and experienced researchers interpret them incorrectly often enough that the errors have names.

Consider the following four statements. Decide what you think of each before reading on.

Statement 1: the probability the mean is inside

> There is a 95% probability that the population mean lies between 24.4 and 27.8.

Incorrect. This is the most common misconception.

Once the sample has been collected, the confidence interval is fixed. The population mean is also fixed. Either the interval contains it or it does not — there is no longer any randomness to attach a probability to.

The 95% refers to the *method*, not to this particular interval. If we repeatedly collected samples and constructed intervals in exactly the same way, about 95% of them would contain the true population mean.

The uncertainty lies in the sampling process, not in the value of the population mean.

Statement 2: the spread of the population

> Ninety-five percent of the population lies within this confidence interval.

Incorrect.

A confidence interval is about the location of the population **mean**, not about the spread of individual observations.

Our interval for average attendance runs from 24.4 to 27.8 classes — a few classes wide. Individual students in the same data attended anywhere from almost none of their classes to nearly all of them.

The clearest way to see that these are different quantities is to watch them respond to sample size. As n increases the interval narrows, because the mean is estimated more precisely. The variability among individual students does not change at all.

Statement 3: two intervals that overlap

> The confidence intervals for two groups overlap, so the groups are not significantly different.

Not necessarily.

Two confidence intervals can overlap even when the difference between the two population means is statistically significant.

The reason is straightforward. Each interval describes uncertainty about a single mean. Whether two groups differ depends on the uncertainty surrounding **their difference**, which is a different quantity with a different standard error.

Later in the book we construct confidence intervals for differences between means. Those intervals, and not visual overlap, are the correct basis for comparison.

Statement 4: a wider interval

> A wider confidence interval means the estimate is worse.

Not necessarily.

A wider interval means the estimate is *less precise*, and precision is not the same thing as quality.

A carefully designed survey with a small sample may produce a wide interval simply because the available information is limited. That interval is an honest report of genuine uncertainty.

A poorly designed survey may produce a very narrow interval by repeatedly sampling the wrong population. Section 2.2 showed exactly this: Survey B, which interviewed only urban households, produced estimates with a *smaller* spread than the correct design while centring on a figure ₹6,800 too high. The interval was precise, and precisely wrong.

Confidence intervals measure uncertainty due to **sampling variation**. They do not detect bias arising from poor sampling, non-response, or measurement error.

> **REMEMBER THIS**
>
> A confidence interval tells us what the data can reasonably support about an unknown population parameter.
> It does **not** give the probability that a particular interval contains the truth, it does **not** describe the spread of individual observations, it does **not** settle whether two groups differ by whether they overlap, and it does

> **not** guarantee that the data were collected well.
> Understanding these limitations is as much a part of using the interval as knowing how to calculate it.

3.9 Summary

> **REMEMBER THIS**
>
> - The **sample variance** $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$ is unbiased for σ^2. The divisor is $n-1$ because $\bar{x}$ was estimated from the same data, leaving $n-1$ free deviations. If μ is known, the divisor is n and the deviations are taken from μ.
> - A **proportion is a mean** — of a variable coded 1 and 0 — so every result about $\bar{X}$ transfers to $\hat{p} = x/n$ without new proof.
> - $\mathrm{se}(\hat{p}) = \sqrt{p(1-p)/n}$, largest at $p = 0.5$. Evenly split populations are the hardest to estimate; near-universal ones are easy.
> - A **confidence interval** is $\bar{x} \pm z_{\alpha/2}\,\sigma/\sqrt{n}$ when σ is known, and $\bar{x} \pm t_{n-1,\,\alpha/2}\,s/\sqrt{n}$ when it is not. For a proportion, $\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$.
> - The t distribution compensates for estimating σ from the same sample. It does not compensate for a non-normal population — two different assumptions, one correction.
> - The confidence level is a property of the **procedure**: across repeated samples, that proportion of intervals contains the parameter. A particular interval either contains it or does not.
> - Higher confidence means a wider interval. Halving the margin of error quadruples the required sample, and $n = (z_{\alpha/2}\sigma/E)^2$ makes that explicit before any fieldwork is commissioned.
> - What determines precision is how many people were asked, not what fraction of the population they are. Population size appears in none of these formulas.

Everything in this unit produced a *range* of values consistent with the data. The next unit asks a sharper question: given a specific claim about the population, is our sample consistent with it?

3.10 Practice problems

YOUR TURN

These are for learning. Work each one before opening the solution. The question bank that follows has no answers at all.

Concept checks

3.1 Why does the sample variance divide by $n - 1$ rather than n?

SOLUTION

Because the deviations are measured from $\bar{x}$, which was itself computed from the same observations. The sample mean is the value that makes $\sum(x_i - \bar{x})^2$ as small as possible, so measuring spread around it gives an answer that is too small.
Formally, the deviations must sum to zero, so only $n - 1$ of them are free to vary. Dividing by $n - 1$ corrects the shortfall exactly, making s^2 unbiased for σ^2.

3.2 A dataset records whether each of 900 households owns a bicycle. Why is no new theory needed to estimate the proportion that do?

SOLUTION

Because coding "owns" as 1 and "does not own" as 0 makes the proportion the mean of that column. Summing ones counts the owners, and dividing by 900 turns the count into a share.
So $\hat{p}$ *is* a sample mean, and everything already established — unbiasedness, standard error falling with $\sqrt{n}$, approximate normality by the Central Limit Theorem — applies without modification.

3.3 Explain why a 99% confidence interval is wider than a 95% one.

SOLUTION

Higher confidence requires the interval to succeed more often across repeated samples, and the only way to succeed more often is to claim less. Widening the interval makes it easier to capture μ.
The trade-off is unavoidable. An interval guaranteed to contain the truth would have to span every possible value, and would carry no information.

3.4 For which value of p is a proportion hardest to estimate precisely, and why does that make sense?

SOLUTION

$p = 0.5$, since $p(1-p)$ is maximised there.
It makes sense because an evenly split population is the one where samples can most easily land either way. If 99% of households have electricity, almost any sample says so. If 50% do, the sample proportion has real room to move.
This is why survey planners use $p = 0.5$ when computing sample sizes: it is the worst case, so it is safe whatever the truth turns out to be.

3.5 A student reports a 95% interval of $[24.4, 27.8]$ and writes: "so 95% of students attend between 24.4 and 27.8 classes." What has gone wrong?

SOLUTION

The interval is about the *population mean*, not about individual students. It says the average attendance is plausibly between 24.4 and 27.8. Individual students vary far more widely — the standard deviation is about 5.5 classes, so many students lie well outside that range.
Note also that the interval narrows as n grows, whereas the spread of students does not change at all. The two quantities behave differently, which is a sign they are measuring different things.

Explain

3.6 Explain what the "95%" in a 95% confidence interval refers to, and why it cannot refer to the particular interval you have computed.

SOLUTION

It refers to the procedure. If samples were drawn repeatedly and an interval constructed from each by this method, about 95% of those intervals would contain the population parameter.
It cannot refer to the interval in front of you because nothing about it is random any more. The population mean is a fixed number, and once the sample is drawn, the two endpoints are fixed numbers too. Either μ is between them or it is not; there is no probability left in the situation.
The randomness lived in which sample was drawn, and that was spent. The 95% is a statement about the method's track record, which is why we report the interval and its confidence level together — the level describes how much such statements can generally be trusted.

3.7 Why is the t distribution needed at all, and why does it stop mattering in large samples?

SOLUTION

When σ is known, the only quantity estimated from the sample is $\bar{x}$. When σ is unknown, s is estimated from the same data, so both the centre and the width of the interval are subject to sampling error. A sample with an unusually small s produces an interval that is narrow *and* poorly positioned, and those errors compound. Using 1.96 anyway gives about 92% coverage at $n = 10$ instead of 95%.
The t distribution has heavier tails, widening the interval by exactly enough to restore the advertised coverage.
It stops mattering in large samples because s becomes an increasingly reliable estimate of σ. The critical value falls from 2.776 at $n = 5$ to 1.984 at $n = 100$, approaching 1.96 as the degrees of freedom grow. Once s is essentially known, there is nothing left to correct for.

3.8 A colleague computes a two-sided 95% interval, notices that it barely includes zero, and recomputes it as a one-sided bound so that zero falls outside. Explain the problem.

SOLUTION

The confidence level describes a procedure fixed in advance. Once the choice between one-sided and two-sided is made *after* seeing where the data fell, the procedure being followed is no longer the one whose 95% property was established.
The real procedure here is "compute a two-sided interval, and if it fails to give the answer I want, switch." That procedure covers the truth far less than 95% of the time, because it has two chances to produce the conclusion sought.
One-sided bounds are legitimate when the question is genuinely one-directional — a regulator who cares only about exceeding a limit. What makes them illegitimate is choosing the direction from the data.

Numerical problems

3.9 A sample of 10 professors reported the following hours worked in a week: 48, 22, 19, 65, 72, 37, 55, 60, 49, 28.

i. Estimate the population mean.
ii. Estimate the population variance and standard deviation.
iii. Estimate them again given that the population mean is known to be 46 hours.

iv. Why does (iii) use a different divisor?

SOLUTION

(i) $\bar{x} = 455/10 = 45.5$ hours.
(ii) With μ unknown, deviations are taken from $\bar{x} = 45.5$ and $\sum(x_i - \bar{x})^2 = 3034.5$:

$$s^2 = \frac{3034.5}{9} = 337.17 \qquad s = 18.36$$

(iii) With $\mu = 46$ known, deviations are taken from 46 and $\sum(x_i - \mu)^2 = 3037$:

$$\hat{\sigma}^2 = \frac{3037}{10} = 303.7 \qquad \hat{\sigma} = 17.43$$

(iv) In (ii) the centre was estimated from the sample, costing one degree of freedom. In (iii) it was known, so none was spent.
Notice that the sum of squares is *larger* in (iii) — 3037 against 3034.5 — because $\bar{x}$ minimises that sum and μ is not $\bar{x}$. The larger total divided by the larger n is what keeps both estimates unbiased.

3.10 The average life of a sample of 10 tyres was 28,400 km. Lifetimes are normally distributed with $\sigma = 3{,}300$ km.

i. Construct 90%, 95% and 99% confidence intervals for the mean life.
ii. Find a value that, with 95% confidence, is larger than the population mean.
iii. Find a value that, with 99% confidence, is smaller than the population mean.

SOLUTION

$\sigma/\sqrt{n} = 3300/\sqrt{10} = 1043.6$.
(i)

$$\begin{aligned} 90\%: &\quad 28400 \pm 1.645(1043.6) = [26683,\ 30117] \\ 95\%: &\quad 28400 \pm 1.960(1043.6) = [26355,\ 30445] \\ 99\%: &\quad 28400 \pm 2.576(1043.6) = [25712,\ 31088] \end{aligned}$$

(ii) A one-sided upper bound at 95% uses $z_{0.05} = 1.645$: $28400 + 1.645(1043.6) = 30{,}117$ km.
(iii) A one-sided lower bound at 99% uses $z_{0.01} = 2.326$: $28400 - 2.326(1043.6) = 25{,}973$ km.
Note that (ii) uses 1.645 rather than 1.960: the whole 5% sits in one tail,

because we are making a claim in one direction only.

3.11 In a random sample of 400 death certificates of college students, 98 recorded a motorcycle accident.

i. Estimate the proportion of such deaths due to motorcycle accidents.
ii. Estimate the standard error of that estimate.
iii. Construct a 95% confidence interval.

SOLUTION

(i) $\widehat{p} = 98/400 = 0.245$.
(ii)

$$\widehat{\text{se}}(\widehat{p}) = \sqrt{\frac{0.245 \times 0.755}{400}} = \sqrt{0.000462} = 0.0215$$

(iii) $0.245 \pm 1.96(0.0215) = [0.203,\ 0.287]$.
So between about 20% and 29%, with 95% confidence.

3.12 In a random sample of 1,000 secondary school teachers in Karnataka, 518 identify as women.

i. Construct a 95% confidence interval for the population proportion.
ii. Construct a 90% upper confidence bound.
iii. Can the state conclude that women are a majority of the profession?

SOLUTION

$\widehat{p} = 0.518$ and $\widehat{\text{se}} = \sqrt{0.518 \times 0.482/1000} = 0.0158$.
(i) $0.518 \pm 1.96(0.0158) = [0.487,\ 0.549]$.
(ii) $0.518 + 1.282(0.0158) = 0.538$.
(iii) No. The 95% interval contains 0.5, so the data are also consistent with women being slightly under half of secondary school teachers. The point estimate is above half, but a sample of a thousand cannot separate 51.8% from 50% with confidence.
This is a good illustration of why the interval is reported rather than the point estimate alone. "51.8% of teachers are women" and "between 48.7% and 54.9% are" support very different headlines.

3.13 A state wants to estimate the proportion of households with a functioning water connection to within 2 percentage points at 95% confidence.

i. How many households must be surveyed?

ii. A pilot suggests the proportion is near 0.85. How many are needed now?
iii. Why is (i) larger, and when should it still be preferred?

SOLUTION

(i) With no prior information, use the worst case $p = 0.5$:

$$n = \frac{1.96^2(0.5)(0.5)}{0.02^2} = \frac{0.9604}{0.0004} = 2401$$

(ii) With $p = 0.85$, $p(1-p) = 0.1275$:

$$n = \frac{1.96^2(0.1275)}{0.0004} = 1225$$

Roughly half as many.
(iii) Because $p(1-p)$ is largest at 0.5, so that choice always gives the biggest requirement and therefore a safe one.
It should still be preferred when the pilot is thin, or drawn from a non-representative area, or when the true proportion might differ across the state. If the real figure turns out to be 0.5 and the survey was sized for 0.85, the margin of error will exceed the 2 points promised.

R exercises

3.14 Using `data/attendance-grades.csv`, compute point estimates of the mean, variance and standard deviation of attendance, semester GPA and final score. Then construct 95% confidence intervals for each mean.

SOLUTION

```r
students <- read.csv("data/attendance-grades.csv")
vars     <- c("attendance", "sem_gpa", "final_score")

est <- t(sapply(vars, function(v) {
  x  <- students[[v]]
  n  <- length(x)
  se <- sd(x) / sqrt(n)
  tc <- qt(0.975, df = n - 1)
  c(mean = mean(x), var = var(x), sd = sd(x), se = se,
    lower = mean(x) - tc * se, upper = mean(x) + tc * se)
}))

round(est, 3)
```

```
#>                  mean    var    sd    se  lower  upper
```

```
#> attendance  26.147 29.757 5.455 0.209 25.736 26.558
#> sem_gpa      6.503  3.391 1.841 0.071  6.364  6.641
#> final_score 90.856 21.455 4.632 0.178 90.507 91.205
```

The intervals are narrow because $n = 680$ is large. Note that `var()` and `sd()` already use the $n-1$ divisor, so no manual correction is needed.

3.15 Verify the $n-1$ correction yourself. Treat attendance as a population, draw 20,000 samples of 8, and compare the average of $\frac{1}{n}\sum(x_i - \bar{x})^2$ with the average of $\frac{1}{n-1}\sum(x_i - \bar{x})^2$.

SOLUTION

```
att <- students$attendance
sigma2 <- mean((att - mean(att))^2)

set.seed(21)
n <- 8
out <- replicate(20000, {
  x <- sample(att, n, replace = TRUE)
  c(over_n = mean((x - mean(x))^2), over_n_1 = var(x))
})

round(c(population_variance = sigma2,
        avg_over_n          = mean(out[1, ]),
        avg_over_n_1        = mean(out[2, ]),
        predicted_over_n    = sigma2 * (n - 1) / n), 3)
```

```
#> population_variance          avg_over_n        avg_over_n_1
 ↪  predicted_over_n
#>               29.71               26.18               29.91
 ↪  26.00
```

Dividing by n understates the population variance, and it does so by exactly the predicted factor $(n-1)/n = 7/8$. Dividing by $n-1$ recovers the right answer on average.
The bias is not small at this sample size — about 12% — which is why the correction is built into `var()` rather than left to the user.

3.16 Treat `wages-india-synthetic.csv` as a population and let p be the proportion of workers in urban areas. Draw 1,000 samples of 400, construct a 95% interval from each, and count how many contain the true p.

SOLUTION

```
wages    <- read.csv("data/wages-india-synthetic.csv")
is_urban <- as.numeric(wages$residence == "Urban")
p_true   <- mean(is_urban)

set.seed(4)
ci <- t(replicate(1000, {
  s  <- sample(is_urban, 400, replace = TRUE)
  ph <- mean(s)
  ph + c(-1, 1) * 1.96 * sqrt(ph * (1 - ph) / 400)
}))

c(true_p = p_true,
  coverage = mean(ci[, 1] <= p_true & ci[, 2] >= p_true))
```

```
#>   true_p coverage
#>   0.2684   0.9580
```

Close to the advertised 95%. The method delivers what it promises for a proportion of this size at this sample size.
It would perform less well for a very rare characteristic — with $p = 0.01$ and $n = 400$ the expected number of successes is 4, the normal approximation is poor, and coverage falls short. A common rule of thumb requires both np and $n(1-p)$ to be at least 10.

3.17 Show that the t correction matters. On a normal population with $\mu = 50$ and $\sigma = 12$, compare the coverage of intervals built with 1.96 against those built with the t critical value, for $n = 5, 10, 30$ and 100.

SOLUTION

```
set.seed(30)
cover <- t(sapply(c(5, 10, 30, 100), function(n) {
  r <- replicate(20000, {
    x  <- rnorm(n, 50, 12)
    se <- sd(x) / sqrt(n)
    c(abs(mean(x) - 50) < 1.96 * se,
      abs(mean(x) - 50) < qt(0.975, n - 1) * se)
  })
  c(n = n, using_z = mean(r[1, ]), using_t = mean(r[2, ]))
}))

round(cover, 4)
```

```
#>        n using_z using_t
#> [1,]   5  0.8792  0.9510
```

```
#> [2,]  10  0.9193  0.9486
#> [3,]  30  0.9404  0.9498
#> [4,] 100  0.9484  0.9508
```

The t intervals hold their 95% at every sample size. The z intervals fall well short when n is small, and the gap closes as n grows — by $n = 100$ the two are nearly indistinguishable.
This is the whole case for the t distribution in one table, and also the reason nobody worries about it in large samples.

3.18 How does interval width respond to sample size and confidence level? Using the attendance population, plot the width of the confidence interval for the mean against n, for the 90%, 95% and 99% levels.

SOLUTION

```
sigma_att <- sqrt(mean((att - mean(att))^2))

grid <- expand.grid(n = seq(10, 500, by = 5),
                    level = c(0.90, 0.95, 0.99))
grid$width <- 2 * qnorm(1 - (1 - grid$level) / 2) * sigma_att /
  sqrt(grid$n)
grid$level <- factor(paste0(100 * grid$level, "%"))

ggplot(grid, aes(n, width, colour = level)) +
  geom_line(linewidth = 0.8) +
  labs(x = "Sample size", y = "Interval width (classes)", colour =
    NULL) +
  theme_book() +
  theme(legend.position = "top")
```

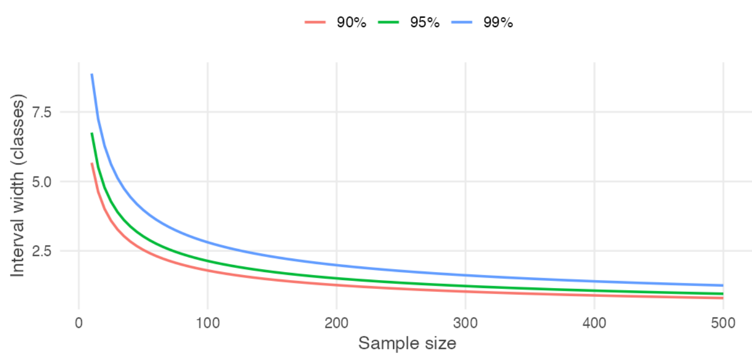

Figure 3.4: Confidence interval width against sample size, at three confidence levels.

Width falls as $1/\sqrt{n}$, steeply at first and then hardly at all. Most of the available precision is bought in the first hundred observations.
The three curves never cross, and the gap between them narrows in absolute terms as n grows. Choosing 99% over 95% is expensive in a small survey and nearly free in a large one.

3.11 Question bank

WATCH OUT

No answers are given for this section, by design. These are for testing yourself once the practice problems are complete.

Multiple choice

QB 3.1 The divisor $n - 1$ in the sample variance is used because

a. it makes the variance larger, which is safer
b. $\bar{x}$ was estimated from the same data, leaving $n - 1$ free deviations
c. the population is finite
d. the sample standard deviation would otherwise be negative

QB 3.2 A 95% confidence interval for μ is $[112, 128]$. Which statement is correct?

a. 95% of the population lies between 112 and 128
b. There is a 95% probability that μ lies between 112 and 128
c. 95% of intervals built this way would contain μ
d. The sample mean has a 95% chance of lying between 112 and 128

QB 3.3 Holding everything else fixed, moving from 95% to 99% confidence

a. narrows the interval
b. widens the interval
c. leaves the width unchanged but shifts the centre
d. requires a larger sample

QB 3.4 For a sample proportion, the standard error is largest when $\widehat{p}$ is closest to

a. 0
b. 0.25
c. 0.5
d. 1

QB 3.5 A survey of 600 households finds 30% own a two-wheeler. The approximate 95% margin of error is

a. 0.4 percentage points
b. 1.9 percentage points
c. 3.7 percentage points
d. 7.5 percentage points

QB 3.6 The t distribution is used instead of the normal when

a. the sample is small
b. σ is estimated from the sample
c. the population is skewed
d. the confidence level exceeds 95%

QB 3.7 To halve the margin of error of a survey, the sample size must be multiplied by

a. $\sqrt{2}$
b. 2
c. 4
d. It depends on the confidence level

QB 3.8 Which of the following would *not* narrow a confidence interval for μ? *(select all that apply)*

a. Increasing the sample size
b. Lowering the confidence level
c. Sampling from a population with smaller σ
d. Sampling only from a subgroup that happens to be homogeneous

Short reasoning

QB 3.9 A report gives a 95% confidence interval for mean household income of [₹18,200, ₹19,600] and concludes: "almost all households in this district earn between ₹18,200 and ₹19,600." Identify the error and state what the interval does say.

QB 3.10 Explain why the population size does not appear in the formula for the margin of error, and why this surprises people.

QB 3.11 Two districts are surveyed. District A's 95% interval for mean income is [₹19,000, ₹23,000]; District B's is [₹22,000, ₹26,000]. A journalist writes that the districts do not differ significantly because the intervals overlap. Evaluate.

QB 3.12 A researcher has a sample of 12 observations from a strongly skewed population and constructs a t interval. State two distinct assumptions being relied on, and say which one the t distribution addresses.

QB 3.13 Explain why s is not an unbiased estimator of σ even though s^2 is unbiased for σ^2, and why this is rarely mentioned in practice.

QB 3.14 A polling organisation reports a margin of error of ±3 points but sampled only households with landline telephones. Explain what the ±3 does and does not cover.

Numerical

QB 3.15 A sample of 16 bags of rice has mean weight 24.6 kg and sample standard deviation 1.2 kg.

i. Construct a 95% confidence interval for the mean weight.
ii. Construct a 99% confidence interval.
iii. The label claims 25 kg. Comment.
iv. How would your answer to (i) change if the standard deviation of 1.2 kg were known to be the population value?

QB 3.16 In a random sample of 250 farm households in a district, 145 reported using irrigation.

i. Estimate the proportion and its standard error.
ii. Construct a 90% confidence interval.
iii. Construct a 95% lower confidence bound.
iv. A scheme guarantees that at least half of households have irrigation. Is the sample consistent with that?

QB 3.17 A state government wishes to estimate mean annual household expenditure to within ₹500 at 95% confidence. Previous rounds suggest $\sigma \approx$ ₹9,000.

i. What sample size is required?
ii. Fieldwork costs ₹1,100 per household. What is the budget?
iii. The budget is capped at ₹15 lakh. What margin of error is achievable?

QB 3.18 Using `data/wages-india-synthetic.csv`, write R code to construct 95% confidence intervals for mean annual earnings separately for urban and rural workers, using samples of 300 from each. Report both intervals, state whether they overlap, and explain carefully what that does and does not tell you about the difference between the two groups.

3.12 Looking ahead

In this unit we learned how to use a sample to estimate an unknown population parameter. Rather than reporting a single number, we reported a **confidence interval** — a range of values consistent with the observed data.

For many problems that is exactly what is needed. Opinion polls report margins of error, medical studies estimate treatment effects, and government surveys publish confidence intervals around unemployment rates, inflation and average income. In each case the question begins with the data:

> Given this sample, what values of the population parameter are plausible?

Many real questions, however, begin somewhere else.

A school claims that its students attend an average of at least 28 classes. A manufacturer advertises that its batteries last 500 hours. A government announces that unemployment has fallen below 5%.

These claims are made *before* any data are collected.

The task is no longer to estimate an unknown quantity. It is to decide whether a sample provides enough evidence to support — or to challenge — a specific claim that someone else has put forward.

This requires turning the question around. Instead of asking

> What values are consistent with the data?

we ask

> Are the data consistent with a particular value?

At first sight these look like different problems. One starts with the data and searches for plausible parameter values; the other starts with a proposed value and asks whether the data agree with it.

As the next unit shows, the two are closely connected. Hypothesis tests and confidence intervals are built from the same ingredients — the sampling distribution, the standard error, and the probability model laid on top of them — but they are pointed in opposite directions.

Confidence intervals ask **what values are plausible**.

Hypothesis tests ask **whether a particular claim is plausible**.

Chapter 4

Hypothesis Testing: One Sample

In the previous unit we learned that every estimate comes with uncertainty. A confidence interval tells us which population values are plausible given our sample.

But many real statistical questions begin with a specific claim. Has average monsoon rainfall changed? Is the unemployment rate below 5%? Does a new drug lower blood pressure?

Rather than asking *which values are plausible*, these questions ask *is this particular claim consistent with the data?* Answering that is the purpose of hypothesis testing.

Consider a claim made about a population. You collect a sample. Your estimate is not exactly what the claim predicted.

Should you conclude that the claim is wrong?

Or is this simply the kind of difference that appears because every random sample is slightly different?

That question is the starting point for hypothesis testing, and it is the engine behind much of modern statistics.

4.1 The problem, before any formulas

The APU administration claims the average height of all students is 165 cm.

You are sceptical. You could settle it by measuring all 3,500 students, but that costs weeks of work. So instead you measure 36 students chosen at random, and you find their average height is **168 cm**.

The claim said 165. Your sample says 168. Is the administration wrong?

> **WATCH OUT**
>
> The tempting answer is "yes, obviously — 168 is not 165." That answer is wrong, and understanding *why* it is wrong is the whole chapter.
> You did not measure everyone. You measured 36 people. Even if the true average really were exactly 165, your particular 36 students would almost never average exactly 165. Some samples come out high, some low. **A gap between the sample and the claim is guaranteed to exist even when the claim is perfectly true.**

Let us actually watch that happen.

We obviously do not know whether the administration is correct. But imagine, just for a moment, that they are. If the true average really were 165 cm, with a standard deviation of 8 cm, what kinds of sample means would we expect to see? Draw a sample of 36 students, ten separate times, and look.

```r
set.seed(42)

ten_samples <- replicate(10, mean(rnorm(n = 36, mean = 165, sd = 8)))
round(ten_samples, 2)
```

```
#>  [1] 165.5 165.1 165.6 163.0 164.8 164.4 165.3 165.4 164.1 164.5
```

Every one of those numbers came from a population whose mean is *exactly* 165. Not one of them is 165. Several are further from 165 than a full centimetre.

So the real question is not "is 168 different from 165?" It obviously is. The real question is:

> **THE IDEA IN PLAIN WORDS**
>
> **If the administration's claim were true, how often would a sample of 36 students come out as far from 165 as ours did?**
> If the answer is "quite often" — then our sample is unremarkable, and we have no grounds to call anyone a liar.
> If the answer is "almost never" — then either something very unusual happened to us, or the administration's claim is not correct. In statistics we usually regard the second explanation as the more convincing one.

We can answer that question by brute force before we do any algebra at all. Let's draw not ten samples but fifty thousand, still assuming the claim is true, and count how many stray at least 3 cm away from 165 (as ours did).

```
set.seed(42)

many_means <- replicate(50000, mean(rnorm(n = 36, mean = 165, sd = 8)))

# How far did our real sample fall from the claim?
observed_gap <- abs(168 - 165)

mean(abs(many_means - 165) >= observed_gap)
```

```
#> [1] 0.02424
```

About 2.4% of the time. So if the administration is telling the truth, a sample like ours turns up roughly once in forty tries.

We have now answered the question using simulation alone, and without naming a single technical term. If the claim were true, a sample mean at least 3 cm away from 165 would appear about once in forty surveys.

That makes our sample unusual. But how unusual is unusual *enough*? Once in forty is rare; would once in ten have been rare enough? Once in a hundred? The rest of this chapter builds a systematic way of answering exactly that — and by the end of it you will be able to obtain the number above exactly, in one line, with no simulation at all.

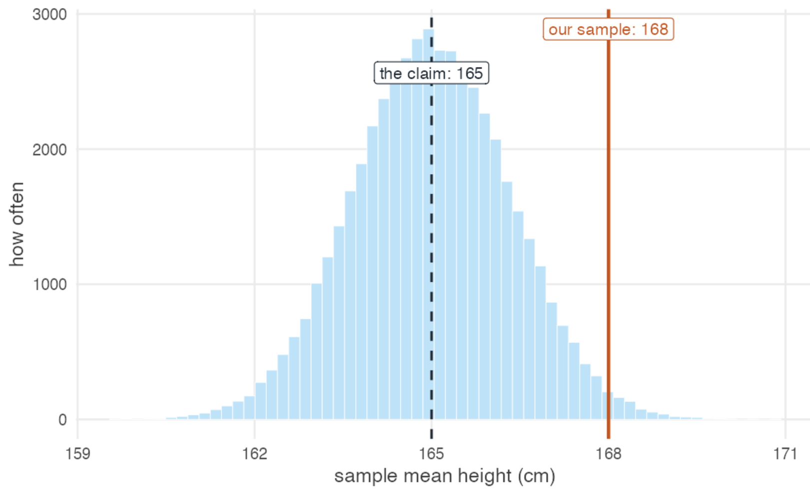

Figure 4.1: Fifty thousand sample means, each from 36 students, drawn from a population whose true mean is 165 cm (dashed line). The solid line marks our observed sample mean of 168 cm. Only about 2.4% of the simulated samples fall at least this far from the claimed value in either direction, which is what makes our sample surprising if the claim is true.

4.2 Naming the two hypotheses

> *We have the reasoning. What are the two positions we are reasoning between?*

The previous section did the whole of a hypothesis test without naming anything. We assumed the administration's claim, asked how often a sample would stray as far from it as ours did, and found the answer was about once in forty.

Let us give the two positions in that argument their names.

We start from a sample $x_1, x_2, \ldots, x_n$ drawn from a population with unknown mean μ and standard deviation σ. The claim under scrutiny is the **null hypothesis**:

$$H_0 : \mu = 165$$

The position we would move to if the claim collapses is the **alternative hypothesis**:

$$H_A : \mu \neq 165$$

Why begin by assuming the claim is true?

At first this looks backwards. If we suspect the claim is wrong, why start by supposing it is right?

Because the assumption is what gives us something to measure against. Notice what we actually did in the previous section: we could only simulate those fifty thousand samples *because* we had a specific value of μ to simulate from. Had we begun with "the mean is not 165," there would have been nothing to draw samples from — not 165 covers 164, and 170, and 3,000, and each would give a different answer.

So we grant the claim, temporarily, and see where it leads. If our sample would be quite ordinary under it, we have no reason to doubt it. If our sample would be extraordinarily unlikely under it, then the claim itself becomes hard to believe.

> **REMEMBER THIS**
>
> The null hypothesis is always the *boring* option — nothing is going on, the claim stands, the difference is zero. We never *prove* it. We only ever fail to knock it down.
> This is why we say "**fail to reject** H_0" and never "accept H_0." A court that acquits has not proven innocence; it has failed to prove guilt.

4.2.1 Which hypothesis does the claim go in?

> *A claim arrives. Does it become H_0 or H_A?*

This is one of the most common places for a beginner to go wrong, and the way out is not to memorise a rule but to ask a question first.

Before writing anything down, ask yourself:

> What result would convince me?

Suppose a report claims that the mean CAT score of applicants is *more than* 499. What would persuade you? Almost everyone answers the same way: a sample mean comfortably **above** 499. A sample averaging 470 would not support the report, and neither would one averaging exactly 499.

That answer has already determined the test. The thing you would find convincing — a mean above 499 — is the thing you are gathering evidence *for*, and a test only ever produces evidence *against* H_0. So the claim has to sit in the alternative:

$$H_0 : \mu \leq 499 \qquad H_A : \mu > 499$$

The direction follows too. You would be convinced by large values, so it is large values that must count against the null, and the test looks in the upper tail.

> **REMEMBER THIS**
>
> Two questions settle every case.
> **What am I trying to gather evidence for?** That goes in H_A. A test can only accumulate evidence against the null, so whatever you hope to establish must be what survives when the null falls.
> **Which direction would count as evidence?** That fixes the inequality in H_A, and with it the tail the test looks in.

Examination questions usually signal this in their phrasing, and the signal is worth recognising.

Table 4.1: How the usual phrasings map onto the two questions above.

The question asks	The claim becomes	Because
"Is there evidence to **support** the claim?"	H_A	The claim is what you are gathering evidence for.
"Can you **reject** the claim?"	H_0	The claim is what you are challenging, and you are asking whether the data can knock it down.

Where the equality goes

One structural rule holds in every case, and it will save you from a good deal of trouble:

> **REMEMBER THIS**
>
> **The equality always belongs in the null hypothesis.**
> H_0 takes $=$, $\leq$ or $\geq$. H_A takes $\neq$, $<$ or $>$, and never equality.

The reason is the one from earlier in this section. Every test begins by assuming one specific value of μ and computing how likely our sample would be under it. Only a hypothesis containing an equality supplies such a value — $\mu \leq 499$ can be tested at its boundary, 499, whereas $\mu > 499$ names no particular number to compute with.

WATCH OUT

A claim phrased with a *strict* inequality — "the mean is more than 499" — therefore cannot go into H_0 as stated. If you are asked for evidence supporting it, the pair is

$$H_0 : \mu \leq 499 \qquad H_A : \mu > 499$$

Writing $H_0 : \mu \geq 499$ is the common error. It puts the claim in the null *and* points the test in the wrong direction, so a sample mean **above** 499 — evidence *for* the claim — ends up producing a p-value near 1 and a "fail to reject." The arithmetic is fine; the conclusion is backwards.

The CAT report, worked through

A report claims the mean CAT score of applicants is more than 499. A sample of 100 applicants has mean 502, with $\sigma = 10.6$. Is there enough evidence at $\alpha = 0.01$ to support the report?

We settled the hypotheses above by asking what would convince us:

$$H_0 : \mu \leq 499 \qquad H_A : \mu > 499$$

Right-tailed, so the critical value is $Z_{0.01} = 2.33$ — positive, not negative.

Before computing anything, check the direction. The sample mean is 502 and the claim is "more than 499." The sample points the *same way* as the claim, so whatever the arithmetic returns, it must come out favourable to the report. If it does not, the tail has been pointed the wrong way.

```
xbar_c <- 502; mu_c <- 499; sigma_c <- 10.6; n_c <- 100

z_c <- (xbar_c - mu_c) / (sigma_c / sqrt(n_c))
c(z_calc = z_c,
  z_crit = qnorm(0.99),
  p_value = pnorm(z_c, lower.tail = FALSE))   # upper tail: H_A is ">"
```

```
#>   z_calc   z_crit  p_value
#> 2.830189 2.326348 0.002326
```

$Z_{calc} = 2.83 > 2.33$, and $p = 0.0023 < 0.01$. We **reject** H_0, which means the data **support** the report's claim — and the answer points the way the sanity check said it would.

> **THE IDEA IN PLAIN WORDS**
>
> Make that check a habit, and make it before you trust any p-value rather than after.
> Ask which way the sample points relative to the claim, and decide what answer you should therefore expect. If the arithmetic hands back something else — a p-value of 0.998 and "no support" when the sample sits above the claimed value — the mistake is almost always in H_A, not in the calculation.

4.3 What a p-value actually says

> *We already computed a probability by simulation. What exactly was it?*

Back in Section 4.1 we simulated fifty thousand samples from a population whose mean really was 165 cm. Only about 2.4% of those samples produced a sample mean at least as far from 165 as the one we actually observed.

Without realising it, we had already computed a p**-value**.

> **REMEMBER THIS**
>
> A p**-value** is the probability of obtaining a sample at least as extreme as the one we observed, *assuming the null hypothesis is true*.

Notice the order carefully.

1. We begin by assuming the null hypothesis is true.
2. We ask what kinds of samples we would expect under that assumption.
3. We compare our observed sample with those possibilities.
4. The p-value is the proportion of those hypothetical samples at least as extreme as ours.

In the height example the p-value was about 0.024. If the true population mean really were 165 cm, a sample as unusual as ours would occur only about 2.4% of the time. That makes the observed sample unusual under the null hypothesis, which is why we would reject it at the 5% significance level.

The p-value does **not** tell us whether the null hypothesis is true. It tells us how surprising our data would be **if** the null hypothesis were true. This distinction is one of the most important ideas in statistics.

A small p-value means the observed data would be difficult to explain if the null hypothesis were correct. That is evidence *against* the null.

A large p-value means the observed data are quite ordinary under the null hypothesis, so we have no convincing reason to reject it. Notice what that does *not* mean: it does not prove the null hypothesis true. It means only that our data are compatible with it.

Table 4.2: Reading a p-value.

p-value	Read it as	At $\alpha = 0.05$
0.60	A result like this is very common if the null hypothesis is true.	Fail to reject H_0
0.18	This would occur roughly once every six samples. Not unusual.	Fail to reject H_0
0.049	This would occur about once every twenty samples. Unusual enough to reject at the 5% level.	Reject H_0
0.0001	This would occur about once every ten thousand samples. Extremely unusual under the null.	Reject H_0

Four ways to read it backwards

p-values are misquoted more often than any other number in empirical work, and the errors are predictable enough to be worth naming.

WATCH OUT

"The p-value is the probability that the null hypothesis is true." False. The null hypothesis is *assumed* true in order to compute the p-value at all. A quantity computed under an assumption cannot also be the probability that the assumption holds.

"The p-value is the probability that we have made the wrong decision." False. It measures how unusual the observed data are under the null, not the probability that our conclusion is incorrect.

"A large p-value proves the null hypothesis." False. It means the data do not provide strong evidence against it, which is a much weaker statement.

"A very small p-value means the effect is important." False. Statistical significance and practical importance are different ideas. A tiny effect

> produces a very small p-value if the sample is large enough.

A useful way to hold the idea:

> A p-value measures how surprising the data are, not how believable the null hypothesis is.

4.4 How far is too far?

> *A gap between the sample and the claim is guaranteed. How large a gap is large enough to act on?*

By now we know the basic logic. We begin by assuming the null hypothesis is true and ask: if that assumption were correct, how unusual would our sample be?

Section 4.1 answered that by simulation. Sampling repeatedly from a population whose mean really was 165 cm, a sample mean at least 3 cm away turned up about 2.4% of the time.

That raises a natural question. **How unusual is unusual enough?**

We need a rule telling us when the difference between the sample and the null is large enough to reject the claim.

Write μ_0 for the value the null claims, so that $H_0 : \mu = \mu_0$. If the sample mean lies close to μ_0, we treat the difference as ordinary sampling variation and keep the null. If it lies far enough away, we reject. That is, we reject whenever

$$|\bar{x} - \mu_0| \geq c$$

for some cutoff c we have yet to determine. The entire problem has reduced to a single question: **how should we choose c?**

Choosing the cutoff

We choose it *before* collecting any data, by deciding how much risk we are willing to run of rejecting a null hypothesis that is in fact true.

> **REMEMBER THIS**
>
> That risk is the **significance level**, written α. It is the probability of rejecting a true null hypothesis, and it is chosen in advance. Common choices

are 0.05 and 0.01.

So c is chosen so that, if the null really is true, only a proportion α of all possible samples would fall beyond it:

$$P[\,|\bar{X} - \mu_0| \geq c\,] = \alpha$$

Everything needed to solve that equation is already in hand from the previous units.

WHERE THIS COMES FROM

When the null is true, the Central Limit Theorem tells us the sampling distribution of the sample mean is approximately normal, centred on μ_0 with standard deviation $\sigma/\sqrt{n}$:

$$\bar{X} \;\sim\; N\left(\mu_0,\, \frac{\sigma^2}{n}\right)$$

Following the usual convention, the second argument is the *variance*, so the standard deviation is its square root, $\sigma/\sqrt{n}$ — the standard error. Standardising, as in Section 3.4, removes the units and the sample size together:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \;\sim\; N(0,1)$$

Dividing both sides of the probability statement by the standard error therefore gives

$$P\left[\,|Z| \geq \frac{c}{\sigma/\sqrt{n}}\,\right] = \alpha$$

The standard normal is symmetric, so that probability is split equally between the two tails:

$$P\left[\,Z \geq \frac{c}{\sigma/\sqrt{n}}\,\right] = \frac{\alpha}{2}$$

The point leaving an upper-tail probability of $\alpha/2$ is the **critical value**, written $Z_{\alpha/2}$. Since $P[Z \geq Z_{\alpha/2}] = \alpha/2$ by definition, the bracketed quantity *is* $Z_{\alpha/2}$:

$$\frac{c}{\sigma/\sqrt{n}} = Z_{\alpha/2} \qquad \Longrightarrow \qquad c = Z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}}$$

Substituting that cutoff back into the rejection rule, and dividing both sides by the standard error, produces the form used in practice:

$$\boxed{\frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}} \geq Z_{\alpha/2}}$$

The quantity on the left is the **test statistic**, usually written Z_{calc}. The quantity on the right is the critical value.

The decision rule

1. Compute the test statistic.
2. Compare it with the critical value.
3. If the statistic is larger, reject the null hypothesis.

Notice what has happened. The complicated question

> Is my sample unusually far from the null hypothesis?

has become the much simpler question

> Is my test statistic larger than the critical value?

That substitution is the whole point of standardising. Every one-sample problem about means — whatever the units, whatever the sample size — now arrives at the same comparison against the same handful of numbers.

DOING IT IN R

Those familiar numbers — 1.645, 1.96 and 2.576 — are simply critical values of the standard normal distribution. There is no need to consult a printed table; `qnorm()` returns them to as many decimals as you like.

```r
alpha <- c(0.10, 0.05, 0.01)

# Two-tailed: we want the value leaving alpha/2 in the upper tail
qnorm(1 - alpha / 2)

#> [1] 1.645 1.960 2.576
```

These are exactly the values defining the rejection regions for the most commonly used significance levels. They are not arbitrary quantities to memorise. They are the points leaving 10%, 5% and 1% of the probability split equally between the two tails of the standard normal distribution.

4.5 The worked example, both ways

Back to the students. We have $\bar{x} = 168$, $\mu_0 = 165$, $\sigma = 8$, $n = 36$, and we will work at $\alpha = 0.05$.

On paper

Step 1 — Hypotheses.

$$H_0 : \mu = 165 \qquad H_A : \mu \neq 165$$

Step 2 — Critical value. Two-tailed at $\alpha = 0.05$, so $Z_{0.025} = 1.96$.

Step 3 — Test statistic.

$$Z_{calc} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{168 - 165}{8/\sqrt{36}} = \frac{3}{8/6} = \frac{3}{1.333} = 2.25$$

Step 4 — Decision. $|2.25| > 1.96$, so we **reject** H_0 at the 5% level.

Step 5 — The p-value.

$$P(|Z| > 2.25) = 2 \times P(Z > 2.25) = 2 \times 0.01222 = 0.0244$$

If the administration's claim were true, only 2.4% of samples would sit as far from 165 as ours. That is unlikely enough to reject the claim.

The same thing in R

Now watch every one of those numbers reappear.

```
xbar  <- 168
mu0   <- 165
sigma <- 8
n     <- 36

se     <- sigma / sqrt(n)          # standard error
z_calc <- (xbar - mu0) / se        # the test statistic
z_crit <- qnorm(1 - 0.05 / 2)      # the critical value
p_val  <- 2 * pnorm(-abs(z_calc))  # the p-value

c(se = se, z_calc = z_calc, z_crit = z_crit, p_value = p_val)
```

```
#>      se  z_calc  z_crit p_value
#> 1.33333 2.25000 1.95996 0.02445
```

The standard error is 1.3333, the test statistic is 2.25, and the p-value is 0.0244 — exactly the hand calculation, and near enough exactly the 0.0242 we got by brute-force simulation at the start of the chapter.

> **THE IDEA IN PLAIN WORDS**
>
> Notice what just happened. The simulation, the algebra, and the R code are three descriptions of one idea. The formula is not a rule handed down from above — it is a shortcut for the counting exercise we did by hand in Section 4.1.

And here is the picture the numbers are describing:

```r
plot_rejection(stat = 2.25, crit = 1.96, tails = "two",
               stat_label = "Z = 2.25")
```

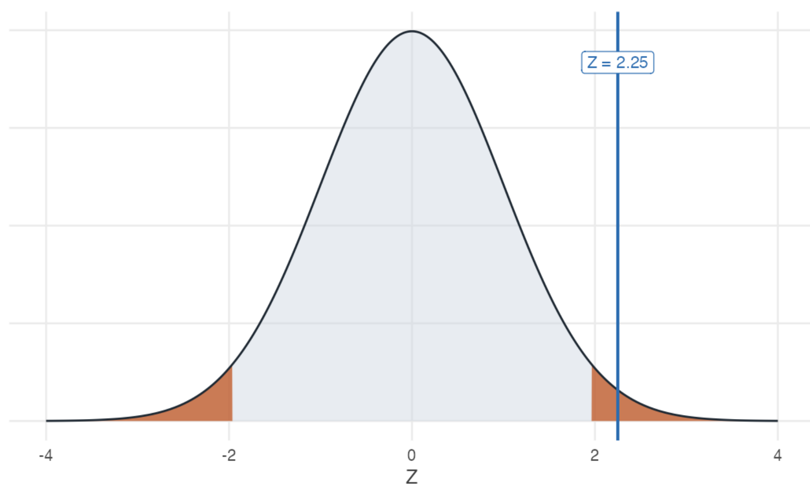

Figure 4.2: The rejection region at $\alpha = 0.05$ (shaded), with our test statistic of 2.25 falling inside it.

Letting a package do it

For a real dataset rather than summary statistics, `z.test()` from the **BSDA** package does the whole thing at once. Since our problem quoted only summary statistics, we first build a sample consistent with them:

```
library(BSDA)

heights <- make_sample(n = 36, mean = 168, sd = 8)

z.test(heights, mu = 165, sigma.x = 8, conf.level = 0.95)
```

```
#>
#>  One-sample z-Test
#>
#> data:  heights
#> z = 2.3, p-value = 0.02
#> alternative hypothesis: true mean is not equal to 165
#> 95 percent confidence interval:
#>  165.4 170.6
#> sample estimates:
#> mean of x
#>       168
```

Read that output against the hand calculation: `z = 2.25`, `p-value = 0.0244`, and a confidence interval that we are about to explain.

> **WATCH OUT**
>
> `z.test()` needs `sigma.x` — the **population** standard deviation. If you do not know it (and in real work you almost never do), you must not use a Z-test. Use `t.test()` instead; see Section 4.7.

4.6 The same test, viewed as an interval

> *Unit 3 built confidence intervals; this unit builds tests. Are they two methods, or one?*

We now have two procedures that use the same sample, the same standard error and the same critical value. It would be strange if they were unrelated, and they are not.

Start by computing the interval for our students, exactly as in the previous unit.

```
xbar + c(-1, 1) * qnorm(0.975) * (8 / sqrt(36))
```

```
#> [1] 165.4 170.6
```

The 95% interval is $[165.39,\ 170.61]$. The claimed value of 165 lies **outside** it — the same verdict the test reached, and not a coincidence.

The reason is visible in the algebra. We fail to reject H_0 when the test statistic is small enough, that is when

$$\left| \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \right| \leq z_{\alpha/2}$$

Multiplying through by $\sigma/\sqrt{n}$ and rearranging for μ_0 gives

$$\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \ \leq\ \mu_0 \ \leq\ \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

which is the confidence interval.

> **REMEMBER THIS**
>
> A confidence interval and a hypothesis test are the same inequality, solved for different unknowns.
> The test fixes μ_0 and asks which sample means are acceptable. The interval fixes $\bar{x}$ and asks which values of μ_0 are acceptable.

Testing every possible claim at once

That equivalence can be checked rather than taken on trust. Rather than testing the single claim $\mu_0 = 165$, test every value on a fine grid and collect the ones that survive.

```r
se   <- 8 / sqrt(36)
grid <- seq(160, 176, by = 0.001)

not_rejected <- grid[abs(xbar - grid) / se <= qnorm(0.975)]

surviving <- range(not_rejected)
interval  <- xbar + c(-1, 1) * qnorm(0.975) * se

rbind(claims_that_survive_the_test = surviving,
      confidence_interval          = interval,
      difference                   = surviving - interval)
```

```
#>                                   [,1]        [,2]
#> claims_that_survive_the_test 165.3870000 170.6130000
#> confidence_interval          165.3867147 170.6132853
#> difference                     0.0002853  -0.0002853
```

The two agree to within a thousandth of a centimetre, which is the spacing of the grid — test a finer grid and the discrepancy shrinks with it. The set of claims a test would not reject *is* the confidence interval. Nothing was assumed to make this happen; it follows from the two being one inequality solved for different unknowns.

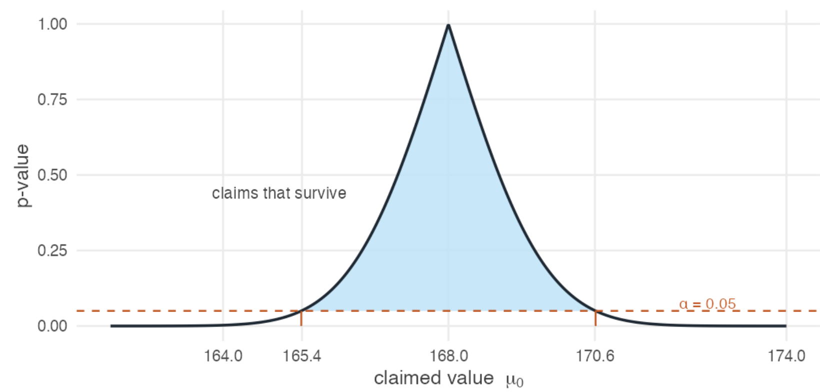

Figure 4.3: The p-value for every possible claimed value of mu. Claims above the 0.05 line survive the test, and the range of surviving claims is exactly the 95% confidence interval.

The curve reaches its maximum at $\mu_0 = \bar{x} = 168$, and it is worth being clear about why the p-value is exactly 1 there.

A p-value asks how often a sample would fall at least as far from the claim as ours did. If the claim *is* our sample mean, then our sample sits at distance zero from it — and every conceivable sample is at least that far away. The proportion is therefore 1. A claim that matches the data exactly leaves nothing to explain.

As the claimed mean moves away from the observed mean in either direction, the sample becomes increasingly surprising under that claim, and the p-value falls steadily — crossing 0.05 precisely at the two endpoints of the interval.

REMEMBER THIS

A $(1-\alpha)$ confidence interval is exactly the set of values μ_0 that a two-tailed test at level α would **fail** to reject.
Test and interval are one object viewed from two sides. If the interval misses the claimed value, the test rejects it. Always.

So why keep both?

Because they carry different information, and the figure above shows what each one throws away.

A confidence interval answers infinitely many tests at once, and reports the answer in the units of the problem. Handed $[165.39,\ 170.61]$, a reader can evaluate any claim about average height — 165, 166, 172 — without recomputing anything. A test answers one question, chosen in advance.

This is why confidence intervals are generally preferred in scientific papers. A single interval communicates the estimate, its uncertainty, and the outcome of every two-sided hypothesis test at that significance level, all at once.

A test, however, reports *how strongly*. Both 165 and 160 fall outside our interval, and the interval says the same thing about each: not plausible. The p-values do not agree — 0.024 for 165, and effectively zero for 160. One claim is marginally implausible; the other is not remotely survivable.

Put precisely: the interval tells us **which claims remain plausible**, and the p-value tells us **how strongly the data disagree with one particular claim**.

> **THE IDEA IN PLAIN WORDS**
>
> Report the interval. It contains the test, states its answer in the units the reader cares about, and does not tempt anyone into treating a single threshold as a verdict.
> Report the p-value alongside it when one specific claim is genuinely at issue, since that is where the interval is silent about degree.

Everything Section 3.8 said about reading a confidence interval applies unchanged here. In particular, the 95% still describes the procedure and not this interval, and an interval that excludes a claimed value has not proved the claim false — it has reported that the data are unlikely under it.

4.7 When σ is unknown: the t-test

Everything so far assumed we knew the population standard deviation σ. We almost never do.

Everything else stays the same. We still compare the sample mean with the value the null hypothesis claims, and we still ask how surprising that gap would be if the claim were true. The only change is that we must first estimate σ from the sample itself.

That estimate is not free. Replacing σ with the sample standard deviation s adds a second source of uncertainty, and the statistic

$$t_{calc} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

no longer follows a normal distribution. It follows Student's t with $n - 1$ **degrees of freedom** — a distribution with fatter tails, which demands more evidence before it will let you reject.

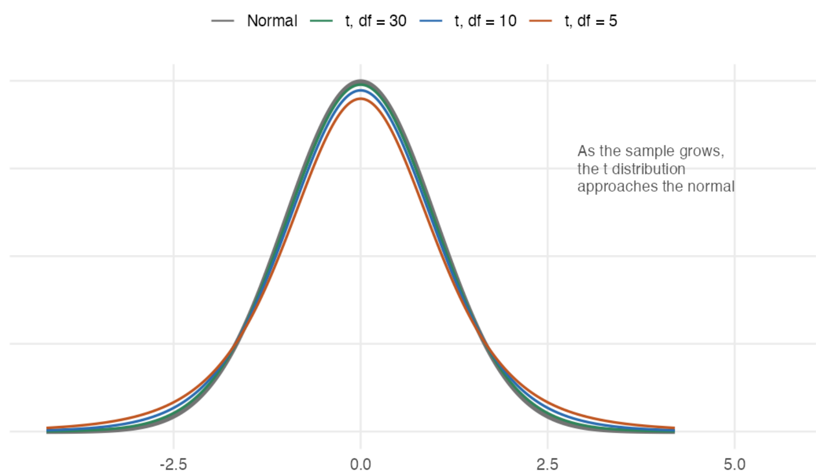

Figure 4.4: Student's t against the standard normal. Fewer degrees of freedom put more probability in the tails, so critical values sit further out and more evidence is needed to reject. By 30 degrees of freedom the two are already hard to tell apart.

Settling the question with the actual class

We opened this chapter with a made-up sample of 36 students. We can now do the real thing. `stats-class.csv` holds the survey this class filled in — names, ages, heights, gender:

```
students <- read.csv("data/stats-class.csv")

str(students)
```

```
#> 'data.frame':	33 obs. of  4 variables:
#>  $ student_id: chr  "S01" "S02" "S03" "S04" ...
#>  $ age       : int  31 19 18 18 19 18 18 21 18 19 ...
#>  $ height    : int  178 161 181 164 170 175 176 186 161 170 ...
#>  $ gender    : chr  "M" "F" "M" "F" ...
```

Thirty-three rows, but look at the last one. One student left age and height blank, and another did not give a gender:

```
students[!complete.cases(students), ]
```

```
#>    student_id age height gender
#> 8         S08  21    186   <NA>
#> 33        S33  NA     NA      F
```

> **WATCH OUT**
>
> **Working with real data.** What follows is not part of hypothesis testing. It is the sort of thing that has to be dealt with the moment you leave textbook datasets behind, and this is the first point in the book where you have.
> **A blank is not always an `NA`.** In a numeric column an empty field is read as `NA`, but in a text column it is read as the empty string `""` — and `""` is a perfectly ordinary value as far as `is.na()`, `complete.cases()` and `na.rm` are concerned. A gap in a text column can therefore hide in plain sight. If a dataset you did not prepare yourself has blanks, say so explicitly when you read it:
>
> ```
> read.csv("somefile.csv", na.strings = c("", "NA"))
> ```
>
> The files in this book are already written so that plain `read.csv()` does the right thing.

> **WATCH OUT**
>
> `mean()` returns `NA` if even one value is missing — it refuses to guess what you meant. Pass `na.rm = TRUE` to drop the gaps, and always know how many you dropped.
>
> ```
> mean(students$height)                # NA — one height is missing
> ```
>
> ```
> #> [1] NA
> ```
>
> ```
> mean(students$height, na.rm = TRUE)    # 32 students
> ```
>
> ```
> #> [1] 169.5
> ```

Now the test. We do **not** know the population standard deviation of student heights, so this is a t-test, not a Z-test:

```r
heights_real <- students$height[!is.na(students$height)]

c(n = length(heights_real),
  mean = mean(heights_real),
  sd = sd(heights_real))
```

```
#>      n   mean     sd
#>  32.00 169.50   8.71
```

```r
t.test(heights_real, mu = 165)
```

```
#>
#>  One Sample t-test
#>
#> data:  heights_real
#> t = 2.9, df = 31, p-value = 0.006
#> alternative hypothesis: true mean is not equal to 165
#> 95 percent confidence interval:
#>  166.4 172.6
#> sample estimates:
#> mean of x
#>     169.5
```

$t = 2.92$ on 31 degrees of freedom, with a p-value of 0.0064. We **reject** the claim that the average is 165 cm. The 95% interval, $[166.4,\ 172.6]$, excludes 165 — the same verdict, as always.

> **THE IDEA IN PLAIN WORDS**
>
> Notice how much narrower this is than the textbook version of the problem. The opening example assumed we somehow knew $\sigma = 8$. Here we had to estimate it from 32 people, and the t distribution charges us for that uncertainty with a wider interval and a higher bar to clear.
> **Statistical validity is not sampling validity.** Notice what the data is not: 32 students in one classroom are not a random sample of all APU students.
> Every number above is correctly computed. The t statistic, the degrees of freedom, the p-value and the interval are all exactly right *for this sample*. Whether they answer a question about **the university** is a separate matter, and a harder one — and no amount of care with the arithmetic addresses it. The formulas cannot tell you which population you were sampling from; only the design can.

Worked example

The dean estimates that full-time faculty teach 11.0 classroom hours per week. You sample eight faculty members. Can you reject the dean's claim at $\alpha = 0.10$? At $\alpha = 0.01$?

```
hours <- c(11.8, 8.6, 10.5, 7.9, 6.4, 10.4, 10.0, 8.2)

c(n = length(hours), mean = mean(hours), sd = sd(hours))
```

```
#>     n  mean    sd
#> 8.000 9.225 1.749
```

> **YOUR TURN**
>
> Before calculating anything, form an expectation.
> The claim is 11.0 hours and the sample averages 9.2 — a gap of nearly two hours. But there are only eight faculty members, and their hours range from 6.4 to 11.8, so the sample is both small and quite spread out.
> Do you expect to reject the claim at the 10% level? At the 1%? Commit to an answer before reading on, and then see whether the arithmetic agrees.

On paper. With $\bar{x} = 9.225$, $s = 1.749$, $n = 8$ and $df = 7$:

$$t_{calc} = \frac{9.225 - 11}{1.749/\sqrt{8}} = \frac{-1.775}{0.618} = -2.87$$

The critical values are $t_{0.05,7} = 1.895$ and $t_{0.005,7} = 3.499$.

- At $\alpha = 0.10$: $2.87 > 1.895 \Rightarrow$ **reject** H_0.
- At $\alpha = 0.01$: $2.87 < 3.499 \Rightarrow$ **fail to reject** H_0.

In R. Both the pieces and the packaged version:

```
t_calc <- (mean(hours) - 11) / (sd(hours) / sqrt(length(hours)))
t_crit <- qt(c(0.95, 0.995), df = 7)     # 10% and 1%, two-tailed

c(t_calc = t_calc, crit_10pct = t_crit[1], crit_1pct = t_crit[2])
```

```
#>     t_calc crit_10pct  crit_1pct
#>     -2.870      1.895      3.499
```

```
t.test(hours, mu = 11, conf.level = 0.90)
```

```
#>
#>  One Sample t-test
#>
#> data:  hours
#> t = -2.9, df = 7, p-value = 0.02
#> alternative hypothesis: true mean is not equal to 11
#> 90 percent confidence interval:
#>   8.053 10.397
#> sample estimates:
#> mean of x
#>     9.225
```

The p-value is 0.024 — below 0.10 but above 0.01, exactly matching the two different verdicts we reached by hand.

> **THE IDEA IN PLAIN WORDS**
>
> The same data can be "significant" at one level and not at another. Nothing has changed about the world between those two lines — only how much risk of a false alarm we were willing to tolerate. This is why reporting the p-value itself is far more informative than reporting a bare "significant / not significant."

4.8 One-tailed tests

> *Sometimes only one direction would worry us. Why spend half the error budget on the other?*

A paper mill discharges effluent into a river. The state pollution control board sets a limit on biochemical oxygen demand of 30 mg per litre, and the mill claims its average discharge is *below* that limit.

Only one direction matters here. Nobody objects if the effluent is far cleaner than required; the concern is entirely about exceeding the limit. So the claim being tested is one-sided:

$$H_0 : \mu \geq 30 \qquad H_A : \mu < 30$$

All the probability we are willing to risk now goes into a single tail, so the critical value moves *inward*: $-Z_{0.05} = -1.645$ rather than ± 1.96.

```
qnorm(0.05)            # left-tailed critical value at 5%
```

```
#> [1] -1.645
```

```
qnorm(0.95)            # right-tailed critical value at 5%
```

```
#> [1] 1.645
```

In R, add the `alternative` argument:

```
bod <- make_sample(n = 20, mean = 29.2, sd = 7)

z.test(bod, mu = 30, sigma.x = 7, alternative = "less", conf.level = 0.95)
```

```
#>
#>  One-sample z-Test
#>
#> data:  bod
#> z = -0.51, p-value = 0.3
#> alternative hypothesis: true mean is less than 30
#> 95 percent confidence interval:
#>     NA 31.77
#> sample estimates:
#> mean of x
#>      29.2
```

$Z_{calc} = -0.51$, which is not below -1.645, so we fail to reject. The mill's claim is not supported by this sample — a mean of 29.2 would arise about 30% of the time even if the true average were exactly at the limit.

Two details in that output are worth noticing. The interval has an *upper* limit of 31.77 for a *left*-tailed test, which looks wrong and is not; that is the subject of the last part of this section. And `z.test()` prints `NA` for the lower limit where it means $-\infty$, whereas `t.test()` prints `-Inf` below. The two packages simply report the same thing differently.

Unknown σ: the one-sided t-test

Nothing about one-tailed testing is special to the Z-test. When σ is unknown, the same `alternative` argument does the same job.

Return to the faculty teaching hours. The dean claims 11.0 hours per week, and suppose we specifically suspect the true figure is *lower* — we are not interested in the possibility that faculty teach more.

$$H_0 : \mu \geq 11 \qquad H_A : \mu < 11$$

```
t.test(hours, mu = 11, alternative = "less", conf.level = 0.90)
```

```
#>
#>  One Sample t-test
#>
#> data:  hours
#> t = -2.9, df = 7, p-value = 0.01
#> alternative hypothesis: true mean is less than 11
#> 90 percent confidence interval:
#>  -Inf 10.1
#> sample estimates:
#> mean of x
#>     9.225
```

Notice that nothing fundamental has changed.

- The hypotheses are written exactly as before.
- The direction comes from `alternative`, not from a different formula.
- The p-value is one-sided — 0.012 here, half the two-sided 0.024 we computed earlier.
- The confidence interval is one-sided, running from $-\infty$ to 10.10.
- The only substantive change is the reference distribution: Student's t with 7 degrees of freedom in place of the normal.

> **REMEMBER THIS**
>
> There are not six different procedures to learn. There are two — the Z-test when σ is known and the t-test when it is not — and each can be pointed in any of three directions.
> Everything else stays the same.

The one-sided interval looks backwards

This trips up nearly everyone. For a left-tailed test, the confidence interval is

$$\left(-\infty,\ \bar{x} + Z_\alpha \frac{\sigma}{\sqrt{n}}\right)$$

An *upper* bound — for a *left*-tailed test. Why?

WHERE THIS COMES FROM

Start from the rejection rule and solve for μ_0 rather than for $\bar{x}$:

$$\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} < -Z_\alpha$$

$$\bar{x} - \mu_0 < -Z_\alpha \frac{\sigma}{\sqrt{n}}$$

$$\mu_0 > \bar{x} + Z_\alpha \frac{\sigma}{\sqrt{n}}$$

The inequality flipped when μ_0 moved across. So we reject exactly when μ_0 is **too large**, and the values we *fail* to reject are those with $\mu \leq \bar{x} + Z_\alpha\, \sigma/\sqrt{n}$.
The result is identical for the t-test. Replace Z_α with $t_{\alpha,\, n-1}$ and σ with s; not a single step of the argument changes. That is why the interval above ran to 10.10 rather than starting there.

THE IDEA IN PLAIN WORDS

The rejection region is defined in Z**-space**; the confidence interval lives in **parameter space**. Solving the inequality for μ reverses the direction.
Put plainly: a left-tailed test rejects when the sample mean comes out too small. So the values of μ it rules out are the ones that are too *large* relative to what we saw. That leaves an upper bound.

4.9 Two ways to be wrong

> *We have spent this unit controlling one kind of mistake. What about the other?*

Every test so far has been built around α, the risk of rejecting a claim that happens to be true. We chose it in advance, usually 0.05, and everything followed.

But rejecting a true claim is only one way to get the answer wrong. We could also fail to reject a false one — conclude that a drug does nothing when it works, that a programme has no effect when it does.

That second error has had no name and no symbol so far. It gets both now, and it turns out to matter at least as much as the first.

The two errors

> **REMEMBER THIS**
>
> A **Type I error** is rejecting H_0 when H_0 is true. A false alarm.
>
> $$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$$
>
> A **Type II error** is failing to reject H_0 when H_A is true. A missed detection.
>
> $$\beta = P(\text{fail to reject } H_0 \mid H_A \text{ true})$$

Think of a smoke alarm.

Sometimes there is no fire, and the alarm goes off anyway — a false alarm, and a Type I error. Setting $\alpha = 0.05$ says we will tolerate this about five times in a hundred when nothing is wrong.

Sometimes there *is* a fire and the alarm stays silent. That is a Type II error, and it is the one that burns the house down.

Table 4.3: The four possible outcomes of a hypothesis test.

	Fail to reject H_0	**Reject H_0**
H_0 is true	Correct	**Type I error** (α)
H_A is true	**Type II error** (β)	Correct

Two cells are mistakes, and they are not the same mistake. A test cannot make both at once — which of them is even possible depends on a fact about the world that we do not know.

The umpire's call

Cricket's Decision Review System is a hypothesis test, and a well-designed one.

$$H_0 : \text{the on-field umpire was right} \qquad H_A : \text{the on-field umpire was wrong}$$

A Type I error overturns a correct decision. A Type II error lets a wrong decision stand.

> **THE IDEA IN PLAIN WORDS**
>
> The system's "umpire's call" rule — where marginal ball-tracking evi-

dence leaves the original decision standing — is exactly a choice to keep α small.
Overturning requires clear evidence, not a coin-flip's worth. The designers decided that wrongly overturning a correct umpire is a worse error than occasionally failing to correct a wrong one, and they built that judgement into the threshold.
Every choice of α is a judgement of the same kind, whether or not it is made deliberately.

Why errors happen at all

Both errors have the same root cause: the sampling distribution under H_0 and the sampling distribution under H_A overlap.

If a sample mean lands in the region where the two overlap, it is consistent with either explanation, and no procedure can tell them apart with certainty.

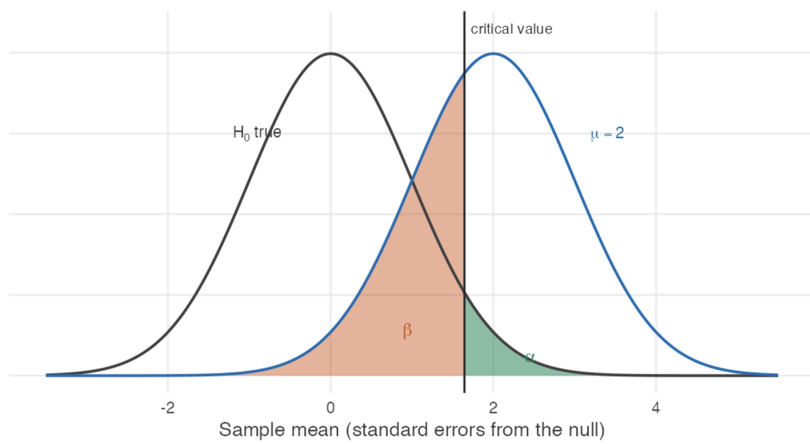

Figure 4.5: Two sampling distributions: the left curve holds if the null is true, the right one if the true mean is 2. The vertical line is the critical value, 1.645. Alpha is the sliver of the null distribution beyond it – rejecting when the null is true. Beta is the much larger part of the alternative distribution below it – failing to reject when the alternative is true. Power is what remains.

Notice the shape of the trade-off. Moving the critical value to the right shrinks the green area and enlarges the red one: fewer false alarms, more missed detections. Moving it left does the reverse.

WATCH OUT

α and β cannot both be made small by choosing a threshold.
The threshold only slides the boundary between them. Reducing α from 0.05 to 0.01 buys fewer false alarms at the direct cost of more missed real effects.
The only way to reduce both at once is to separate the two distributions — which means a larger sample, a bigger effect, or less noise. That is a matter of study design, not of choosing a number.

Power

REMEMBER THIS

The **power** of a test is the probability of detecting an effect that is really there:

$$\text{Power} = 1 - \beta = P(\text{reject } H_0 \mid H_A \text{ true})$$

α is chosen. Power is a *consequence* — of the sample size, the size of the effect, and the noise in the data.

Unlike α, power cannot be stated without saying *how large* the effect is. A test that easily detects a large effect may be hopeless against a small one, so "the power of this test" is always shorthand for "the power against an effect of this particular size."

Working it out

A new drug is believed to lower blood pressure. We test

$$H_0 : \mu \leq 0 \qquad H_A : \mu > 0$$

where μ is the average reduction. Suppose $\sigma = 4$, the trial has $n = 16$ patients, and we work at $\alpha = 0.05$.

Step 1 — the rejection rule. One-sided at 5%, so the critical value is $z_{0.05} = 1.645$, and we reject when $Z > 1.645$.

Step 2 — suppose the drug works. Say the true average reduction is $\mu_1 = 2$ points. The standard error is

$$\text{se} = \frac{\sigma}{\sqrt{n}} = \frac{4}{\sqrt{16}} = 1$$

Step 3 — where the statistic now sits. Under H_0 the statistic $Z = \bar{X}/\text{se}$ is centred on zero. If the truth is $\mu_1 = 2$, it is centred instead on

$$\delta = \frac{\mu_1 - \mu_0}{\text{se}} = \frac{2-0}{1} = 2$$

Step 4 — the probability of missing it. We fail to reject when $Z \leq 1.645$. Under the alternative, Z is centred on 2, so

$$\beta = P(Z \leq 1.645 \mid \mu = 2) = P(Z - 2 \leq -0.355) = \Phi(-0.355)$$

```r
se    <- 4 / sqrt(16)
delta <- (2 - 0) / se
crit  <- qnorm(0.95)

c(critical_value = crit,
  shift          = delta,
  beta           = pnorm(crit - delta),
  power          = 1 - pnorm(crit - delta))
```

```
#> critical_value          shift           beta          power
#>         1.6449         2.0000         0.3612         0.6388
```

So $\beta = 0.36$ and the power is **0.64**.

Read that carefully. The drug genuinely works — it lowers blood pressure by two points on average — and this trial has a better than one-in-three chance of failing to notice.

> **WATCH OUT**
>
> A trial like this is **underpowered**. If it returns "no significant effect", that tells us very little: we already knew the study would miss a real two-point effect about 36% of the time.
> This is the sharpest version of a point made in Section 4.3. Failing to reject H_0 is not evidence that H_0 is true, and when power is low it is barely evidence of anything at all.

What determines power

Four things, and it is worth seeing which of them are under our control.

```r
power_at <- function(n, effect = 2, sigma = 4, alpha = 0.05) {
  se <- sigma / sqrt(n)
  round(1 - pnorm(qnorm(1 - alpha) - effect / se), 3)
}

data.frame(n          = c(16, 25, 50, 100, 200),
           power_vs_2 = sapply(c(16, 25, 50, 100, 200), power_at),
           power_vs_1 = sapply(c(16, 25, 50, 100, 200), power_at, effect
            ↪  = 1))
```

```
#>     n power_vs_2 power_vs_1
#> 1  16      0.639      0.260
#> 2  25      0.804      0.346
#> 3  50      0.971      0.549
#> 4 100      1.000      0.804
#> 5 200      1.000      0.971
```

REMEMBER THIS

Power rises when

- the **effect is larger** — big effects are easy to find;
- the **sample is larger** — the two distributions pull apart;
- the **noise is smaller** — the distributions are narrower;
- α **is larger** — a lower bar to clear, at the cost of more false alarms.

The first is a fact about the world. The last is a choice with a price. Only the middle two are improvements in the ordinary sense.

The table shows the practical consequence. Against a two-point effect this trial needs about fifty patients for respectable power. Against a one-point effect, even two hundred patients leave it detecting the effect barely two-thirds of the time.

THE IDEA IN PLAIN WORDS

This is why power should be computed *before* a study, not after. Deciding how many patients to recruit means asking: what is the smallest effect worth detecting, and how many observations does it take to have a fair chance of detecting it? A study that cannot answer its question is not worth running, and that can be known in advance.
Computing power after a null result is a different and largely futile exercise. The honest statement is the one the design already implied.

4.10 Reference tables

Table 4.4: Every one-sample test of a mean in this unit. Use `z.test()` when σ is known and `t.test()` when it is not; the argument below selects the direction. There are only two procedures, each run three ways.

σ	Direction	Hypotheses	Reject when	R argument
Known	Two-sided	$H_0 : \mu = \mu_0$ vs $H_A : \mu \neq \mu_0$	$\lvert Z_{calc} \rvert > Z_{\alpha/2}$	`alternative = "two.sided"`
Known	Left	$H_0 : \mu \geq \mu_0$ vs $H_A : \mu < \mu_0$	$Z_{calc} < -Z_{\alpha}$	`alternative = "less"`
Known	Right	$H_0 : \mu \leq \mu_0$ vs $H_A : \mu > \mu_0$	$Z_{calc} > Z_{\alpha}$	`alternative = "greater"`
Unknown	Two-sided	$H_0 : \mu = \mu_0$ vs $H_A : \mu \neq \mu_0$	$\lvert t_{calc} \rvert > t_{\alpha/2,\, n-1}$	`alternative = "two.sided"`
Unknown	Left	$H_0 : \mu \geq \mu_0$ vs $H_A : \mu < \mu_0$	$t_{calc} < -t_{\alpha,\, n-1}$	`alternative = "less"`
Unknown	Right	$H_0 : \mu \leq \mu_0$ vs $H_A : \mu > \mu_0$	$t_{calc} > t_{\alpha,\, n-1}$	`alternative = "greater"`

Read down the first column rather than across all six rows. The distribution changes when σ becomes unknown; the direction changes where the rejection region sits. Nothing else differs.

Table 4.5: Standard normal critical values. Every figure below is `qnorm()` evaluated inline, not copied from a printed table.

α	Two-tailed, $Z_{\alpha/2}$	One-tailed, Z_{α}
10%	1.645	1.282
5%	1.96	1.645
1%	2.576	2.326

4.11 Summary

REMEMBER THIS

1. A gap between a sample and a claim proves nothing on its own — sampling noise guarantees some gap. The test asks whether the gap is *bigger than noise*.
2. $Z_{calc} = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ when σ is known; $t_{calc} = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ with $df = n - 1$ when it is not.
3. Reject when the statistic beats the critical value, or equivalently when the p-value falls below α. The two rules never disagree.
4. A confidence interval is the set of null values a test would *not* reject. Interval and test are the same statement.
5. A p-value is $P(\text{data} \mid H_0)$, never $P(H_0 \mid \text{data})$.
6. The equals sign always lives in H_0. "Can you reject the claim?" puts the claim in H_0; "is there evidence supporting the claim?" puts it in H_A.
7. A **Type I error** rejects a true null, with probability α. A **Type II error** fails to reject a false one, with probability β. Lowering one raises the other; only better data lowers both.
8. **Power** $= 1 - \beta$ is the chance of detecting an effect that is really there. It is not chosen but inherited — from the sample size, the size of the effect and the noise. A trial with 64% power misses a real effect more than once in three.
9. `qnorm()` and `qt()` replace every printed table you own.

4.12 Practice problems

YOUR TURN

Work each one on paper first, then check yourself in R. The solutions are one click away, which makes them worth nothing unless you try first. These are for learning, so the workings are shown in full. The question bank that follows is for testing yourself, and has no answers.

11.1 A consumer research organisation states that the mean caffeine content per 12-ounce bottle of a caffeinated soft drink is 37.7 mg. A random sample of 36 bottles has a mean of 35.28 mg. The population standard deviation is 10.8 mg. At $\alpha = 0.01$, can you reject the organisation's claim?

SOLUTION

$H_0 : \mu = 37.7$ against $H_A : \mu \neq 37.7$. Two-tailed at $\alpha = 0.01$, so the critical value is $Z_{0.005} = 2.58$.

$$Z_{calc} = \frac{35.28 - 37.7}{10.8/\sqrt{36}} = \frac{-2.42}{1.8} = -1.34$$

```
caffeine <- make_sample(n = 36, mean = 35.28, sd = 10.8)
z.test(caffeine, mu = 37.7, sigma.x = 10.8, conf.level = 0.99)
```

```
#>
#>  One-sample z-Test
#>
#> data:  caffeine
#> z = -1.3, p-value = 0.2
#> alternative hypothesis: true mean is not equal to 37.7
#> 99 percent confidence interval:
#>  30.64 39.92
#> sample estimates:
#> mean of x
#>     35.28
```

$|-1.34| < 2.58$ and the p-value of 0.179 is far above 0.01, so we **fail to reject**. If the true mean really were 37.7 mg, a sample mean this far away would turn up about 18% of the time — thoroughly unremarkable.

11.2 Historical data show household water use is normally distributed with mean 360 gallons and standard deviation 40 gallons per day. A sample of 200 households averages 374 gallons. Are these data consistent with the historical distribution at the 5% level? What is the p-value?

SOLUTION

$$Z_{calc} = \frac{374 - 360}{40/\sqrt{200}} = \frac{14}{2.828} = 4.95$$

```
water <- make_sample(n = 200, mean = 374, sd = 40)
z.test(water, mu = 360, sigma.x = 40, conf.level = 0.95)
```

```
#>
#>  One-sample z-Test
#>
#> data:  water
#> z = 4.9, p-value = 0.0000007
#> alternative hypothesis: true mean is not equal to 360
#> 95 percent confidence interval:
```

```
#>  368.5 379.5
#> sample estimates:
#> mean of x
#>       374
```

$4.95 \gg 1.96$, and the p-value is about 0 — under one in a million. **Reject** H_0. Water use has moved away from the historical mean.

11.3 A study claims the average monthly salary of full professors in India is ₹87,800. Students at one university sample ten salaries (in thousands):

```
91.0, 79.8, 102.0, 93.5, 82.0, 88.6, 90.0, 98.6, 101.0, 84.0
```

They believe their professors earn *more*. Test at the 10% and 5% levels.

SOLUTION

The word "more" makes this **right-tailed**: $H_0 : \mu \leq 87.8$ against $H_A : \mu > 87.8$. Since σ is unknown and $n = 10$, use a t-test with $df = 9$.

```
salary <- c(91.0, 79.8, 102.0, 93.5, 82.0, 88.6, 90.0, 98.6, 101.0,
 ↪  84.0)

c(mean = mean(salary), sd = sd(salary))
```

```
#>   mean     sd
#> 91.050  7.797
```

```
t.test(salary, mu = 87.8, alternative = "greater")
```

```
#>
#>  One Sample t-test
#>
#> data:  salary
#> t = 1.3, df = 9, p-value = 0.1
#> alternative hypothesis: true mean is greater than 87.8
#> 95 percent confidence interval:
#>  86.53   Inf
#> sample estimates:
#> mean of x
#>     91.05
```

$t_{calc} = 1.318$, against critical values $t_{0.10,\,9} = 1.383$ and $t_{0.05,\,9} = 1.833$. The statistic falls short of both. **Fail to reject** at either level — there is not enough evidence that these professors are paid above the national figure.

11.4 A Department of Transportation claims mean wait time is *at most* 6 minutes. A sample of 34 locations has mean 10.3 minutes, standard deviation 8.0. Test at 5% and 1%.

SOLUTION

"At most" makes this right-tailed: $H_0 : \mu \leq 6$ against $H_A : \mu > 6$, t-test with $df = 33$.

```
waits <- make_sample(n = 34, mean = 10.3, sd = 8.0)
t.test(waits, mu = 6, alternative = "greater")
```

```
#>
#>  One Sample t-test
#>
#> data:  waits
#> t = 3.1, df = 33, p-value = 0.002
#> alternative hypothesis: true mean is greater than 6
#> 95 percent confidence interval:
#>  7.978   Inf
#> sample estimates:
#> mean of x
#>      10.3
```

$t_{calc} = 3.14$, well beyond $t_{0.05,33} = 1.692$ and $t_{0.01,33} = 2.445$. **Reject at both levels** — wait times are longer than claimed.

11.5 `marriage-ages.csv` records the ages of both partners in ten couples. Is there evidence that one partner tends to be older? Test at the 5% level.

Hint: the two columns are not independent samples — they are ten couples. What is the one number per couple that captures the question?

SOLUTION

The two columns are **paired**: each row is one couple. Treating them as two independent samples throws away that pairing and inflates the standard error. The right move is to reduce each couple to a single number — the age gap — and run a **one-sample** test on it against zero.

```
couples <- read.csv("data/marriage-ages.csv")

gap <- couples$partner1 - couples$partner2
gap
```

```
#>  [1] 1 5 4 1 1 4 0 8 1 0
```

```r
c(mean = mean(gap), sd = sd(gap), n = length(gap))
```

```
#>   mean     sd      n
#>  2.500  2.635 10.000
```

```r
t.test(gap, mu = 0)
```

```
#>
#>  One Sample t-test
#>
#> data:  gap
#> t = 3, df = 9, p-value = 0.01
#> alternative hypothesis: true mean is not equal to 0
#> 95 percent confidence interval:
#>  0.6149 4.3851
#> sample estimates:
#> mean of x
#>       2.5
```

The mean gap is 2.5 years, $t = 3.0$ on 9 degrees of freedom, $p = 0.015$. Since $0.015 < 0.05$, we **reject** H_0: partner 1 is reliably older.
R will do the pairing for you if you ask:

```r
t.test(couples$partner1, couples$partner2, paired = TRUE)
```

```
#>
#>  Paired t-test
#>
#> data:  couples$partner1 and couples$partner2
#> t = 3, df = 9, p-value = 0.01
#> alternative hypothesis: true mean difference is not equal to 0
#> 95 percent confidence interval:
#>  0.6149 4.3851
#> sample estimates:
#> mean difference
#>             2.5
```

Identical output — `paired = TRUE` computes exactly the differences we formed by hand. Compare what happens if you forget:

```r
t.test(couples$partner1, couples$partner2, paired = FALSE)$p.value
```

```
#> [1] 0.607
```

The same ten couples, a p-value of 0.607 instead of 0.015, and the opposite conclusion. **Pairing is not a detail.**

11.6 *(No computation.)* A classmate reports "$p = 0.03$, so there is a 3% chance the null hypothesis is true." Explain what is wrong, and state what $p = 0.03$ does mean.

SOLUTION

The p-value is computed **assuming the null is true**. It cannot then also be the probability that the null is true — that would be circular. Formally, your classmate has swapped $P(\text{data} \mid H_0)$ for $P(H_0 \mid \text{data})$, which are different quantities entirely.
The correct reading: *if* H_0 were true, samples at least as extreme as this one would arise 3% of the time. Since that is unusual, we treat it as evidence against H_0 — but the statement is about how surprising the data are, not about how probable the hypothesis is.

Type I and Type II errors

4.15 A district administration tests whether a new teaching programme raised average test scores. State, in the language of the problem, what a Type I error and a Type II error would be, and say which you would consider more serious.

SOLUTION

A **Type I error** is concluding the programme raised scores when it did not. The district would scale up an ineffective programme, spending money and teaching time that could have gone elsewhere, and would believe a problem had been solved when it had not.
A **Type II error** is concluding the programme did not work when it did. A genuinely effective intervention would be abandoned.
Which is worse is not a statistical question. If the programme is cheap and the alternative is doing nothing, a missed effect is the costlier error and a larger α might be defensible. If scaling up is expensive and irreversible, the false alarm is worse.
The point is that α encodes such a judgement whether or not anyone makes it deliberately. Choosing 0.05 out of habit is still choosing.

4.16 Explain why setting $\alpha = 0.001$ does not make a test "more accurate".

SOLUTION

It makes the test less likely to raise a false alarm and *more* likely to miss a real effect. Lowering α moves the critical value further out, which shrinks the rejection region — and the rejection region is where real effects get detected too.

Accuracy in any useful sense involves both errors. Reducing one by moving the threshold always increases the other, because the threshold is the boundary between them.
Both fall together only when the two sampling distributions move apart, which requires a larger sample, a bigger effect or less noise. That is a matter of design, not of choosing a smaller number.

4.17 A trial with $\sigma = 4$ and $n = 16$ tests $H_0 : \mu \leq 0$ against $H_A : \mu > 0$ at $\alpha = 0.05$. The true effect is $\mu_1 = 2$.

i. Compute the critical value and the standard error.
ii. Compute β and the power.
iii. Recompute the power at $\alpha = 0.01$.
iv. What does the comparison show?

SOLUTION

(i) One-sided at 5%: $z_{0.05} = 1.645$. Standard error $= 4/\sqrt{16} = 1$.
(ii) Under the alternative the statistic is centred on $\delta = (2-0)/1 = 2$, so

$$\beta = P(Z \leq 1.645 - 2) = \Phi(-0.355) = 0.361$$

and the power is $1 - 0.361 = 0.639$.
(iii) At $\alpha = 0.01$ the critical value is $z_{0.01} = 2.326$, so $\beta = \Phi(2.326 - 2) = \Phi(0.326) = 0.628$ and the power falls to **0.372**.
(iv) Tightening α from 5% to 1% cut the chance of detecting a real two-point effect from 64% to 37%. The stricter test is not the better test; it has simply traded one error for the other. Nothing about the drug changed.

4.18 How large must the trial in the previous question be to detect a two-point effect with 80% power?

SOLUTION

Power of 80% means $\beta = 0.20$, and the critical value must sit $z_{0.20}$ below the alternative's centre. Since $z_{0.20} = 0.842$, we need

$$\delta = z_\alpha + z_\beta = 1.645 + 0.842 = 2.487$$

With $\delta = (\mu_1 - \mu_0)\sqrt{n}/\sigma$,

$$\sqrt{n} = \frac{2.487 \times 4}{2} = 4.97 \qquad \Longrightarrow \qquad n = 24.7$$

so 25 patients. Checking against the table in Section 4.9, $n = 25$ gives power 0.804.

Note the shape of this calculation: it is done *before* the trial, and it requires naming the smallest effect worth detecting. That is a judgement about medicine, not statistics, and it has to be made either way.

4.19 Plot the power of the trial in question 4.17 against sample size, for true effects of one point and two points.

SOLUTION

```
power_at <- function(n, effect, sigma = 4, alpha = 0.05) {
  1 - pnorm(qnorm(1 - alpha) - effect / (sigma / sqrt(n)))
}

grid <- expand.grid(n = seq(5, 250, by = 5), effect = c(1, 2))
grid$power <- power_at(grid$n, grid$effect)
grid$effect <- factor(grid$effect, labels = c("1 point", "2 points"))

ggplot(grid, aes(n, power, colour = effect)) +
  geom_hline(yintercept = 0.8, linetype = "dashed", colour =
  ↪  "grey45") +
  geom_line(linewidth = 0.9) +
  scale_colour_manual(values = c(book_palette$reject,
  ↪  book_palette$accent),
                      name = "True effect") +
  labs(x = "Sample size", y = "Power") +
  theme_book() +
  theme(legend.position = "top")
```

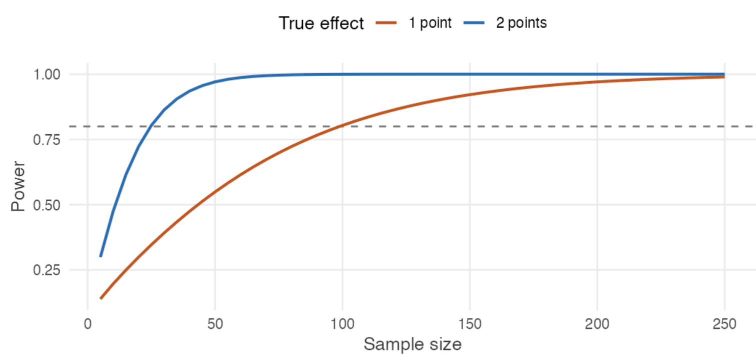

Figure 4.6: Power against sample size for two effect sizes. Small effects need far more data.

Both curves rise towards 1, and the dashed line marks the conventional

80% target. Detecting a two-point effect takes about 25 patients; detecting a one-point effect takes about 100.
Halving the effect roughly quadruples the sample needed, which is the $\sqrt{n}$ rule appearing once more: precision improves with the square root of the sample, so detecting an effect half the size requires four times the data.

4.13 Question bank

WATCH OUT

No solutions are given for this section, by design. These questions are drawn from past quizzes and exams. If you can work them without checking an answer key, you are ready; if you cannot, the section to reread is named beside each one.
They run from easiest to hardest. Do them in order.

Multiple choice

QB 11.1 The level of significance, α, is the risk of *(select all that apply)*

a. Rejecting the null hypothesis when it is true
b. Rejecting the null hypothesis when the alternative hypothesis is true
c. Rejecting the alternative hypothesis when the null hypothesis is true
d. Rejecting the alternative hypothesis when it is true

See Section 4.4.

QB 11.2 An economist sets the alternative hypothesis that the average CGPA of undergraduate students is less than 8. What type of test should be used, and what is the critical value at $\alpha = 5\%$?

a. One-tailed test and $+1.96$
b. One-tailed test and -1.96
c. One-tailed test and $+1.645$
d. One-tailed test and -1.645

See Section 4.8.

QB 11.3 The margin of error for a confidence interval is $\pm Z_{\alpha/2}\,\sigma/\sqrt{n}$. Which statements are correct? *(select all that apply)*

a. Increasing the sample size will decrease the margin of error
b. The margin of error decreases as α rises
c. The larger the margin of error, the less confident one is about the estimate
d. The margin of error increases with the population standard deviation σ

See Section 4.6.

QB 11.4 A 95% confidence interval for mean student height is $(165.41,\ 170.95)$. You now compute a 99% interval from the same data. Which statements are likely correct? *(select all that apply)*

a. The 90%, 95% and 99% intervals are the same
b. The 99% interval is $(164.54,\ 171.83)$
c. The 99% interval is $(165.86,\ 170.51)$
d. The answer cannot be determined from the information given

See Section 4.6. Ask yourself which way an interval must move when you demand more confidence.

QB 11.5 A company tests the null hypothesis that the average time to accept a job offer is at most 7 days. The test concludes that the average exceeds 7 days. It is later learned that the true mean is exactly 7 days. What error occurred?

a. Type II error
b. Type I error
c. A correct decision was made
d. Cannot be determined from the information provided

Short reasoning

QB 11.6 "A high p-value is evidence that the null hypothesis is true." Do you agree? Why or why not?

See Section 4.3.

QB 11.7 Two economists, Amit and Anand, collect data on Indian agricultural wages to test the effect of MNREGA. Amit concludes that wages after the policy are higher than before, when in fact wages have not changed. Anand concludes that wages before and after are the same — also incorrectly.

It is claimed that Amit has committed a Type I error and Anand a Type II error. Do you agree? Why or why not?

Numerical

QB 11.8 A researcher wants to estimate the average weekly study hours of university students. The population standard deviation is 6 hours. What sample size is needed to estimate the mean with a margin of error of 1 hour at 90% confidence?

a. $n = 98$
b. $n = 64$
c. $n = 121$
d. $n = 82$

QB 11.9 The numbers of people taking the morning university shuttle on 16 randomly chosen days are:

```
27, 46, 35, 33, 29, 45, 28, 24, 30, 40, 38, 35, 41, 27, 38, 29
```

The administration claims the daily average is 30. Taking $\sigma = 7$:

a. Construct a 99% confidence interval for the mean number of riders.
b. Use that interval to test the administration's claim.
c. Would your conclusion change at the 95% level? Explain *why* before you compute it.

Part (c) is the one that separates understanding from arithmetic.

Errors and power

QB 4.18 A Type II error is

a. rejecting a true null hypothesis
b. failing to reject a false null hypothesis
c. choosing the wrong significance level
d. computing the p-value incorrectly

QB 4.19 Reducing α from 0.05 to 0.01, holding everything else fixed,

a. reduces both error probabilities
b. reduces Type I error and increases Type II error
c. increases Type I error and reduces Type II error
d. leaves power unchanged

QB 4.20 The power of a test is

a. α
b. $1 - \alpha$
c. β
d. $1 - \beta$

QB 4.21 Power increases when *(select all that apply)*

a. the sample size grows
b. the true effect is larger
c. the population standard deviation is larger
d. α is made smaller

QB 4.22 A study reports "no significant effect" and had 30% power against the effect it was looking for. The most accurate conclusion is

a. there is no effect
b. the effect is small
c. the study was unlikely to detect the effect even if it existed
d. the null hypothesis has been confirmed

QB 4.23 In the Decision Review System, "umpire's call" on a marginal ball-tracking decision reflects a choice to

a. keep α small
b. keep β small
c. maximise power
d. eliminate both errors

QB 4.24 Explain why power cannot be stated for a test without also stating an effect size, whereas α can.

QB 4.25 A colleague computes the power of a study after it has returned a null result, using the effect size the study itself estimated. Explain what is wrong with this.

QB 4.26 A quality inspector tests whether a machine's mean fill weight has drifted from 500 g. With $\sigma = 8$ g, $n = 25$ and $\alpha = 0.05$ two-sided:

i. State the critical values.
ii. Compute the power against a true mean of 503 g.
iii. Compute the power against a true mean of 505 g.
iv. How large a sample would give 90% power against a 3 g drift?

QB 4.27 Two studies test the same hypothesis. Study A has 40 observations, Study B has 400. Both report $p = 0.04$.

i. Which provides stronger evidence that the effect is *large*?
ii. Which had more power against a given effect size?
iii. Explain why these two questions have different answers.

4.14 Looking ahead

We have now completed the basic toolkit of statistical inference for a single mean. Given one sample, we can

- estimate an unknown population mean,
- quantify the uncertainty around that estimate with a confidence interval, and
- test a specific claim about the population with a hypothesis test.

The last two, as Section 4.6 showed, are the same object seen from two sides.

Every test in this unit compared a sample against a *claimed* number — a figure from a government report, a manufacturer's specification, a previous year's estimate. That number had to come from somewhere outside the data.

Most questions worth asking do not arrive in that form.

Did the households that received the transfer end up better off than those that did not? Do girls in the scheme attend school more than girls outside it? Did the districts that got the road see wages rise more than the districts that did not? In each case there is no external claim to test against. There are two groups, each with its own sample mean, and the question is whether the difference between them is larger than sampling variation alone would produce.

That is the subject of the next unit. It is also where the difficulty changes character. Deciding whether two groups differ is a statistical problem, and we will solve it. Deciding whether one group differs *because* of the treatment is a question about how the groups came to be formed — and no test statistic can answer it.

Chapter 5

Hypothesis Testing: Two Samples

Every test so far compared a sample against a number that came from outside the data — a government figure, a manufacturer's specification, a dean's estimate.

Most questions worth asking are not shaped like that. They compare two groups.

Do men and women earn different wages for the same work? Do students from government schools perform differently from those from private schools? Does the high-yielding variety of wheat out-yield the conventional one? Did the villages that received the road see incomes rise more than those that did not?

In each case there is no external claim to test against. There are two groups, each with its own sample mean, and the question is whether the difference between them is larger than sampling variation alone would produce.

This unit builds that test. It also brings us to a distinction that will occupy the rest of the book: the difference between establishing that two groups differ, and establishing that something *caused* them to differ.

5.1 Comparing two groups

> *Two groups, two sample means, and they are not equal. What have we learned?*

Suppose a claim is made that two populations have the same mean. Male and female students at a university consume the same number of cigarettes on average; conventional and high-yielding wheat produce the same yield per hectare; two districts have the same average household income.

We could settle it by measuring everyone in both populations. That is expensive in exactly the way Unit 2 described, and nobody does it.

So we draw a sample from each group and compute both sample means. Unlike the one-sample case, we now have *two* numbers, each carrying its own sampling variation.

Suppose they differ. Have we shown the populations differ?

> **WATCH OUT**
>
> No — and for the same reason as in Unit 4, only more so.
> Each sample mean would miss its population mean even if we had measured perfectly. Now there are two of them, each free to land high or low. A difference between two sample means is *guaranteed* to appear even when the two populations are identical.

So the question is the one we have asked all along, adapted:

> If the two populations really had the same mean, how often would two samples come out as far apart as ours did?

If that would happen often, we have learned nothing about the populations. If it would almost never happen, then either something unusual happened to us or the two populations are not the same.

Everything that follows is machinery for answering that question. And as before, the machinery needs a sampling distribution — this time of a *difference*.

5.2 What does a difference of means do across samples?

> *We know how $\bar{X}$ behaves. How does $\bar{X} - \bar{Y}$ behave?*

Take a sample $x_1, \ldots, x_n$ from a population with mean μ_x and standard deviation σ_x, and an independent sample $y_1, \ldots, y_m$ from a population with mean μ_y and standard deviation σ_y. The sample sizes need not be equal.

The null hypothesis is that the two populations have the same mean:

$$H_0 : \mu_x = \mu_y \qquad H_A : \mu_x \neq \mu_y$$

It is more useful to write the same thing as a statement about the difference:

$$H_0 : \mu_x - \mu_y = 0$$

because that turns a comparison of two quantities into a claim about one, and we already know how to test a claim about one quantity.

The quantity we will study is therefore $\bar{X} - \bar{Y}$. Two things need establishing: where its distribution sits, and how wide it is.

WHERE THIS COMES FROM

Where it sits. Expectation passes through a difference:

$$E[\bar{X} - \bar{Y}] = E[\bar{X}] - E[\bar{Y}] = \mu_x - \mu_y$$

So the difference of sample means is an unbiased estimator of the difference of population means. Under H_0 it is centred on zero.
How wide it is. Variance does *not* pass through a difference in the same way. For independent quantities,

$$\begin{aligned} \text{Var}[\bar{X} - \bar{Y}] &= \text{Var}[\bar{X}] + \text{Var}[-\bar{Y}] \\ &= \text{Var}[\bar{X}] + (-1)^2\text{Var}[\bar{Y}] \\ &= \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m} \end{aligned}$$

using $\text{Var}(aX) = a^2\text{Var}(X)$ from Unit 2, with $a = -1$. Taking the square root gives the standard error of the difference:

$$\text{se}(\bar{X} - \bar{Y}) = \sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}$$

Independence is needed for the first line, exactly as it was in Unit 2. It is what allows the variance of a sum to be the sum of the variances.

REMEMBER THIS

The variances add, even though the means subtract.
This is the single most important line in the unit. Squaring -1 gives $+1$, so subtracting a second noisy quantity makes the result *more* variable, not less.
Comparing two groups is therefore harder than measuring one. Both estimates wobble, and both wobbles end up in the answer.

That has a practical consequence worth seeing now rather than later. Suppose two groups each have $\sigma = 10$ and each is sampled $n = 100$ times. The standard error of either mean alone is $10/\sqrt{100} = 1$. The standard error of their difference is

$$\sqrt{\frac{100}{100} + \frac{100}{100}} = \sqrt{2} \approx 1.41$$

about 40% larger. Detecting a difference between two groups needs more data than estimating either group's mean to the same precision.

5.3 Testing the difference when σ is known

> *We have a sampling distribution. What is the test?*

The argument is the one from Section 4.4, with the new standard error in place of the old one. We reject when the observed difference is too large:

$$|\bar{x} - \bar{y}| \geq c$$

and we choose c so that, if the null were true, a gap that large would arise with probability only α.

WHERE THIS COMES FROM

Standardising the difference gives a quantity with no unknowns in it:

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_y)}{\sqrt{\dfrac{\sigma_x^2}{n} + \dfrac{\sigma_y^2}{m}}} \sim N(0,1)$$

Under H_0 the term $(\mu_x - \mu_y)$ is zero and drops out. Setting

$$P[\,|\bar{X} - \bar{Y}| \geq c\,] = \alpha$$

and dividing through by the standard error,

$$P\left[\,|Z| \geq \frac{c}{\sqrt{\sigma_x^2/n + \sigma_y^2/m}}\,\right] = \alpha$$

The two tails carry $\alpha/2$ each, so the bracketed quantity is $Z_{\alpha/2}$, giving

$$c = Z_{\alpha/2}\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}$$

REMEMBER THIS

The test statistic and the interval for the difference are

$$Z_{calc} = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \qquad (\bar{x} - \bar{y}) \pm Z_{\alpha/2}\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}$$

Reject H_0 when $|Z_{calc}| > Z_{\alpha/2}$, which happens exactly when the interval excludes zero — the same equivalence as Section 4.6, now about a difference rather than a mean.

Conventional against high-yielding wheat

Twelve plots are sown with conventional seed and fourteen with a high-yielding variety. The conventional plots average 45.2 kg per hectare and the HYV plots 48.6 kg. From long agronomic experience the standard deviations are known: $\sigma_x = 0.8$ for conventional and $\sigma_y = 1.0$ for HYV.

Are these data consistent, at the 5% level, with the two varieties yielding the same on average?

```
xbar <- 45.2; sx <- 0.8; n <- 12      # conventional
ybar <- 48.6; sy <- 1.0; m <- 14      # high-yielding

se_diff <- sqrt(sx^2 / n + sy^2 / m)
z_calc  <- (xbar - ybar) / se_diff

c(difference = xbar - ybar,
  se         = se_diff,
  z_calc     = z_calc,
  p_value    = 2 * pnorm(-abs(z_calc)))
```

```
#>                   difference                           se
#> -3.39999999999999985789145285  0.3532165125838608865649348
#>                       z_calc                      p_value
#> -9.62582404522429690985 06364  0.00000000000000000000006221
```

$Z_{calc} = -9.63$, far beyond -1.96, and the p-value is smaller than any conventional threshold. We reject the claim that the two varieties yield the same.

The interval says how much the varieties differ, which is the more useful statement:

```
(xbar - ybar) + c(-1, 1) * qnorm(0.975) * se_diff
```

```
#> [1] -4.092 -2.708
```

The HYV seed yields between 2.7 and 4.1 kg per hectare more than the conventional variety, with 95% confidence. Note that the interval lies entirely below zero and so excludes it — the same verdict the test reached.

> **THE IDEA IN PLAIN WORDS**
>
> Report the interval, not just the rejection. "The varieties differ significantly" is a much weaker statement than "the HYV variety yields 2.7 to 4.1 kg per hectare more," and a farmer deciding what to sow needs the second.

5.4 When σ is unknown

> *As always, we do not know σ_x or σ_y. Now there are two of them.*

In practice both population standard deviations must be estimated from the samples, using s_x and s_y. As in Section 4.7, that costs something, and what it costs depends on the situation. Three cases arise, and they differ only in what goes underneath the square root.

Large samples

If both samples are reasonably large — a common rule of thumb is at least 20 each — then s_x and s_y are reliable enough that substituting them changes little, and the normal distribution remains a good approximation:

$$Z_{calc} = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}}}$$

This is the same statistic with hats on the standard deviations.

Equal variances: the pooled estimator

If we are willing to assume the two populations share a common standard deviation, $\sigma_x = \sigma_y = \sigma$, then both samples are estimating the *same* quantity and it is wasteful to estimate it twice. We combine them.

> **REMEMBER THIS**
>
> The **pooled variance estimator** is a weighted average of the two sample variances, each weighted by its degrees of freedom:
>
> $$s_p^2 = \frac{(n-1)s_x^2 + (m-1)s_y^2}{n+m-2}$$
>
> and the test statistic becomes
>
> $$t_{calc} = \frac{\bar{x} - \bar{y}}{\sqrt{s_p^2\left(\frac{1}{n} + \frac{1}{m}\right)}}$$
>
> compared against t with $n+m-2$ degrees of freedom.

The degrees of freedom are worth pausing on. We began with $n+m$ observations, and spent one estimating each of the two sample means, leaving $n+m-2$. The same accounting as Section 3.2, applied twice.

Pooling buys precision — $n+m-2$ degrees of freedom rather than something closer to the smaller sample's — but only if the assumption holds. If one group really is more variable than the other, the pooled estimate is a weighted average of two different things, and the test is wrong in a way the output will not reveal.

Unequal variances: Welch's test

The safer option makes no assumption about the variances. It uses the separate estimates,

$$t_{calc} = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}}}$$

and adjusts the degrees of freedom downwards to compensate for the extra uncertainty. The adjustment is not a whole number and is not worth memorising; R computes it.

WATCH OUT

R's `t.test()` uses Welch by default. If you have derived a pooled statistic by hand and R disagrees with you, this is usually why.
To pool, pass `var.equal = TRUE` explicitly.
Modern practice is to prefer Welch. It costs very little when the variances really are equal, and it protects you when they are not — and you can never verify equal variances, only fail to reject the claim that they are equal.

Do North and South Indian students differ in height?

`student-survey-height.csv` records 100 students, each classified by region.

```
heights <- read.csv("data/student-survey-height.csv")

aggregate(height ~ region, data = heights,
          function(x) round(c(n = length(x), mean = mean(x), sd = sd(x)),
          ↪  2))
```

```
#>         region height.n height.mean height.sd
#> 1 North Indian    57.00      165.16      7.29
#> 2 South Indian    43.00      163.67      9.85
```

The two sample means differ by about 1.5 cm, and the two sample standard deviations are not equal — 7.3 against 9.9. That is a case for Welch.

```
t.test(height ~ region, data = heights)
```

```
#>
#>  Welch Two Sample t-test
#>
#> data:  height by region
#> t = 0.83, df = 74, p-value = 0.4
#> alternative hypothesis: true difference in means between group North
 ↪  Indian and group South Indian is not equal to 0
#> 95 percent confidence interval:
#>  -2.074  5.043
#> sample estimates:
#> mean in group North Indian mean in group South Indian
#>                      165.2                      163.7
```

$t = 0.83$ on 74.4 degrees of freedom, with $p = 0.41$. We fail to reject. The 95% interval for the difference runs from -2.07 to 5.04 cm, and it **contains zero** — the two statements agree, as they must.

So a 1.5 cm gap in the samples is entirely unremarkable. With these sample sizes and this much variation within each group, differences of that size turn up constantly when the populations are identical.

> **GOING A LITTLE FURTHER**
>
> Compare the degrees of freedom the two approaches give here.
>
> ```
> c(welch  = t.test(height ~ region, data = heights)$parameter,
>   pooled = t.test(height ~ region, data = heights, var.equal =
>    ↪  TRUE)$parameter)
> ```
>
> ```
> #>  welch.df pooled.df
> #>     74.36     98.00
> ```
>
> Pooling claims 98 degrees of freedom; Welch allows only 74.4. The difference is the price of not assuming the two groups are equally variable — and here, where one sample's standard deviation is a third larger than the other's, it is a price worth paying.
> Both give the same verdict in this case. They do not always.

5.5 When two samples are not really two samples

> *Ten couples. Two columns of ages. Is that two samples?*

Suppose we record the ages of both partners in ten married couples.

```
couples <- read.csv("data/marriage-ages.csv")
couples
```

```
#>    partner1 partner2
#> 1        36       35
#> 2        72       67
#> 3        37       33
#> 4        36       35
#> 5        51       50
#> 6        50       46
#> 7        47       47
#> 8        50       42
#> 9        37       36
#> 10       41       41
```

Our question is simple: **is partner 1, on average, older than partner 2?**

At first glance this looks like a two-sample problem. There are two columns. There are ten observations in each. So why not perform the test we have just learned?

```
t.test(couples$partner1, couples$partner2)
```

```
#>
#>  Welch Two Sample t-test
#>
#> data:  couples$partner1 and couples$partner2
#> t = 0.52, df = 18, p-value = 0.6
#> alternative hypothesis: true difference in means is not equal to 0
#> 95 percent confidence interval:
#>  -7.537 12.537
#> sample estimates:
#> mean of x mean of y
#>      45.7      43.2
```

The test finds essentially nothing: $t = 0.52$ with $p = 0.61$. On this evidence the two are the same age on average, and the interval for the difference runs from -7.5 to 12.5 years.

Something is wrong

Look carefully at the data. In every couple, which partner is older?

Couple	1	2	3	4	5	6	7	8	9	10
Older	P1	P1	P1	P1	P1	P1	same	P1	P1	same

Partner 1 is older in eight couples. In the other two the partners are exactly the same age. **Partner 2 is never older.**

How can a statistical test look at that and return $p = 0.61$?

The answer is that we analysed the wrong problem.

What the two-sample test assumed

The two-sample t-test assumes the observations in one sample have nothing to do with the observations in the other. It treats the data as two unrelated collections of people:

$$\text{Partner 1:} \quad 36 \quad 72 \quad 37 \quad 36 \quad 51 \quad \ldots$$

Partner 2: 35 67 33 35 50 ...

But they are not unrelated. Each observation belongs to a *married couple*, and within a couple the two ages are naturally linked. Someone who marries at 70 is likely to have a spouse around 70; someone who marries at 35 is likely to have a spouse around 35. The ages within a couple are strongly correlated.

Two sources of variation

There are two quite different sources of variation in this dataset, and only one of them is of any interest.

Variation between couples. Some couples are in their thirties, some in their fifties, one in their seventies. These differences are enormous.

Variation within couples. Partners usually differ by only a few years. This is the variation we actually care about.

The two-sample test mixes them together. The between-couple variation dominates the calculation and hides the small but remarkably consistent within-couple differences.

Seen properly, the structure is obvious.

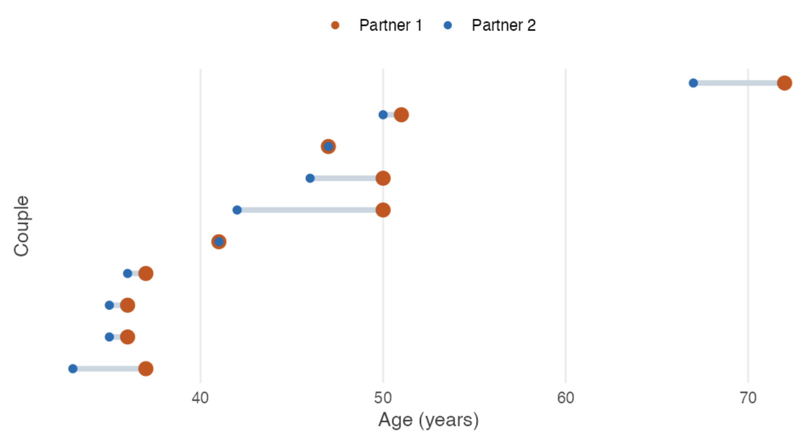

Figure 5.1: Ten couples, ordered by the age of partner 2. Each segment joins the two partners in one couple. The couples span nearly forty years between them; within any couple the gap is at most eight. Two couples are the same age, and show as a red ring around a blue centre.

The figure explains both results at once. Vertically, the couples are spread across nearly forty years — that is the variation the unpaired test measures the gap against, and it is overwhelming. Horizontally, within each couple, the red point is at or to the right of the blue one every single time, and never more than eight years away. That small, utterly consistent gap is what a paired test measures.

Removing what we do not care about

Instead of comparing all the partner 1s with all the partner 2s, compare each person with *their own* partner. Compute the difference for every couple.

```
d <- couples$partner1 - couples$partner2
d
```

```
#>  [1] 1 5 4 1 1 4 0 8 1 0
```

Now something worth pausing on has happened. The ages of the couples have disappeared completely.

The couple aged 72 and 67 contributes $72 - 67 = 5$. The couple aged 36 and 35 contributes $36 - 35 = 1$. It no longer matters whether a couple is young or old — subtracting removes the between-couple variation automatically, because it was common to both columns.

Only the within-couple difference remains, and that is exactly the quantity we wanted to study.

The statistical model

Let

$$d_i = x_i - y_i$$

where x_i is the age of partner 1 in couple i and y_i is the age of partner 2 in the *same* couple.

Instead of two samples, we now have one sample of differences $d_1, d_2, \ldots, d_n$. Our parameter of interest is

$$\mu_d = E(d)$$

the population mean difference.

Setting up the hypotheses

If we simply want to know whether the average ages differ,

$$H_0 : \mu_d = 0 \qquad H_A : \mu_d \neq 0$$

If we specifically suspect partner 1 is older, the alternative is one-sided, as in Section 4.8:

$$H_0 : \mu_d \leq 0 \qquad H_A : \mu_d > 0$$

> **REMEMBER THIS**
>
> Notice that the hypotheses are no longer about two population means. We are not testing $H_0 : \mu_1 = \mu_2$.
> We are testing whether the average *difference* equals zero — a claim about a single parameter, which is why the machinery of Unit 4 applies unchanged.

Estimating the mean difference

The sample mean difference is

$$\bar{d} = \frac{1}{n} \sum_{i=1}^{n} d_i$$

and the sample standard deviation of the differences is the ordinary one from Section 3.2, applied to the d_i:

$$s_d = \sqrt{\frac{\sum_{i=1}^{n} (d_i - \bar{d})^2}{n-1}}$$

```r
c(n = length(d), d_bar = mean(d), s_d = sd(d), se = sd(d) /
  sqrt(length(d)))
```

```
#>       n   d_bar     s_d      se
#> 10.0000  2.5000  2.6352  0.8333
```

So $\bar{d} = 2.5$ years and $s_d = 2.635$ years.

The test statistic

Under the null hypothesis $\mu_d = 0$,

$$t = \frac{\bar{d} - 0}{s_d/\sqrt{n}}$$

This is exactly the one-sample t statistic from Section 4.7, with d in place of x. Substituting,

$$t = \frac{2.5}{2.635/\sqrt{10}} = \frac{2.5}{0.833} = 3.00$$

and under H_0 this follows t_{n-1}, so with $n = 10$ we have $df = 9$.

```
t_calc <- mean(d) / (sd(d) / sqrt(length(d)))

c(t_calc  = t_calc,
  df      = length(d) - 1,
  t_crit  = qt(0.975, df = length(d) - 1),
  p_value = 2 * pt(-abs(t_calc), df = length(d) - 1))
```

```
#>  t_calc      df  t_crit p_value
#> 3.00000 9.00000 2.26216 0.01496
```

Since $3.00 > 2.262$, we reject H_0 at the 5% level. The two-sided p-value is 0.015.

Confidence interval

A 95% interval for the mean difference is

$$\bar{d} \pm t_{\alpha/2,\, n-1} \frac{s_d}{\sqrt{n}}$$

```
mean(d) + c(-1, 1) * qt(0.975, df = 9) * sd(d) / sqrt(10)
```

```
#> [1] 0.6149 4.3851
```

Partner 1 is between about 0.6 and 4.4 years older, on average. The interval excludes zero, agreeing with the test as always.

Doing it in R

Everything above can be had two ways. Either compute the differences yourself and run a one-sample test,

```r
t.test(d, mu = 0)
```

```
#>
#>  One Sample t-test
#>
#> data:  d
#> t = 3, df = 9, p-value = 0.01
#> alternative hypothesis: true mean is not equal to 0
#> 95 percent confidence interval:
#>  0.6149 4.3851
#> sample estimates:
#> mean of x
#>       2.5
```

or let R take the differences for you:

```r
t.test(couples$partner1, couples$partner2, paired = TRUE)
```

```
#>
#>  Paired t-test
#>
#> data:  couples$partner1 and couples$partner2
#> t = 3, df = 9, p-value = 0.01
#> alternative hypothesis: true mean difference is not equal to 0
#> 95 percent confidence interval:
#>  0.6149 4.3851
#> sample estimates:
#> mean difference
#>             2.5
```

The two are the same test, because `paired = TRUE` does nothing more than subtract the columns and run the one-sample test on the result.

Why did the answer change so much?

Notice what changed, and what did not.

Table 5.1: The same ten couples, analysed twice.

Analysis	Treats the observations as	p-value
Two-sample t-test	Twenty unrelated people	0.61
Paired t-test	Ten differences within couples	0.015

The data never changed. The same ten couples were analysed twice. The only difference was whether the analysis respected the way the data were collected.

WHERE THIS COMES FROM

The algebra shows exactly what pairing removes. Write each age as

$$x_i = \mu + \alpha_i + \delta_i \qquad\qquad y_i = \mu + \alpha_i + \varepsilon_i$$

where μ is the overall average age, α_i captures whether couple i is generally young or old, and δ_i and ε_i are the remaining individual departures. The term α_i is what makes the columns so variable: it ranges across nearly forty years in these data. Crucially, it is *the same* α_i in both expressions, because both partners belong to the same couple.

Taking differences,

$$\begin{aligned} d_i &= x_i - y_i \\ &= (\mu + \alpha_i + \delta_i) - (\mu + \alpha_i + \varepsilon_i) \\ &= \delta_i - \varepsilon_i \end{aligned}$$

Both μ and α_i cancel. The large between-couple effect disappears entirely, and the differences carry only the within-couple departures.

That is why the paired test is so much more powerful. It is not measuring a smaller quantity more cleverly; it is measuring the same quantity against a much smaller standard error.

The standard deviations make the point numerically:

```
round(c(sd_partner1 = sd(couples$partner1),
        sd_partner2 = sd(couples$partner2),
        sd_of_diffs = sd(d)), 3)
```

```
#> sd_partner1 sd_partner2 sd_of_diffs
#>      11.156      10.174       2.635
```

Each column varies by about eleven years. The differences vary by 2.6. The unpaired test measured a 2.5-year gap against eleven years of noise and saw nothing; the paired test measures it against 2.6 and finds it clearly.

When should we use a paired test?

A paired test is appropriate whenever the two measurements belong to the same unit. Typical cases:

- before and after measurements on the same patients;
- yields from the same plots under two fertilisers in successive seasons;
- examination scores of the same students before and after a training programme;
- the same households interviewed in two survey rounds;
- twins, or matched pairs, receiving different treatments.

The common feature is that every observation in one column has a natural partner in the other. The analysis should therefore focus on the difference within each pair, not on the two columns separately.

> **WATCH OUT**
>
> The reverse error is also possible. If the two columns are *not* paired — 45 government school students and 52 private school students, say — then there is no natural partner for anyone, `paired = TRUE` will pair them by row number, which is meaningless, and R will simply refuse if the lengths differ.
> Pairing is a fact about how the data were collected, not an option to try.

> **REMEMBER THIS**
>
> **A paired t-test is not a special kind of two-sample t-test. It is a one-sample t-test applied to the differences between paired observations.**
> If you remember one sentence from this section, it should be that one.

Table 5.2: A paired test is a one-sample test wearing different clothes.

	One-sample t-test	Paired t-test
Data	$x_1, \dots, x_n$	$d_1, \dots, d_n$ where $d_i = x_i - y_i$
Parameter	μ	μ_d
Null	$H_0 : \mu = \mu_0$	$H_0 : \mu_d = 0$
Statistic	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$	$t = \dfrac{\bar{d}}{s_d/\sqrt{n}}$
Degrees of freedom	$n - 1$	$n - 1$

Read across the rows. There is no new mathematics here at all — only a new first step, in which two columns become one.

5.6 A difference is not an effect

> *The test says two groups differ. Does it say why?*

We can now establish that two groups differ by more than sampling variation. That is a genuine achievement, and it is not what most interesting questions actually ask.

Consider the attendance data from this course. Split the 680 students at the median attendance and compare their final examination scores.

```
students <- read.csv("data/attendance-grades.csv")
students$group <- ifelse(students$attendance >=
↪  median(students$attendance),
                         "High attendance", "Low attendance")

aggregate(final_score ~ group, data = students,
          function(x) round(c(n = length(x), mean = mean(x)), 2))
```

```
#>              group final_score.n final_score.mean
#> 1 High attendance         346.0             91.3
#> 2  Low attendance         334.0             90.4
```

```
t.test(final_score ~ group, data = students)
```

```
#>
#>  Welch Two Sample t-test
#>
#> data:  final_score by group
#> t = 2.5, df = 671, p-value = 0.01
#> alternative hypothesis: true difference in means between group High
↪  attendance and group Low attendance is not equal to 0
#> 95 percent confidence interval:
#>  0.2063 1.5927
#> sample estimates:
#> mean in group High attendance  mean in group Low attendance
#>                          91.3                          90.4
```

Students who attended more scored higher, and $p = 0.011$ rejects the claim that the two groups have the same mean.

It is tempting to conclude that attending class raises examination scores by about a point. Two separate problems stand in the way.

The difference is significant and trivial

The interval for the difference runs from about 0.21 to 1.59 marks out of 100.

A gap that small would not change anyone's grade, and would not justify any policy. Yet it is "statistically significant", because 680 students is a large enough sample to detect very small differences reliably.

> **REMEMBER THIS**
>
> Statistical significance answers "is this difference bigger than noise?" It does not answer "is this difference big enough to matter?" That is a question about the subject, not about the statistics, and no p-value speaks to it.
> Always read the interval, in the units of the problem, before deciding whether a result is important.

The groups were not assigned

The deeper problem is that nobody allocated students to high and low attendance. They chose, and the reasons they chose are not in the dataset.

A student who attends regularly may be more motivated, better prepared, less burdened by paid work or family obligations, or living closer to campus. Any of those could raise examination scores on its own. The two groups differ in attendance *and* in everything that attendance is correlated with, and the test cannot separate them.

> **WATCH OUT**
>
> The two-sample test compares two groups as they arrive. It has nothing to say about *why* they came to be different.
> If the groups formed themselves, a difference in outcomes may reflect the treatment, or the reasons for selection, or both — and the arithmetic is identical either way. This is the problem of **selection**, and it is the central obstacle in all of empirical economics.

It is worth being precise about what has and has not been shown here.

We have shown that students who attended more scored slightly higher. That is a fact about the world and the test establishes it properly. We have not shown that attending more *causes* higher scores, and no amount of additional care with the t statistic would help.

What would help is a different design. If attendance had been assigned by something unrelated to the students themselves — a lottery, a timetable clash, a bus route — then the groups would be comparable in expectation and the difference could be read as an effect. Creating such comparisons, or finding them in the world, is the subject of the second part of this book.

> **THE IDEA IN PLAIN WORDS**
>
> The question "do these groups differ?" is a statistical one, and this unit answers it.
> The question "does this treatment work?" is a question about how the groups were formed. It is answered by the design of the study, not by the choice of test.

5.7 Summary

> **REMEMBER THIS**
>
> - Two groups produce two sample means, and a difference between them is guaranteed to appear even when the populations are identical. The question is whether it is larger than sampling variation would produce.
> - $E[\bar{X} - \bar{Y}] = \mu_x - \mu_y$: the difference of sample means is unbiased for the difference of population means.
> - **The variances add even though the means subtract**, so $\mathrm{se}(\bar{X} - \bar{Y}) = \sqrt{\sigma_x^2/n + \sigma_y^2/m}$. Comparing two groups is noisier than measuring one, and needs more data.
> - With σ known the test is a Z-test on the difference; the interval is $(\bar{x} - \bar{y}) \pm Z_{\alpha/2}$ se, and it excludes zero exactly when the test rejects.
> - With σ unknown, three cases differ only under the square root: large samples use s_x and s_y with the normal; **pooling** assumes equal variances and buys $n + m - 2$ degrees of freedom; **Welch** assumes nothing and adjusts the degrees of freedom down. R's `t.test()` uses Welch unless you pass `var.equal = TRUE`, and Welch is the safer default.
> - A **paired** design gives one number per unit. Take differences and run a one-sample test. Treating paired data as two independent samples discards the comparison the design was built to make — in the marriage data it turned $p = 0.015$ into $p = 0.61$.
> - A significant difference need not be an important one. Read the interval in the units of the problem.
> - A difference between two groups that formed themselves is not an effect. The test compares the groups as they arrive and is silent

> about why they differ.

5.8 Practice problems

> **YOUR TURN**
>
> Work each one before opening the solution. The question bank that follows has no answers.

Concept checks

5.1 Why is the standard error of a difference of means larger than the standard error of either mean on its own?

SOLUTION

Because the two sample means each carry sampling variation, and both end up in the difference. Formally, $\text{Var}(aX) = a^2\text{Var}(X)$ with $a = -1$ gives $(-1)^2 = +1$, so the variances add:

$$\text{Var}[\bar{X} - \bar{Y}] = \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}$$

Subtracting a noisy quantity does not cancel noise; it adds more. This is why detecting a difference between two groups takes more data than estimating one group's mean to the same precision.

5.2 A study reports that two groups differ with $p = 0.003$. A colleague concludes the difference is large. What is wrong?

SOLUTION

The p-value measures how incompatible the data are with the claim of no difference. It says nothing about the size of the difference.
A very small p-value can come from a tiny difference measured in a large sample. The size of the difference is reported by the confidence interval, in the units of the problem, and that is what has to be inspected before calling a result large or important.

5.3 Why does R's `t.test()` report fractional degrees of freedom?

SOLUTION

Because it performs Welch's test by default, which does not assume the two populations have equal variances. The correction for that extra uncertainty adjusts the degrees of freedom downwards, and the adjustment is generally not a whole number.
The pooled test, obtained with `var.equal = TRUE`, uses $n+m-2$ degrees of freedom, which always is.

5.4 The same twenty households are surveyed before and after a cash transfer. Why is a two-sample test the wrong tool?

SOLUTION

Because the two columns are not independent samples — each household appears in both. The design produces one number per household, the change.
Households differ enormously among themselves, and that variation is present identically in both columns. A two-sample test treats it as noise, and it swamps the change we are trying to detect. Taking differences within households removes it, leaving only the quantity of interest.

5.5 Villages that received a road show higher incomes than villages that did not. State two distinct reasons this need not mean the road raised incomes.

SOLUTION

First, the villages may have differed beforehand. Roads are not built at random — they tend to reach places that are more accessible, more populous, more politically connected or already better off. Those same features raise incomes on their own.
Second, the two groups may differ in things that changed alongside the road: electrification, a market, a school. The road is correlated with a bundle of improvements, and a comparison of means attributes the whole bundle to it.
In both cases the arithmetic is correct and the causal reading is not.

Explain

5.6 Explain the trade-off between the pooled and Welch tests, and why modern practice prefers Welch.

SOLUTION

Pooling assumes the two populations share a standard deviation. If true, both samples estimate the same quantity, they can be combined, and the resulting test has $n+m-2$ degrees of freedom — more than Welch allows, so more power.
If false, the pooled estimate averages two different quantities and the test is wrong in a way nothing in the output reveals. Welch avoids the assumption entirely, at the cost of fewer degrees of freedom.
The reason to prefer Welch is asymmetric risk. When the variances really are equal, Welch loses very little. When they are not, pooling misleads. And the assumption cannot be verified — a test of equal variances can only fail to reject, which is not the same as establishing it.

5.7 A researcher tests whether men and women in a firm are paid differently, finds $p = 0.02$, and concludes the firm discriminates. Evaluate.

SOLUTION

The test establishes that average pay differs by more than sampling variation would explain. That is not the same as discrimination.
Men and women in the firm may differ in role, seniority, years of experience, hours, or field of training, and any of those could account for a pay gap without anyone being treated differently for the same work. The comparison of raw means attributes the whole difference to sex because sex is the only variable in the analysis.
Establishing discrimination requires comparing like with like — people doing the same work with the same experience — which is a question of design and, in later units, of holding other variables constant. The two-sample test is not capable of it.

Numerical problems

5.8 Two irrigation methods are trialled. Drip irrigation on 15 plots yields a mean of 38.4 quintals per hectare; flood irrigation on 18 plots yields 35.9. Population standard deviations are known to be 3.2 and 4.1 respectively.

i. Test at the 5% level whether the mean yields differ.
ii. Construct a 95% confidence interval for the difference.
iii. Does the interval agree with the test?

SOLUTION

(i) The standard error of the difference is

$$\sqrt{\frac{3.2^2}{15}+\frac{4.1^2}{18}}=\sqrt{0.6827+0.9339}=\sqrt{1.6166}=1.271$$

$$Z_{calc}=\frac{38.4-35.9}{1.271}=\frac{2.5}{1.271}=1.967$$

Since $1.967 > 1.96$, we reject H_0 at the 5% level — but only just.
(ii) $2.5 \pm 1.96(1.271) = 2.5 \pm 2.491 = [0.009,\ 4.991]$.
(iii) Yes. The interval excludes zero, and the test rejects. Both do so by the narrowest possible margin, which is the same fact stated twice: the interval endpoint sits at 0.009 because Z_{calc} sits at 1.967.
A result this marginal should be reported as marginal. Nothing about it justifies confidence that drip irrigation is better.

5.9 Two samples give $n = 10$, $s_x = 4$ and $m = 14$, $s_y = 6$.

i. Compute the pooled variance estimate.
ii. How many degrees of freedom does the pooled test use?
iii. Why is it not $n + m$?

SOLUTION

(i)

$$s_p^2=\frac{(10-1)16+(14-1)36}{10+14-2}=\frac{144+468}{22}=\frac{612}{22}=27.82$$

so $s_p = 5.27$, lying between 4 and 6 and closer to 6 because the larger sample carries more weight.
(ii) $n + m - 2 = 22$.
(iii) Because two quantities were estimated from the data before the spread could be measured — one sample mean in each group. Each costs a degree of freedom, exactly as in Section 3.2, where estimating a single mean cost one.

5.10 A programme is evaluated by comparing 200 participants with 200 non-participants. Participants' mean income is ₹4,200 higher, with a standard error of ₹1,400 for the difference.

i. Test whether the difference is zero at the 5% level.
ii. Construct a 95% interval.
iii. The programme was open to anyone who applied. What can be concluded about its effect?

SOLUTION

(i) $Z_{calc} = 4200/1400 = 3.0$, which exceeds 1.96. We reject the claim that the two groups have the same mean income.
(ii) $4200 \pm 1.96(1400) = [₹1{,}456,\ ₹6{,}944]$.
(iii) Nothing about its effect. Because the programme was open to applicants, participants selected themselves. People who apply to income programmes may be more motivated, better informed, or better connected — or, equally plausibly, more desperate and worse off to begin with. The direction of the bias is not even clear.
The interval is a correct statement about the difference between two groups of people. It is not an estimate of what the programme did to anybody.

R exercises

5.11 Using `data/student-survey-sleep.csv`, test whether urban and rural students report different average hours of sleep. Report the interval and say what you conclude.

SOLUTION

```
sleep <- read.csv("data/student-survey-sleep.csv")

aggregate(sleep ~ residence, data = sleep,
          function(x) round(c(n = length(x), mean = mean(x), sd =
           ↪  sd(x)), 2))
```

```
#>   residence sleep.n sleep.mean sleep.sd
#> 1     Rural   21.00       6.52     1.53
#> 2     Urban   85.00       6.60     1.12
```

```
t.test(sleep ~ residence, data = sleep)
```

```
#>
#>  Welch Two Sample t-test
#>
#> data:  sleep by residence
#> t = -0.21, df = 26, p-value = 0.8
#> alternative hypothesis: true difference in means between group
 ↪  Rural and group Urban is not equal to 0
#> 95 percent confidence interval:
#>  -0.8064  0.6540
#> sample estimates:
#> mean in group Rural mean in group Urban
```

```
#>              6.524              6.600
```

The two group means are close and the interval for the difference contains zero, so we fail to reject. There is no evidence here that urban and rural students sleep different amounts on average.
Note what "no evidence of a difference" is not: evidence of no difference. With these sample sizes, a real difference of well under an hour would be difficult to detect at all, and the interval says as much by being wide.

5.12 Test whether social media use differs between North and South Indian students in the same dataset, using both Welch and the pooled test. Do they agree, and which would you report?

SOLUTION

```r
rbind(
  welch  = unlist(t.test(socialmediahours ~ region, data =
    sleep)[c("statistic", "parameter", "p.value")]),
  pooled = unlist(t.test(socialmediahours ~ region, data = sleep,
    var.equal = TRUE)[c("statistic", "parameter", "p.value")])
)
```

```
#>        statistic.t parameter.df p.value
#> welch       0.9172        50.63  0.3634
#> pooled      0.9845       104.00  0.3271
```

The two agree closely here, which is what happens when the sample standard deviations are similar — the pooled and separate estimates of the variance are then nearly the same number.
Report Welch. It gives the same answer when the assumption holds and a correct one when it does not, so there is no situation in which pooling is the safer choice.

5.13 Take the wage data as a population and draw independent samples of 200 urban and 200 rural workers. Test whether mean annual earnings differ, and report the interval. Then say what the interval does and does not establish.

SOLUTION

```r
wages <- read.csv("data/wages-india-synthetic.csv")

set.seed(9)
urban <- sample(wages$annual_earnings[wages$residence == "Urban"],
 ↪  200)
rural <- sample(wages$annual_earnings[wages$residence == "Rural"],
 ↪  200)

t.test(urban, rural)
```

```
#>
#>  Welch Two Sample t-test
#>
#> data:  urban and rural
#> t = 6.5, df = 223, p-value = 0.0000000005
#> alternative hypothesis: true difference in means is not equal to 0
#> 95 percent confidence interval:
#>  47095 88103
#> sample estimates:
#> mean of x mean of y
#>    100811     33212
```

The difference is large and the interval is nowhere near zero, so urban workers in this population earn substantially more than rural workers. What the interval establishes is the size of that gap. What it does not establish is that moving a worker from a village to a city would raise their earnings by that amount. Urban and rural workers differ in education, occupation and industry, and the sample says nothing about which part of the gap belongs to the place and which to the people.

5.14 Simulate the point that variances add. Draw two independent samples of 100 from a population with $\sigma = 10$, and compare the observed standard deviation of $\bar{x}$ with that of $\bar{x} - \bar{y}$ across 5,000 repetitions. Check both against theory.

SOLUTION

```r
set.seed(4)
sims <- replicate(5000, {
  x <- rnorm(100, mean = 50, sd = 10)
  y <- rnorm(100, mean = 50, sd = 10)
  c(one_mean = mean(x), difference = mean(x) - mean(y))
})

round(c(sd_of_one_mean       = sd(sims["one_mean", ]),
        theory_one_mean      = 10 / sqrt(100),
        sd_of_difference     = sd(sims["difference", ]),
        theory_difference    = sqrt(10^2/100 + 10^2/100)), 4)
```

```
#>    sd_of_one_mean   theory_one_mean  sd_of_difference
 ↪  theory_difference
#>          0.9989          1.0000          1.4260          1.4142
```

The difference is about $\sqrt{2}$ times as variable as a single mean, exactly as $\sqrt{\sigma^2/n + \sigma^2/m}$ predicts. Note also that the difference is centred on zero: both populations have mean 50, and the estimator is unbiased for a difference of zero.

5.15 Demonstrate the cost of ignoring pairing. Using `marriage-ages.csv`, report the paired and unpaired p-values, and explain the gap by comparing the standard deviation of the differences with the standard deviations of the two columns.

SOLUTION

```r
couples <- read.csv("data/marriage-ages.csv")

c(paired   = t.test(couples$partner1, couples$partner2, paired =
 ↪  TRUE)$p.value,
  unpaired = t.test(couples$partner1, couples$partner2)$p.value)
```

```
#>   paired unpaired
#>  0.01496  0.60699
```

```r
round(c(sd_partner1  = sd(couples$partner1),
        sd_partner2  = sd(couples$partner2),
        sd_of_diffs  = sd(couples$partner1 - couples$partner2)), 3)
```

```
#> sd_partner1 sd_partner2 sd_of_diffs
#>      11.156      10.174       2.635
```

The two columns each have a standard deviation of about 11 years, because the couples range from their mid-thirties to their seventies. The differences have a standard deviation of about 2.6 years.
That is the whole explanation. The unpaired test measures the small gap against the huge between-couple variation and sees nothing. The paired test removes that variation — it is common to both columns and cancels on subtraction — and measures the gap against what is left.

Paired designs

5.16 A paired test on n pairs has $n - 1$ degrees of freedom, not $2n - 2$. Why, when there are $2n$ numbers in the dataset?

SOLUTION

Because the analysis is not performed on the $2n$ numbers. It is performed on the n differences.
Taking differences reduces $2n$ observations to n, one per pair. From those n we estimate one quantity, $\bar{d}$, before we can measure their spread, so one degree of freedom is spent and $n - 1$ remain — exactly as in the one-sample case of Section 3.2.
The $2n - 2$ of the pooled two-sample test is right for a design in which the $2n$ observations really are $2n$ independent pieces of information. In a paired design they are not: knowing one member of a pair tells you a great deal about the other.

5.17 What feature of the data determines whether pairing was worth doing?

SOLUTION

The correlation between the two columns.
Pairing removes whatever is common to both members of a pair. If the two measurements are strongly related — the same household in two rounds, the same plot in two seasons — then a great deal is common, differencing removes it, and the standard error falls sharply.
If the two columns happen to be unrelated, there is nothing common to remove. Differencing then leaves the variability essentially unchanged, and the paired test performs about as well as the unpaired one, losing only the few degrees of freedom.
So pairing rarely costs much and can gain a great deal. What determines which happens is not a choice; it is a property of the design.

5.18 A researcher measures productivity at the same 30 firms before and after a policy change, then runs a two-sample t-test. Describe what will happen to the p-value and why.

SOLUTION

It will be too large — often dramatically so — and the researcher will under-report a real effect.
Firms differ enormously among themselves in productivity: a large firm and a small one may differ by a factor of ten, while the policy might move any single firm by a few per cent. In a two-sample test that between-firm variation is counted as noise, and it swamps the change being looked for.
The pairing is exactly what removes it. Each firm's before-measurement acts as its own comparison, so the firm's general productivity level cancels and only its change remains. Writing $x_i = \mu + \alpha_i + \delta_i$ and $y_i = \mu + \alpha_i + \varepsilon_i$, the large firm effect α_i appears in both and disappears on subtraction.
The error is conservative rather than anti-conservative, which is why it often goes unnoticed: the researcher simply concludes there is no effect.

5.19 A new irrigation technique is trialled on eight plots. Yields in quintals per hectare before and after are

Plot	1	2	3	4	5	6	7	8
Before	32	28	35	30	33	29	31	34
After	35	33	37	36	37	30	36	40

i. Explain why this is a paired design.
ii. Compute the differences, $\bar{d}$ and s_d.
iii. Test at the 5% level whether the technique changed yields.
iv. Construct a 95% confidence interval for the mean change.

SOLUTION

(i) Each plot is measured twice. The two columns are not eight plots and eight other plots; they are the same eight plots under two conditions, so every observation in one column has a natural partner in the other. Plots differ in soil, slope and drainage, and those differences are present identically in both rounds.
(ii) Differences (after − before): $3, 5, 2, 6, 4, 1, 5, 6$.

$$\bar{d} = \frac{32}{8} = 4.0$$

Deviations from 4: $-1, 1, -2, 2, 0, -3, 1, 2$, whose squares sum to 24, so

$$s_d^2 = \frac{24}{7} = 3.4286 \qquad s_d = 1.852$$

and the standard error is $1.852/\sqrt{8} = 0.655$.
(iii)

$$t = \frac{4.0}{0.655} = 6.11$$

with $df = 7$. The critical value is $t_{0.025,\,7} = 2.365$, and $6.11 > 2.365$, so we reject $H_0 : \mu_d = 0$. The p-value is 0.0005.
(iv) $4.0 \pm 2.365(0.655) = [2.45,\ 5.55]$ quintals per hectare.
Note the interval is in the units of the problem and excludes zero, agreeing with the test. Note also that the estimated gain of 4 quintals per hectare is large in agronomic terms as well as statistically detectable — the two questions are separate, and here both answers happen to be favourable.

5.20 Seven workers are timed on a task, in minutes, before and after training.

Worker	1	2	3	4	5	6	7
Before	14	12	15	13	16	11	14
After	11	10	13	12	13	11	12

i. State the hypotheses for a test that training *reduced* completion time.
ii. Compute the test statistic and its degrees of freedom.
iii. Test at the 5% level.
iv. Construct the appropriate one-sided 95% bound and interpret it.

SOLUTION

(i) Working with $d_i = \text{before}_i - \text{after}_i$, so that a reduction in time gives a positive difference:

$$H_0 : \mu_d \leq 0 \qquad H_A : \mu_d > 0$$

The equality lives in H_0, as always, and the claim we want evidence *for* sits in H_A.
(ii) Differences: $3, 2, 2, 1, 3, 0, 2$. So $\bar{d} = 13/7 = 1.857$ and $s_d = 1.069$, giving a standard error of $1.069/\sqrt{7} = 0.404$ and

$$t = \frac{1.857}{0.404} = 4.60$$

with $df = 6$.

(iii) Right-tailed at 5%, so the critical value is $t_{0.05,\,6} = 1.943$ — not 2.447. Since $4.60 > 1.943$ we reject H_0; the one-sided p-value is 0.0019.
(iv) A one-sided lower bound spends all of α in one tail:

$$\bar{d} - t_{0.05,\,6}\,\frac{s_d}{\sqrt{n}} = 1.857 - 1.943(0.404) = 1.07$$

so the interval is $[1.07,\ \infty)$. With 95% confidence, training reduced completion time by at least 1.07 minutes. We have deliberately given up any statement about how large the reduction might be, because the question only asked whether there was one.

5.21 Using `marriage-ages.csv`, verify by direct computation that `paired = TRUE` does nothing more than take differences. Then run the one-sided test of whether partner 1 is older.

SOLUTION

```
couples <- read.csv("data/marriage-ages.csv")
d <- couples$partner1 - couples$partner2

# Three routes to the same number
c(by_hand    = mean(d) / (sd(d) / sqrt(length(d))),
  one_sample = t.test(d, mu = 0)$statistic,
  paired     = t.test(couples$partner1, couples$partner2, paired =
   ↪  TRUE)$statistic)
```

```
#>      by_hand one_sample.t     paired.t
#>            3            3            3
```

Identical, as they must be: `paired = TRUE` subtracts the columns and calls the one-sample routine.

```
t.test(couples$partner1, couples$partner2,
       paired = TRUE, alternative = "greater")
```

```
#>
#>  Paired t-test
#>
#> data:  couples$partner1 and couples$partner2
#> t = 3, df = 9, p-value = 0.007
#> alternative hypothesis: true mean difference is greater than 0
#> 95 percent confidence interval:
#>  0.9724    Inf
#> sample estimates:
#> mean difference
#>             2.5
```

The one-sided p-value is half the two-sided one, 0.0075 against 0.015, and the interval is now a lower bound: partner 1 is older by at least 1.02 years. The one-sided test is defensible here only because the direction was decided before looking — we asked whether partner 1 is older, not whether the two differ. Choosing the direction after seeing that eight of ten differences were positive would not be a test at all.

5.22 Show that the benefit of pairing depends on the correlation between the two columns. Simulate pairs of measurements with correlation 0, 0.5 and 0.9, a true difference of 4, and $n = 10$, and compare how often each test rejects.

SOLUTION

```r
set.seed(7)

power_at <- function(rho, reps = 4000) {
  out <- replicate(reps, {
    a <- rnorm(10)
    b <- rho * a + sqrt(1 - rho^2) * rnorm(10)
    x <- 50 + 10 * a
    y <- 50 + 10 * b + 4                          # true difference is 4
    c(paired   = t.test(x, y, paired = TRUE)$p.value < 0.05,
      unpaired = t.test(x, y)$p.value              < 0.05,
      se_paired   = sd(x - y) / sqrt(10),
      se_unpaired = sqrt(var(x)/10 + var(y)/10))
  })
  c(rho = rho, rowMeans(out))
}

round(t(sapply(c(0, 0.5, 0.9), power_at)), 3)
```

```
#>      rho paired unpaired se_paired se_unpaired
#> [1,] 0.0  0.125    0.126     4.348       4.408
#> [2,] 0.5  0.212    0.073     3.069       4.402
#> [3,] 0.9  0.717    0.014     1.374       4.389
```

Read the two standard error columns first, because they explain everything else. The paired standard error falls from about 4.35 to 1.37 as the correlation rises. The unpaired standard error sits near 4.40 throughout — it is computed from the two columns separately and so cannot notice that they are related at all.

The rejection rates follow. At $\rho = 0$ the two tests are equivalent, and the paired one is marginally worse because it has 9 degrees of freedom against 18. At $\rho = 0.9$ the paired test rejects about 72% of the time and the unpaired test almost never does.

That last figure deserves a moment. The unpaired test does not merely fail to improve as the correlation rises; it gets *worse*. With strongly correlated columns the observed difference is tightly concentrated near its true value of 4, while the standard error the unpaired test uses stays at 4.4 — so its statistic sits reliably near $4/4.4 \approx 0.9$ and almost never strays past the critical value. Ignoring a strong pairing produces a test that is consistently, confidently unable to see anything.

5.9 Question bank

WATCH OUT

No answers are given for this section, by design.

Multiple choice

QB 5.1 The standard error of $\bar{X} - \bar{Y}$ for independent samples is

a. $\dfrac{\sigma_x}{\sqrt{n}} - \dfrac{\sigma_y}{\sqrt{m}}$
b. $\sqrt{\dfrac{\sigma_x^2}{n} + \dfrac{\sigma_y^2}{m}}$
c. $\sqrt{\dfrac{\sigma_x^2}{n} - \dfrac{\sigma_y^2}{m}}$
d. $\dfrac{\sigma_x + \sigma_y}{\sqrt{n+m}}$

QB 5.2 The pooled two-sample t-test uses how many degrees of freedom?

a. $n + m$
b. $n + m - 1$
c. $n + m - 2$
d. $\min(n, m) - 1$

QB 5.3 A 95% confidence interval for $\mu_x - \mu_y$ is $[-1.2,\ 4.8]$. At the 5% level we

a. reject $H_0 : \mu_x = \mu_y$
b. fail to reject $H_0 : \mu_x = \mu_y$

c. cannot tell without the p-value
d. conclude the two means are equal

QB 5.4 `t.test(x, y)` in R by default

a. assumes equal variances
b. performs Welch's test
c. performs a paired test
d. performs a one-tailed test

QB 5.5 Which of the following are paired designs? *(select all that apply)*

a. Blood pressure of 40 patients before and after a drug
b. Test scores of 40 boys and 45 girls
c. Yields on 20 plots under one seed and 20 different plots under another
d. Household expenditure in the same 200 households in 2019 and 2023

QB 5.6 Ignoring pairing in a genuinely paired design usually

a. makes it easier to reject the null
b. makes it harder to reject the null
c. has no effect on the p-value
d. reverses the sign of the difference

QB 5.7 Two groups differ with $p < 0.001$ in a sample of 50,000. This implies

a. the difference is large
b. the difference is important
c. the difference is unlikely to be sampling variation alone
d. one group caused the other's outcome

Short reasoning

QB 5.8 Explain why subtracting one sample mean from another increases variability rather than reducing it, and give the algebraic reason.

QB 5.9 A researcher has two samples with $s_x = 2$ and $s_y = 9$, and uses the pooled test. State the assumption being made, why it is doubtful here, and what the consequence is.

QB 5.10 A newspaper reports that graduates of a coaching institute score 40 marks higher on average than non-attendees, with a very small p-value. Explain what the comparison can and cannot support.

QB 5.11 In a paired design, the two columns are strongly positively correlated. Explain, without algebra, why this makes the paired test more powerful than the unpaired one.

QB 5.12 A trial assigns a new fertiliser to plots by lottery. Explain why the difference in mean yields can be read as an effect here, when the same comparison in Section 5.6 could not.

QB 5.13 Two studies estimate the same difference. One reports $[0.1,\ 9.9]$ and the other $[4.8,\ 5.2]$. Both exclude zero. Explain what distinguishes them and which is more useful.

Numerical

QB 5.14 Government school students score a mean of 62.4 with $s = 11.2$ in a sample of 45. Private school students score 66.1 with $s = 9.8$ in a sample of 52.

i. Compute the standard error of the difference.
ii. Compute the test statistic.
iii. Test at the 5% level whether the means differ.
iv. Construct a 95% confidence interval, and state whether it agrees with (iii).

QB 5.15 A sample of 8 plots under a new seed yields a mean of 41.2 quintals with $s = 3.1$. A sample of 10 plots under the old seed yields 37.8 with $s = 2.9$.

i. Compute the pooled variance estimate and s_p.
ii. Compute the pooled test statistic and its degrees of freedom.
iii. Test at the 5% level whether the new seed yields more, stating your hypotheses carefully.
iv. Would Welch's test use more or fewer degrees of freedom here, and why?

QB 5.16 Twelve workers are timed on a task before and after training. The mean reduction is 4.2 minutes with a standard deviation of differences of 5.1 minutes.

i. State the hypotheses for a test that training reduced completion time.
ii. Compute the test statistic and its degrees of freedom.
iii. Test at the 5% level.
iv. Explain why the standard deviation of the *differences* is the right quantity here rather than the standard deviations of the two rounds.

QB 5.17 Using `data/attendance-grades.csv`, write R code to compare cumulative GPA between students above and below the median attendance. Report the interval, then write a short paragraph stating precisely what your result establishes and what it does not.

Paired designs

QB 5.18 A paired t-test on 15 pairs has how many degrees of freedom?

a. 13
b. 14
c. 28
d. 30

QB 5.19 The paired t-test is equivalent to

a. a two-sample t-test with `var.equal = TRUE`
b. a one-sample t-test on the differences
c. Welch's test with adjusted degrees of freedom
d. a Z-test on the difference of means

QB 5.20 Pairing gives the largest gain in precision when

a. the two sample sizes are equal
b. the two columns are strongly correlated
c. the two columns are independent
d. the sample is large

QB 5.21 Which analysis is appropriate? *(match each design to a test)*

i. Yields on 20 plots under seed A and on 20 *different* plots under seed B
ii. Yields on the same 20 plots in two consecutive seasons, one seed each
iii. Blood pressure of 40 patients, half given a drug and half a placebo
iv. Blood pressure of 40 patients, each measured before and after a drug

a. Two-sample t-test
b. Paired t-test

QB 5.22 A researcher analyses genuinely paired data as two independent samples. The most likely consequence is

a. a p-value that is too small
b. a p-value that is too large
c. a confidence interval that is too narrow
d. no change, since the sample means are the same either way

Paired reasoning

QB 5.23 Explain why the paired test uses the standard deviation of the differences rather than the standard deviations of the two columns, and what this achieves.

QB 5.24 A colleague argues that pairing should always be used because it is "more powerful". Under what circumstances is that claim false, and what is the cost of pairing when the two columns are unrelated?

QB 5.25 A study reports before-and-after measurements on the same 12 villages and finds a significant improvement. A critic replies that there was no comparison group, so nothing has been shown. Explain what the paired test does establish here and what the critic is right about.

Paired numerical

QB 5.26 Nine households report monthly food expenditure, in rupees, before and after a ration card reform.

Household	Before	After
1	4200	4050
2	3800	3750
3	5100	4800
4	4600	4500
5	3900	3700
6	4400	4250
7	5000	4650
8	4100	4050
9	4700	4400

i. Compute the differences, $\bar{d}$ and s_d.
ii. Test at the 5% level whether expenditure changed.
iii. Construct a 95% confidence interval for the mean change.
iv. State one reason the change should not be attributed to the reform.

QB 5.27 A paired study on $n = 16$ pairs gives $\bar{d} = 3.5$ and $s_d = 6.4$.

i. Compute the test statistic and its degrees of freedom.
ii. Test at the 5% and 1% levels.
iii. How large would n have to be for the same $\bar{d}$ and s_d to reject at the 1% level?

QB 5.28 Two analyses of the same twelve paired observations report $s_d = 2.1$ and, from the two columns separately, $s_x = 9.4$ and $s_y = 9.1$.

i. Compute the standard error the paired test would use.
ii. Compute the standard error an unpaired test would use.
iii. Explain the ratio between them in terms of what pairing removes.

QB 5.29 Using `data/student-survey-sleep.csv`, write R code that deliberately performs a paired test on two variables that are *not* paired in any meaningful sense. Report what R does, and explain why the output is meaningless even though no error is raised.

5.10 Looking ahead

We can now compare two groups properly. We can also say exactly what such a comparison does not deliver, which turns out to be the more important lesson.

Two limitations remain, and both point the same way.

The first is that comparing two groups uses the data wastefully. Splitting the 680 students at the median attendance threw away everything we knew about *how much* each student attended, reducing a rich variable to a coin flip. A student who attended 32 classes and one who attended 20 were treated as identical.

The second is selection. The attendance groups differed in attendance and in everything correlated with it, and a two-sample test cannot separate them because it knows about nothing else.

Both problems have the same solution. Instead of splitting a variable in two and comparing averages, we can model the relationship between the whole of one variable and the whole of another — and, crucially, do so while holding other variables fixed.

That is regression, and it occupies the rest of this book.

Chapter 6

The Simple Regression Model

Throughout this book we have asked whether differences between groups are larger than we would expect from chance alone. Regression asks a broader question: **how is one variable related to another?**

That shift matters because economic theory usually makes qualitative predictions — higher prices reduce demand, more education increases earnings — and qualitative predictions are hard to act on. Regression lets us estimate *how much* one variable moves with another, express it in units someone can use, and attach a measure of uncertainty to the answer.

Almost every empirical question in economics has this shape. How do wages change with education? How does consumption change with income? How does crop yield change with rainfall? How does a student's GPA change with attendance?

Answering such questions turns out to be surprisingly difficult, and three problems arise immediately.

Other things matter. Attendance is not the only determinant of GPA, and education is not the only determinant of wages. The relationship is never exact, so we need a way of describing a pattern that holds on average while individuals depart from it.

We do not know the functional form. The relationship between two variables might be a straight line, or a curve, or something that rises and then flattens. Economic theory rarely specifies which.

An observed relationship need not be causal. Even if attendance and grades move together, attendance may not be responsible for the difference.

REMEMBER THIS

The rest of this unit develops a model for studying relationships between variables. Along the way it answers four questions.

1. What relationship are we trying to describe?
2. Why do individuals not lie on that relationship?
3. Under what conditions can we interpret it causally?
4. How do we estimate it from a sample?

Each section that follows answers one of them.

We build all of it from ideas we already have — variation, sampling and statistical inference. Nothing genuinely new is required; what changes is the object of study, from a mean or a difference of means to a relationship.

The result is the central tool of **econometrics** — the field concerned with measuring economic relationships from data — though the tool itself is used across the social sciences and beyond.

6.1 From comparing groups to modelling relationships

> *We can compare two groups. Why is that not enough?*

Unit 5 compared students above and below the median attendance. Those who attended more had higher average grades, by more than sampling variation alone would explain.

But knowing that two groups differ is only the beginning. We still do not know how grades change *as* attendance changes, or whether attendance itself is responsible for the difference.

That second gap is the deeper one, and it is worth seeing clearly.

The experiment we cannot run

What we would really like is a thought experiment. Take one student and observe them twice: once attending 20 classes and once attending 30, with everything else about them unchanged — the same ability, the same motivation, the same teacher, the same week. The difference in their grades would isolate the effect of attendance itself.

That experiment is impossible. A student cannot attend both 20 and 30 classes. Instead, we observe *different* students who differ in many ways at once. The challenge of empirical economics is to separate the effect of attendance from all the other differences between those students.

> **WATCH OUT**
>
> Simple regression does not solve that problem. Section 7.8 returns to this point in detail. If a reader finishes this unit believing that regression automatically uncovers causal effects, then this book has failed them.
> What regression does is change the question we ask of the data, and build the machinery that later units extend towards the thought experiment above.

The question has changed

Instead of asking whether students with high attendance differ from those with low attendance, we now ask whether grades tend to rise as attendance rises.

Attendance is no longer a group label. It becomes a **numerical variable whose entire range carries information.**

That is the conceptual step from a t-test to a regression, and everything technical in this unit follows from it.

The price of the old question

To compare two groups, Unit 5 took a variable recording how many classes each of 680 students attended — a number running from 2 to 32 — and replaced it with a label saying "high" or "low". A student who attended 32 classes and one who attended 28 were treated as identical, while one who attended 28 and one who attended 27 were placed in opposite groups.

We can quantify the price of that simplification.

```
students <- read.csv("data/attendance-grades.csv")
students$high <- students$attendance >= median(students$attendance)

c(split_t      = abs(t.test(sem_gpa ~ high, data = students)$statistic),
  regression_t = summary(lm(sem_gpa ~ attendance, data =
  ↪  students))$coefficients[2, 3])
```

```
#>    split_t.t regression_t
#>        11.80        17.59
```

The same data. The same outcome. Almost the same question. Splitting attendance into two groups produces a t statistic of 11.8; using the full attendance variable produces one of 17.6.

The relationship was always present — the median split simply discarded much of the evidence for it.

REMEMBER THIS

Regression does not merely give us a better answer. It asks a different question — and in doing so improves on a two-sample comparison in two ways.

- It uses the **full** explanatory variable rather than reducing it to two groups.
- It estimates a **slope** — the expected change in y for a one-unit increase in x — instead of merely comparing two averages.

In the next unit we will add more explanatory variables, allowing us to compare observations while holding other measured characteristics fixed. That brings us closer to the counterfactual comparison we would like to make, although — as we shall see — it does not eliminate the problem entirely.

6.2 The population regression function

> *What exactly is regression trying to estimate?*

The previous section argued that regression asks a different question from a two-sample comparison. The natural next question is: what is the object we are trying to estimate?

It is tempting to think that regression is trying to draw a straight line through a cloud of points. That is not quite right.

Start with the individual outcomes, because they are noisier than students expect.

Forty-nine students in our data attended exactly 25 classes. They did not receive the same semester GPA.

```r
at25 <- students[students$attendance == 25, ]

round(c(n = nrow(at25), average_gpa = mean(at25$sem_gpa),
        lowest = min(at25$sem_gpa), highest = max(at25$sem_gpa)), 3)
```

```
#>           n average_gpa      lowest     highest
#>      49.000       6.098       1.575       9.075
```

One scored 1.58 and another 9.08, and forty-seven others lay in between — all of them having attended the same twenty-five classes. Attendance alone plainly does not determine anyone's grade.

So a line through the individual points cannot be what we are looking for. There is no line that passes near 1.58 and 9.08 at the same value of x.

Instead of looking at every individual student, suppose we ask a different question:

> Among students who attended 25 classes, what is the average GPA?

Now repeat the exercise for students who attended 26 classes, then 27, then 28, and so on.

Those averages are called **conditional means**, because they average GPA conditional on a particular attendance level. We write the conditional mean as

$$E(Y \mid X = x)$$

which simply means "the average value of Y among all members of the population whose X equals x."

The table below approximates these conditional means by grouping attendance into bands.

```
students$band <- cut(students$attendance,
                     breaks = c(0, 20, 24, 27, 30, 33),
                    labels = c("2–20", "21–24", "25–27", "28–30", "31–32"))

aggregate(sem_gpa ~ band, data = students,
          function(x) round(c(n = length(x), mean_gpa = mean(x)), 2))
```

```
#>    band sem_gpa.n sem_gpa.mean_gpa
#> 1  2–20     94.00             4.38
#> 2 21–24     85.00             6.18
#> 3 25–27    155.00             6.30
#> 4 28–30    209.00             6.98
#> 5 31–32    137.00             7.67
```

Two patterns immediately emerge.

First, students with the same attendance still obtain very different grades. Attendance is therefore not enough to predict an individual's GPA.

Second, the average GPA rises steadily as attendance increases — from 4.38 to 7.67. Individual outcomes are noisy, but the centre of the distribution shifts upwards.

Regression is interested in this second pattern.

If we computed the conditional mean for every possible attendance level and plotted those averages, we would obtain a curve describing how the average GPA changes with attendance throughout the population.

> **REMEMBER THIS**
>
> That curve is called the **population regression function** (PRF):
>
> $$\text{PRF} \;=\; E(Y \mid X = x)$$
>
> The population regression function is therefore not a fitted line. It exists whether or not we ever estimate it. It is simply the true relationship between attendance and the population's average GPA.

Figure 6.1 illustrates the idea. The grey points are individual students. The orange points are estimates of the conditional mean within each attendance band. Regression is concerned with those averages, not with explaining every individual observation.

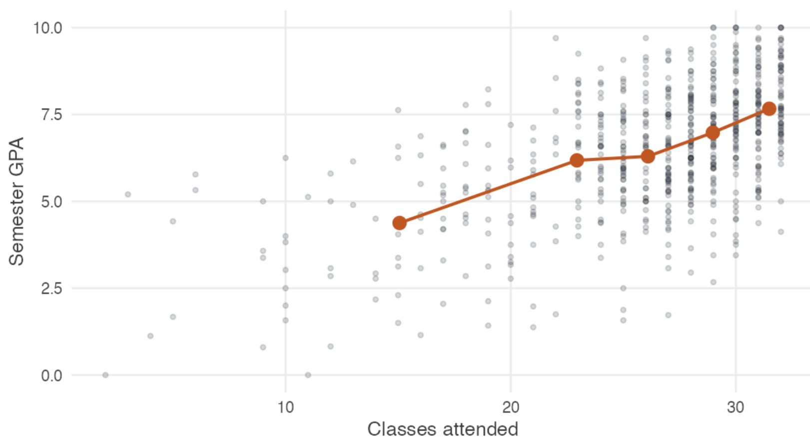

Figure 6.1: Semester GPA against attendance for 680 students. Grey points are individual students; orange points are the average GPA within each attendance band. Regression describes the orange points.

Each conditional mean $E(Y \mid X = x)$ is a function of x, so the PRF is also called the **conditional expectation function** (CEF):

$$E(Y \mid X = x) = f(x)$$

WATCH OUT

The CEF tells us how the *average* value of y changes with x. It does **not** say that $y = \beta_0 + \beta_1 x$ for every member of the population.
Individual students depart from the average for their attendance level, often by a great deal. Confusing the conditional mean with the individual outcome is the most common misreading of a regression.

Notice that the orange points do not lie exactly on a straight line. Nothing in the definition of the population regression function requires linearity. The conditional mean could be curved, flat over some ranges, or highly nonlinear.

Only now do we make a modelling choice.

The PRF is the object we want to estimate. Unfortunately it is an unknown *function*, and functions can take infinitely many shapes. To make estimation possible at all, we approximate it with a simple mathematical form.

In this unit we assume that the population regression function is approximately linear,

$$E(Y \mid X = x) = \beta_0 + \beta_1 x$$

because a straight line is simple to estimate, easy to interpret, and often provides a useful approximation to more complicated relationships.

Whether a straight line is a good approximation is an empirical question rather than a mathematical one. Section 7.2 returns to how well it performs for these data.

The population regression function gives us the target. Two questions remain.

First, why do individual students differ from the conditional mean? Second, how can we estimate the population regression function when we observe only a sample rather than the entire population?

The next two sections answer those questions.

6.3 From averages to observations

The PRF describes averages. Individual students are not averages. How do we write down the difference?

Return to the forty-nine students who attended exactly 25 classes. Their conditional mean was 6.10, and two of them scored 1.58 and 9.08.

The conditional expectation function accounts for the 6.10. It has nothing to say about why one student was four and a half points below it and another three points above.

So an individual's grade is the conditional mean *plus* a departure from it. Give that departure a name.

> **REMEMBER THIS**
>
> For each individual i,
>
> $$y_i = E(Y \mid X = x_i) + u_i$$
>
> where u_i is the **error term** or **stochastic disturbance** — the amount by which individual i departs from the average for their value of x. Substituting the linear conditional mean from Section 6.2 gives the **simple linear regression model**:
>
> $$y_i = \beta_0 + \beta_1 x_i + u_i$$

This is not a new model. It is simply another way of writing the population regression function once we give the unexplained part a name.

The two students above become

$$1.58 = \beta_0 + \beta_1(25) + u_1 \qquad 9.08 = \beta_0 + \beta_1(25) + u_2$$

with a large negative u_1 and a large positive u_2. Both students attended the same twenty-five classes, so the first two terms are identical for each of them; everything that distinguishes them sits in u.

We now have a model. Before asking how to estimate it, we should be clear about what that last term contains.

6.4 The error term

> *What is u?*

Students meet the word "error" and assume something has gone wrong. Nothing has.

For the grades example, u contains everything other than attendance that affects a student's GPA:

- how hard they worked outside class,
- how well prepared they were when the course began,

- the quality of teaching they happened to receive,
- whether they were ill during the examinations,
- whether they held a job that semester,
- how much they happened to like the subject,
- and luck on the day.

None of these is a mistake, and none of them is measurement noise. They are real influences on real outcomes that our model does not include — because they are not in the dataset, and in some cases could not be.

> **REMEMBER THIS**
>
> **The error term is not a mistake and not measurement noise. It is the rest of the world.**

That is why u is sometimes called the **unobserved** variable, and why econometricians spend so much of their time worrying about it. It is the part of the story we did not measure.

Whether those omitted influences are related to attendance is the central question of causal inference.

Systematic and stochastic

The model divides y into two parts, and it is worth naming them.

$$y_i = \underbrace{\beta_0 + \beta_1 x_i}_{\text{systematic}} + \underbrace{u_i}_{\text{stochastic}}$$

The **systematic** or deterministic part, $\beta_0 + \beta_1 x$, is what attendance accounts for. It is identical for every student with the same attendance.

The **stochastic** part, u, is everything else. It differs from person to person and is not observed.

What the two parameters mean

In the linear PRF $E(Y \mid X = x) = \beta_0 + \beta_1 x$:

β_0 is the conditional mean when $x = 0$ — the average y among individuals with no x at all. It is called the **intercept**, and it is rarely central to the analysis; sometimes $x = 0$ does not even occur in the data.

β_1 is the amount by which the conditional mean changes when x increases by one unit, $\partial y/\partial x$. It is called the **slope**, and it is almost always the parameter of interest.

The remaining names are worth having in one place. y is the **dependent** variable — semester GPA. x is the **independent** variable, also called the explanatory or control variable — attendance. u is the **error** or disturbance term.

For the grades example,

$$GPA = \beta_0 + \beta_1\, attendance + u$$

and β_1 is the change in GPA associated with attending one more class.

Notice the phrase *associated with*. Whether β_1 is the change attendance would *cause* is a separate question, and it is the subject of the next section.

6.5 When does β1 measure an effect?

> *We have a model and a slope. What has to be true for that slope to mean what we want it to mean?*

The model so far is a decomposition. It says y can be split into a part that moves with x and a part that does not, which is true of any two variables whatever and therefore not yet useful.

To read β_1 as the effect of x on y, we need an assumption about how u and x are related. Two assumptions are usually stated; only the second restricts the world.

The error averages zero

$$E(u) = 0$$

This costs nothing. If the unobserved factors had a non-zero average, we could absorb that average into the intercept and start again.

> GOING A LITTLE FURTHER
>
> Suppose $u = e + c$ where c is some non-zero constant — a fixed effect of, say, generous marking. Then
>
> $$\begin{aligned} y &= \beta_0 + \beta_1 x + (e + c) \\ &= (\beta_0 + c) + \beta_1 x + e \\ &= \beta_0^* + \beta_1 x + e \end{aligned}$$
>
> with $E(e) = 0$. The intercept has changed and the slope has not. Since

> β_1 is what we want, assuming $E(u) = 0$ restricts nothing of importance.

The error is unrelated to x

We now ask a simple question. Among students with the same attendance, what is the average value of everything contained in u?

The assumption is that this average is the *same* at every value of x, and equal to the overall average of zero:

$$E(u \mid x) = E(u) = 0$$

In that case u is said to be **mean independent** of x. Taken together with $E(u) = 0$, this is the **zero conditional mean assumption**.

> **REMEMBER THIS**
>
> $$E(u \mid x) = 0$$
>
> This is the key assumption for interpreting the slope as the effect of x, and the only way to judge it is to state it in the language of the problem.
> For the grades example it says
>
> $$E(\textit{effort} \mid \textit{attendance} = \text{low}) = E(\textit{effort} \mid \textit{attendance} = \text{high}) = 0$$
>
> — that students who attend more classes are, on average, no more and no less able, motivated, or well-taught than students who attend fewer.

Note carefully what is *not* required. The assumption does not say that students have identical effort — obviously they do not. It says only that effort does not systematically differ with attendance. Equal averages, not equal individuals.

Stated that concretely, the assumption is clearly doubtful here. Students who attend regularly are plausibly working harder in other ways too, and effort raises grades on its own.

We cannot check it. Effort is not in the dataset, which is precisely what "unobserved" means. There is no diagnostic, no test, and no amount of additional data that settles it — only an argument about how the data came to look the way it does.

WATCH OUT

When $E(u \mid x) \neq 0$, β_1 does not measure the effect of x. It measures the effect of x plus the influence of whatever else differs between high and low values of x.
This is **omitted variable bias**. Section 7.8 returns to it once we have the algebra to show exactly how large it is and in which direction it runs.

With the model written down and its assumptions stated, we can turn to the practical question: how do we estimate β_0 and β_1 from data?

6.6 From the population to a sample

> *The PRF describes a population we cannot see. What do we actually have?*

The population regression function is a population object. So are β_0 and β_1: they are fixed, unknown numbers describing every member of the population, exactly like μ in Unit 2.

We never observe the population. What we observe is a sample of y values with their corresponding x values.

We therefore estimate. From the sample we compute $\hat{\beta}_0$ and $\hat{\beta}_1$, and these define the **sample regression function**:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

where $\hat{y}$ estimates $E(Y \mid X = x)$, and $\hat{\beta}_0$ and $\hat{\beta}_1$ estimate β_0 and β_1.

Two sections, two lines

Suppose two sections of thirty students each are surveyed, and a line is fitted to each. Which one is the population regression function?

```
set.seed(12)
shuffle <- sample(nrow(students))

section_a <- students[shuffle[1:30], ]
section_b <- students[shuffle[31:60], ]

round(c(section_a   = coef(lm(sem_gpa ~ attendance, section_a))[2],
        section_b   = coef(lm(sem_gpa ~ attendance, section_b))[2],
        whole_class = coef(lm(sem_gpa ~ attendance, students))[2]), 4)
```

```
#>    section_a.attendance   section_b.attendance whole_class.attendance
#>                  0.3004                 0.1621                 0.1890
```

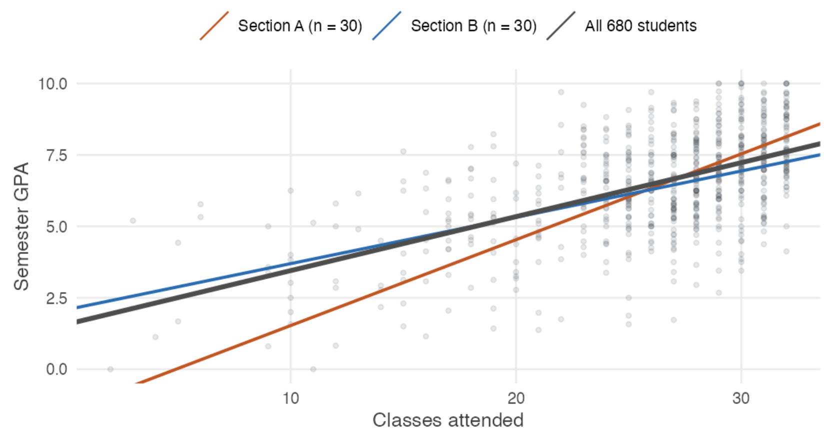

Figure 6.2: Two sections of thirty students, each with its own fitted line, against the line fitted to all 680. Neither section's line is the population regression function, and from a single section there would be no way to tell how far off it was.

There is no way to tell. Section A's slope is 0.300 and Section B's is 0.162 — one is nearly twice the other, and both come from the same class. A third section would give a third answer.

This is the situation of Unit 2 in a new costume. The estimate varies from sample to sample, and our task is to understand how much and why. Notice too that the two sections of thirty scatter far more widely around the full-sample line than two halves of a large sample would — which is the $\sqrt{n}$ lesson of Unit 2 arriving in a new form.

REMEMBER THIS

Population (PRF)	Sample (SRF)
$y = \beta_0 + \beta_1 x + u$	$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \hat{u}_i$
$y = E(Y \mid X = x) + u$	$y_i = \hat{y}_i + \hat{u}_i$
β_0 is the intercept	$\hat{\beta}_0$ estimates β_0
β_1 is the slope	$\hat{\beta}_1$ estimates β_1
u is the error, and is unobservable	**$\hat{u}$ is the residual, and is computed**

> The distinction between u and $\hat{u}$ is the one to hold onto. The error is the deviation from the true line, which nobody sees. The residual is the deviation from our fitted line, which we can print.

Because every sample gives a different line, $\hat{\beta}_1$ is a random variable with a sampling distribution — and every question from Units 2 to 4 applies to it. Is it unbiased? What is its standard error? How do we test a claim about it?

Before any of that, though, we need to decide *which* line to fit.

6.7 Which line should we fit?

> *Infinitely many lines pass through a scatter of points. Which one?*

Every sample admits infinitely many possible lines. We therefore need a rule telling us which one deserves to be called *the* regression line.

A residual measures how badly a candidate line misses observation i:

$$\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

We want the residuals small. The obvious criterion is to make their sum as close to zero as possible.

That criterion fails, and it is worth seeing why.

> **WATCH OUT**
>
> Consider two lines. The first misses four points by $+1, -1, +1, -1$. The second misses them by $+8, -8, +8, -8$.
> Both have residuals summing to exactly zero. By that criterion the two lines are equally good, which is absurd — the second is dreadful.
> The problem is that positive and negative misses cancel. A criterion that lets errors offset one another can be satisfied by a line that is nowhere near the data.

The fix is the one from Section 1.2: square the deviations first, so that they cannot cancel and large misses count for more than small ones. We choose $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimise

$$L = \sum_{i=1}^{n} \hat{u}_i^2 = \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2$$

This is the **method of ordinary least squares**, or OLS.

Notice that the criterion depends only on quantities we can observe. It is built from residuals, not errors — we could not minimise $\sum u_i^2$ even if we wanted to, because nobody sees u.

We now have a criterion. Finding the line that satisfies it is a matter of calculus, and it is where the next unit begins.

REMEMBER THIS

Regression chooses the line making the sum of *squared* vertical distances from the points to the line as small as possible.
Squaring does two things. It stops positive and negative misses from cancelling, and it makes one large miss count for more than several small ones.

6.8 Summary

REMEMBER THIS

- Regression asks how one variable is related to another rather than whether two groups differ. Splitting attendance at the median gave $t = 11.8$; using the whole variable gave $t = 17.6$.
- Individuals with the same x obtain very different y. Forty-nine students attended exactly 25 classes and scored between 1.58 and 9.08, so no line can pass through the individual points.
- The **population regression function** is the conditional mean $E(Y \mid X = x)$. It is not a fitted line, and it exists whether or not we ever estimate it.
- Since each conditional mean is a function of x, the PRF is also the **conditional expectation function**. It says how the *average* of y moves with x, not what any individual obtains.
- Taking the CEF to be linear is a **modelling choice**, made after the target is defined rather than before it.
- Writing $y_i = E(Y \mid X = x_i) + u_i$ and substituting the linear CEF gives $y_i = \beta_0 + \beta_1 x_i + u_i$ — the PRF with a name attached to the part it does not describe, not a new assumption.
- The **error term** is not a mistake and not measurement noise. It is the rest of the world: effort, ability, teaching, health, luck.
- $E(u) = 0$ costs nothing, since a non-zero average can be absorbed into the intercept. $E(u \mid x) = 0$ costs everything, cannot be tested, and is what separates a description from an effect.

- β_0 and β_1 are population parameters; $\hat{\beta}_0$ and $\hat{\beta}_1$ are sample estimates defining the **sample regression function**. Two sections of thirty students gave slopes of 0.300 and 0.162.
- The **error** u is the deviation from the true line and is unobservable. The **residual** $\hat{u}$ is the deviation from the fitted line and can be printed.
- Minimising $\sum \hat{u}_i$ fails because positive and negative misses cancel. **Ordinary least squares** minimises $\sum \hat{u}_i^2$.

6.9 Practice problems

YOUR TURN

Work each one before opening the solution. The question bank that follows has no answers.

Concept checks

6.1 Why does regression estimate a conditional mean rather than predict individual outcomes?

SOLUTION

Because y is not determined by x. At any given attendance level, students obtain a wide range of grades, because grades also depend on effort, ability and much else — all of which sit in the error term.
What is stable across the population is the *average* of y at each value of x. That is what $E(y \mid x)$ describes and what the regression line estimates.
The scatter around the line is therefore not a failure of the method. It is the part of y that x does not account for, and a model claiming otherwise would be claiming that attendance alone determines grades.

6.2 Why is minimising the sum of residuals a bad criterion?

SOLUTION

Because positive and negative residuals cancel, so a line that misses badly in both directions can have residuals summing to zero.
Misses of $+8, -8, +8, -8$ sum to zero, and so do misses of $+1, -1, +1, -1$. By that criterion the two lines are equally good,

though the first is far worse.
Squaring removes the cancellation and makes large misses count more than small ones, which is what we want from a measure of fit.

6.3 A student writes: "the regression line shows what GPA a student who attended 25 classes will get." Correct the statement.

SOLUTION

The line gives the *average* GPA among students who attended 25 classes, which in these data is 6.10. It does not give what any particular such student will get.
Forty-nine students attended exactly 25 classes, scoring between 1.58 and 9.08. A prediction for an individual would be wrong for almost all of them.
The confusion is between the conditional mean and the individual outcome — which is the difference the error term exists to describe.

Explain

6.4 State the zero conditional mean assumption for a regression of earnings on years of schooling, in concrete terms, and say whether you believe it.

SOLUTION

The model is $\textit{earnings} = \beta_0 + \beta_1\,\textit{schooling} + u$, where u holds everything else affecting earnings — ability, family connections, school quality, motivation, region.
$E(u \mid \textit{schooling}) = 0$ says that people with more schooling are, on average, no more able, no better connected, and no better taught than people with less.
That is not believable. Schooling is not distributed at random: children of wealthier and better-connected families obtain more of it, and those advantages raise earnings on their own. So u is positively related to schooling, and $\hat{\beta}_1$ overstates the return to education.
This does not make the regression useless. It correctly describes how average earnings differ across schooling levels. It simply does not measure what an extra year of schooling would do to a given person.

6.5 Distinguish the error u_i from the residual $\hat{u}_i$.

SOLUTION

The error is the deviation of y_i from the *population* regression line: $u_i = y_i - \beta_0 - \beta_1 x_i$. It involves the true parameters and is therefore unobservable.
The residual is the deviation from our *fitted* line: $\hat{u}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$. It is computed from the data and can be printed.
The relationship mirrors μ against $\bar{x}$. Residuals are estimates of errors, and confusing the two leads to claims like "the errors are uncorrelated with x, so CLRM4 holds" — which is wrong, because the *residuals* are uncorrelated with x by construction, whatever the errors do.

6.6 Explain why $y_i = \beta_0 + \beta_1 x_i + u_i$ is not an extra assumption once the conditional expectation function has been defined.

SOLUTION

Because u_i is *defined* as the gap between what individual i obtained and the average for their value of x:

$$u_i = y_i - E(Y \mid X = x_i)$$

Rearranged, that reads $y_i = E(Y \mid X = x_i) + u_i$, which is an identity. It says only that an observation equals its conditional mean plus however far it sits from that mean.
Substituting the linear CEF gives $y_i = \beta_0 + \beta_1 x_i + u_i$. The genuine assumption was made earlier, when we chose to treat the conditional mean as linear. Writing the model down adds nothing but a name for the remainder.

6.7 Two sections of the same class produce fitted slopes of 0.300 and 0.162. A student concludes that one section must contain an arithmetic error. Evaluate.

SOLUTION

No error is implied. Each section is a different sample from the same population, and every sample produces a different sample regression function — exactly as every sample produced a different $\bar{x}$ in Unit 2.
Neither slope is β_1. Both estimate it, and with thirty students each they estimate it imprecisely. The whole class of 680 gives 0.189, which is also an estimate rather than the truth.
The useful response is not to hunt for a mistake but to ask how much $\hat{\beta}_1$ varies from sample to sample — a question about its sampling distribution, and the subject of the next unit.

Numerical and conceptual

6.8 A population has these conditional means of y given x:

x	1	2	3	4
$E(Y \mid X = x)$	5	9	13	17

i. Is the population regression function linear here?
ii. Write down β_0 and β_1.
iii. An individual with $x = 3$ has $y = 11$. What is their u_i?
iv. What is $E(u \mid X = 3)$, and why?

SOLUTION

(i) Yes. The conditional mean rises by exactly 4 for each unit increase in x, so one straight line passes through all four points.
(ii) $\beta_1 = 4$. Working back from $E(Y \mid X = 1) = 5$ gives $\beta_0 = 5-4 = 1$, so the PRF is $E(Y \mid X = x) = 1 + 4x$.
(iii) $u_i = y_i - E(Y \mid X = 3) = 11 - 13 = -2$. This individual sits two units below the average for their value of x.
(iv) Zero. The conditional mean *is* the average of y among all individuals with $X = 3$, so deviations from it must average to zero within that group. That is why $E(u \mid x) = 0$ is natural when the linear CEF is correct.

6.9 Explain why a regression of y on x cannot be estimated when every observation has the same value of x — using the definition of the conditional mean rather than any formula.

SOLUTION

The PRF describes how the average of y changes *as x changes*. If x never changes, there is exactly one conditional mean to observe and no information about any other value.
Concretely: if all 680 students had attended 25 classes, we could compute the average GPA of students attending 25 classes and nothing else. Whether GPA would be higher at 30 classes is unanswerable from such data — not because the arithmetic breaks down, but because the comparison the question requires was never observed.
Variation in the explanatory variable is not a technical requirement. It is the evidence.

6.10 Question bank

WATCH OUT

No answers are given for this section, by design.

Multiple choice

QB 6.1 In the model $y = \beta_0 + \beta_1 x + u$, the term u represents

a. measurement error in y
b. all factors other than x that affect y
c. the residual from the fitted line
d. the difference between $\hat{\beta}_1$ and β_1

QB 6.2 The population regression function is

a. the line fitted by ordinary least squares
b. the conditional mean $E(Y \mid X = x)$
c. the average of y in the sample
d. the line that minimises squared residuals

QB 6.3 $E(u \mid x) = 0$ says that

a. the residuals sum to zero
b. the unobserved factors average zero at every value of x
c. x and y are uncorrelated
d. the model is linear

QB 6.4 Which of the following are unobservable? *(select all that apply)*

a. u_i
b. $\hat{u}_i$
c. β_1
d. $\hat{\beta}_1$

QB 6.5 Minimising $\sum \hat{u}_i$ rather than $\sum \hat{u}_i^2$ fails because

a. the sum is always negative
b. positive and negative residuals cancel
c. it cannot be differentiated

d. it requires a larger sample

QB 6.6 Treating the conditional expectation function as linear is

a. implied by the definition of the PRF
b. a modelling choice
c. required for $E(u) = 0$
d. a consequence of random sampling

Short reasoning

QB 6.7 Explain the difference between the population regression function and the sample regression function, and say which one changes when a new sample is drawn.

QB 6.8 A colleague says the error term captures "measurement error in the data". Explain what is wrong with that, and give three examples of what u actually contains in a regression of wages on years of schooling.

QB 6.9 Why is $E(u) = 0$ described as costing nothing, while $E(u \mid x) = 0$ is the assumption on which everything rests?

QB 6.10 State the zero conditional mean assumption in words for a regression of crop yield on fertiliser use, and give one concrete reason it might fail.

QB 6.11 A researcher regresses district crop yield on rainfall and finds a strong positive slope. Give two reasons the slope may not measure the effect of rainfall on yield.

QB 6.12 Explain why the least squares criterion penalises one residual of 8 more heavily than eight residuals of 1.

Numerical

QB 6.13 A population's conditional means are $E(Y \mid X = x) = 12 - 0.5x$.

i. State β_0 and β_1.
ii. What is the average y among individuals with $x = 10$?
iii. An individual with $x = 10$ has $y = 9$. Find u_i.
iv. Interpret β_1 in words.

QB 6.14 Four individuals with $x = 6$ have y values 14, 18, 11 and 21.

i. Estimate $E(Y \mid X = 6)$.
ii. Compute the four implied values of u_i.
iii. Verify that they sum to zero, and explain why that had to happen.
iv. Explain why this does *not* demonstrate that $E(u \mid x) = 0$ in the population.

6.11 Looking ahead

We have a target, a model and a criterion.

The target is the population regression function $E(Y \mid X = x)$, which we have chosen to approximate with a straight line. The model $y_i = \beta_0 + \beta_1 x_i + u_i$ adds a name for the part of y that attendance does not explain. The criterion says to pick the line minimising the sum of squared residuals.

What we do not yet have is the line.

The next unit finds it. Two derivatives set to zero give $\hat{\beta}_0$ and $\hat{\beta}_1$ in closed form, and the formulas turn out to say something interpretable: the slope is the covariance of x and y divided by the variance of x.

With estimates in hand we can then ask what Units 2 to 4 taught us to ask of any estimator. Is it centred on the truth? How much does it vary from sample to sample? How do we test a claim about it? And how much of the variation in y has it actually accounted for?

Chapter 7

Ordinary Least Squares

The previous unit concluded without providing a complete answer. It identified the target, the population regression function, introduced a model for individual deviations, and established a criterion for selecting among candidate lines: minimising the sum of squared residuals.

This unit implements these steps by deriving the two estimators, clarifying their guarantees and limitations, evaluating their precision, and testing related claims.

> **REMEMBER THIS**
>
> The unit answers six questions.
>
> 1. Which line minimises the sum of squared residuals?
> 2. What does that line guarantee, and what does it merely appear to guarantee?
> 3. How much of the variation in y has it accounted for?
> 4. Is $\hat{\beta}_1$ centred on β_1, and how much does it vary?
> 5. What happens when the errors are not equally spread?
> 6. Is the slope we found distinguishable from zero?
>
> And then the question the mechanics cannot answer: what does that slope actually mean?

7.1 Deriving the estimators

> *Which values of $\hat{\beta}_0$ and $\hat{\beta}_1$ make the sum of squared residuals as small as possible?*

Recall the criterion from Section 6.7. We choose $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimise

$$L = \sum_{i=1}^{n} \hat{u}_i^2 = \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2$$

This is a minimisation problem in two variables, and it is solved the usual way: differentiate with respect to each, set both derivatives to zero, and solve the pair of equations that results. Those two equations are called the **normal equations**.

WHERE THIS COMES FROM

Minimising L means setting both partial derivatives to zero.
With respect to the intercept:

$$\begin{aligned}
\frac{\partial L}{\partial \hat{\beta}_0} &= \sum_{i=1}^{n} -2\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right) = 0 \\
0 &= \sum_{i=1}^{n} y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^{n} x_i && \text{divide by } -2 \text{ and expand} \\
0 &= n\bar{y} - n\hat{\beta}_0 - \hat{\beta}_1 n\bar{x} && \text{since } \textstyle\sum x_i = n\bar{x} \\
\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} && \text{divide by } n \text{ and rearrange}
\end{aligned}$$

With respect to the slope:

$$\begin{aligned}
\frac{\partial L}{\partial \hat{\beta}_1} &= \sum_{i=1}^{n} -2x_i\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right) = 0 \\
0 &= \sum x_i y_i - \hat{\beta}_0 \sum x_i - \hat{\beta}_1 \sum x_i^2 \\
0 &= \sum x_i y_i - \left(\bar{y} - \hat{\beta}_1 \bar{x}\right) n\bar{x} - \hat{\beta}_1 \sum x_i^2 && \text{substitute } \hat{\beta}_0 \\
0 &= \sum x_i y_i - n\bar{x}\bar{y} - \hat{\beta}_1 \left(\sum x_i^2 - n\bar{x}^2\right) \\
\hat{\beta}_1 &= \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2}
\end{aligned}$$

Both parts of that fraction can be written more meaningfully. Expanding the numerator,

$$\begin{aligned}\sum(x_i-\bar{x})(y_i-\bar{y}) &= \sum x_i y_i - \bar{y}\sum x_i - \bar{x}\sum y_i + n\bar{x}\bar{y} \\ &= \sum x_i y_i - n\bar{x}\bar{y}\end{aligned}$$

and the denominator,

$$\sum(x_i-\bar{x})^2 = \sum x_i^2 - 2\bar{x}\sum x_i + n\bar{x}^2 = \sum x_i^2 - n\bar{x}^2$$

so that

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2}$$

REMEMBER THIS

The OLS estimators are

$$\hat{\beta}_1 = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2} \qquad\qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$$

The slope is the **sample covariance of x and y divided by the sample variance of x**. It answers: when x is above its mean, is y also above its mean, and by how much per unit?

THE IDEA IN PLAIN WORDS

That reading explains several things at once.
If x and y move together, the numerator is positive and so is the slope. If they move oppositely, it is negative. If they are unrelated, the products cancel and the slope is near zero.
And the denominator explains why x must vary. If every student attended the same number of classes, $\sum(x_i-\bar{x})^2 = 0$ and the slope is undefined — there is no way to learn how GPA responds to attendance from data in which attendance never changes. This is the same requirement as Section 1.1: no variation, nothing to learn.

Doing it by hand and in R

```
x <- students$attendance
```

```
y <- students$sem_gpa

b1 <- sum((x - mean(x)) * (y - mean(y))) / sum((x - mean(x))^2)
b0 <- mean(y) - b1 * mean(x)

c(b0 = b0, b1 = b1)
```

```
#>    b0    b1
#> 1.562 0.189
```

```
fit <- lm(sem_gpa ~ attendance, data = students)
coef(fit)
```

```
#> (Intercept)  attendance
#>       1.562       0.189
```

Identical. `lm()` solves the same two equations.

So $\widehat{\beta}_1 = 0.189$: each additional class attended is associated with about 0.19 of a GPA point. Ten more classes, roughly 1.9 points — which on a ten-point scale is a great deal.

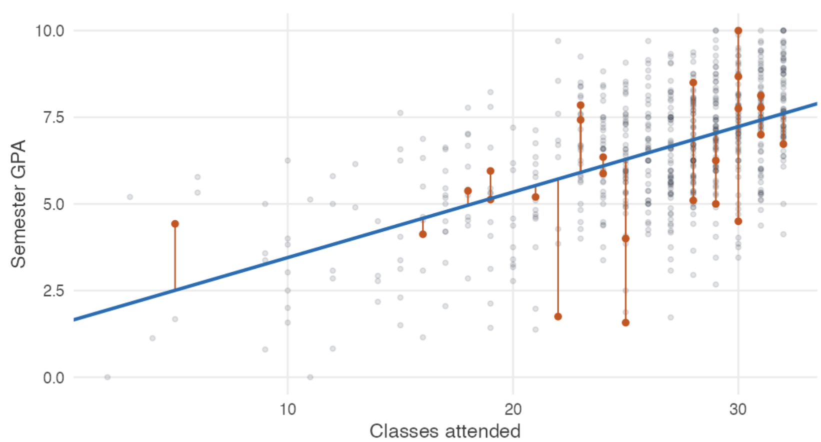

Figure 7.1: The fitted line, with the vertical distances it misses by shown for a handful of students. Ordinary least squares chooses the line making the sum of those squared distances as small as possible.

7.2 What the fitted line guarantees

> *Having minimised the squares, what do we get for free?*

Three algebraic facts follow directly from the two equations we just solved. They hold for *every* OLS regression, in every dataset, whether or not the model is any good.

> **REMEMBER THIS**
>
> 1. **The residuals sum to zero**: $\sum \hat{u}_i = 0$. This is the first normal equation, restated.
> 2. **The residuals are uncorrelated with** x: $\sum x_i \hat{u}_i = 0$. Whatever pattern is left over, it is not linearly related to x — OLS has already extracted all of that.
> 3. **The line passes through the means**: $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$. The point $(\bar{x}, \bar{y})$ always lies on the fitted line.

```r
u_hat <- residuals(fit)

c(sum_residuals    = sum(u_hat),
  sum_x_times_u    = sum(x * u_hat),
  predicted_at_xbar = coef(fit)[1] + coef(fit)[2] * mean(x),
  mean_of_y         = mean(y))
```

```
#>                 sum_residuals                sum_x_times_u
#>           -0.0000000000000111           0.0000000000009965
#> predicted_at_xbar.(Intercept)                    mean_of_y
#>            6.5025367647058836           6.5025367647058827
```

The first two are zero up to floating-point error, and the last two agree.

> **WATCH OUT**
>
> These are *algebraic* properties, not evidence. They hold by construction, so finding them in your output tells you the arithmetic worked and nothing more.
> In particular, $\sum x_i \hat{u}_i = 0$ is true whether or not $E(u \mid x) = 0$. The first is forced by the estimator; the second is an assumption about the world that no computation can verify.

How much of the variation has the line accounted for?

The fitted line runs through the data, but how much of what we see does it actually capture? Answering that begins with a single observation, and requires no algebra at all.

Take one student — number 554 in the data, who attended 9 classes and finished with a semester GPA of 0.8. The class average is 6.50, so this student sits a long way below it.

That gap can be split in two.

Part of it is anticipated by the line. This student attended only 9 classes, and the line predicts a GPA of 3.26 for such a student — already well below average. The rest is not anticipated at all: the line said 3.26 and the student scored 0.8.

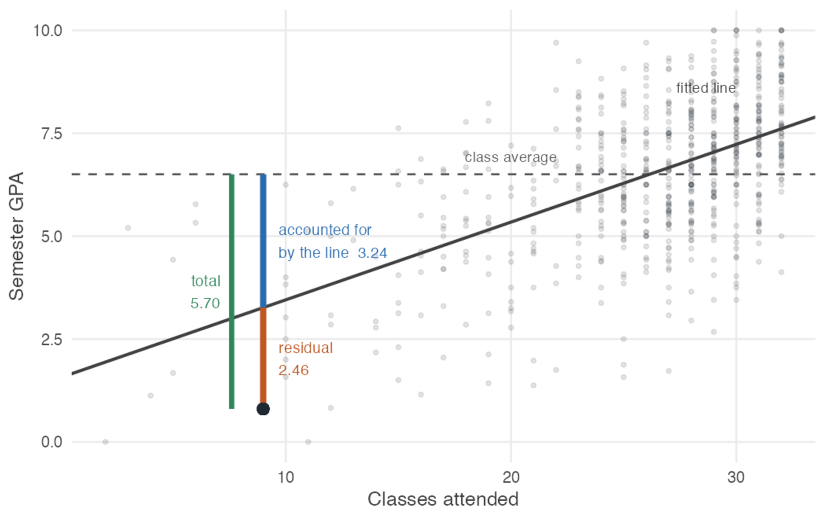

Figure 7.2: One student's distance from the class average, split in two. The green segment is the total gap between this student and the mean. The blue segment is the part the fitted line accounts for – this student attended few classes, and the line predicts a low grade. The orange segment is what the line does not account for. The two pieces add exactly to the whole.

Read the three segments and the arithmetic is immediate:

$$\underbrace{(y_i - \bar{y})}_{\text{total}} = \underbrace{(\hat{y}_i - \bar{y})}_{\text{accounted for}} + \underbrace{(y_i - \hat{y}_i)}_{\text{residual}}$$

For this student, $-5.70 = -3.24 + (-2.46)$.

REMEMBER THIS

Every observation's distance from the mean splits into two parts: the part the fitted line accounts for, and the part it does not.

$$\text{total} = \text{accounted for by the line} + \text{residual}$$

That identity is not an approximation. Adding $\hat{y}_i$ and subtracting it again changes nothing, so it holds exactly, for every observation, in every dataset.

From one observation to all of them

Now do the same for all 680 students. Square each distance so that departures above and below the mean do not cancel — the same reason we squared residuals in Section 6.7 — and add:

$$\sum (y_i - \bar{y})^2 = \sum (\hat{y}_i - \bar{y})^2 + \sum (y_i - \hat{y}_i)^2$$

The cross-term that ought to appear when squaring a sum vanishes exactly, and it does so because of property 2 above: the residuals are uncorrelated with x, and therefore with $\hat{y}$, which is a linear function of x. That algebraic identity is what makes this decomposition work at all.

Only now are the names worth having.

REMEMBER THIS

Quantity	Name	What it measures
$\sum (y_i - \bar{y})^2$	**SST**, total sum of squares	All the variation in y
$\sum (\hat{y}_i - \bar{y})^2$	**SSE**, explained sum of squares	The part the line accounts for
$\sum \hat{u}_i^2$	**SSR**, residual sum of squares	The part it does not

$$\text{SST} = \text{SSE} + \text{SSR}$$

WATCH OUT

"Explained" is conventional and overstates the case. The line has *accounted for* that variation in an arithmetic sense; whether it has explained anything depends on whether the model is appropriate and the relationship causal.

Nothing in SSE establishes that attendance produces grades. A regression of grades on shoe size would produce a perfectly respectable SSE in a sample of children, where both rise with age.

The coefficient of determination

Dividing through by SST turns the decomposition into shares.

REMEMBER THIS

$$R^2 = \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\text{SSR}}{\text{SST}}$$

R^2 is the proportion of the variation in y accounted for by the regression. It lies between 0 and 1, and in a simple regression it is exactly the square of the correlation between x and y.

```r
sst <- sum((y - mean(y))^2)
sse <- sum((fitted(fit) - mean(y))^2)
ssr <- sum(u_hat^2)

round(c(SST = sst, SSE = sse, SSR = ssr, SSE_plus_SSR = sse + ssr), 2)
```

```
#>          SST          SSE          SSR SSE_plus_SSR
#>       2302.2        721.4       1580.9       2302.2
```

```r
c(r_squared    = sse / sst,
  one_minus    = 1 - ssr / sst,
  from_summary = summary(fit)$r.squared,
  cor_squared  = cor(x, y)^2)
```

```
#>    r_squared    one_minus from_summary  cor_squared
#>       0.3133       0.3133       0.3133       0.3133
```

SSE and SSR add to SST exactly, and all four routes to R^2 agree.

So attendance accounts for about 31% of the variation in semester GPA. The other 69% is everything else — ability, effort, subject, luck — which is to say, the error term of Section 6.4 measured as a share.

> **WATCH OUT**
>
> A low R^2 does not mean a regression is useless, and a high one does not mean it is right.
> R^2 measures how tightly the points cluster around the line. It says nothing about whether β_1 is estimated accurately, and nothing whatever about whether the relationship is causal.
> In economics, R^2 values of 0.1 or 0.2 are common and unremarkable. Human behaviour is not tidy, and a model that accounted for most of the variation in something like earnings would be more suspicious than impressive.

7.3 Every sample gives a different line

> *We have one sample and one line. What would a different sample have given?*

Section 6.6 showed two sections of thirty students producing slopes of 0.300 and 0.162. That was a demonstration with two samples. Let us do it properly.

Draw forty students at random from the 680, fit a line, record the slope. Then do it again. And again, five hundred times.

```
set.seed(1)

draw_line <- function(n = 40) {
  samp <- students[sample(nrow(students), n, replace = TRUE), ]
  coef(lm(sem_gpa ~ attendance, data = samp))
}

many <- t(replicate(100, draw_line()))
colnames(many) <- c("intercept", "slope")

ggplot(students, aes(attendance, sem_gpa)) +
  geom_point(alpha = 0.08, colour = book_palette$ink, size = 0.9) +
  geom_abline(data = as.data.frame(many),
              aes(intercept = intercept, slope = slope),
              colour = "grey55", linewidth = 0.3, alpha = 0.6) +
  geom_abline(intercept = coef(fit)[1], slope = coef(fit)[2],
              colour = book_palette$reject, linewidth = 1.2) +
  labs(x = "Classes attended", y = "Semester GPA") +
  theme_book()
```

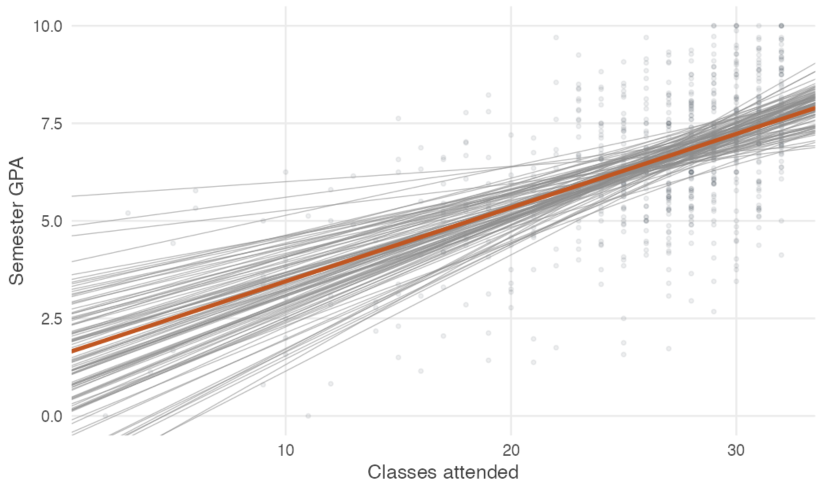

Figure 7.3: One hundred fitted lines, each from a different random sample of forty students, in grey. The line fitted to all 680 students is in orange. Every sample gives a different answer.

Every grey line came from the same population. None of them is the population regression function, and a researcher holding any one of them would have no way of knowing how far off it was.

> **REMEMBER THIS**
>
> This is the situation of Unit 2, with a slope in place of a mean. $\hat{\beta}_1$ is computed from a sample, so it is a **random variable**. It has a distribution across the samples that might have been drawn, and that distribution is what tells us how much to trust the one estimate we have.

Looking only at the slopes

Discard the lines and keep the numbers. Five hundred samples give five hundred slopes.

```
set.seed(1)
slopes <- replicate(5000, draw_line()["attendance"])

ggplot(data.frame(slope = slopes), aes(slope)) +
  geom_histogram(bins = 45, fill = book_palette$fill, colour = "white",
                 linewidth = 0.2) +
  geom_vline(xintercept = coef(fit)[2], colour = book_palette$reject,
             linewidth = 1) +
```

```
  labs(x = expression(hat(beta)[1]), y = "Number of samples") +
  theme_book()
```

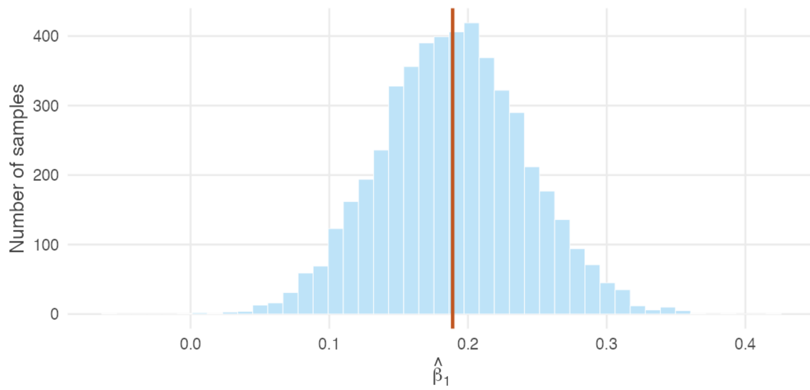

Figure 7.4: The sampling distribution of the slope: 5,000 slopes, each from a fresh sample of forty students. The orange line marks the slope from all 680 students. The distribution is centred on it and roughly symmetric.

That histogram is the **sampling distribution of** $\hat{\beta}_1$, and it is the central object of this section. Two features of it matter.

```
round(c(centre_of_distribution = mean(slopes),
        spread_of_distribution = sd(slopes),
        slope_from_all_680     = coef(fit)[2]), 4)
```

```
#>        centre_of_distribution        spread_of_distribution
#>                        0.1895                        0.0526
#> slope_from_all_680.attendance
#>                        0.1890
```

It is centred on the right place. The average of the five thousand slopes is 0.190, against 0.189 from the full data. No individual sample got it exactly right, and the average of them did.

It has a spread, and that spread is 0.0526.

REMEMBER THIS

The **standard error** of $\hat{\beta}_1$ is the standard deviation of its sampling distribution.
That is the same definition as in Unit 2, where the standard error of $\bar{X}$

> was the standard deviation of *its* sampling distribution. The estimator has changed; the idea has not.

The parallel is worth setting out in full, because everything that follows in this unit is the second column of a table whose first column the reader already knows.

Table 7.1: Regression inference is the machinery of Units 2 to 4 applied to a different estimator.

Unit 2–4	This unit
One sample gives one $\bar{x}$	One sample gives one $\hat{\beta}_1$
Repeat the sampling	Repeat the sampling
Sampling distribution of $\bar{X}$	Sampling distribution of $\hat{\beta}_1$
$E(\bar{X}) = \mu$	$E(\hat{\beta}_1) = \beta_1$
$\mathrm{Var}(\bar{X}) = \sigma^2/n$	$\mathrm{Var}(\hat{\beta}_1) = \ ?$
$\mathrm{se}(\bar{X}) = \sigma/\sqrt{n}$	$\mathrm{se}(\hat{\beta}_1) = \ ?$
Confidence interval for μ	Confidence interval for β_1
t-test of $H_0 : \mu = \mu_0$	t-test of $H_0 : \beta_1 = 0$

Nothing in the right-hand column is a new idea. What remains is to fill in the two question marks.

What makes the distribution narrow?

Before deriving anything, we can find out by experiment. Three things plausibly matter: how many students we draw, how spread out their attendance is, and how noisy grades are around the line.

Take the first two in turn.

```r
set.seed(1)
sd_at <- function(n) sd(replicate(2000, draw_line(n)["attendance"]))

round(sapply(c(20, 40, 80, 160, 320), sd_at), 4)
```

```
#> [1] 0.0796 0.0525 0.0363 0.0242 0.0179
```

Larger samples give a tighter distribution, and the pattern is familiar: each quadrupling of n roughly halves the spread. That is the $\sqrt{n}$ rule of Unit 2, arriving again.

Now hold the sample size fixed at forty and change the *range of attendance* instead. Restrict the pool to students who attended between 24 and 30 classes, so that x varies far less.

```
narrow <- students[students$attendance >= 24 & students$attendance <= 30, ]

set.seed(2)
sd_narrow <- sd(replicate(2000, {
  samp <- narrow[sample(nrow(narrow), 40, replace = TRUE), ]
  coef(lm(sem_gpa ~ attendance, data = samp))[2]
}))

round(c(sd_x_full    = sd(students$attendance),
        sd_x_narrow  = sd(narrow$attendance),
        sd_b1_full   = sd_at(40),
        sd_b1_narrow = sd_narrow), 4)
```

```
#>    sd_x_full  sd_x_narrow   sd_b1_full sd_b1_narrow
#>       5.4550       1.8949       0.0525       0.1329
```

This is the striking one. The sample size is identical. What changed is that attendance now varies by 1.9 classes instead of 5.5 — and the slope estimate became about two and a half times more variable.

THE IDEA IN PLAIN WORDS

A slope is a rate of change, and rates of change are measured by observing change.

Comparing students who attended 27 classes with students who attended 28 tells you very little about how attendance matters. Comparing those who attended 5 with those who attended 30 tells you a great deal. Spread in the explanatory variable is *information*, and a study that observes x over a narrow range will estimate its effect poorly however many observations it collects.

This is also why the median split of Section 6.1 performed so badly. It discarded most of the spread in attendance.

The third determinant, the noise in y around the line, needs no simulation to believe: if grades were entirely determined by attendance, every sample would recover the same line exactly.

REMEMBER THIS

Three things make the sampling distribution of $\hat{\beta}_1$ narrow.

1. A **large sample**.
2. **Wide variation in x**.

3. **Little noise** in y around the line.

We have found all three by experiment. The next two sections establish them algebraically, which requires being explicit about what we are assuming.

7.4 Why does OLS get the right answer on average?

> *The histogram was centred on the truth. Was that luck?*

The histogram in Figure 7.4 had two striking features.

First, the slopes varied from sample to sample — sometimes a good deal.

Second, they fluctuated *around* the full-sample value rather than sitting systematically above or below it. The average of five thousand slopes was 0.190 against 0.189 from all 680 students.

Was that simply good fortune with this population, or does OLS always behave this way?

Showing that it always does requires assumptions, and this is a convenient place to name the set they belong to.

Regression rests on a standard list, known collectively as the assumptions of the **classical linear regression model**, or **CLRM**. There are six. Four of them deliver unbiasedness, and they are the four we need here.

REMEMBER THIS

CLRM1 — linear in parameters. The population model is $y = \beta_0 + \beta_1 x + u$.
CLRM2 — random sampling. The observations are a random sample from that population.
CLRM3 — variation in x. The x_i are not all equal.
CLRM4 — zero conditional mean. $E(u \mid x) = 0$.
Nothing about the *spread* of the errors is assumed yet, and nothing about their shape.

GOING A LITTLE FURTHER

Textbooks label these differently. Wooldridge calls them SLR1 to SLR6 in the simple regression case and MLR1 to MLR6 with several explanatory variables; Gujarati and others speak of the classical linear regression model. The assumptions are the same and only the labels move.

WHERE THIS COMES FROM

Write the slope in deviation form and substitute the model. With $\text{SST}_x = \sum(x_i - \bar{x})^2$,

$$\begin{aligned}
\hat{\beta}_1 &= \frac{\sum(x_i - \bar{x})y_i}{\text{SST}_x} && \text{since } \sum(x_i - \bar{x})\bar{y} = 0 \\
&= \frac{\sum(x_i - \bar{x})(\beta_0 + \beta_1 x_i + u_i)}{\text{SST}_x} && \text{substituting CLRM1} \\
&= \beta_1 + \frac{\sum(x_i - \bar{x})u_i}{\text{SST}_x}
\end{aligned}$$

because $\sum(x_i - \bar{x}) = 0$ kills the β_0 term and $\sum(x_i - \bar{x})x_i = \text{SST}_x$ leaves β_1.

That last line deserves to be pulled out and looked at on its own.

REMEMBER THIS

$$\hat{\beta}_1 \quad = \quad \underbrace{\beta_1}_{\text{truth}} \quad + \quad \underbrace{\frac{\sum(x_i - \bar{x})u_i}{\text{SST}_x}}_{\text{sampling error}}$$

Estimate = truth + sampling error.

The equation separates two quite different things.

The first term, β_1, is the quantity we are trying to estimate. It is a fixed feature of the population and it is identical in every sample.

The second term is the price of observing only one sample. It depends entirely on the unobserved errors u_i — on which particular students happened to be drawn and how far each of them sat from the line. It is different in every sample, and it is the whole reason the grey lines in Figure 7.3 fan out.

So the question of whether OLS is centred correctly comes down to one thing: does that second term average out to zero?

Taking expectations conditional on the x's, and using CLRM4 to set $E(u_i \mid x) = 0$,

$$E(\hat{\beta}_1) = \beta_1 + \frac{\sum(x_i - \bar{x})E(u_i \mid x)}{\text{SST}_x} = \beta_1$$

It does. Across repeated samples $\hat{\beta}_1$ is centred on β_1, which is the first question mark in Table 7.1 filled in, and what the histogram was showing.

> **THE IDEA IN PLAIN WORDS**
>
> The proof is worth having for a second reason. It shows that unbiasedness hangs on exactly one place — the second term — and on exactly one condition, that the average error does not change with x.
> If students who attend more classes also happen to be more motivated, then the errors are systematically larger at larger x. The second term no longer vanishes, and the estimator is biased by precisely its expectation. That single term is why omitted variables matter, and Section 7.8 returns to it with this equation in hand.

7.5 How precise is OLS?

> *The histogram had a width as well as a centre. Can we know that width from one sample?*

The sampling distribution in Section 7.3 had a standard deviation of 0.0526. That number is exactly what we want: it says how far a slope from a single sample of forty is liable to fall from the truth.

But look at how we obtained it. We drew five thousand samples.

Nobody does that. A survey is run once, at considerable expense, and yields one sample and one slope. The standard deviation of the distribution those samples came from is not something we can watch.

> **REMEMBER THIS**
>
> So the question is this. Can we recover, from a *single* sample, the spread of a distribution we could otherwise see only by drawing thousands? Remarkably, yes.

The formula

The width of the sampling distribution turns out to depend on two things, and both are things we can measure.

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum(x_i - \bar{x})^2}$$

The **numerator** is the noise: how far individual observations scatter around the line. Noisier relationships are harder to pin down.

The **denominator** is the total variation in x — the same sum of squares we met in Section 7.2, computed on the explanatory variable rather than the outcome. It measures how much the data actually tell us about x.

Since σ^2 is unknown, it is estimated from the residuals. Two parameters were estimated on the way, so the divisor is $n-2$ — the same accounting as Section 3.2, where estimating one mean cost one degree of freedom:

$$\hat{\sigma}^2 = \frac{\sum \hat{u}_i^2}{n-2} \qquad \mathrm{se}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum (x_i - \bar{x})^2}}$$

```r
n <- length(y)
sigma_hat <- sqrt(sum(u_hat^2) / (n - 2))
se_b1     <- sigma_hat / sqrt(sum((x - mean(x))^2))

c(sigma_hat = sigma_hat, se_b1 = se_b1,
  se_from_lm = summary(fit)$coefficients[2, 2])
```

```
#>  sigma_hat      se_b1 se_from_lm
#>    1.52697    0.01074    0.01074
```

Does it work?

A formula claiming to reconstruct a distribution we never observe deserves to be checked rather than believed.

Build a population where the assumptions hold exactly — a genuinely linear relationship, errors of constant variance — then draw from it repeatedly and compare what the formula says against what the samples actually do.

```r
set.seed(4)
x_fixed <- runif(40, 2, 32)                   # forty students, attendance
↪  2 to 32

check <- replicate(5000, {
  y_sim <- 1.56 + 0.189 * x_fixed + rnorm(40, sd = 1.53)
  m <- lm(y_sim ~ x_fixed)
  c(slope = coef(m)[2], se = summary(m)$coefficients[2, 2])
})

round(c(simulated_sd     = sd(check["slope.x_fixed", ]),
        average_formula = mean(check["se", ]),
       exact_formula   = 1.53 / sqrt(sum((x_fixed - mean(x_fixed))^2))), 4)
```

```
#>    simulated_sd average_formula   exact_formula
#>          0.0287          0.0286          0.0287
```

The three agree to three decimal places. The formula, computed from one sample of forty, recovers the spread that five thousand samples reveal.

> **REMEMBER THIS**
>
> In other words: the standard error printed by `lm()` is an estimate of the standard deviation of the sampling distribution we simulated in Section 7.3.
> That sentence connects the whole unit. The histogram is the thing; the standard error is our estimate of how wide it is, computed without ever drawing it.

> **GOING A LITTLE FURTHER**
>
> Notice the words *where the assumptions hold exactly*. They were doing work.
> Repeat the comparison on the real attendance data and the agreement is good rather than perfect: the formula gives about 0.046 for a sample of forty, while resampling gives about 0.053.
> The formula is not wrong. Its assumptions are only approximately true here — Section 6.2 already showed the conditional mean of GPA is not exactly linear — and a formula inherits whatever inexactness its assumptions carry.
> Which assumption is chiefly responsible in this particular dataset is not obvious and we shall not pretend otherwise. The general point stands regardless, and is worth holding on to whenever software prints a standard error to four decimal places: the number is only as good as the model behind it.

Why the denominator matters

Most readers remember that larger samples give smaller standard errors. Far fewer remember the other half, which is at least as important.

Look at the denominator again and ask what happens in extreme cases.

What if every student attended exactly 25 classes? Then $x_i = \bar{x}$ for everyone, $\sum(x_i - \bar{x})^2 = 0$, and the variance is undefined. There is no slope to estimate — we have observed grades at one attendance level and nothing about any other. This is assumption CLRM3, arriving as a consequence rather than a stipulation.

What if attendance varies only between 24 and 26? The denominator is small and the standard error is large. We are trying to infer a rate of change from almost no change.

What if attendance ranges from 2 to 32? The denominator is large and the slope is pinned down tightly.

The following figure makes the point with two datasets built from the *same* true line and the *same* error variance. Only the spread of x differs.

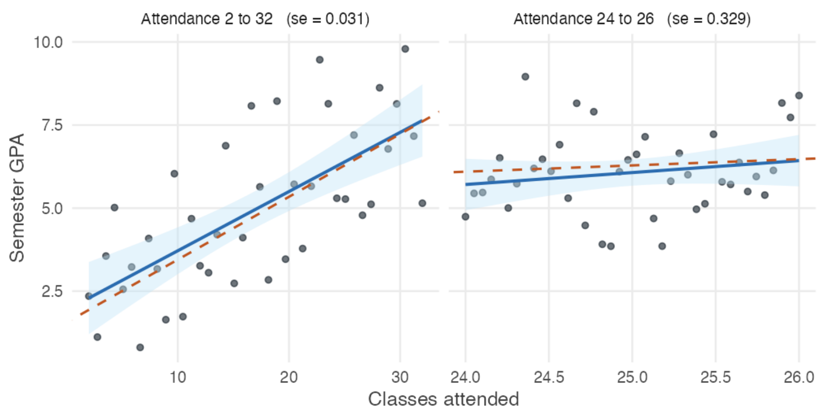

Figure 7.5: Two samples of forty from the same model: the same true slope of 0.189, the same noise, and the same sample size. Only the range of attendance differs. On the left the line is pinned down; on the right, where attendance varies by two classes rather than thirty, the same data are consistent with wildly different slopes.

The dashed orange line is the truth in both panels, and it is the same line.

On the left the fitted line sits almost on top of it: an estimated slope of 0.178 against a true 0.189. On the right the fitted slope is 0.360 — nearly twice the truth — and the shaded band around it is so wide as to be useless.

Note that the two panels have different horizontal scales, so compare the fitted lines to the dashed one within each panel rather than by eye across them.

Nothing about the relationship changed. Only the range over which we observed it.

THE IDEA IN PLAIN WORDS

A slope is a rate of change, and rates of change are measured by observing change.
Comparing students who attended 24 classes with students who attended 26 tells you very little about how attendance matters. Comparing those who attended 5 with those who attended 30 tells you a great deal.

Spread in the explanatory variable is *information*. A study that observes x over a narrow range will estimate its effect poorly however many observations it collects — which is also why the median split of Section 6.1 did so badly.

Where the formula comes from

Having seen that the formula works and what its parts mean, we can derive it.

One further assumption is needed, and it is worth seeing why. The derivation adds up the contributions of the individual errors, and to add them we must know how variable each one is. So far we have said nothing about that at all.

REMEMBER THIS

CLRM5 — homoskedasticity. $\text{Var}(u \mid x) = \sigma^2$, the same at every value of x.
This is what allows a single number σ^2 to stand in for the variability of every error. It played no part in unbiasedness, and Section 7.6 examines what happens when it fails.

WHERE THIS COMES FROM

Start from the expression that gave unbiasedness, and treat the x's as fixed:

$$\begin{aligned}
\text{Var}(\hat{\beta}_1) &= \text{Var}\left(\beta_1 + \frac{\sum(x_i - \bar{x})u_i}{\text{SST}_x}\right) & \\
&= \text{Var}\left(\frac{\sum(x_i - \bar{x})u_i}{\text{SST}_x}\right) & \beta_1 \text{ is a constant} \\
&= \frac{1}{\text{SST}_x^2}\sum(x_i - \bar{x})^2\,\text{Var}(u_i) & \text{independence} \\
&= \frac{\sigma^2\,\text{SST}_x}{\text{SST}_x^2} & \text{homoskedasticity, CLRM5} \\
&= \frac{\sigma^2}{\text{SST}_x}
\end{aligned}$$

REMEMBER THIS

Why is the slope estimated more precisely?
The standard error becomes smaller when

- the relationship is **less noisy** — small σ;
- the sample is **larger** — more terms in $\sum(x_i - \bar{x})^2$;
- the explanatory variable **varies over a wider range** — bigger terms in that same sum.

The last two both enter through the denominator, which is why n and the spread of x are not really two separate considerations. What matters is the total variation in x, and either more observations or a wider range will increase it.

Those are the three findings of Section 7.3, now derived rather than observed. The second question mark in Table 7.1 is filled in.

7.6 When the error variance is not constant

> *The standard error formula rested on one new assumption. Is it true?*

The previous section derived a formula for the standard error, and that derivation needed CLRM5: the errors have the same variance at every value of x.

$$\mathrm{Var}(u \mid x) = \sigma^2 \quad \text{for every } x$$

Read as a claim about the world, that says the unobserved factors are just as spread out among students who attended 5 classes as among those who attended 30, and just as spread out among poor households as among rich ones.

Often they are not.

THE IDEA IN PLAIN WORDS

Consider annual earnings against years of schooling.
Someone with two years of schooling has little scope to be unusual. Their earnings are constrained from above by the work available to them, and there is not much room between the floor and that ceiling.
Someone with sixteen years of schooling might be a clerk, a teacher, a doctor or a company director. The same qualification is consistent with an enormous range of outcomes, because what happens after school depends on far more than school.
So the spread of the errors grows with x. This is **heteroskedasticity**, and

in economic data it is the normal case rather than the exception.

What it looks like

The diagnostic is a **residual plot**: the residuals against the fitted values. Homoskedasticity appears as a band of roughly constant width, heteroskedasticity as a fan.

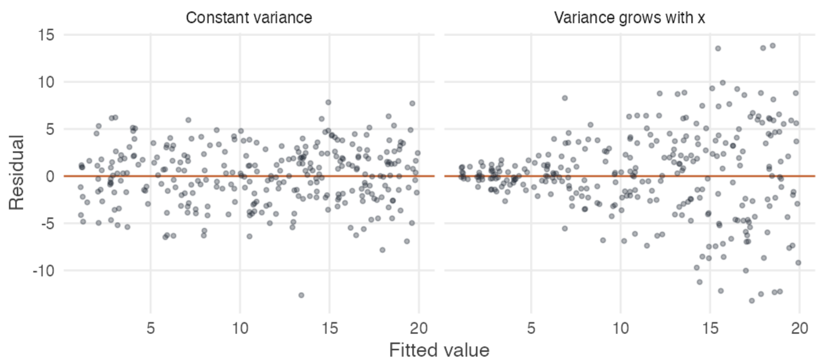

Figure 7.6: Residual plots from two simulated datasets. On the left the error variance is constant and the band has an even width. On the right it grows with x, and the band fans out. This is the shape to look for.

Now the real thing. Regress annual earnings on years of education for the 20,000 wage earners and look at how the residuals are spread.

```
wages <- read.csv("data/wages-india-synthetic.csv")
fit_w <- lm(annual_earnings ~ education, data = wages)

u_w <- residuals(fit_w)
f_w <- fitted(fit_w)

quartile <- cut(f_w, quantile(f_w, 0:4/4), include.lowest = TRUE)
round(tapply(u_w, quartile, sd))
```

```
#> [2.34e+04,2.88e+04] (2.88e+04,5.04e+04] (5.04e+04,7.19e+04]
↪ (7.19e+04,1.04e+05]
#>               42504               61998               93743
↪ 153304
```

The residuals in the top quarter of fitted values are more than three times as spread out as those in the bottom quarter.

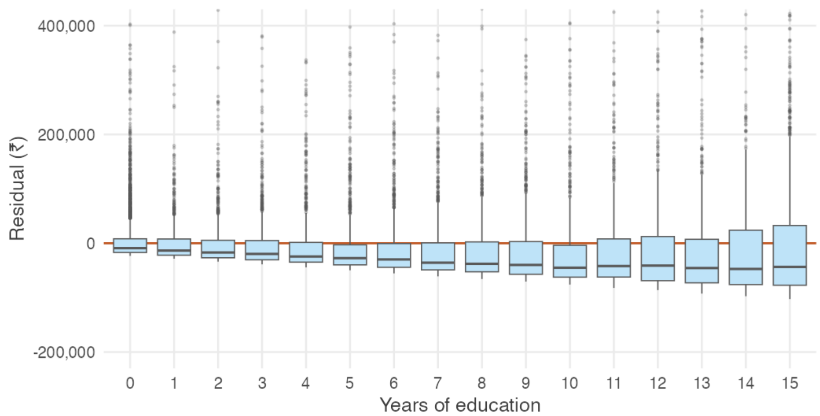

Figure 7.7: Residuals from a regression of annual earnings on years of schooling, shown as a boxplot at each level of education. The boxes widen steadily to the right. A few extreme residuals lie above the range plotted.

Education takes only whole-number values, so a plain scatter of residuals against fitted values collapses into vertical stripes. A box at each level of education shows the same thing more legibly, and the widening is unmistakable.

> **GOING A LITTLE FURTHER**
>
> Formal tests exist. The Breusch–Pagan and White tests both regress squared residuals on the explanatory variables and ask whether anything is there.
>
> They are worth knowing about, and they are not what this book recommends reaching for first. A plot settles the matter faster and tells you the *shape* of the problem rather than merely announcing its presence — and, as the next section shows, a fan-shaped residual plot sometimes means the functional form is wrong rather than the variance non-constant. A test statistic cannot tell you which.

Does heteroskedasticity matter?

Here is the surprise.

> **REMEMBER THIS**
>
> **The fitted line is unaffected.**
>
> The proof that $E(\hat{\beta}_1) = \beta_1$ used CLRM1 to CLRM4 and never mentioned CLRM5. The estimate is still centred on the truth, and heteroskedasticity

> introduces no bias whatever.
> **The standard error is wrong.**
> $\text{Var}(\hat{\beta}_1) = \sigma^2/\text{SST}_x$ was obtained by replacing every $\text{Var}(u_i)$ with a common σ^2. If no common σ^2 exists, that step is invalid — and with it the standard errors, the t statistics, the p-values and the confidence intervals.

So the coefficient is fine and the *reported uncertainty around it* is not. That is why this matters: not because the arithmetic of the line changes, but because everything we say about how much to trust the line is computed from a formula that no longer applies.

How wrong?

We can measure it, by simulating from a model where we know both the truth and the severity of the problem.

```r
robust_se <- function(m) {
  X <- model.matrix(m); u <- residuals(m)
  n <- nrow(X); k <- ncol(X)
  XtXi <- solve(crossprod(X))
  V <- XtXi %*% (t(X) %*% diag(u^2) %*% X) %*% XtXi * n / (n - k)
  sqrt(diag(V))[2]
}

set.seed(8)
out <- replicate(4000, {
  x <- rexp(60, rate = 1/5)             # skewed x, as economic data often
↪  is
  y <- 2 + 1 * x + rnorm(60, sd = x)     # sd of the error grows with x
  m <- lm(y ~ x)
  c(b1 = coef(m)[2], classical = coef(summary(m))[2, 2], robust =
   ↪  robust_se(m))
})

round(c(actual_sd_of_b1  = sd(out[1, ]),
        avg_classical_se = mean(out[2, ]),
        avg_robust_se    = mean(out[3, ])), 4)
```

```
#>  actual_sd_of_b1 avg_classical_se    avg_robust_se
#>           0.4301           0.1764           0.3293
```

The true slope is 1. Across four thousand samples $\hat{\beta}_1$ has a standard deviation of 0.43 — that is the actual variability of the estimator, the thing a standard error is supposed to report.

The classical standard error reports 0.18. Software printing 0.18 is claiming a precision the estimator does not have, by a factor of more than two.

```
round(c(classical = mean(abs(out[1, ] - 1) < qt(0.975, 58) * out[2, ]),
        robust    = mean(abs(out[1, ] - 1) < qt(0.975, 58) * out[3, ])), 3)
```

```
#> classical    robust
#>     0.588     0.851
```

This is the finding that matters, and it is worth stating slowly.

A confidence interval advertised as 95% should contain the true slope in roughly 95 of every 100 repeated samples. Here it succeeds **59** times.

The problem is not the estimator. $\hat{\beta}_1$ is unbiased and behaving exactly as it should. The problem is that we underestimated its variability, and so drew an interval far too narrow around a perfectly good estimate.

Robust standard errors

The repair does not touch $\hat{\beta}_1$, which needs no repairing. It replaces the formula for the variance with one that does not assume a common σ^2, using each observation's own squared residual in place of a single estimate.

These are called **White heteroskedasticity-robust standard errors**, after Halbert White's influential work, and in ordinary conversation simply **robust standard errors**. All three names refer to the same idea.

> **REMEMBER THIS**
>
> **Robust standard errors leave the fitted line unchanged.** They alter only how the uncertainty around it is calculated.
> The coefficients you report are identical either way. What changes is the second column of the output, and everything computed from it.

In the simulation they report 0.33 against a true 0.43, and lift coverage from 59% to 85%. A large improvement, and visibly not a complete repair at this sample size and this severity.

```
round(c(classical_se = coef(summary(fit_w))[2, 2],
        robust_se    = robust_se(fit_w)), 2)
```

```
#>       classical_se robust_se.education
#>              137.4               175.2
```

For the earnings regression the robust standard error is 175 against a classical 137 — a quarter larger again. Any conclusion resting on the difference between those two numbers was resting on an assumption the residual plot had already refused.

DOING IT IN R

The calculation above is written out longhand so that nothing is hidden. In practice one reaches for a package.

```
library(sandwich)   # variance estimators
library(lmtest)     # tests that accept them

coeftest(fit_w, vcov = vcovHC(fit_w, type = "HC1"))
```

Stata users will recognise the same thing as `reg y x, vce(robust)`. The variants `HC0` through `HC3` differ only in how the variance is adjusted for finite samples. `HC1` is the most common default and matches Stata's; for moderate and large samples the choice rarely matters.

GOING A LITTLE FURTHER

White standard errors are the first member of a family, and it is worth knowing that the family exists.

- **White (heteroskedasticity-robust)** standard errors relax the assumption of constant error variance.
- **Cluster-robust** standard errors also allow errors to be correlated within groups — students within a school, households within a village.
- **HAC** or **Newey–West** standard errors allow correlation across time.

All three estimate the variance of the *same* OLS estimator in different circumstances. None of them changes the coefficients, and none of them is a different method of regression. They are different answers to the single question of how much the estimator would vary across repeated samples.

WATCH OUT

Three things robust standard errors do not do.
They do not fix bias. There was none to fix.
They do not fix a wrong functional form. A fan-shaped residual plot sometimes means the relationship is not linear rather than that the variance is non-constant, and robust standard errors would paper over that. A later unit takes this up.

They do not work miracles in small samples, as the simulation shows.

THE IDEA IN PLAIN WORDS

Modern practice is to report robust standard errors by default in cross-sectional work.
This is exactly the logic of Section 5.4. Classical standard errors assume equal variances, as the pooled two-sample t-test did; robust standard errors relax that assumption, as Welch's test did. In both cases the relaxed version costs very little when the assumption holds and protects you when it does not — and in both cases the assumption can never be established, only left un-rejected.

The wider lesson

Heteroskedasticity is a good example of a broader principle, and the principle is worth more than the topic.

REMEMBER THIS

Regression coefficients answer one question: what is the relationship?
Standard errors answer another: how sure are we?
The first can be correct while the second is badly wrong.

Students often assume that if a regression is right, everything about it is right. This section is the first place where that fails cleanly — the line is unbiased, the residuals sum to zero, R^2 is computed correctly, and the confidence interval still misses the truth two times in five.

Estimation and inference are separate problems, and they can fail separately.

7.7 Inference about the slope

Is the slope we found distinguishable from zero?

$\hat{\beta}_1$ is an estimator with a standard error, so everything in Unit 4 applies without modification.

One assumption is quietly added at this point, and it is the last of the six.

REMEMBER THIS

CLRM6 — normality. The errors are normally distributed: $u \sim N(0, \sigma^2)$, independently of x.
This is what makes the t statistic below follow a t distribution *exactly*, in any sample however small. CLRM1 to CLRM5 gave us an unbiased estimator and a correct standard error; CLRM6 gives us the distribution needed to turn them into a p-value.

Unlike the others, this assumption is one to hold lightly.

GOING A LITTLE FURTHER

Why normality matters less than it appears to.
Nothing established so far depends on it. Unbiasedness came from CLRM1 to CLRM4. The variance formula came from CLRM5. The estimator, the standard error and the interpretation are all untouched by the shape of the error distribution.
Normality earns its keep only in small samples, where it makes the t distribution exact rather than approximate. In larger samples the Central Limit Theorem of Section 1.6 does the work instead: $\hat{\beta}_1$ is a weighted sum of the errors, and sums tend towards normality whatever they are summing. The t statistic is then approximately correct regardless of how the errors are distributed.
So the practical position is this. With a few dozen observations and visibly skewed errors, treat small p-values with some caution. With a few hundred, the assumption is close to irrelevant.
That is a far weaker requirement than CLRM4, which cannot be relaxed by collecting more data, and which no amount of sample size will repair. It is worth keeping the two firmly apart: normality is a convenience that large samples make unnecessary, while zero conditional mean is the assumption everything rests on.

The hypothesis of interest is almost always that x has no effect at all:

$$H_0 : \beta_1 = 0 \qquad H_A : \beta_1 \neq 0$$

and the test statistic has the form it always has — estimate minus hypothesised value, over standard error:

$$t = \frac{\hat{\beta}_1 - 0}{\text{se}(\hat{\beta}_1)}$$

compared against t with $n - 2$ degrees of freedom.

```r
summary(fit)$coefficients
```

```
#>             Estimate Std. Error t value
#> (Intercept)    1.562    0.28692   5.444
#> attendance     0.189    0.01074  17.589
#>                                                                         Pr(>|t|)
#> (Intercept)
↪   0.000000072863221377091288988811794338079641519811957550700754
#> attendance
↪   0.000000000000000000000000000000000000000000000000000000002468
```

$\widehat{\beta}_1 = 0.189$ with a standard error of 0.0107, giving $t = 17.6$ on 678 degrees of freedom and a p-value indistinguishable from zero. We reject the claim that attendance is unrelated to GPA.

The confidence interval is the more informative report:

```r
confint(fit, level = 0.95)
```

```
#>              2.5 % 97.5 %
#> (Intercept) 0.9987  2.125
#> attendance  0.1679  0.210
```

Each additional class is associated with between 0.168 and 0.210 of a GPA point, with 95% confidence. The interval excludes zero, agreeing with the test as it must.

> **WATCH OUT**
>
> "Statistically significant" is a claim about distinguishability from zero, not about magnitude. With 680 observations, very small slopes clear the bar easily.
> Always ask what the slope means in the units of the problem. Here 0.19 GPA points per class is large. In the regression of final examination score on attendance in the same dataset, the slope is 0.117 marks per class and equally "significant" — and 0.117 marks is nothing at all.

7.8 Two students

> *The slope says grades rise with attendance. Does attending raise grades?*

Regression tells us how average outcomes differ across observations. It does not, by itself, tell us *why* they differ.

Student A attends thirty classes. Student B attends twenty.

Student A also studies harder outside class, enjoys the subject more, and came into the course better prepared.

If Student A earns a higher GPA, how much of the difference is due to attendance?

Regression has no difficulty describing the difference between these two students. The fitted line tells us that students attending more classes tend to earn higher grades on average, and it says so with a standard error and a confidence interval.

The harder question is whether attendance itself produced the difference.

What must be true

To read the slope as the effect of attendance, something more is required. The remaining influences on grades must not systematically differ across students with different attendance.

That is the assumption we met in Section 6.5:

$$E(u \mid X) = 0$$

For attendance and grades it is a demanding one. Students who attend regularly are also likely to be more motivated, better organised and more interested in the subject. Those differences belong in u. If they are related to attendance, then

$$E(u \mid X) \neq 0$$

and the regression slope reflects both attendance and the omitted differences between students. This is **omitted variable bias**.

Section 7.4 showed where it enters mathematically: in the part of the estimator that depends on the errors. When the errors are related to x, that term no longer averages away, and its expectation is the bias.

None of this appears in the output

The coefficient may be estimated very precisely. The standard error may be small. The p-value may be close to zero. The residuals sum to zero, the algebra checks out, and every diagnostic in this unit passes.

Every calculation in the output may be correct.

> **REMEMBER THIS**
>
> What is wrong is not the computation but the interpretation.

Regression answers a statistical question:

> How do average outcomes differ as X changes?

Causal inference asks a different one:

> What would happen if X were changed?

The first can be answered from the data alone. The second cannot.

Where to go from here

Multiple regression is the first attempt at the causal question, and it works by comparing observations that are similar in other *observed* respects. If motivation can be measured, students can be compared at the same level of it.

When the important differences cannot be observed, a different strategy is needed. The second half of this book develops those strategies.

7.9 Summary

> **REMEMBER THIS**
>
> - **OLS** minimises $\sum \hat{u}_i^2$. Setting both partial derivatives to zero gives the normal equations, whose solution is $\hat{\beta}_1 = \sum(x_i - \bar{x})(y_i - \bar{y})/\sum(x_i - \bar{x})^2$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$.
> - The slope is the **sample covariance of x and y over the sample variance of x**. That reading explains its sign, and it explains why x must vary: with no variation in x the denominator is zero.
> - By construction $\sum \hat{u}_i = 0$, $\sum x_i\hat{u}_i = 0$, and the fitted line passes through $(\bar{x}, \bar{y})$. These are algebraic identities and are not evidence about the model.
> - SST = SSE+SSR, and R^2 = SSE/SST = 1−SSR/SST is the share of the variation in y the regression accounts for. In a simple regression it equals the squared correlation between x and y.
> - R^2 measures fit, not correctness. Attendance accounts for 31% of the variation in semester GPA; values of 0.1 or 0.2 are unremarkable

in economics.

- The six standard assumptions are known collectively as the **classical linear regression model**. Under CLRM1–CLRM4, $E(\widehat{\beta}_1) = \beta_1$. The derivation runs through $\widehat{\beta}_1 = \beta_1 + \sum(x_i - \bar{x})u_i/\text{SST}_x$, which is the most useful line in the unit.
- Adding CLRM5, $\text{Var}(\widehat{\beta}_1) = \sigma^2/\sum(x_i - \bar{x})^2$. Small errors and widely spread x both buy precision, which is why spread in the explanatory variable is worth seeking out.
- $\widehat{\sigma}^2 = \sum \widehat{u}_i^2/(n-2)$, the divisor reflecting the two parameters estimated on the way.
- Testing $H_0 : \beta_1 = 0$ uses $t = \widehat{\beta}_1/\text{se}(\widehat{\beta}_1)$ on $n-2$ degrees of freedom, exactly as in Unit 4. The interval is the more informative report.
- **CLRM6**, normality of the errors, makes that t distribution exact in small samples. It is the weakest of the six: the Central Limit Theorem makes it unnecessary in large ones, whereas no sample size repairs a failure of CLRM4.
- When $E(u \mid x) \neq 0$ the second term in the unbiasedness expression does not vanish. That is **omitted variable bias**, and nothing in the regression output reveals it.

7.10 Practice problems

YOUR TURN

Work each one before opening the solution. The question bank that follows has no answers.

Numerical problems

7.1 For five plots, fertiliser applied (x, kg) and yield (y, quintals):

x	2	4	6	8	10
y	4	7	8	12	14

i. Compute $\bar{x}$, $\bar{y}$, $\widehat{\beta}_1$ and $\widehat{\beta}_0$.
ii. Write the fitted line and interpret the slope.
iii. Compute the fitted values and residuals, and verify they sum to zero.
iv. Compute R^2.

SOLUTION

(i) $\bar{x} = 6$, $\bar{y} = 9$.

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	product	$(x_i - \bar{x})^2$
2	4	-4	-5	20	16
4	7	-2	-2	4	4
6	8	0	-1	0	0
8	12	2	3	6	4
10	14	4	5	20	16
				50	**40**

$$\hat{\beta}_1 = \frac{50}{40} = 1.25 \qquad \hat{\beta}_0 = 9 - 1.25(6) = 1.5$$

(ii) $\hat{y} = 1.5 + 1.25x$. Each additional kilogram of fertiliser is associated with 1.25 more quintals of yield.
(iii) Fitted values: $4.0, 6.5, 9.0, 11.5, 14.0$. Residuals: $0, 0.5, -1, 0.5, 0$, which sum to zero as they must.
(iv) SSR $= 0+0.25+1+0.25+0 = 1.5$ and SST $= 25+4+1+9+25 = 64$, so

$$R^2 = 1 - \frac{1.5}{64} = 0.977$$

An unusually tight fit, because these are made-up numbers. Real data rarely behaves so well.

7.2 Using the previous question's results:

i. Compute $\hat{\sigma}^2$ and $\hat{\sigma}$.
ii. Compute se$(\hat{\beta}_1)$.
iii. Test $H_0 : \beta_1 = 0$ at the 5% level.
iv. Construct a 95% confidence interval for β_1.

SOLUTION

(i) With $n = 5$ and two parameters estimated,

$$\hat{\sigma}^2 = \frac{\text{SSR}}{n-2} = \frac{1.5}{3} = 0.5 \qquad \hat{\sigma} = 0.707$$

(ii) se$(\hat{\beta}_1) = \dfrac{0.707}{\sqrt{40}} = 0.1118$.
(iii) $t = 1.25/0.1118 = 11.18$ on $n - 2 = 3$ degrees of freedom. The

critical value is $t_{0.025,3} = 3.182$, and $11.18 > 3.182$, so we reject H_0.
(iv) $1.25 \pm 3.182(0.1118) = [0.894,\ 1.606]$.
Note how wide that interval is despite the excellent fit — three degrees of freedom buy very little. The critical value of 3.182 rather than 1.96 is doing most of the damage.

7.3 A regression of household expenditure on income in a sample of 200 gives $\hat{\beta}_1 = 0.62$ with $\text{se}(\hat{\beta}_1) = 0.05$.

i. Test $H_0 : \beta_1 = 0$.
ii. Test $H_0 : \beta_1 = 1$, and explain why an economist might care about that hypothesis specifically.
iii. Construct a 95% interval and interpret it.

SOLUTION

(i) $t = 0.62/0.05 = 12.4$ on 198 degrees of freedom. Reject overwhelmingly.
(ii) $t = (0.62 - 1)/0.05 = -7.6$, which also rejects.
An economist cares because β_1 is the marginal propensity to consume, and $\beta_1 = 1$ would mean households spend every additional rupee of income. The test says they do not: expenditure rises by about 62 paise per additional rupee, and the remainder is saved.
This is worth noticing as a matter of method. The interesting null hypothesis is not always zero. Testing against a value that theory suggests is often far more informative.
(iii) $0.62 \pm 1.96(0.05) = [0.522,\ 0.718]$. The interval excludes both 0 and 1, so the data are consistent with a marginal propensity to consume of roughly two-thirds and with neither extreme.

Concept checks

7.4 What goes wrong if every observation has the same value of x?

SOLUTION

The slope is undefined. Its denominator is $\sum(x_i - \bar{x})^2$, which is zero when every x_i equals $\bar{x}$, and the estimate would require dividing by zero.
The statistical content is more interesting than the algebra. If attendance never varies, the data contain no information about how GPA responds to attendance — there is nothing to compare. This is assumption CLRM3, and it is the same point as Section 1.1: where nothing varies, nothing can

be learned.

7.5 A regression reports $R^2 = 0.04$ and a slope significant at the 1% level. Is that a contradiction?

SOLUTION

No. They measure different things.
R^2 says the regression accounts for 4% of the variation in y — the points scatter widely around the line. The p-value says the slope is distinguishable from zero, which depends on the standard error and therefore on the sample size.
With a large sample, a real but small relationship produces exactly this pattern: a reliably estimated slope explaining little of the variation. Both statements are true and neither is more important; they answer "how much does x account for?" and "is there a relationship at all?"

7.6 Explain why $\text{Var}(\hat{\beta}_1)$ falls as the spread of x increases, and what this implies for research design.

SOLUTION

The formula is $\text{Var}(\hat{\beta}_1) = \sigma^2 / \sum(x_i - \bar{x})^2$, so spread in x appears in the denominator and larger spread means a smaller variance.
The intuition is that a slope is a rate of change, and rates of change are measured by observing change. If every student attended between 27 and 29 classes, we would be trying to infer the effect of attendance from almost no variation in attendance, and small errors in y would translate into wild swings in the fitted slope. Observing attendance from 5 to 30 pins the line down.
For design, this says that variation in the explanatory variable is valuable and should be sought rather than avoided. A study of the effect of class size that only observes classes of 39 to 41 students will struggle no matter how many schools it covers. It also explains why the median split did worse than the regression: collapsing attendance to two categories discards most of its spread.

R exercises

7.7 Using `data/car-mileage.csv`, regress fuel efficiency on weight. Report the slope with its interval, interpret it in the units of the data, and plot the fit.

SOLUTION

```
cars <- read.csv("data/car-mileage.csv")

fit_cars <- lm(mileage_kmpl ~ weight_kg, data = cars)
summary(fit_cars)$coefficients
```

```
#>              Estimate Std. Error t value
#> (Intercept) 39.440284  1.6140031   24.44
#> weight_kg   -0.006009  0.0005179  -11.60
#>                                            Pr(>|t|)
#> (Intercept) 0.000000000000000000000000000000000001385
#> weight_kg   0.000000000000000003798485936567012204243
```

```
confint(fit_cars)["weight_kg", ]
```

```
#>     2.5 %    97.5 %
#> -0.007041 -0.004976
```

```
ggplot(cars, aes(weight_kg, mileage_kmpl)) +
  geom_point(colour = book_palette$ink, alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE, colour = book_palette$accent,
              fill = book_palette$fill) +
  labs(x = "Weight (kg)", y = "Fuel efficiency (km per litre)") +
  theme_book()
```

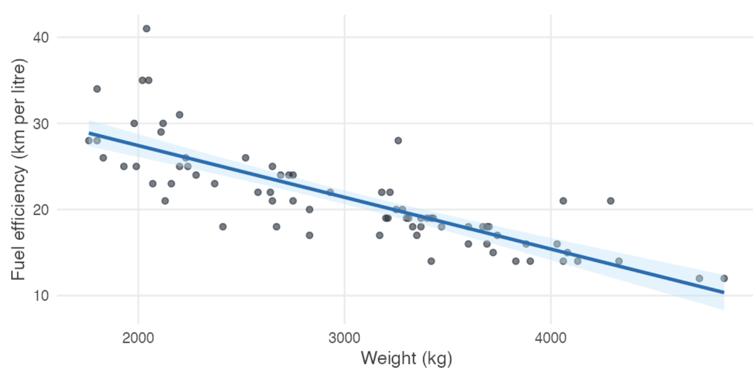

Figure 7.8: Fuel efficiency against weight for 74 cars, with the fitted line.

The slope is -0.00601 km per litre per kilogram, which is awkward to read at that scale. Multiply by 100: each additional 100 kg costs about 0.6 km per litre, with an interval of roughly 0.50 to 0.70.

R^2 is 0.65, much higher than the attendance regression, because weight is a genuinely dominant determinant of fuel consumption in a way that attendance is not of grades.

7.8 Verify the algebra. For the car regression, check that the residuals sum to zero, that they are uncorrelated with weight, and that the line passes through the point of means.

SOLUTION

```
u <- residuals(fit_cars)
w <- cars$weight_kg

round(c(sum_u        = sum(u),
        sum_w_u      = sum(w * u),
        cor_w_u      = cor(w, u),
        fitted_at_wbar = coef(fit_cars)[1] + coef(fit_cars)[2] *
         ↪  mean(w),
        mean_mileage   = mean(cars$mileage_kmpl)), 6)
```

```
#>                    sum_u                  sum_w_u
#>                      0.0                      0.0
#>                  cor_w_u fitted_at_wbar.(Intercept)
#>                      0.0                     21.3
#>             mean_mileage
#>                     21.3
```

All three hold. The first two are zero to within floating-point error, and the last two agree exactly.
None of this is evidence that the model is correct. These identities are forced by the two equations OLS solves, and they would hold just as exactly if we regressed mileage on something absurd.

7.9 Demonstrate that OLS is unbiased, and that spread in x buys precision. Simulate from a known model with $\beta_1 = 2$, using x spread widely in one case and narrowly in the other.

SOLUTION

```
simulate_slope <- function(x_sd, reps = 3000) {
  b1 <- replicate(reps, {
    x <- rnorm(50, mean = 20, sd = x_sd)
    y <- 5 + 2 * x + rnorm(50, sd = 3)          # truth: beta_1 = 2
    coef(lm(y ~ x))[2]
  })
  c(x_sd = x_sd, mean_b1 = mean(b1), sd_b1 = sd(b1))
}

set.seed(11)
round(t(sapply(c(1, 5), simulate_slope)), 4)
```

```
#>      x_sd mean_b1  sd_b1
#> [1,]    1   1.996 0.4384
#> [2,]    5   1.999 0.0875
```

Both rows are centred on 2, so OLS is unbiased at either spread — that is what CLRM1–CLRM4 deliver, and the spread of x has nothing to do with it.
What the spread changes is precision. With x five times as spread out, the standard deviation of $\hat{\beta}_1$ falls by roughly a factor of five, exactly as $\sigma^2 / \sum(x_i - \bar{x})^2$ predicts.

7.10 Show what omitted variable bias looks like. Simulate data where y depends on both x and a variable z that is correlated with x, then estimate a regression that omits z.

SOLUTION

```
set.seed(2)

biased <- replicate(3000, {
  z <- rnorm(200)                           # unobserved: motivation, say
  x <- 10 + 2 * z + rnorm(200)              # x is correlated with z
  y <- 5 + 1 * x + 3 * z + rnorm(200)   # truth: the effect of x is 1
  c(omitting_z  = coef(lm(y ~ x))[2],
    including_z = coef(lm(y ~ x + z))[2])
})

round(rowMeans(biased), 4)
```

```
#>  omitting_z.x including_z.x
#>        2.1997        0.9996
```

The true effect of x on y is 1. Omitting z returns about 2.2 — more than double.

The reason is exactly the term in the unbiasedness derivation. With z left out, it sits inside the error, and since z is correlated with x the errors are correlated with x too, so $E(u \mid x) \neq 0$ and the bias does not vanish. Including z recovers 1.
Note that the biased regression is not badly behaved in any visible way. Its residuals still sum to zero, its standard errors are still computed correctly, and its p-value is still tiny. It is confidently wrong, and nothing in the output says so.

7.11 Question bank

WATCH OUT

No answers are given for this section, by design.

Multiple choice

QB 7.1 The OLS slope estimator equals

a. the correlation between x and y
b. the sample covariance of x and y divided by the sample variance of y
c. the sample covariance of x and y divided by the sample variance of x
d. the ratio of the two sample means

QB 7.2 Which of the following hold by construction in every OLS regression? *(select all that apply)*

a. $\sum \hat{u}_i = 0$
b. $\sum x_i \hat{u}_i = 0$
c. $E(u \mid x) = 0$
d. The line passes through $(\bar{x}, \bar{y})$

QB 7.3 In a simple regression, $\hat{\sigma}^2$ divides the sum of squared residuals by

a. n
b. $n - 1$
c. $n - 2$
d. $n - 3$

QB 7.4 Var($\hat{\beta}_1$) decreases when

a. σ^2 increases
b. the spread of x increases
c. the spread of y increases
d. R^2 decreases

QB 7.5 A regression of wages on schooling gives $R^2 = 0.11$. This means

a. the model is misspecified
b. schooling accounts for 11% of the variation in wages
c. 11% of the coefficient is reliable
d. schooling causes 11% of wage differences

QB 7.6 Omitted variable bias arises when the omitted variable

a. is correlated with y only
b. is correlated with x only
c. is correlated with both x and y
d. has a large variance

Short reasoning

QB 7.7 Explain why $\sum x_i \hat{u}_i = 0$ is not evidence that $E(u \mid x) = 0$.

QB 7.8 A student reports that their regression "explains 95% of the variation, so the model must be correct." Evaluate.

QB 7.9 Why does the divisor in $\hat{\sigma}^2$ change from $n-1$ in Unit 3 to $n-2$ here?

QB 7.10 In the fertiliser example the slope was 1.25 quintals per kilogram. Explain what would happen to that number, and to its standard error, if fertiliser were measured in grams instead.

QB 7.11 Explain why a regression can have a correctly computed standard error, a valid p-value, and a completely misleading interpretation.

QB 7.12 For six observations, $\sum(x_i - \bar{x})(y_i - \bar{y}) = 84$, $\sum(x_i - \bar{x})^2 = 56$, $\bar{x} = 9$, $\bar{y} = 21$ and $\sum \hat{u}_i^2 = 12$.

i. Compute $\hat{\beta}_1$ and $\hat{\beta}_0$.
ii. Compute $\hat{\sigma}^2$ and se($\hat{\beta}_1$).
iii. Test $H_0 : \beta_1 = 0$ at the 5% level.
iv. Construct a 95% confidence interval for β_1.

Numerical

QB 7.13 A regression on 30 observations gives $\hat{\beta}_1 = 4.8$, se$(\hat{\beta}_1) = 1.5$, SST $=$ 900 and SSR $= 400$.

i. Compute R^2.
ii. Test $H_0 : \beta_1 = 0$ at the 1% level.
iii. Test $H_0 : \beta_1 = 8$ at the 5% level.
iv. Compute $\hat{\sigma}^2$.

QB 7.14 Yields and rainfall for five districts:

Rainfall (cm)	60	80	100	120	140
Yield (quintals)	18	23	24	30	35

i. Compute the OLS estimates.
ii. Compute all five residuals and verify they sum to zero.
iii. Compute R^2.
iv. Predict yield at 110 cm, and say why predicting at 300 cm would be unwise.

7.12 Looking ahead

The next unit takes the first step described above: measuring the variables that create the bias, and comparing observations that are alike in those respects.

Remarkably little new machinery is required. The same least squares criterion, the same normal equations, the same t statistics, the same standard errors — all of it carries over with an extra column of data attached.

What changes is the meaning of a coefficient. In a simple regression $\hat{\beta}_1$ answered "how does average y differ across values of x?" With more than one explanatory variable it answers something considerably more useful, and considerably easier to misread.

Chapter 8

Multiple Regression: Holding Other Things Equal

Unit 7 ended with two students. One attended thirty classes, the other twenty, and the first earned the higher grade. The difficulty was that the two students differed in many other ways as well. One may have been better prepared, more motivated, more interested in the subject, or simply worked harder outside class. The regression slope faithfully described the difference in their average grades, but it could not tell us how much of that difference was due to attendance itself.

A natural response is to measure some of those other differences and compare students who are alike in every observed respect except attendance.

That is the idea behind multiple regression.

Conceptually, very little changes. We still model the conditional mean, fit the model by least squares, and measure uncertainty with standard errors, confidence intervals and hypothesis tests. The machinery is familiar.

What changes is the interpretation.

In a simple regression, the slope describes how average outcomes change as one variable changes. In a multiple regression, each coefficient describes how the outcome changes as one variable changes while the others are held fixed. That seemingly small change is what makes multiple regression one of the most useful tools in empirical economics.

It is also where many misunderstandings begin.

What does it really mean to "hold other things equal"? How does regression compare observations that are never exactly identical? When does controlling for additional variables reduce bias, and when can it make matters worse? Most importantly, does adding more variables bring us any closer to the causal comparison we wanted in Unit 7?

Those are the questions this unit takes up.

8.1 Why one variable is not enough

> *The simple regression of grades on attendance was estimated precisely. What is missing?*

Unit 7 ended with a simple regression of semester GPA on attendance.

```
students <- read.csv("data/attendance-grades.csv")

simple <- lm(sem_gpa ~ attendance, data = students)
round(coef(summary(simple)), 4)
```

```
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)    1.562     0.2869   5.444        0
#> attendance     0.189     0.0107  17.589        0
```

The estimate is clear. Students who attend one additional class earn, on average, a semester GPA that is 0.189 points higher. The standard error is small, the confidence interval is narrow, and there is overwhelming evidence that attendance and grades are associated.

The question is whether this is the comparison we wanted to make.

Suppose we compare two students. One attended thirty classes and the other twenty. Their attendance differs by ten classes, but that is unlikely to be the only difference between them. The student who attended more may also have entered the semester better prepared, have stronger study habits, or simply be more academically able.

Our data contain measures of two such characteristics: each student's cumulative GPA before the semester began, and their admission score.

```
round(cor(students[, c("attendance", "sem_gpa",
                       "cum_gpa", "admission_score")]), 3)
```

```
#>                 attendance sem_gpa cum_gpa admission_score
#> attendance           1.000   0.560   0.427          -0.156
#> sem_gpa              0.560   1.000   0.653           0.246
#> cum_gpa              0.427   0.653   1.000           0.354
#> admission_score     -0.156   0.246   0.354           1.000
```

One number immediately stands out. Attendance and prior GPA have a correlation of 0.43. Students who attend more classes also tend to have performed better before this semester even began.

Prior GPA, in turn, is strongly related to semester GPA.

So when the simple regression compares students with different attendance, it is also comparing students with different academic histories. Part of the estimated relationship may reflect attendance, and part may simply reflect the fact that students who were already performing well continue to do so.

This is exactly the problem discussed at the end of Unit 7. Variables that affect the outcome and move with attendance remain in the error term, and their influence becomes mixed with the coefficient on attendance.

The comparison we really want is more specific.

Instead of asking,

> Do students who attend more classes earn higher grades?

we would like to ask,

> Among students with the same prior GPA and the same admission score, do those who attend more classes earn higher grades?

That is a different comparison. We are no longer comparing every student with every other student. We are comparing students who are similar in their observed academic background and differ primarily in attendance.

Simple regression assumes that every relevant difference between students is either unrelated to attendance or hidden inside the error term. Multiple regression takes the first step towards relaxing that assumption by bringing some of those differences into the model.

8.2 Holding other things equal

> *We add a second explanatory variable. What does β_1 now measure?*

The motivation for multiple regression was straightforward. We wanted to compare students who differed in attendance but were otherwise similar.

To do that we introduce additional explanatory variables.

With two explanatory variables, the population model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

and with k explanatory variables,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$$

Nothing fundamental has changed. There is still a population regression function that we cannot observe, an error term containing everything left out of the model, and a sample from which we estimate the unknown coefficients.

What has changed is the meaning of a coefficient.

A different comparison

In the simple regression of Unit 7, the coefficient on attendance compared students who attended different numbers of classes.

Now the comparison is narrower.

The coefficient on attendance compares students who attended different numbers of classes but who had the same values of the other explanatory variables.

That is the meaning of holding other things equal.

Mathematically,

$$\frac{\partial E(Y \mid X_1, X_2)}{\partial X_1} = \beta_1$$

The notation looks new, but the idea is not. A partial derivative asks how the conditional mean changes when one explanatory variable changes while the others remain fixed.

> **REMEMBER THIS**
>
> β_1 is the change in the average value of Y associated with a one-unit increase in X_1, holding all the other explanatory variables constant.
> This *ceteris paribus* interpretation is the entire reason for using multiple regression.

An example

Adding explanatory variables in R is as simple as separating them with +.

```r
full <- lm(sem_gpa ~ attendance + cum_gpa + admission_score,
           data = students)

round(coef(summary(full)), 4)
```

```
#>                 Estimate Std. Error t value Pr(>|t|)
#> (Intercept)      -7.6603     1.2944  -5.918        0
#> attendance        0.1360     0.0104  13.040        0
#> cum_gpa           0.5748     0.0441  13.026        0
#> admission_score   0.0835     0.0158   5.299        0
```

Compare the coefficient on attendance with the simple regression.

```r
round(c(simple   = coef(simple)["attendance"],
        multiple = coef(full)["attendance"]), 4)
```

```
#>   simple.attendance multiple.attendance
#>               0.189               0.136
```

The estimated association falls from 0.189 to 0.136.

The interpretation has changed. In the simple regression, the comparison was between students who differed in attendance. In the multiple regression, the comparison is between students who have the same prior GPA and the same admission score, but who differ in attendance.

Some of the relationship that the simple regression attributed to attendance is now explained by differences in students' previous academic performance. The coefficient on attendance therefore becomes smaller.

> **THE IDEA IN PLAIN WORDS**
>
> Notice that nothing has happened to the data. The students are the same. Only the comparison has changed.

Reading the other coefficients

Every coefficient is interpreted in exactly the same way.

The coefficient on cumulative GPA says that among students with the same attendance and the same admission score, a one-point increase in prior GPA is associated with an increase of about 0.575 points in semester GPA.

The coefficient on admission score compares students with the same attendance and the same prior GPA.

Each coefficient asks what happens when one explanatory variable changes while all the others remain fixed.

Another example

The same interpretation applies outside education. This is a synthetic dataset of 20,000 Indian wage earners, built to resemble the structure of household survey data.

```r
wages <- read.csv("data/wages-india-synthetic.csv")

wages$male  <- as.integer(wages$sex == "Male")
wages$urban <- as.integer(wages$residence == "Urban")

w_full <- lm(annual_earnings ~ education + age + male + urban,
             data = wages)

round(coef(summary(w_full)), 1)
```

```
#>              Estimate Std. Error t value Pr(>|t|)
#> (Intercept) -28175.8     2612.7   -10.8        0
#> education     4281.2      134.7    31.8        0
#> age            559.8       57.4     9.8        0
#> male         31999.1     1442.8    22.2        0
#> urban        51287.4     1517.5    33.8        0
```

The coefficient on education is interpreted as the association between an additional year of schooling and annual earnings holding age, sex and place of residence constant.

The variables `male` and `urban` take only the values 0 and 1, so their coefficients compare two groups rather than describing the effect of a one-unit increase. We return to indicator variables in the next unit.

What "holding fixed" does not mean

The phrase *holding other things fixed* often gives the wrong impression.

Regression does not perform an experiment.

Nobody took the same student, kept their prior GPA unchanged and forced them to attend one more class. Nor did anyone move a rural worker to an urban area while leaving everything else unchanged.

> **WATCH OUT**
>
> The comparison is entirely statistical. The regression compares observations that already exist and are similar in the variables included in the model.
> Whether that comparison is enough to recover a causal effect depends on

> whether the important differences between individuals have been measured.

That question returns in Section 8.7.

8.3 What does “holding constant” actually mean?

> *Nothing in the data was held anywhere. So what does the regression do?*

The phrase *holding other variables constant* is used constantly in economics. It is also one of the least understood.

Nothing in the data was actually held fixed. Every student has a different attendance, a different prior GPA and a different admission score. Nobody reran the semester while changing only one variable.

So what does the regression actually do?

Predicting attendance

The answer can be described as a simple two-step procedure.

1. Predict attendance using the control variables.
2. Keep only what cannot be predicted.
3. Ask whether that remaining variation predicts grades.

Take the first step slowly, because everything rests on it.

Imagine predicting every student’s attendance using only their previous GPA and admission score. Some students attend exactly as much as we would expect. Some attend much more. Some attend much less.

Those unexpected differences are the residuals.

```
step1 <- lm(attendance ~ cum_gpa + admission_score, data = students)
r1    <- residuals(step1)

round(c(r2_of_step1      = summary(step1)$r.squared,
        cor_with_cumgpa  = cor(r1, students$cum_gpa),
        cor_with_admscr  = cor(r1, students$admission_score)), 4)
```

```
#>     r2_of_step1 cor_with_cumgpa cor_with_admscr
#>          0.2906          0.0000          0.0000
```

Prior GPA and admission score together account for 29% of the variation in attendance. The residual carries the rest, and it is uncorrelated with both — to machine precision, because that is what least squares guarantees.

That last point is the one that matters. Whatever these residuals contain, they contain nothing about prior GPA or admission score.

Using only what could not be predicted

Now regress semester GPA on that residual alone.

```
step2 <- lm(students$sem_gpa ~ r1)

noquote(format(c(from_two_steps   = unname(coef(step2)[2]),
                 from_lm_directly = unname(coef(full)["attendance"])),
               digits = 10))
```

```
#>   from_two_steps from_lm_directly
#>     0.1360121381     0.1360121381
```

The same number, to ten significant figures.

This result is known as the **Frisch–Waugh–Lovell theorem**, and it holds for every coefficient in every multiple regression, not just this one.

> **REMEMBER THIS**
>
> Multiple regression does not compare students with the same attendance. It compares students who differ in the part of attendance that cannot already be explained by the other variables.

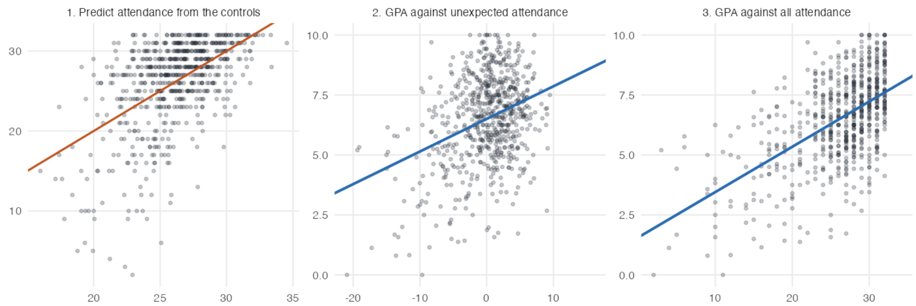

Figure 8.1: The two steps, and what they change. Panel 1 predicts attendance from prior GPA and admission score; the orange line is where a student would sit if the prediction were exact, and the vertical distances from it are the residuals. Panel 2 regresses GPA on those residuals alone. Panel 3 regresses GPA on attendance itself. The horizontal axes in panels 2 and 3 are drawn to the same width, so the two slopes can be compared directly.

THE IDEA IN PLAIN WORDS

Two students with the same prior GPA and admission score are predicted to attend roughly the same number of classes.
If one attends more than predicted and the other less, that difference is unrelated to the observed controls.
Multiple regression asks whether those unexpected differences in attendance are associated with differences in grades.

Why the coefficient changes

The coefficient changes because the variation changes.

The simple regression used all the variation in attendance — the third panel of the figure. The multiple regression uses only the part that prior record cannot account for — the second panel. Those are different questions asked of different variation, and there is no reason for them to give the same answer.

This is worth separating from the more familiar account, which says that controls remove bias. They may well do so, and Section 8.7 takes that up. But the mechanism is not that something contaminating has been subtracted from the answer. It is that the estimator is now working with a different set of comparisons.

Why the standard error usually grows

The residual $\hat{r}_1$ has less spread than attendance itself: a standard deviation of 4.6 classes against 5.5.

```
round(c(sd_of_attendance = sd(students$attendance),
        sd_of_residual   = sd(r1)), 3)
```

```
#> sd_of_attendance   sd_of_residual
#>            5.455            4.595
```

Section 7.5 established that the precision of a slope depends on the spread of the explanatory variable. Discarding variation must therefore cost something, and Section 8.4 works out the bill.

That is the trade the whole method rests on. A control buys a narrower, more credible comparison, and pays for it in precision.

8.4 The assumptions of multiple regression

> *One explanatory variable has become several. Does that require a completely new set of assumptions?*

Surprisingly, very little changes.

Everything that made OLS work in Unit 7 still matters. We still need a model that is linear in the parameters, a random sample, a conditional mean that is correctly specified, and — if we want the classical standard errors — constant error variance.

The only genuinely new assumption is one that could not arise before: the explanatory variables cannot perfectly explain one another.

> WHERE THIS COMES FROM
>
> **CLRM1 — linear in parameters.** $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u$.
> **CLRM2 — random sampling.** The n observations are a random sample from that population.
> **CLRM3 — variation in the explanatory variables.** No x_j takes the same value for every observation.
> **CLRM4 — zero conditional mean.** $E(u \mid x_1, \dots, x_k) = 0$.
> **CLRM5 — homoskedasticity.** $\mathrm{Var}(u \mid x_1, \dots, x_k) = \sigma^2$.
> **CLRM6 — normality.** The errors are normally distributed. As before, this is needed only for exact small-sample inference.

> **CLRM7 — no perfect collinearity.** No explanatory variable is an exact linear combination of the others.

Under CLRM1–CLRM4, OLS remains unbiased, exactly as before. The proof is identical to Section 7.4, written in matrix notation rather than scalar notation; Section 8.12 gives the compact version.

Three assumptions deserve comment.

Zero conditional mean

The assumption has not changed. The error term must still be unrelated to the explanatory variables,

$$E(u \mid x_1, \dots, x_k) = 0$$

What has changed is what remains inside u.

In the simple regression, prior GPA and admission score were omitted variables. Because they were correlated with attendance, they threatened the assumption directly.

In the multiple regression they are no longer omitted. They have become explanatory variables.

> **REMEMBER THIS**
>
> That is the sense in which multiple regression improves on simple regression. It does not make the assumption true. It removes some of the reasons why it might fail.

Whatever is still omitted — motivation, ability, family background — remains inside u, and can still bias the coefficients if it is related to the explanatory variables.

No perfect collinearity

This assumption is not about economics at all. It is a requirement for the regression to be mathematically identifiable.

Our data contain a violation, and it is instructive.

```
head(students$attendance + students$classes_missed)
```

```
#> [1] 32 32 32 32 32 32
```

The course had 32 classes, so attendance and classes missed always sum to 32. Knowing one is knowing the other. Ask R to use both.

```
coef(lm(sem_gpa ~ attendance + classes_missed, data = students))
```

```
#>    (Intercept)     attendance classes_missed
#>          1.562          0.189             NA
```

R returns NA for the second coefficient. It has not failed; it is reporting that the question has no answer.

Why there is no answer

The reason can be seen with one substitution. Since missed $= 32 -$ attendance, the fitted part of the model is

$$\begin{aligned}
\beta_0 + \beta_1 \,\text{att} + \beta_2 \,\text{missed} &= \beta_0 + \beta_1 \,\text{att} + \beta_2(32 - \text{att}) \\
&= \underbrace{(\beta_0 + 32\beta_2)}_{\text{intercept}} + \underbrace{(\beta_1 - \beta_2)}_{\text{slope}} \,\text{att}
\end{aligned}$$

The two coefficients enter only through their **difference**. Any pair with the same $\beta_1 - \beta_2$ produces exactly the same fitted values, and therefore exactly the same sum of squared residuals.

Least squares chooses the coefficients that minimise that sum. Here infinitely many pairs minimise it equally well, so there is nothing to choose between them.

```
att <- students$attendance
mis <- students$classes_missed

## Three quite different pairs, all with the same difference b1 - b2.
fit_at <- function(b0, b1, b2) b0 + b1 * att + b2 * mis

round(c(pair1_vs_pair2 = max(abs(fit_at(1.562, 0.189, 0) -
                                 fit_at(1.562 -  32, 1.189, 1))),
        pair1_vs_pair3 = max(abs(fit_at(1.562, 0.189, 0) -
                                 fit_at(1.562 - 160, 5.189, 5)))), 10)
```

```
#> pair1_vs_pair2 pair1_vs_pair3
#>              0              0
```

A slope of 0.189 on attendance with 0 on classes missed, a slope of 1.189 with 1, and a slope of 5.189 with 5 all fit the data identically. The differences are zero to machine precision.

The data cannot distinguish between them, and neither can any estimator.

> **GOING A LITTLE FURTHER**
>
> The same fact appears in the two-step procedure of Section 8.3.
> Regress attendance on classes missed, and the fit is perfect: $R^2 = 1$, and every residual is zero. Step two would then regress GPA on a variable that does not vary at all.
> That is assumption CLRM3 failing, and it is also the variance formula of the next subsection with $R_j^2 = 1$, so that $(1 - R_j^2) = 0$ and the denominator vanishes. Perfect collinearity is not a separate problem from the one the next subsection describes. It is the same problem at its limit.

> **THE IDEA IN PLAIN WORDS**
>
> The coefficient on attendance is supposed to answer: what happens to the grade when attendance rises by one, holding classes missed fixed? Nothing happens, because nothing *can* happen. Attending one more class means missing one fewer. There is no student for whom one changed and the other did not, so the data contain no information about that comparison, and no amount of cleverness can extract it.

When controls tell us the same thing

Perfect collinearity is rare. The important case is when two explanatory variables contain much of the same information.

Attendance and prior GPA are not identical. But students with high prior GPAs also tend to attend more classes. Part of the variation in attendance can therefore already be predicted from prior GPA.

That is exactly what Section 8.3 removed during the residualisation step.

The more predictable attendance becomes, the less independent variation remains for estimating its coefficient. Less variation means less information. Less information means larger standard errors.

The variance formula puts a number on it:

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2 \, (1 - R_j^2)}$$

Compare it with the simple regression formula of Section 7.5: everything is the same except the extra factor $(1 - R_j^2)$ in the denominator.

> **REMEMBER THIS**
>
> R_j^2 is not the R^2 from the regression you care about.
> It is the R^2 from regressing one explanatory variable on all the others. It tells us how much of that variable is already predictable before the regression even begins.
> If $R_j^2 = 0$, the variable contributes completely new information.
> If $R_j^2 = 0.95$, almost everything it contains was already available from the remaining explanatory variables. OLS is then trying to estimate a coefficient using only the tiny amount of variation left over.
> That is why the standard error grows.

About 29% of attendance can already be predicted from prior GPA and admission score. The coefficient therefore relies on the remaining 71% — the unexpected part of attendance. There is still plenty of independent variation left, so the increase in the standard error is modest.

As before, estimating additional coefficients consumes degrees of freedom, so the residual variance is now divided by $n - k - 1$.

The trade

Multiple regression always involves a trade-off.

Every additional control makes the comparison more credible by removing another potential source of bias.

But every additional control also consumes variation, making the coefficient less precise.

Good empirical work is largely about deciding which controls are worth that price.

8.5 Testing one coefficient

> *Does the machinery of Unit 7 still work?*

It does, unchanged apart from the degrees of freedom. Before using it, it is worth recalling why it works at all.

Every coefficient is an estimate

The four numbers in the first column of a regression output are not facts about the population. They were computed from one sample of 680 students.

A different sample of 680 students would have produced four different numbers.

> **REMEMBER THIS**
>
> Each $\hat{\beta}_j$ is a **statistic** — a quantity computed from a sample — and so each has a sampling distribution, exactly as the sample mean did in Unit 2. Section 7.3 drew that distribution for the simple regression slope by taking five thousand samples and plotting the result. The same is true of every coefficient here, and of all of them jointly.

That is what makes the rest of this unit possible. Unbiasedness says the distribution of $\hat{\beta}_j$ is centred on β_j. The standard error estimates its width. And the Central Limit Theorem of Unit 1 says that in reasonably large samples the distribution is approximately normal, whatever the distribution of the errors — which is why CLRM6 matters so little in practice.

So the problem is the one this book has been solving since Unit 2. We have a single draw from a distribution centred on the quantity we want, and we know roughly how wide that distribution is. Everything that followed from that — standard errors, confidence intervals, t statistics, p-values — applies here without modification.

The test

To test whether a single population coefficient equals zero,

$$H_0 : \beta_j = 0 \qquad\qquad t = \frac{\hat{\beta}_j}{\mathrm{se}(\hat{\beta}_j)} \sim t_{n-k-1}$$

and confidence intervals follow as before.

```r
round(coef(summary(full)), 4)
```

```
#>                 Estimate Std. Error t value Pr(>|t|)
#> (Intercept)      -7.6603     1.2944  -5.918        0
#> attendance        0.1360     0.0104  13.040        0
#> cum_gpa           0.5748     0.0441  13.026        0
#> admission_score   0.0835     0.0158   5.299        0
```

```r
round(confint(full), 4)
```

```
#>                    2.5 %  97.5 %
#> (Intercept)     -10.2019 -5.1188
#> attendance        0.1155  0.1565
#> cum_gpa           0.4881  0.6614
#> admission_score   0.0526  0.1144
```

Every coefficient is significant at any conventional level, and the interval for attendance runs from 0.116 to 0.157.

> **WATCH OUT**
>
> The null tested is $\beta_j = 0$ **in the model as specified**.
> Rejecting it says attendance is related to grades among students alike in prior GPA and admission score. It says nothing about a model with different controls, where the coefficient, its standard error and its significance may all differ.
> A t statistic is a statement about one column of one regression.

8.6 Testing several coefficients at once

> *We can test one coefficient. Why not just do it repeatedly?*

Suppose we want to know whether prior record matters at all — whether prior GPA and admission score, taken together, belong in the model.

$$H_0 : \beta_2 = 0 \text{ and } \beta_3 = 0$$

The obvious approach is two t-tests. It does not work, and seeing why is more useful than the test that replaces it.

Why separate tests fail

Each t-test asks about one coefficient while the others stay in the model. The joint null asks about both coefficients simultaneously. These are different claims, and they can disagree.

Here is a small constructed example where the disagreement is stark. Two explanatory variables both genuinely affect y, and they are highly correlated with each other.

```r
set.seed(11)
n  <- 60
z  <- rnorm(n)
x1 <- z + rnorm(n, sd = 0.25)
x2 <- z + rnorm(n, sd = 0.25)
y  <- 1 + 0.8 * x1 + 0.8 * x2 + rnorm(n, sd = 2)

round(cor(x1, x2), 3)
```

```
#> [1] 0.912
```

```r
round(coef(summary(lm(y ~ x1 + x2))), 4)
```

```
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   1.0765     0.2577   4.177   0.0001
#> x1            0.9376     0.7197   1.303   0.1979
#> x2            0.7246     0.6890   1.052   0.2974
```

Neither variable is close to significant. The p-values are 0.20 and 0.30, and a reader taking each test at face value would drop both.

Now test them together.

```r
anova(lm(y ~ 1), lm(y ~ x1 + x2))
```

```
#> Analysis of Variance Table
#>
#> Model 1: y ~ 1
#> Model 2: y ~ x1 + x2
#>   Res.Df RSS Df Sum of Sq    F    Pr(>F)
#> 1     59 334
#> 2     57 216  2       119 15.7 0.0000038 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Jointly, they are overwhelmingly significant.

> **THE IDEA IN PLAIN WORDS**
>
> There is no contradiction, and the resolution is Section 8.3.
> The t-test on x_1 uses only the part of x_1 that is unrelated to x_2. Since the two are correlated at 0.91, almost nothing is left, and that remainder tells us very little. The same is true of x_2.
> The data cannot say which of the two is responsible. They are perfecty clear that one of them is.

> Two individually insignificant coefficients therefore do not license dropping both. What can be said about them separately, and what can be said about them together, are different things.

The restricted and unrestricted models

The joint test compares two models.

The **unrestricted** model is the one we believe might be appropriate:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u$$

The **restricted** model is what remains after imposing the null. If the null sets q coefficients to zero, the restricted model simply omits those q variables.

Imposing a restriction can only make the fit worse: the unrestricted model was free to choose those coefficients and chose them to minimise the sum of squared residuals. So $\text{SSR}_r \geq \text{SSR}_{ur}$ always.

> **REMEMBER THIS**
>
> The question is therefore not *whether* the fit got worse, but **whether it got worse by more than dropping irrelevant variables would**.
> If the restricted variables truly have zero coefficients, the deterioration should be no larger than sampling noise would produce.

Building the statistic

Two quantities are needed, and both are sums of squared residuals.

The **deterioration** is $\text{SSR}_r - \text{SSR}_{ur}$. Larger values are evidence against the null. But it must be judged relative to something, for two reasons: dropping more variables must be expected to hurt more, and the raw size depends on how noisy the data are.

So divide the deterioration by q, the number of restrictions, and divide the remaining unexplained variation by its degrees of freedom, $n - k - 1$. The ratio is

$$F = \frac{(\text{SSR}_r - \text{SSR}_{ur}) \big/ q}{\text{SSR}_{ur} \big/ (n - k - 1)}$$

REMEMBER THIS

Read the two pieces separately.
The numerator is **the variation the restricted variables account for, per restriction**.
The denominator is **the variation nothing accounts for, per degree of freedom** — an estimate of σ^2.
F asks whether the variables we are testing explain more per variable than noise does. Under the null, they explain nothing systematic, so both pieces estimate the same thing and the ratio averages about one.

An equivalent form uses R^2, which is often more convenient because the sums of squares depend on the units of y and R^2 does not:

$$F = \frac{(R^2_{ur} - R^2_r)\big/q}{(1 - R^2_{ur})\big/(n-k-1)}$$

Why an F distribution

Every test in this book compares a statistic to the distribution it would follow if the null were true. So what distribution does this ratio follow?

The reasoning is one we have used before. In Section 3.2 a sample variance built from normal errors, scaled appropriately, followed a chi-squared distribution with degrees of freedom equal to the number of independent squared deviations. That is exactly what both parts of F are.

WHERE THIS COMES FROM

Under CLRM1–CLRM6, the numerator and denominator are each a sum of squared normal deviations, scaled by σ^2:

$$\frac{\text{SSR}_r - \text{SSR}_{ur}}{\sigma^2} \sim \chi^2_q \qquad\qquad \frac{\text{SSR}_{ur}}{\sigma^2} \sim \chi^2_{n-k-1}$$

They are independent, and dividing each by its degrees of freedom and taking the ratio gives, by definition, an F **distribution** with q and $n-k-1$ degrees of freedom.
The unknown σ^2 appears in both and cancels — the same cancellation that made a t statistic computable in Section 3.5.

That is the formal answer. The useful one is shorter: **an F distribution is what the ratio of two independent variance estimates looks like when both are estimating the same thing.** Under the null they are, which is why F averages about one and why large values are evidence against it.

The distribution is skewed rather than symmetric, and heavily so when q is small. Most of its mass sits below one — a ratio can fall no lower than zero but has no ceiling — which is why the average and the typical value are not the same number, and why the critical value of 3.12 lies so far out.

Since only large values count as evidence, the test is **always one-tailed on the right**, whatever the alternative.

We can watch this happen. Generate data in which two variables genuinely have no effect, compute F thousands of times, and compare the results with the distribution the theory predicts.

```
set.seed(3)
xa <- rnorm(80); xb <- rnorm(80); xc <- rnorm(80)

f_stats <- replicate(4000, {
  yy <- 2 + 1.5 * xa + rnorm(80)                    # xb and xc have no effect
  anova(lm(yy ~ xa), lm(yy ~ xa + xb + xc))$F[2]
})

round(rbind(simulated = quantile(f_stats, c(0.50, 0.90, 0.95, 0.99)),
            theory    = qf(c(0.50, 0.90, 0.95, 0.99), df1 = 2, df2 = 76)), 3)
```

```
#>             50%   90%   95%   99%
#> simulated 0.716 2.401 3.183 5.019
#> theory    0.700 2.374 3.117 4.896
```

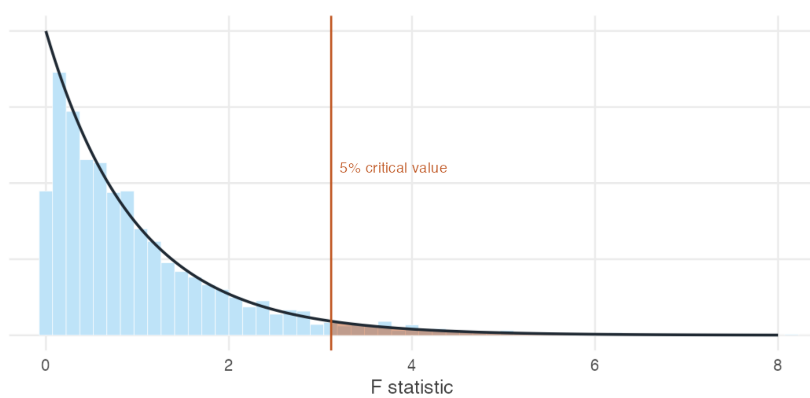

Figure 8.2: Four thousand F statistics computed from data in which the two tested variables have no effect whatever, against the F distribution with 2 and 76 degrees of freedom. The shaded region beyond the 5% critical value is where the test rejects – and since the null is true here, every rejection in it is a Type I error.

The simulated quantiles match the theoretical ones closely, and the test rejects a true null about 5% of the time, which is what a 5% test is supposed to do.

Does prior record belong in the model?

Back to the question that opened this section.

```
restricted   <- lm(sem_gpa ~ attendance, data = students)
unrestricted <- lm(sem_gpa ~ attendance + cum_gpa + admission_score,
                   data = students)

anova(restricted, unrestricted)
```

```
#> Analysis of Variance Table
#>
#> Model 1: sem_gpa ~ attendance
#> Model 2: sem_gpa ~ attendance + cum_gpa + admission_score
#>   Res.Df  RSS Df Sum of Sq   F               Pr(>F)
#> 1    678 1581
#> 2    676 1054  2       527 169 <0.0000000000000002 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

By hand, from the two sums of squared residuals:

```
ssr_r  <- sum(residuals(restricted)^2)
ssr_ur <- sum(residuals(unrestricted)^2)
q <- 2; n <- nrow(students); k <- 3

f_stat <- ((ssr_r - ssr_ur) / q) / (ssr_ur / (n - k - 1))

round(c(ssr_restricted = ssr_r, ssr_unrestricted = ssr_ur,
        F = f_stat, critical_5pct = qf(0.95, q, n - k - 1),
        p_value = 1 - pf(f_stat, q, n - k - 1)), 4)
```

```
#>   ssr_restricted ssr_unrestricted                F    critical_5pct
#>         1580.858         1054.233          168.842            3.009
#>          p_value
#>            0.000
```

$F = 168.8$ against a critical value of 3.01. Prior GPA and admission score belong in the model, jointly and emphatically.

The F statistic R prints by itself

Every regression summary ends with a line reporting an F statistic. It is this same test applied to one particular null: that **all** the slope coefficients are zero at once.

```r
summary(full)$fstatistic
```

```
#> value numdf dendf
#> 266.8   3.0 676.0
```

The restricted model in that comparison has no explanatory variables at all — it predicts $\bar{y}$ for everyone — so $R_r^2 = 0$ and the formula reduces to

$$F = \frac{R^2/k}{(1 - R^2)/(n - k - 1)}$$

> **WATCH OUT**
>
> This is the least interesting test in the output.
> It asks whether the model explains anything whatever. With reasonable data and sensible variables it is almost always rejected, and rejecting it establishes very little. It is worth noticing chiefly when it *fails* to reject.

A joint test that does not reject

Not every set of variables earns its place. The data also record which year of the programme each student is in.

```r
with_years <- lm(sem_gpa ~ attendance + cum_gpa + admission_score + year1
 ↪  + year2,
                 data = students)

anova(unrestricted, with_years)
```

```
#> Analysis of Variance Table
#>
#> Model 1: sem_gpa ~ attendance + cum_gpa + admission_score
#> Model 2: sem_gpa ~ attendance + cum_gpa + admission_score + year1 +
 ↪  year2
#>   Res.Df  RSS Df Sum of Sq    F Pr(>F)
#> 1    676 1054
#> 2    674 1051  2       3.2 1.02   0.36
```

$F = 1.02$ with a p-value of 0.36. Once attendance and prior record are accounted for, year of study adds nothing detectable.

> **WATCH OUT**
>
> Failing to reject is not proof that year of study is irrelevant, for the reason given in Section 4.9: a test that does not reject may simply lack the power to detect a small effect.
> The honest statement is that these data provide no evidence that year of study matters once the other variables are held fixed.

8.7 Does this give us causality?

> *We have controlled for prior record. Is 0.136 the effect of attending class?*

Not necessarily, and the reason is worth setting out precisely rather than as a caution.

Where bias comes from

Suppose the true population model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

but we omit x_2 and estimate

$$y = \alpha_0 + \alpha_1 x_1 + e$$

What does $\hat{\alpha}_1$ estimate? Write the relationship between the two explanatory variables as a regression of its own,

$$x_2 = \delta_0 + \delta_1 x_1 + \eta$$

and substitute into the true model:

> **WHERE THIS COMES FROM**
>
> $$\begin{aligned} y &= \beta_0 + \beta_1 x_1 + \beta_2(\delta_0 + \delta_1 x_1 + \eta) + u \\ &= (\beta_0 + \beta_2\delta_0) + (\beta_1 + \beta_2\delta_1)x_1 + (\beta_2\eta + u) \end{aligned}$$

Matching this against $y = \alpha_0 + \alpha_1 x_1 + e$ gives

$$\alpha_1 = \beta_1 + \beta_2 \delta_1$$

So the short regression estimates the true coefficient **plus a bias term**:

$$\text{bias} = \beta_2 \delta_1 = (\text{effect of } x_2 \text{ on } y) \times (\text{relationship between } x_2 \text{ and } x_1)$$

This is **omitted variable bias**, and the formula says something immediately useful. The bias is a product of two things, so it vanishes if *either* is zero. An omitted variable is harmless unless it both affects y and moves with x_1.

REMEMBER THIS

The sign of the bias is the product of the two signs.

β_2	δ_1	Bias in $\hat{\alpha}_1$
$+$	$+$	Upward
$+$	$-$	Downward
$-$	$+$	Downward
$-$	$-$	Upward

This can often be signed from knowledge of the subject even when the variable cannot be measured, which makes it possible to say which way an estimate is likely to be wrong.

Checking the arithmetic

The decomposition is exact in a sample, not merely a statement about expectations. We can verify it on the grades data.

```r
long <- coef(unrestricted)

d_cum <- coef(lm(cum_gpa ~ attendance, data = students))[2]
d_adm <- coef(lm(admission_score ~ attendance, data = students))[2]

round(c(short_regression = coef(restricted)["attendance"],
        long_coefficient = long["attendance"],
        bias_via_cum_gpa = long["cum_gpa"] * d_cum,
        bias_via_adm     = long["admission_score"] * d_adm,
        reassembled      = long["attendance"] +
                           long["cum_gpa"] * d_cum +
                           long["admission_score"] * d_adm), 6)
```

```
#>  short_regression.attendance  long_coefficient.attendance
#>                     0.188952                     0.136012
#>     bias_via_cum_gpa.cum_gpa bias_via_adm.admission_score
#>                     0.061295                    -0.008356
#>       reassembled.attendance
#>                     0.188952
```

The simple regression coefficient reassembles exactly: 0.136 of attendance, plus 0.062 arriving through prior GPA, less a small amount through admission score. The gap between 0.189 and 0.136 is not a mystery. It is a quantity we can name and compute.

> **THE IDEA IN PLAIN WORDS**
>
> Prior GPA biased the simple regression *upward* because both signs were positive: better students attend more ($\delta_1 > 0$) and better students earn higher grades ($\beta_2 > 0$).
> Admission score worked the other way. Its coefficient is positive, but in these data students with higher admission scores attend slightly *fewer* classes, so its contribution is negative and small.

What the controls did and did not do

The controls removed the bias from prior GPA and admission score. They did nothing about anything else.

> **WATCH OUT**
>
> Motivation is not in the model. Neither is interest in the subject, nor health, nor how far a student lives from campus, nor whether they hold a job.
> Every one of these plausibly affects grades, and every one plausibly moves with attendance. Each contributes its own $\beta_2\delta_1$ to the estimate, and the formula does not care that we cannot measure it.

So 0.136 is a better estimate than 0.189 in a specific and limited sense: two named sources of bias have been removed. It is not the causal effect of attendance, and the regression output cannot tell us how far from it we are.

> **REMEMBER THIS**
>
> Multiple regression controls for what is **measured**. Omitted variable bias is caused by what is **not**.
> Adding controls narrows the gap between the estimate and the causal effect without ever closing it, and nothing in the output reports how much

remains.

The other three ways CLRM4 fails

Omission is the most common threat, and it is not the only one. The zero conditional mean assumption also fails when:

- **The functional form is wrong.** If earnings rise with experience and then flatten, a model linear in experience is misspecified, and the misspecification lands in u.
- **Variables are measured with error.** A variable recorded imperfectly is not the variable the model refers to, and the difference goes into the error term.
- **y and x are determined together.** Regressing quantity on price estimates neither the demand curve nor the supply curve, since the observed pairs are where the two meet.

Each has its own remedy, and this book returns to them.

THE IDEA IN PLAIN WORDS

It is worth being clear about what has and has not changed since Unit 7. Multiple regression is a real advance. It is the reason economists can report returns to education that are not simply the fact that educated people differ from uneducated ones in a dozen other ways.
It is also not enough. The comparison it makes is between observations alike in the variables we happened to measure, and the variables we most want to hold fixed — ability, motivation, the quality of a school — are usually the ones no survey records.
Closing that gap requires a different kind of idea, and the second half of this book is about where such ideas come from.

8.8 Summary

REMEMBER THIS

- The **multiple regression model** is $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u$. Taking conditional expectations, $E[y \mid x_1, \ldots, x_k]$ is the population regression function, now a plane rather than a line.
- Each β_j is a **partial derivative**: the change in average y for a one-

unit rise in x_j, holding the other explanatory variables fixed. This is the *ceteris paribus* interpretation, and it is why multiple regression exists.

- "Holding fixed" is an operation on the data, not an experiment. By the **Frisch–Waugh–Lovell** result, $\hat{\beta}_j$ can be obtained by regressing x_j on the other regressors, keeping the residuals, and regressing y on those residuals alone.
- It follows that the coefficient uses only the variation in x_j that the other variables cannot explain. Controls remove variation, and removing variation costs precision.
- The assumptions are those of Unit 7 with the conditioning extended to all regressors, plus **CLRM7, no perfect collinearity**. A violation is not a statistical problem but an arithmetic one: the comparison the coefficient refers to does not exist in the data, and R returns `NA`.
- $\text{Var}(\hat{\beta}_j) = \sigma^2/[\sum(x_{ij} - \bar{x}_j)^2(1 - R_j^2)]$, where R_j^2 comes from regressing x_j on the other regressors. The larger R_j^2, the more the control costs. Also $\hat{\sigma}^2 = \sum \hat{u}_i^2/(n - k - 1)$.
- Testing a single coefficient is unchanged: $t = \hat{\beta}_j/\text{se}(\hat{\beta}_j)$ on $n-k-1$ degrees of freedom.
- Testing several at once requires the **F test**, which compares a restricted model against an unrestricted one: $F = [(\text{SSR}_r - \text{SSR}_{ur})/q]\big/[\text{SSR}_{ur}/(n - k - 1)]$. It is always a right-tailed test.
- Separate t-tests are not a substitute. Correlated regressors can be individually insignificant and jointly overwhelming, because each t-test uses only the part of its variable that the others cannot explain.
- F is the ratio of two independent variance estimates, each divided by its degrees of freedom. Under the null both estimate σ^2, so F averages about one. Its distribution is right-skewed, so most of the mass lies below that average.
- **Omitted variable bias** is $\beta_2\delta_1$ — the effect of the omitted variable on y, times its relationship with the included one. It vanishes if either factor is zero, and its sign is the product of their signs.
- Controlling for measured variables removes their contribution to the bias and nothing else. Multiple regression narrows the gap to a causal effect without closing it, and reports nothing about what remains.

8.9 Practice problems

YOUR TURN

Work each one before opening the solution. The question bank that follows has no answers.

Numerical problems

8.1 A researcher estimates the effect of fertiliser on rice yield across farms, first alone and then controlling for irrigation:

$$\widehat{\text{yield}} = 12.4 + 0.85\,\text{fert} \qquad\qquad \widehat{\text{yield}} = 9.1 + 0.52\,\text{fert} + 6.3\,\text{irrig}$$

where `irrig` is hectares under irrigation.

i. Interpret 0.52 precisely.
ii. Using the omitted variable bias formula, what must be the sign of the relationship between fertiliser use and irrigation?
iii. Compute the implied slope from a regression of irrigation on fertiliser.

SOLUTION

(i) Among farms with the same irrigated area, one additional unit of fertiliser is associated with 0.52 more quintals of yield on average.
(ii) The short regression gives 0.85 and the long one 0.52, so the bias is $0.85 - 0.52 = +0.33$. Since $\beta_2 = 6.3 > 0$, the bias formula $\beta_2\delta_1 = 0.33$ requires $\delta_1 > 0$. Farms using more fertiliser also have more irrigation — plausible, since both are investments made by better-resourced farms.
(iii) $\delta_1 = 0.33/6.3 = 0.052$ hectares of irrigation per unit of fertiliser.

8.2 A regression of household consumption on income and household size gives $R^2 = 0.68$ with $n = 240$.

i. State the null hypothesis of the overall F test and compute the statistic.
ii. The 5% critical value with 2 and 237 degrees of freedom is 3.03. What do you conclude?

SOLUTION

(i) $H_0 : \beta_1 = \beta_2 = 0$; both slopes are zero. With $k = 2$ and $n - k - 1 = 237$,

$$F = \frac{0.68/2}{(1-0.68)/237} = \frac{0.34}{0.00135} = 251.8$$

(ii) $251.8 \gg 3.03$, so we reject decisively. Income and household size together explain a substantial share of variation in consumption. As Section 8.6 notes, this is a low bar and rejecting it establishes little beyond the model not being empty.

8.3 An unrestricted model with 5 explanatory variables and $n = 90$ has $\text{SSR}_{ur} = 148$. Dropping three of them gives $\text{SSR}_r = 176$.

i. Compute the F statistic.
ii. With 3 and 84 degrees of freedom the 5% critical value is 2.71. Conclude.
iii. Why must SSR_r exceed SSR_{ur}?

SOLUTION

(i) $q = 3$, $n - k - 1 = 90 - 5 - 1 = 84$.

$$F = \frac{(176-148)/3}{148/84} = \frac{9.333}{1.762} = 5.30$$

(ii) $5.30 > 2.71$: reject. At least one of the three dropped variables has a non-zero coefficient.
(iii) The unrestricted model can always reproduce the restricted one by setting those three coefficients to zero, and least squares chooses the values that minimise SSR. So the unrestricted SSR can never be larger.

8.4 In a wage regression, R_j^2 from regressing years of education on the other explanatory variables is 0.10 in one specification and 0.90 in another. Holding σ^2 and $\sum(x_{ij} - \bar{x}_j)^2$ fixed, by what factor does the standard error of the education coefficient differ between the two?

SOLUTION

The variance is proportional to $1/(1 - R_j^2)$, so the standard error is proportional to $1/\sqrt{1 - R_j^2}$.

$$\frac{\sqrt{1/(1-0.90)}}{\sqrt{1/(1-0.10)}} = \sqrt{\frac{0.90}{0.10}} = 3$$

The standard error is three times larger in the second specification. The point estimate may be no better; the reported uncertainty is three times worse.

Concept checks

8.5 A student says: "The coefficient on attendance fell from 0.189 to 0.136 when I added controls, so the controls made the estimate worse." Respond.

SOLUTION

The two numbers answer different questions, so neither is a corrupted version of the other. 0.189 compares all students to all students; 0.136 compares students alike in prior record.
If the goal is to describe how grades vary with attendance in the population, 0.189 is the correct answer to that question. If the goal is to isolate what attendance contributes, 0.136 is closer, because one identified source of contamination has been removed.
"Worse" presumes a target. The change is evidence that prior record was contributing to the simple slope — which is exactly what we suspected.

8.6 Explain why a variable that is strongly correlated with y can be insignificant in a multiple regression.

SOLUTION

By Section 8.3, the coefficient uses only the part of the variable that the other regressors cannot explain. If the variable is closely related to the other regressors, that remaining part is small, and a small amount of variation supports only an imprecise estimate.
The variable may be strongly related to y while contributing nothing the other variables have not already contributed. That is a statement about what the variable adds, not about whether it matters.

8.7 A regression includes both `attendance` and `classes_missed`, and R reports `NA` for the second. Explain what has happened and what should be done.

SOLUTION

The two sum to 32 for every student, so one is an exact linear function of the other and CLRM7 fails. There is no observation for which attendance changes while classes missed does not, so the *ceteris paribus* comparison the coefficient refers to does not exist in the data.

R drops one variable and reports NA rather than failing. The remedy is to include only one of them: they carry identical information.

8.8 Two variables are individually insignificant at 5% but jointly significant at 1%. What should be concluded, and what should not?

SOLUTION

Conclude that at least one of the two has a non-zero coefficient, and that the data cannot say which — a signature of strongly correlated regressors. Do not conclude that both are irrelevant, and do not drop both on the strength of the two t-tests. Do not conclude that both matter either: the joint test is silent about which.

R exercises

8.9 Using `attendance-grades.csv`, verify the Frisch–Waugh–Lovell result for `cum_gpa` rather than `attendance`: regress `cum_gpa` on the other two regressors, keep the residuals, regress `sem_gpa` on them, and compare the slope with the multiple regression coefficient.

SOLUTION

```
students <- read.csv("data/attendance-grades.csv")
full <- lm(sem_gpa ~ attendance + cum_gpa + admission_score, data =
 ↪  students)

r <- residuals(lm(cum_gpa ~ attendance + admission_score, data =
 ↪  students))

c(two_step = coef(lm(students$sem_gpa ~ r))[2],
  direct   = coef(full)["cum_gpa"])
```

The two agree to machine precision. The result holds for every coefficient, not just the one of interest — each is the slope from a regression on the part of its own variable that the others cannot explain.

8.10 Using `wages-india-synthetic.csv`, test whether sex and residence jointly belong in a regression of annual earnings on education and age. Report the F statistic, its degrees of freedom and your conclusion.

SOLUTION

```
wages <- read.csv("data/wages-india-synthetic.csv")
wages$male  <- as.integer(wages$sex == "Male")
wages$urban <- as.integer(wages$residence == "Urban")

r  <- lm(annual_earnings ~ education + age, data = wages)
ur <- lm(annual_earnings ~ education + age + male + urban, data =
↪  wages)

anova(r, ur)
```

$F = 900.9$ on 2 and 19,995 degrees of freedom, with a p-value below any printable threshold. Sex and residence belong in the model.
That they belong does not make the model causal. It says earnings differ systematically by sex and residence among people of the same education and age — a fact about the labour market, not an explanation of it.

8.11 Show by simulation that omitted variable bias behaves as the formula predicts. Generate x_2 correlated with x_1, construct $y = 1+2x_1+3x_2+u$, estimate the short regression 2,000 times, and compare the average $\hat{\alpha}_1$ with $\beta_1 + \beta_2\delta_1$.

SOLUTION

```
set.seed(20)
n <- 200

results <- replicate(2000, {
  x1 <- rnorm(n)
  x2 <- 0.5 * x1 + rnorm(n)              # delta_1 = 0.5 by
↪  construction
  y  <- 1 + 2 * x1 + 3 * x2 + rnorm(n, sd = 2)
  c(short = coef(lm(y ~ x1))[2], long = coef(lm(y ~ x1 + x2))[2])
})

c(mean_short = mean(results["short.x1", ]),
  predicted  = 2 + 3 * 0.5,
  mean_long  = mean(results["long.x1", ]))
```

The short regression averages about 3.5, matching $\beta_1+\beta_2\delta_1 = 2+3(0.5)$.
The long regression averages about 2, the true value.
The bias does not shrink with sample size. Increasing n to 20,000 leaves it where it is — bias is not sampling error, and no amount of data removes it.

8.12 Investigate the cost of a redundant control. Regress `sem_gpa` on `attendance` alone, then add `cum_gpa`, then add a variable constructed to be almost identical to `attendance`. Compare the standard errors and explain.

SOLUTION

```
students <- read.csv("data/attendance-grades.csv")
set.seed(5)
students$attendance_copy <- students$attendance +
  ↪ rnorm(nrow(students), sd = 0.4)

se <- function(m) round(coef(summary(m))["attendance", 2], 4)

c(alone      = se(lm(sem_gpa ~ attendance, data = students)),
  plus_cum   = se(lm(sem_gpa ~ attendance + cum_gpa, data =
   ↪ students)),
  plus_copy  = se(lm(sem_gpa ~ attendance + attendance_copy, data =
   ↪ students)))
```

Adding `cum_gpa` barely moves the standard error and removes a real source of bias. Adding the near-copy inflates it enormously while contributing no information at all: R_j^2 for attendance is now close to one, so $(1 - R_j^2)$ is close to zero.
This is the practical content of Section 8.4. Controls are not free, and a control that duplicates the variable of interest is all cost.

8.10 Question bank

Multiple choice

QB 8.1 In the model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$, the coefficient β_1 measures

a. the total change in y when x_1 rises by one unit
b. the change in average y when x_1 rises by one unit, holding x_2 fixed
c. the correlation between x_1 and y
d. the share of variation in y explained by x_1

QB 8.2 The degrees of freedom for a t-test on a single coefficient in a regression with $n = 150$ and four explanatory variables is

a. 146 b. 145 c. 148 d. 149

QB 8.3 Perfect collinearity between two explanatory variables means

a. the OLS estimates are biased

b. the OLS estimates have large standard errors
c. the OLS estimates cannot be computed at all
d. the R^2 will be exactly one

QB 8.4 An omitted variable causes no bias if

a. it is uncorrelated with the dependent variable
b. it is uncorrelated with the included explanatory variables
c. its coefficient is large
d. the sample is large

QB 8.5 The F test for exclusion restrictions is always

a. two-tailed
b. left-tailed
c. right-tailed
d. tailed according to the alternative hypothesis

Short reasoning

QB 8.6 Explain, without using the phrase "holding constant", what the Frisch–Waugh–Lovell result says a multiple regression coefficient is computed from.

QB 8.7 A colleague argues that since adding controls always reduces precision, one should include as few as possible. Assess this.

QB 8.8 Why does the numerator of the F statistic divide by q and the denominator by $n - k - 1$? What would go wrong if neither division were performed?

QB 8.9 In a study of the effect of an irrigation scheme on farm incomes, the researcher controls for landholding, soil quality and district. Name two variables likely still omitted, and use the bias formula to argue in which direction each would push the estimate.

QB 8.10 A regression of infant mortality on the number of doctors per thousand people across Indian districts returns a *positive* coefficient. Explain using the omitted variable bias formula, and suggest a control.

QB 8.11 The overall F statistic in a regression is significant but no individual coefficient is. What does this suggest about the explanatory variables, and what would you do next?

QB 8.12 Explain why an omitted variable that strongly affects y but is unrelated to the explanatory variable of interest costs precision but causes no bias.

Numerical

QB 8.13 A researcher estimates $\widehat{\text{wage}} = 4200 + 380\, educ$ and then $\widehat{\text{wage}} = 3100 + 240\, educ + 95\, exper$. Compute the implied regression slope of experience on education, and comment on its sign.

QB 8.14 An unrestricted model with $n = 120$, $k = 6$ has $R^2_{ur} = 0.54$. Imposing four exclusion restrictions gives $R^2_r = 0.41$. Compute the F statistic and state its degrees of freedom.

QB 8.15 A regression with $n = 500$ and $k = 3$ reports $\sum \hat{u}_i^2 = 892$. Compute $\hat{\sigma}^2$ and $\hat{\sigma}$, and state why the divisor is not n.

QB 8.16 For a regression of yield on fertiliser, irrigation and labour, the R^2 from regressing fertiliser on the other two is 0.76. By what percentage is the standard error on fertiliser inflated relative to a hypothetical case in which fertiliser were uncorrelated with the other regressors?

8.11 Looking ahead

This unit took the assumptions and the variables as given. Both deserve questioning.

The model has assumed throughout that a one-unit rise in x has the same consequence wherever it occurs — that the tenth year of schooling is worth what the second was, and that the effect of experience never flattens. Economics rarely believes this. The next unit shows how logarithms, squares and interaction terms let a linear model describe relationships that are not linear at all, at no cost to any of the machinery developed here.

The unit also treated the choice of controls as settled. It is not. Some variables belong in a regression, some are harmless, and some make the estimate worse than leaving them out — a possibility that surprises most readers, since the instinct is that controlling for more is always safer. Distinguishing good controls from bad ones is the second theme of the next unit, and it depends on exactly the bias formula derived here.

8.12 Mathematical note: multiple regression in matrix form

GOING A LITTLE FURTHER

Nothing in this unit depends on what follows. It is here for readers who would like to see the general case written compactly, and it can be skipped without consequence.

The estimator. With k explanatory variables, writing out $k+1$ normal equations and solving them by hand is unpleasant. Matrix notation does the same work in three lines.

Collect the observations into an $n \times 1$ vector Y and an $n \times (k+1)$ matrix X whose first column is all ones and whose remaining columns are the explanatory variables. The model is

$$Y = X\beta + u$$

The sum of squared residuals is $(Y - X\beta)'(Y - X\beta)$. Differentiating with respect to β and setting the result to zero,

$$-2X'Y + 2X'X\beta = 0 \qquad \Longrightarrow \qquad X'X\hat{\beta} = X'Y$$

which is the whole system of normal equations in one line. Provided $X'X$ can be inverted,

$$\hat{\beta} = (X'X)^{-1}X'Y$$

The variance. Substituting $Y = X\beta + u$ gives $\hat{\beta} = \beta + (X'X)^{-1}X'u$. The second term has expectation zero under CLRM4, which is unbiasedness. Taking the variance and using $\text{Var}(Au) = A\,\text{Var}(u)\,A'$, then imposing homoskedasticity,

$$\text{Var}(\hat{\beta}) = \sigma^2(X'X)^{-1}$$

This is a $(k+1) \times (k+1)$ matrix. Its diagonal holds the variances of the individual coefficients — the square roots of which are the standard errors R prints — and its off-diagonal entries hold the covariances between them, which is what the F test of Section 8.6 needs in order to test several coefficients jointly.

Checking it. The formula is short enough to compute directly.

```
X <- cbind(1, students$attendance, students$cum_gpa,
  ↪  students$admission_score)
y <- students$sem_gpa

b_hat <- solve(t(X) %*% X) %*% t(X) %*% y
round(drop(b_hat), 5)
```

```
#> [1] -7.6603  0.1360  0.5748  0.0835
```

The same four numbers `lm()` reported in Section 8.2.

Where collinearity appears. The requirement in Section 8.4 that no explanatory variable be an exact linear combination of the others is exactly the requirement that $X'X$ be invertible. If one column of X can be written in terms of the others, X is not of full rank, $X'X$ is singular, and there is no $\hat{\beta}$ to compute — which is what R is reporting when it returns `NA`.